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WESTERN INTERNATIONAL SCHOOL

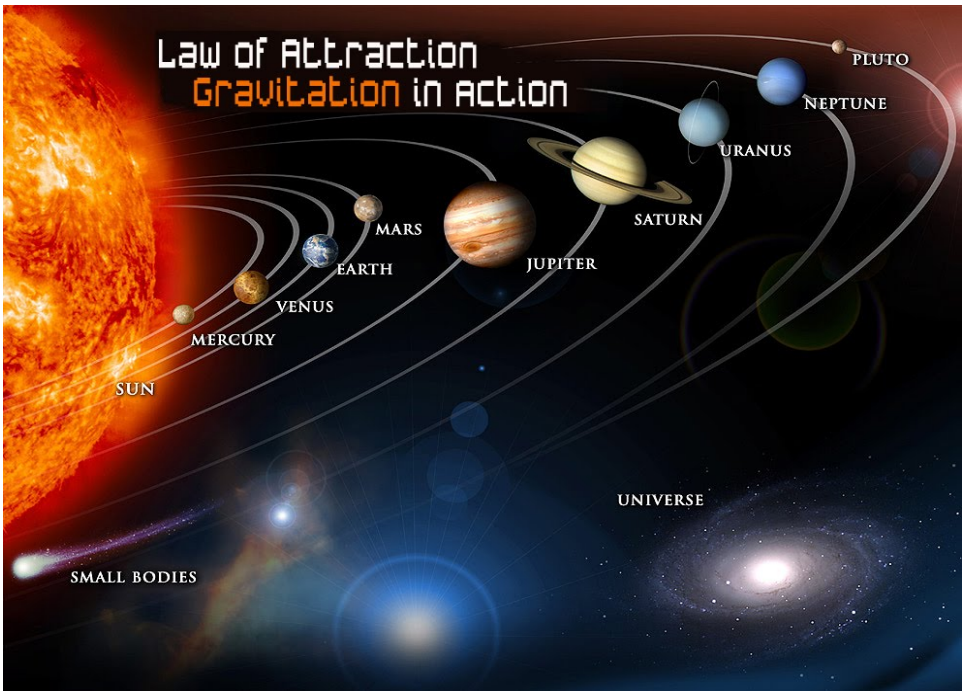
Physics



Grade 11

Chapter 1

Gravitation



- You will learn the nature of gravitational force.
- You will relate Kepler's laws of planetary motion to Newton's laws of motion.
- You will describe the orbits of planets and satellites using the law of universal gravitation.

Why It's Important

Kepler's laws and the law of universal gravitation will help you understand the motion of planets and satellites.

Comets Comet Hale-Bopp was discovered by Alan Hale and Thomas Bopp in 1995. The comet entered the inner solar system in 1997 and was visible from Joshua Tree National Park in California, providing spectacular views of its white dust tail and blue ion tail.

LAUNCH Lab

Can you model Mercury's motion?

Question

Do planets in our solar system have circular orbits or do they travel in some other path?

Procedure

1. Use the data table to plot the orbit of Mercury using the scale 10 cm = 1 AU. Note that one astronomical unit, AU, is Earth's distance from the Sun. 1 AU is equal to 1.5×10^8 km.
2. Calculate the distance in cm for each distance measured in AU.
3. Mark the center of your paper and draw a horizontal zero line and a vertical zero line going through it.
4. Place your protractor on the horizontal line and center it on the center point. Measure the degrees and place a mark.
5. Place a ruler connecting the center and the angle measurement. Mark the distance in centimeters for the corresponding angle. You will need to place the protractor on the vertical zero line for certain angle measurements.
6. Once you have marked all the data points, draw a line connecting them.

Analysis

Describe the shape of Mercury's orbit. Draw a line going through the Sun that represents the longest axis of the orbit, called the major axis.

Critical Thinking

How does the orbit of Mercury compare to the orbit of comet Hale-Bopp, shown on page 170?

Mercury's Orbit

θ (°)	d (AU)
4	0.35
61	0.31
122	0.32
172	0.38
209	0.43
239	0.46
266	0.47
295	0.44
330	0.40
350	0.37

1. Planetary Motion and Gravitation

Since ancient times, the Sun, Moon, planets, and stars had been assumed to revolve around Earth. Nicholas Copernicus, a Polish astronomer, noticed that the best available observations of the movements of planets and stars did not fully agree with the Earth-centered model. The results of his many years of work were published in 1543, when Copernicus was on his deathbed. His book showed that the motion of planets is much more easily understood by assuming that Earth and other planets revolve around the Sun.

Tycho Brahe was born a few years after the death of Copernicus. As a boy of 14 in Denmark, Brahe observed an eclipse of the Sun on August 21, 1560, and vowed to become an astronomer.

Brahe studied astronomy as he traveled throughout Europe for five years. He did not use telescopes. Instead, he used huge instruments that he designed and built in his own shop on the Danish island of Hven. He spent the next 20 years carefully recording the exact positions of the planets and stars. Brahe concluded that the Sun and the Moon orbit Earth and that all other planets orbit the Sun.

Objectives

- **Relate** Kepler's laws to the law of universal gravitation.
- **Calculate** orbital speeds and periods.
- **Describe** the importance of Cavendish's experiment.

Vocabulary

Kepler's first law
Kepler's second law
Kepler's third law
gravitational force
law of universal gravitation



Figure 7-1 Among the huge astronomical instruments that Tycho Brahe constructed to use on Hven **(a)** were an astrolabe **(b)** and a sextant **(c).**

Kepler's Laws

Johannes Kepler, a 29-year-old German, became one of Brahe's assistants when he moved to Prague. Brahe trained his assistants to use instruments, such as those shown in **Figure 7-1.** Upon his death in 1601, Kepler inherited 30 years' worth of Brahe's observations. He studied Brahe's data and was convinced that geometry and mathematics could be used to explain the number, distance, and motion of the planets. Kepler believed that the Sun exerted a force on the planets and placed the Sun at the center of the system. After several years of careful analysis of Brahe's data on Mars, Kepler discovered the laws that describe the motion of every planet and satellite.

Kepler's first law states that the paths of the planets are ellipses, with the Sun at one focus. An ellipse has two foci, as shown in **Figure 7-2.** Like planets and stars, comets also orbit the Sun in elliptical orbits. Comets are divided into two groups—long-period comets and short-period comets—based on orbital periods, each of which is the time it takes the comet to complete one revolution. Long-period comets have orbital periods longer than 200 years, and short-period comets have orbital periods shorter than 200 years. Comet Hale-Bopp, with a period of 2400 years, is an example of a long-period comet. Comet Halley, with a period of 76 years, is an example of a short-period comet.

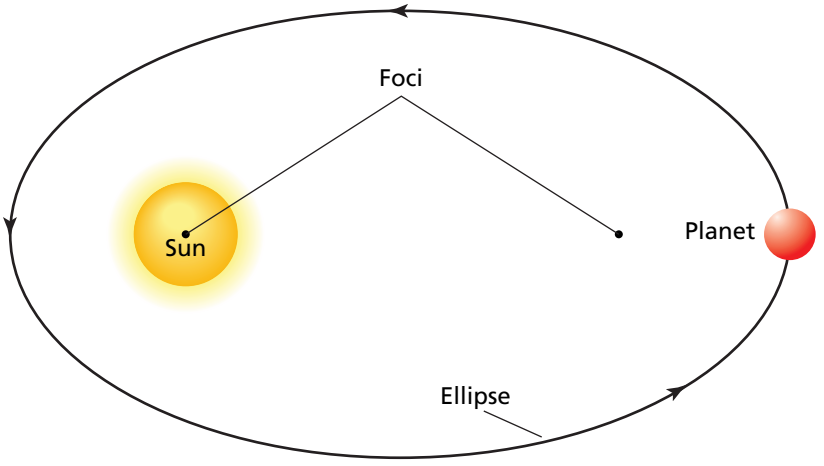


Figure 7-2 Planets orbit the Sun in elliptical orbits with the Sun at one focus. (Illustration not to scale)

Kepler found that the planets move faster when they are closer to the Sun and slower when they are farther away from the Sun. Thus, **Kepler's second law** states that an imaginary line from the Sun to a planet sweeps out equal areas in equal time intervals, as illustrated in **Figure 7-3.**

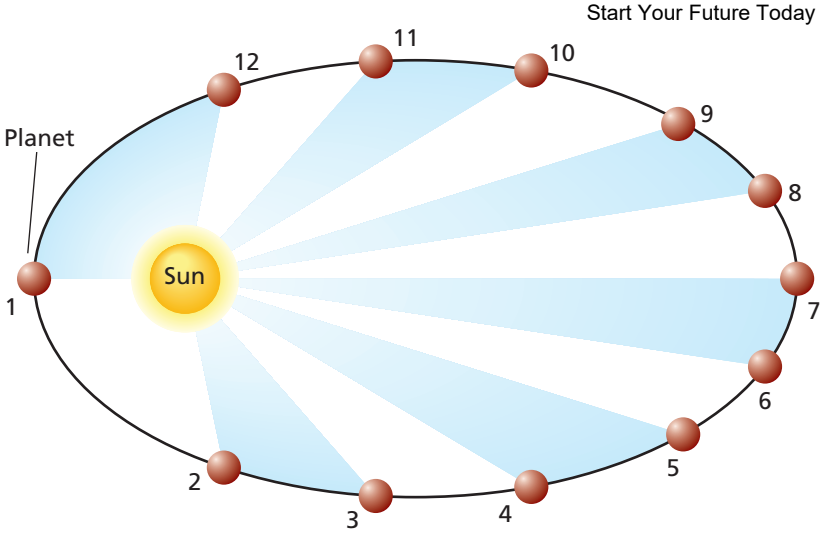


Figure 7-3 A planet moves fastest when it is close to the Sun and slowest when it is farther from the Sun. Equal areas are swept out in equal amounts of time. (Illustration not to scale)

Kepler also found that there is a mathematical relationship between periods of planets and their mean distances away from the Sun. **Kepler's third law** states that the square of the ratio of the periods of any two planets revolving about the Sun is equal to the cube of the ratio of their average distances from the Sun. Thus, if the periods of the planets are T_A and T_B, and their average distances from the Sun are r_A and r_B, Kepler's third law can be expressed as follows.

Kepler's Third Law $\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_A}{r_B}\right)^3$

The squared quantity of the period of object A divided by the period of object B, is equal to the cubed quantity of object A's average distance from the Sun, divided by object B's average distance from the Sun.

Note that the first two laws apply to each planet, moon, and satellite individually. The third law, however, relates the motion of several objects about a single body. For example, it can be used to compare the planets' distances from the Sun, shown in **Table 7-1,** to their periods about the Sun. It also can be used to compare distances and periods of the Moon and artificial satellites orbiting Earth.

Table 7-1
Planetary Data

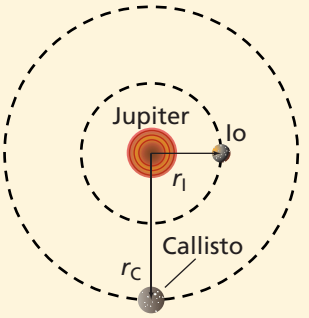
Name	Average Radius (m)	Mass (kg)	Mean Distance From Sun (m)
Sun	6.96×10^{8}	1.99×10^{30}	–
Mercury	2.44×10^{6}	3.30×10^{23}	5.79×10^{10}
Venus	6.05×10^{6}	4.87×10^{24}	1.08×10^{11}
Earth	6.38×10^{6}	5.98×10^{24}	1.50×10^{11}
Mars	3.40×10^{6}	6.42×10^{23}	2.28×10^{11}
Jupiter	7.15×10^{7}	1.90×10^{27}	7.78×10^{11}
Saturn	6.03×10^{7}	5.69×10^{26}	1.43×10^{12}
Uranus	2.56×10^{7}	8.68×10^{25}	2.87×10^{12}
Neptune	2.48×10^{7}	1.02×10^{26}	4.50×10^{12}
Pluto	1.20×10^{6}	1.25×10^{22}	5.87×10^{12}

EXAMPLE Problem 1

Callisto's Distance from Jupiter Galileo measured the orbital sizes of Jupiter's moons using the diameter of Jupiter as a unit of measure. He found that Io, the closest moon to Jupiter, had a period of 1.8 days and was 4.2 units from the center of Jupiter. Callisto, the fourth moon from Jupiter, had a period of 16.7 days. Using the same units that Galileo used, predict Callisto's distance from Jupiter.

1 Analyze and Sketch the Problem

- Sketch the orbits of Io and Callisto.
- Label the radii.



Known:

$T_C = 16.7$ days

$T_I = 1.8$ days

$r_I = 4.2$ units

Unknown:

$r_C = ?$

2 Solve for the Unknown

Solve Kepler's third law for r_C.

$$\left(\frac{T_C}{T_I}\right)^2 = \left(\frac{r_C}{r_I}\right)^3$$

$$r_C{}^3 = r_I{}^3 \left(\frac{T_C}{T_I}\right)^2$$

$$r_C = \sqrt[3]{r_I{}^3 \left(\frac{T_C}{T_I}\right)^2}$$

$$= \sqrt[3]{(4.2 \text{ units})^3 \left(\frac{16.7 \text{ days}}{1.8 \text{ days}}\right)^2}$$

Substitute $r_I = 4.2$ units, $T_C = 16.7$ days, $T_I = 1.8$ days

$$= \sqrt[3]{6.4 \times 10^3 \text{ units}^3}$$

$$= 19 \text{ units}$$

Math Handbook
Isolating a Variable
page 845

3 Evaluate the Answer

- **Are the units correct?** r_C should be in Galileo's units, like r_I.
- **Is the magnitude realistic?** The period is large, so the radius should be large.

PRACTICE Problems

Additional Problems, Appendix B

1. If Ganymede, one of Jupiter's moons, has a period of 32 days, how many units are there in its orbital radius? Use the information given in Example Problem 1.
2. An asteroid revolves around the Sun with a mean orbital radius twice that of Earth's. Predict the period of the asteroid in Earth years.
3. From Table 7-1, on page 173, you can find that, on average, Mars is 1.52 times as far from the Sun as Earth is. Predict the time required for Mars to orbit the Sun in Earth days.
4. The Moon has a period of 27.3 days and a mean distance of 3.90×10^5 km from the center of Earth.
   a. Use Kepler's laws to find the period of a satellite in orbit 6.70×10^3 km from the center of Earth.
   b. How far above Earth's surface is this satellite?
5. Using the data in the previous problem for the period and radius of revolution of the Moon, predict what the mean distance from Earth's center would be for an artificial satellite that has a period of exactly 1.00 day.

Newton's Law of Universal Gravitation

In 1666, 45 years after Kepler published his work, Newton began his studies of planetary motion. He found that the magnitude of the force, F, on a planet due to the Sun varies inversely with the square of the distance, r, between the centers of the planet and the Sun. That is, F is proportional to $1/r^2$. The force, ***F,*** acts in the direction of the line connecting the centers of the two objects.

It is quoted that the sight of a falling apple made Newton wonder if the force that caused the apple to fall might extend to the Moon, or even beyond. He found that both the apple's and Moon's accelerations agreed with the $1/r^2$ relationship. According to his own third law, the force Earth exerts on the apple is exactly the same as the force the apple exerts on Earth. The force of attraction between two objects must be proportional to the objects' masses, and is known as the **gravitational force.**

Newton was confident that the same force of attraction would act between any two objects, anywhere in the universe. He proposed his **law of universal gravitation,** which states that objects attract other objects with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. This can be represented by the following equation.

Law of Universal Gravitation $F = G\frac{m_1 m_2}{r^2}$

The gravitational force is equal to the universal gravitational constant, times the mass of object 1, times the mass of object 2, divided by the distance between the centers of the objects, squared.

According to Newton's equation, F is directly proportional to m_1 and m_2. Thus, if the mass of a planet near the Sun were doubled, the force of attraction would be doubled. Use the Connecting Math to Physics feature below to examine how changing one variable affects another. **Figure 7-4** illustrates the inverse square law graphically.

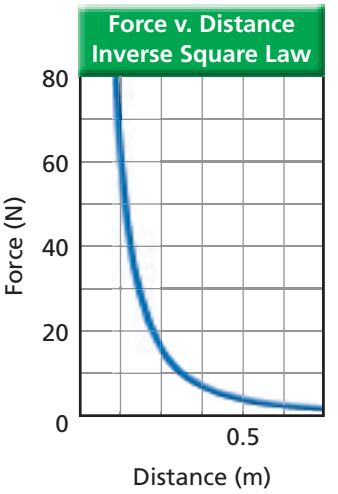


Figure 7-4 The change in gravitational force with distance follows the inverse square law.

Connecting Math to Physics

Direct and Inverse Relationships Newton's law of universal gravitation has both direct and inverse relationships.

$F \propto m_1 m_2$		$F \propto \frac{1}{r^2}$	
Change	**Result**	**Change**	**Result**
$2m_1 m_2$	$2F$	$2r$	$\frac{1}{4}F$
$3m_1 m_2$	$3F$	$3r$	$\frac{1}{9}F$
$2m_1\ 3m_2$	$6F$	$\frac{1}{2}r$	$4F$
$\frac{1}{2}\ m_1 m_2$	$\frac{1}{2}F$	$\frac{1}{3}r$	$9F$

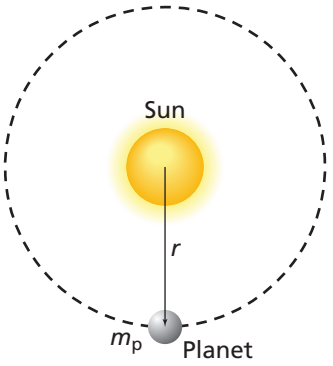


Figure 7-5 A planet with mass m_p and orbital radius r orbits the Sun with mass m_S. (Illustration not to scale)

Universal Gravitation and Kepler's Third Law

Newton stated his law of universal gravitation in terms that applied to the motion of planets about the Sun. This agreed with Kepler's third law and confirmed that Newton's law fit the best observations of the day.

Consider a planet orbiting the Sun, as shown in **Figure 7-5.** Newton's second law of motion, $F_{net} = ma$, can be written as $F_{net} = m_p a_c$, where F is the gravitational force, m_p is the mass of the planet, and a_c is the centripetal acceleration of the planet. For simplicity, assume circular orbits. Recall from your study of uniform circular motion in Chapter 6 that, for a circular orbit, $a_c = 4\pi^2 r/T^2$. This means that $F_{net} = m_p a_c$ may now be written as $F_{net} = m_p 4\pi^2 r/T^2$. In this equation, T is the time required for the planet to make one complete revolution about the Sun. If you set the right side of this equation equal to the right side of the law of universal gravitation, you arrive at the following result:

$$G\frac{m_S m_p}{r^2} = \frac{m_p 4\pi^2 r}{T^2}$$

$$T^2 = \left(\frac{4\pi^2}{Gm_S}\right) r^3$$

$$\text{Thus, } T = \sqrt{\left(\frac{4\pi^2}{Gm_S}\right) r^3}$$

The period of a planet orbiting the Sun can be expressed as follows.

Period of a Planet Orbiting the Sun $T = 2\pi\sqrt{\frac{r^3}{Gm_S}}$

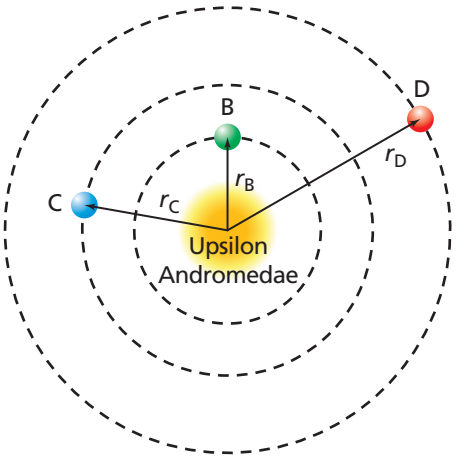
The period of a planet orbiting the Sun is equal to 2π times the square root of the orbital radius cubed, divided by the product of the universal gravitational constant and the mass of the Sun.

Squaring both sides makes it apparent that this equation is Kepler's third law of planetary motion: the square of the period is proportional to the cube of the distance that separates the masses. The factor $4\pi^2/Gm_S$ depends on the mass of the Sun and the universal gravitational constant. Newton found that this derivation applied to elliptical orbits as well.

CHALLENGE PROBLEM

Astronomers have detected three planets that orbit the star Upsilon Andromedae. Planet B has an average orbital radius of 0.059 AU and a period of 4.6170 days. Planet C has an average orbital radius of 0.829 AU and a period of 241.5 days. Planet D has an average orbital radius of 2.53 AU and a period of 1284 days. (Distances are given in astronomical units (AU)—Earth's average distance from the Sun. The distance from Earth to the Sun is 1.00 AU.)

1. Do these planets obey Kepler's third law?
2. Find the mass of the star Upsilon Andromedae in units of the Sun's mass.



Measuring the Universal Gravitational Constant

How large is the constant, G? As you know, the force of gravitational attraction between two objects on Earth is relatively small. The slightest attraction, even between two massive bowling balls, is difficult to detect. In fact, it took 100 years from the time of Newton's work for scientists to develop an apparatus that was sensitive enough to measure the force of gravitational attraction.

Cavendish's experiment In 1798, Englishman Henry Cavendish used equipment similar to the apparatus shown in **Figure 7-6** to measure the gravitational force between two objects. The apparatus had a horizontal rod with two small lead spheres attached to each end. The rod was suspended at its midpoint by a thin wire so that it could rotate. Because the rod was suspended by a thin wire, the rod and spheres were very sensitive to horizontal forces. To measure G, Cavendish placed two large lead spheres in a fixed position, close to each of the two small spheres, as shown in **Figure 7-7.** The force of attraction between the large and the small spheres caused the rod to rotate. When the force required to twist the wire equaled the gravitational force between the spheres, the rod stopped rotating. By measuring the angle through which the rod turned, Cavendish was able to calculate the attractive force between the objects. The angle through which the rod turned is measured using the beam of light that is reflected from the mirror. He measured the masses of the spheres and the distance between their centers. Substituting these values for force, mass, and distance into Newton's law of universal gravitation, he found an experimental value for G: when m_1 and m_2 are measured in kilograms, r in meters, and F in newtons, then $G = 6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2$.



Figure 7-6 Modern Cavendish balances are used to measure the gravitational forces between two objects.

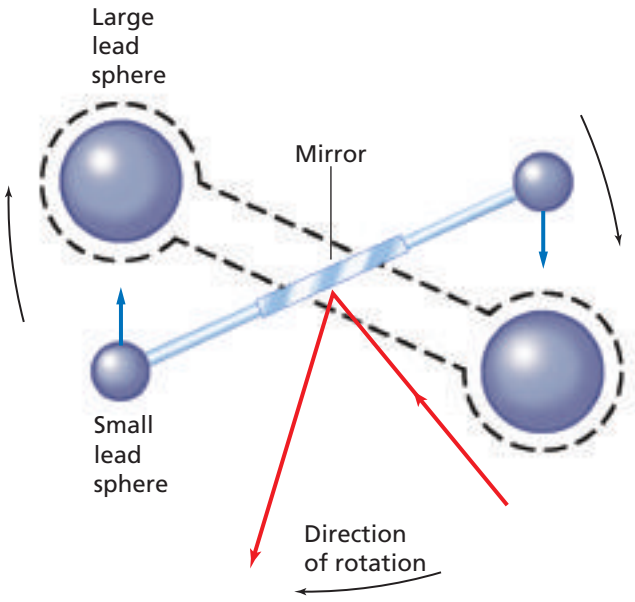


Figure 7-7 When the large lead spheres are placed near the small lead spheres, the gravitational attraction between the spheres causes the rod to rotate. The rotation is measured with the help of the reflected light ray.

The importance of *G* Cavendish's experiment often is called "weighing Earth," because his experiment helped determine Earth's mass. Once the value of G is known, not only the mass of Earth, but also the mass of the Sun can be determined. In addition, the gravitational force between any two objects can be calculated using Newton's law of universal gravitation. For example, the attractive gravitational force, F_g, between two bowling balls of mass 7.26 kg, with their centers separated by 0.30 m, can be calculated as follows:

$$F_g = \frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(7.26\ \text{kg})(7.26\ \text{kg})}{(0.30\ \text{m})^2} = 3.9\times10^{-8}\ \text{N}$$

You know that on Earth's surface, the weight of an object of mass m is a measure of Earth's gravitational attraction: $F_g = mg$. If Earth's mass is represented by m_E and Earth's radius is represented by r_E, the following is true:

$$F_g = G\frac{m_E m}{r_E^2} = mg, \text{ and so } g = G\frac{m_E}{r_E^2}$$

This equation can be rearranged to solve for m_E.

$$m_E = \frac{gr_E^2}{G}$$

Using $r_E = 6.38\times10^6$ m, $g = 9.80$ m/s^2, and $G = 6.67\times10^{-11}$ N·m^2/kg^2, the following result is obtained for Earth's mass:

$$m_E = \frac{(9.80\ \text{m/s}^2)(6.38\times10^6\ \text{m})^2}{6.67\times10^{-11}\ \text{N}\cdot\text{m/kg}^2} = 5.98\times10^{24}\ \text{kg}$$

When you compare the mass of Earth to that of a bowling ball, you can see why the gravitational attraction between everyday objects is not easily observed. Cavendish's experiment determined the value of G, confirmed Newton's prediction that a gravitational force exists between two objects, and helped calculate the mass of Earth.

1. Section Review

6. Neptune's Orbital Period Neptune orbits the Sun with an orbital radius of 4.495×10^{12} m, which allows gases, such as methane, to condense and form an atmosphere, as shown in **Figure 7-8.** If the mass of the Sun is 1.99×10^{30} kg, calculate the period of Neptune's orbit.

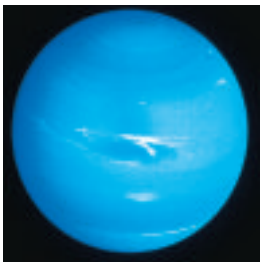


Figure 7-8

7. Gravity If Earth began to shrink, but its mass remained the same, what would happen to the value of g on Earth's surface?

8. Gravitational Force What is the gravitational force between two 15-kg packages that are 35 cm apart? What fraction is this of the weight of one package?

9. Universal Gravitational Constant Cavendish did his experiment using lead spheres. Suppose he had replaced the lead spheres with copper spheres of equal mass. Would his value of G be the same or different? Explain.

10. Laws or Theories? Kepler's three statements and Newton's equation for gravitational attraction are called "laws." Were they ever theories? Will they ever become theories?

11. Critical Thinking Picking up a rock requires less effort on the Moon than on Earth.

a. How will the weaker gravitational force on the Moon's surface affect the path of the rock if it is thrown horizontally?

b. If the thrower accidentally drops the rock on her toe, will it hurt more or less than it would on Earth? Explain.

2. Using the Law of Universal Gravitation

The planet Uranus was discovered in 1781. By 1830, it was clear that the law of gravitation didn't correctly predict its orbit. Two astronomers proposed that Uranus was being attracted by the Sun and by an undiscovered planet. They calculated the orbit of such a planet in 1845, and, one year later, astronomers at the Berlin Observatory found the planet now called Neptune. How do planets, such as Neptune, orbit the Sun?

Objectives

- **Solve** orbital motion problems.
- **Relate** weightlessness to objects in free fall.
- **Describe** gravitational fields.
- **Compare** views on gravitation.

Vocabulary

gravitational field
inertial mass
gravitational mass

Orbits of Planets and Satellites

Newton used a drawing similar to the one shown in **Figure 7-9** to illustrate a thought experiment on the motion of satellites. Imagine a cannon, perched high atop a mountain, firing a cannonball horizontally with a given horizontal speed. The cannonball is a projectile, and its motion has both vertical and horizontal components. Like all projectiles on Earth, it would follow a parabolic trajectory and fall back to the ground.

If the cannonball's horizontal speed were increased, it would travel farther across the surface of Earth, and still fall back to the ground. If an extremely powerful cannon were used, however, the cannonball would travel all the way around Earth, and keep going. It would fall toward Earth at the same rate that Earth's surface curves away. In other words, the curvature of the projectile would continue to just match the curvature of Earth, so that the cannonball would never get any closer or farther away from Earth's curved surface. The cannonball would, therefore, be in orbit.

Newton's thought experiment ignored air resistance. For the cannonball to be free of air resistance, the mountain on which the cannon is perched would have to be more than 150 km above Earth's surface. By way of comparison, the mountain would have to be much taller than the peak of Mount Everest, the world's tallest mountain, which is only 8.85 km in height. A cannonball launched from a mountain that is 150 km above Earth's surface would encounter little or no air resistance at an altitude of 150 km, because the mountain would be above most of the atmosphere. Thus, a cannonball or any object or satellite at or above this altitude could orbit Earth for a long time.

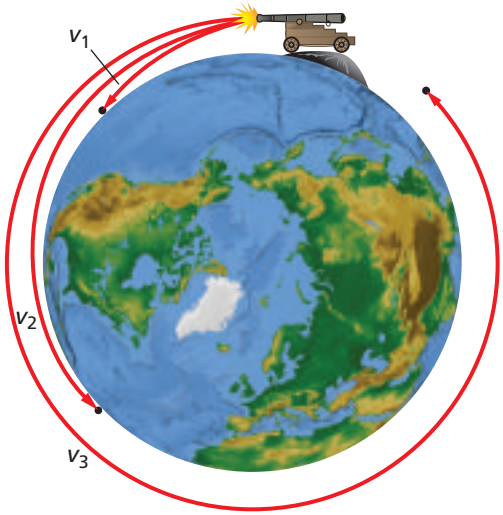


Figure 7-9 The horizontal speed v_1 is not high enough and the cannonball falls to the ground. With a higher speed, v_2, the cannonball travels farther. The cannonball travels all the way around Earth when the horizontal speed, v_3, is high enough.

APPLYING PHYSICS

Geosynchronous Orbit The *GOES-12* weather satellite orbits Earth once a day at an altitude of 35,785 km. The orbital speed of the satellite matches Earth's rate of rotation. Thus, to an observer on Earth, the satellite appears to remain above one spot on the equator. Satellite dishes on Earth can be directed to one point in the sky and not have to change position as the satellite orbits.

A satellite in an orbit that is always the same height above Earth moves in uniform circular motion. Recall that its centripetal acceleration is given by $a_c = v^2/r$. Newton's second law, $F_{net} = ma_c$, can thus be rewritten as $F_{net} = mv^2/r$. If Earth's mass is m_E, then this expression combined with Newton's law of universal gravitation produces the following equation:

$$G\frac{m_E m}{r^2} = \frac{mv^2}{r}$$

Solving for the speed of a satellite in circular orbit about Earth, v, yields the following.

Speed of a Satellite Orbiting Earth $v = \sqrt{\frac{Gm_E}{r}}$

The speed of a satellite orbiting Earth is equal to the square root of the universal gravitational constant times the mass of Earth, divided by the radius of the orbit.



Figure 7-10 *Landsat 7*, a remote sensing satellite, has a mass of about 2200 kg and orbits Earth at an altitude of about 705 km.

A satellite's orbital period A satellite's orbit around Earth is similar to a planet's orbit about the Sun. Recall that the period of a planet orbiting the Sun is expressed by the following equation:

$$T = 2\pi\sqrt{\frac{r^3}{Gm_S}}$$

Thus, the period for a satellite orbiting Earth is given by the following equation.

Period of a Satellite Orbiting Earth $T = 2\pi\sqrt{\frac{r^3}{Gm_E}}$

The period for a satellite orbiting Earth is equal to 2π times the square root of the radius of the orbit cubed, divided by the product of the universal gravitational constant and the mass of Earth.

The equations for the speed and period of a satellite can be used for any object in orbit about another. The mass of the central body will replace m_E in the equations, and r will be the distance between the centers of the orbiting body and the central body. If the mass of the central body is much greater than the mass of the orbiting body, then r is equal to the distance between the centers of the orbiting body and the central body. Orbital speed, v, and period, T, are independent of the mass of the satellite. Are there any factors that limit the mass of a satellite?

Geology Connection

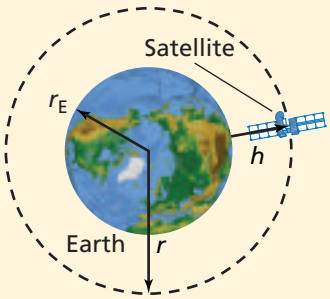
A satellite's mass *Landsat 7*, shown in **Figure 7-10,** is an artificial satellite that provides images of Earth's continental surfaces. *Landsat* images have been used to create maps, study land use, and monitor resources and global changes. The *Landsat 7* system enables researchers to monitor small-scale processes, such as deforestation, on a global scale. Satellites, such as *Landsat 7*, are accelerated to the speeds necessary for them to achieve orbit by large rockets, such as shuttle-booster rockets. Because the acceleration of any mass must follow Newton's second law of motion, $F_{net} = ma$, more force is required to launch a more massive satellite into orbit. Thus, the mass of a satellite is limited by the capability of the rocket used to launch it.

EXAMPLE Problem 2

Orbital Speed and Period Assume that a satellite orbits Earth 225 km above its surface. Given that the mass of Earth is 5.97×10^{24} kg and the radius of Earth is 6.38×10^{6} m, what are the satellite's orbital speed and period?

1 Analyze and Sketch the Problem

- Sketch the situation showing the height of the satellite's orbit.



Known:

$h = 2.25\times10^{5}$ m

$r_E = 6.38\times10^{6}$ m

$m_E = 5.97\times10^{24}$ kg

$G = 6.67\times10^{-11}$ N·m²/kg²

Unknown:

$v = ?$

$T = ?$

2 Solve for the Unknown

Determine the orbital radius by adding the height of the satellite's orbit to Earth's radius.

$$r = h + r_E$$

$$= 2.25\times10^{5}\text{ m} + 6.38\times10^{6}\text{ m} = 6.61\times10^{6}\text{ m}$$

Substitute $h = 2.25\times10^{5}$ m, $r_E = 6.38\times10^{6}$ m

Solve for the speed.

$$v = \sqrt{\frac{Gm_E}{r}}$$

$$= \sqrt{\frac{(6.67\times10^{-11}\text{ N·m}^2/\text{kg}^2)(5.97\times10^{24}\text{ kg})}{6.61\times10^{6}\text{ m}}}$$

Substitute $G = 6.67\times10^{-11}$ N·m²/kg², $m_E = 5.97\times10^{24}$ kg, $r = 6.61\times10^{6}$ m

$$= 7.76\times10^{3}\text{ m/s}$$

Math Handbook
Square and Cube Roots
pages 839–840

Solve for the period.

$$T = 2\pi\sqrt{\frac{r^3}{Gm_E}}$$

$$= 2\pi\sqrt{\frac{(6.61\times10^{6}\text{ m})^3}{(6.67\times10^{-11}\text{ N·m}^2/\text{kg}^2)(5.97\times10^{24}\text{ kg})}}$$

Substitute $r = 6.61\times10^{6}$ m, $G = 6.67\times10^{-11}$ N·m²/kg², $m_E = 5.97\times10^{24}$ kg

$$= 5.35\times10^{3}\text{ s}$$

This is approximately 89 min, or 1.5 h.

3 Evaluate the Answer

- **Are the units correct?** The unit for speed is m/s and the unit for period is s.

PRACTICE Problems

Additional Problems, Appendix B

For the following problems, assume a circular orbit for all calculations.

12. Suppose that the satellite in Example Problem 2 is moved to an orbit that is 24 km larger in radius than its previous orbit. What would its speed be? Is this faster or slower than its previous speed?

13. Use Newton's thought experiment on the motion of satellites to solve the following.

a. Calculate the speed that a satellite shot from a cannon must have to orbit Earth 150 km above its surface.

b. How long, in seconds and minutes, would it take for the satellite to complete one orbit and return to the cannon?

14. Use the data for Mercury in Table 7-1 on page 173 to find the following.

a. the speed of a satellite that is in orbit 260 km above Mercury's surface

b. the period of the satellite

MINI LAB

Weightless Water

This activity is best done outdoors. Use a pencil to poke two holes through a foam or paper cup: one on the bottom and the other on the side. Hold your fingers over the two holes to block them as your lab partner pours colored water into the cup until it is two-thirds full.

1. Predict what will happen as the cup is allowed to fall.

2. Test your prediction: drop the cup and watch closely.

Analyze and Conclude

3. Describe your observations.

4. Explain your results.

Acceleration Due To Gravity

The acceleration of objects due to Earth's gravity can be found by using Newton's law of universal gravitation and his second law of motion. For a free-falling object, m, the following is true:

$$F = G\frac{m_E m}{r^2} = ma, \text{ so } a = G\frac{m_E}{r^2}$$

Because $a = g$ and $r = r_E$ on Earth's surface, the following equation can be written:

$$g = G\frac{m_E}{r_E^2}, \text{ thus, } m_E = \frac{gr_E^2}{G}$$

You found above that $a = G\frac{m_E}{r^2}$ for a free-falling object. Substituting the above expression for m_E yields the following:

$$a = G\frac{\left(\frac{gr_E^2}{G}\right)}{r^2}$$

$$a = g\left(\frac{r_E}{r}\right)^2$$

This shows that as you move farther from Earth's center, that is, as r becomes larger, the acceleration due to gravity is reduced according to this inverse square relationship. What happens to your weight, m_g, as you move farther and farther from Earth's center?

Weight and weightlessness You probably have seen photos similar to the one in **Figure 7-11** in which astronauts are on the space shuttle in an environment often called "zero-g" or "weightlessness." The shuttle orbits about 400 km above Earth's surface. At that distance, $g = 8.7$ m/s^2, only slightly less than on Earth's surface. Thus, Earth's gravitational force is certainly not zero in the shuttle. In fact, gravity causes the shuttle to orbit Earth. Why, then, do the astronauts appear to have no weight?

Remember that you sense weight when something, such as the floor or your chair, exerts a contact force on you. But if you, your chair, and the floor all are accelerating toward Earth together, then no contact forces are exerted on you. Thus, your apparent weight is zero and you experience weightlessness. Similarly, the astronauts experience weightlessness as the shuttle and everything in it falls freely toward Earth.



Figure 7-11 Astronaut Chiaki Mukai experiences weightlessness on board the space shuttle *Columbia*, as the shuttle and everything in it falls freely toward Earth.

The Gravitational Field

Recall from Chapter 6 that many common forces are contact forces. Friction is exerted where two objects touch, for example, when the floor and your chair or desk push on you. Gravity, however, is different. It acts on an apple falling from a tree and on the Moon in orbit. It even acts on you in midair as you jump up or skydive. In other words, gravity acts over a distance. It acts between objects that are not touching or that are not close together. Newton was puzzled by this concept. He wondered how the Sun could exert a force on planet Earth, which is hundreds of millions of kilometers away.

The answer to the puzzle arose from a study of magnetism. In the 19th century, Michael Faraday developed the concept of a field to explain how a magnet attracts objects. Later, the field concept was applied to gravity. Any object with mass is surrounded by a **gravitational field** in which another object experiences a force due to the interaction between its mass and the gravitational field, $\boldsymbol{g}$, at its location. This is expressed by the following equation.

Gravitational Field $\boldsymbol{g} = \frac{GM}{r^2}$

The gravitational field is equal to the universal gravitational constant times the object's mass, divided by the square of the distance from the object's center. The direction is toward the mass's center.

Suppose the gravitational field is created by the Sun. Then a planet of mass m has a force exerted on it that depends on its mass and the magnitude of the gravitational field at its location. That is, $\boldsymbol{F} = m\boldsymbol{g}$, toward the Sun. The force is caused by the interaction of the planet's mass with the gravitational field at its location, not with the Sun millions of kilometers away. To find the gravitational field caused by more than one object you would calculate both gravitational fields and add them as vectors.

The gravitational field can be measured by placing an object with a small mass, m, in the gravitational field and measuring the force, F, on it. The gravitational field can be calculated using $g = F/m$. The gravitational field is measured in N/kg, which is also equal to m/s^2.

On Earth's surface, the strength of the gravitational field is 9.80 N/kg, and its direction is toward Earth's center. The field can be represented by a vector of length g pointing toward the center of the object producing the field. You can picture the gravitational field of Earth as a collection of vectors surrounding Earth and pointing toward it, as shown in **Figure 7-12.** The strength of the field varies inversely with the square of the distance from the center of Earth. The gravitational field depends on Earth's mass, but not on the mass of the object experiencing it.



Figure 7-12 Vectors representing Earth's gravitational field all point toward Earth's center. The field is weaker farther from Earth.

Two Kinds of Mass

Recall that when the concept of mass was discussed in Chapter 4, it was defined as the slope of a graph of force versus acceleration. That is, mass is equal to the ratio of the net force exerted on an object to its acceleration. This kind of mass, related to the inertia of an object, is called **inertial mass** and is represented by the following equation.

Inertial Mass $m_{\text{inertial}} = \frac{F_{\text{net}}}{a}$

Inertial mass is equal to the net force exerted on the object divided by the acceleration of the object.

The inertial mass of an object is measured by exerting a force on the object and measuring the object's acceleration using an inertial balance, such as the one shown in **Figure 7-13.** The more inertial mass an object has, the less it is affected by any force—the less acceleration it undergoes. Thus, the inertial mass of an object is a measure of the object's resistance to any type of force.

Figure 7-13 An inertial balance allows you to calculate the inertial mass of an object from the period (T) of the back-and-forth motion of the object. Calibration masses, such as the cylindrical ones shown here, are used to create a graph of T^2 versus the mass. The period of the unknown mass is then measured, and the inertial mass is determined from the calibration graph.





Figure 7-14 The platform balance shown here allows you to measure the force on a mass due to Earth's gravity.

Newton's law of universal gravitation, $F = Gm_1m_2/r^2$, also involves mass, but a different kind of mass. Mass as used in the law of universal gravitation determines the size of the gravitational force between two objects and is called **gravitational mass.** It can be measured using a simple balance, such as the one shown in **Figure 7-14.** If you measure the attractive force exerted on an object by another object of mass, m, at a distance, r, then you can define the gravitational mass in the following way.

Gravitational Mass $m_{grav} = \frac{r^2F_{grav}}{Gm}$

The gravitational mass of an object is equal to the distance between the objects squared, times the gravitational force, divided by the product of the universal gravitational constant, times the mass of the other object.

How different are these two kinds of mass? Suppose you have a watermelon in the trunk of your car. If you accelerate the car forward, the watermelon will roll backwards, relative to the trunk. This is a result of its inertial mass—its resistance to acceleration. Now, suppose your car climbs a steep hill at a constant speed. The watermelon will again roll backwards. But this time, it moves as a result of its gravitational mass. The watermelon is being attracted downward toward Earth. Newton made the claim that inertial mass and gravitational mass are equal in magnitude. This hypothesis is called the principle of equivalence. All experiments conducted so far have yielded data that support this principle. Albert Einstein also was intrigued by the principle of equivalence and made it a central point in his theory of gravity.

Einstein's Theory of Gravity

Newton's law of universal gravitation allows us to calculate the gravitational force that exists between two objects because of their masses. The concept of a gravitational field allows us to picture the way gravity acts on objects that are far away. Einstein proposed that gravity is not a force, but rather, an effect of space itself. According to Einstein, mass changes the space around it. Mass causes space to be curved, and other bodies are accelerated because of the way they follow this curved space.

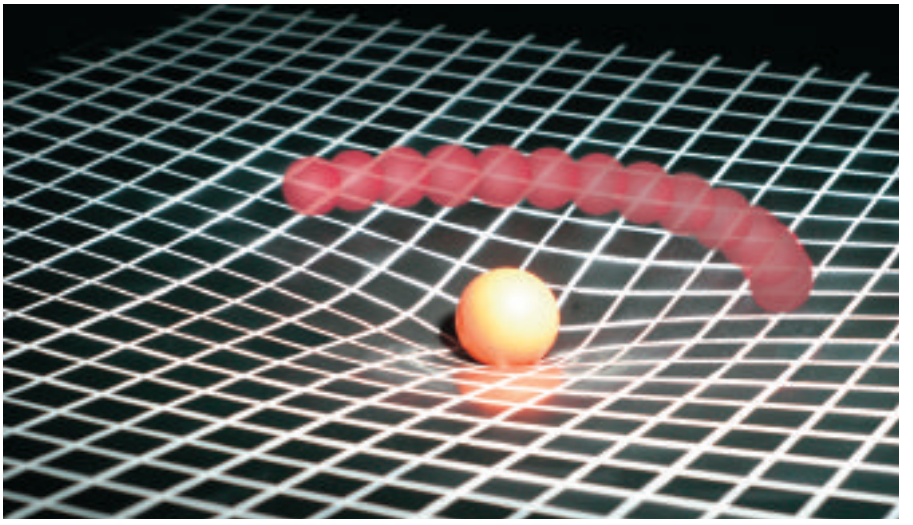


Figure 7-15 Matter causes space to curve just as an object on a rubber sheet curves the sheet around it. Moving objects near the mass follow the curvature of space. The red ball is moving clockwise around the center mass.

One way to picture how space is affected by mass is to compare space to a large, two-dimensional rubber sheet, as shown in **Figure 7-15.** The yellow ball on the sheet represents a massive object. It forms an indentation. A red ball rolling across the sheet simulates the motion of an object in space. If the red ball moves near the sagging region of the sheet, it will be accelerated. In the same way, Earth and the Sun are attracted to one another because of the way space is distorted by the two objects.

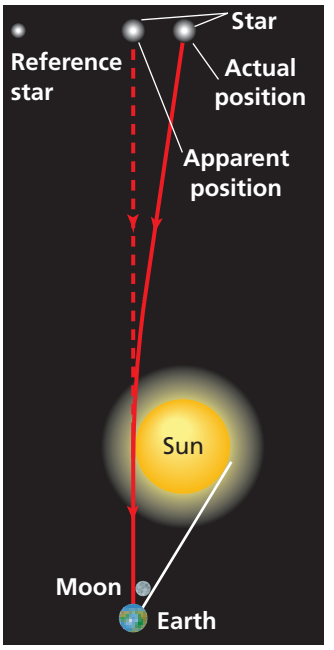
Einstein's theory, called the general theory of relativity, makes many predictions about how massive objects affect one another. In every test conducted to date, Einstein's theory has been shown to give the correct results.

Deflection of light Einstein's theory predicts the deflection or bending of light by massive objects. Light follows the curvature of space around the massive object and is deflected, as shown in **Figure 7-16.** In 1919, during an eclipse of the Sun, astronomers found that light from distant stars that passed near the Sun was deflected in agreement with Einstein's predictions.

Another result of general relativity is the effect on light from very massive objects. If an object is massive and dense enough, the light leaving it will be totally bent back to the object. No light ever escapes the object. Objects such as these, called black holes, have been identified as a result of their effects on nearby stars. The radiation produced when matter is pulled into black holes has also been helpful in their detection.

While Einstein's theory provides very accurate predictions of gravity's effects, it is still incomplete. It does not explain the origin of mass or how mass curves space. Physicists are working to understand the deeper meaning of gravity and the origin of mass itself.

■ **Figure 7-16** The light from a distant star bends due to the Sun's gravitational field, thereby changing the apparent position of the star. (Illustration not to scale)



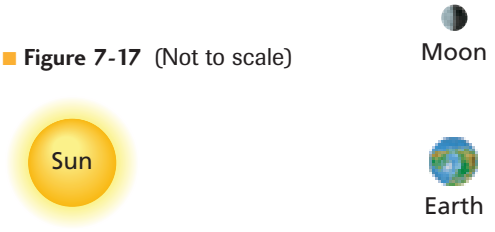
2. Section Review

15. Gravitational Fields The Moon is 3.9×10^5 km from Earth's center and 1.5×10^8 km from the Sun's center. The masses of Earth and the Sun are 6.0×10^{24} kg and 2.0×10^{30} kg, respectively.

a. Find the ratio of the gravitational fields due to Earth and the Sun at the center of the Moon.

b. When the Moon is in its third quarter phase, as shown in **Figure 7-17,** its direction from Earth is at right angles to the Sun's direction. What is the net gravitational field due to the Sun and Earth at the center of the Moon?

■ **Figure 7-17** (Not to scale)



16. Gravitational Field The mass of the Moon is 7.3×10^{22} kg and its radius is 1785 km. What is the strength of the gravitational field on the surface of the Moon?

17. A Satellite's Mass When the first artificial satellite was launched into orbit by the former Soviet Union in 1957, U.S. president Dwight D. Eisenhower asked his scientific advisors to calculate the mass of the satellite. Would they have been able to make this calculation? Explain.

18. Orbital Period and Speed Two satellites are in circular orbits about Earth. One is 150 km above the surface, the other 160 km.

a. Which satellite has the larger orbital period?

b. Which one has the greater speed?

19. Theories and Laws Why is Einstein's description of gravity called a "theory," while Newton's is a "law?"

20. Weightlessness Chairs in an orbiting spacecraft are weightless. If you were on board such a spacecraft and you were barefoot, would you stub your toe if you kicked a chair? Explain.

21. Critical Thinking It is easier to launch a satellite from Earth into an orbit that circles eastward than it is to launch one that circles westward. Explain.

PHYSICS LAB

Modeling the Orbits of Planets and Satellites

In this experiment, you will analyze a model that will show how Kepler's first and second laws of motion apply to orbits of objects in space. Kepler's first law states that orbits of planets are ellipses, with the Sun at one focus. Kepler's second law states that an imaginary line from the Sun to a planet sweeps out equal areas in equal time intervals.

The shape of the elliptical orbit is defined by eccentricity, *e*, which is the ratio of the distance between the foci to the length of the major axis. When an object is at its farthest distance from the Sun along the major axis, it is at aphelion. When the object is at its closest distance from the Sun along the major axis, it is at perihelion.

QUESTION

What is the shape of the orbits of planets and satellites in the solar system?

Objectives

- **Formulate models** to infer the shape of orbits of planets and satellites.
- **Collect and organize data** for aphelion distances and perihelion distances of objects as they orbit the Sun.
- **Draw conclusions** about Kepler's first and second laws of motion.

Safety Precautions



- **Pins are sharp and can puncture the skin.**

Materials

piece of cardboard
sheet of blank, white paper
two push pins
metric ruler
sharp pencil or pen
four small pieces of tape
string (25 cm)



Procedure

1. Place a piece of paper on a piece of cardboard using tape at the four corners.
2. Draw a line across the center of the paper, along the length of the paper. This line represents the major axis.
3. Mark the center of the line and label it *C*.
4. Use the string to tie a loop, which, when stretched, has a length of 10 cm. For each object listed in the data table, calculate the distance between the foci, *d*, using the following equation:

$$d = \frac{2e(10.0\text{ cm})}{e + 1}$$

5. For the circle, place a pin at C. Put the loop of string over the pin and pull it tight with your pencil. Move the pencil in a circular fashion around the center, letting the string guide it.
6. For the next object, place one pin a distance of *d*/2 from C along the major axis.
7. Place a second pin a distance of *d*/2 on the opposite side of C. The two pins represent the foci. One focus is the location of the Sun.
8. Put the loop of string over both pins and pull it tight with your pencil. Move the pencil in a circular fashion, letting the string guide it.
9. Using the same paper, repeat steps 6-8 for each of the listed objects.
10. After all of the orbits are plotted, label each orbit with the name and eccentricity of the object plotted.

Data Table

Object	Eccentricity (e)	d (cm)	Measured A	Measured P	Experimental e	% Error
Circle	0					
Earth	0.017					
Pluto	0.25					
Comet	0.70					

Analyze

1. Measure the aphelion distance, A, by measuring the distance between one focus and the farthest point in the orbit along the major axis. Record your data in the data table.
2. Measure the perihelion distance, P, by measuring the closest distance between one focus and the closest point in the orbit along the major axis. Record the data in the data table.
3. Calculate the experimental eccentricity for each of the objects and record your data in the data table. Use the following equation:

$$e = \frac{A - P}{A + P}$$

4. **Error Analysis** Calculate the percent error for each object using the experimental eccentricities compared to the known eccentricities. Record your values in the data table.
5. **Analyze** Why is the shape of the orbit with $e = 0$ a circle?
6. **Compare** How does Earth's orbit compare to a circle?
7. **Observe** Which of the orbits truly looks elliptical?

Conclude and Apply

1. Does the orbit model you constructed obey Kepler's first law? Explain.
2. Kepler studied the orbit data of Mars ($e = 0.093$) and concluded that planets move about the Sun in elliptical orbits. What would Kepler have concluded if he had been on Mars and studied Earth's orbit?
3. Where does a planet travel fastest: at aphelion or perihelion? Why?
4. Kepler's second law helps to determine the ratio between Pluto's velocity at aphelion and perihelion (v_A/v_P). To determine this ratio, first calculate the area swept out by Pluto's orbit. This area is approximately equal to the area of a triangle: Area $= \frac{1}{2}$ (distance to the Sun) current velocity $\times$ time. If the area that the orbit sweeps out in a fixed amount of time, such as 30 days, is the same at aphelion and perihelion, this relationship can be written

$$\frac{1}{2}Pv_P t = \frac{1}{2}Av_A t$$

What is the ratio v_P/v_A for Pluto?

5. Pluto's minimum orbital velocity is 3.7 km/s. What are the values for v_P and v_A?

Going Further

1. You used rough approximations to look at Kepler's second law. Suggest an experiment to obtain precise results to confirm the second law.
2. Design an experiment to verify Kepler's third law.

Real-World Physics

Does a communications or weather satellite that is orbiting Earth follow Kepler's laws? Collect data to verify your answer.

Physics Online

To find out more about gravitation, visit the Web site: physicspp.com

Black Holes

What would happen if you were to travel to a black hole? Your body would be stretched, flattened, and eventually pulled apart. What is a black hole? What is known about black holes?

A black hole is one of the possible final stages in the evolution of a star. When fusion reactions stop in the core of a star that is at least 20 times more massive than the Sun, the core collapses forever, compacting matter into an increasingly smaller volume. The infinitely small, but infinitely dense, object that remains is called a singularity. The force of gravity is so immense in the region around the singularity that nothing, not even light, can escape it. This region is called a black hole.

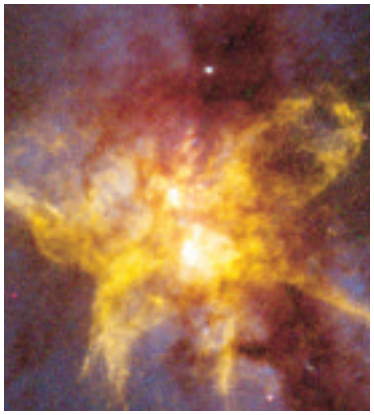
Nothing Can Escape In 1917, German mathematician Karl Schwarzschild verified, mathematically, that black holes could exist. Schwarzschild used solutions to Einstein's theory of general relativity to describe the properties of black holes. He derived an expression for a radius, called the Schwarzschild radius, within which neither light nor matter escapes the force of gravity of the singularity. The Schwarzschild radius is represented by the following equation:

$$R_s = \frac{2GM}{c^2}$$

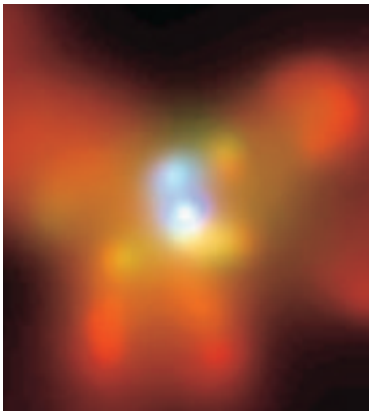
In this equation, G is Newton's universal gravitational constant, M is the mass of the black hole, and c is the speed of light. The edge of the sphere defined by the Schwarzschild radius is called the event horizon. At the event horizon, the escape velocity equals the speed of light. Because nothing travels faster than the speed of light, objects that cross the event horizon can never escape.

Indirect and Direct Evidence Black holes have three physical properties that can theoretically be measured—mass, angular momentum, and electric charge. A black hole's mass can be determined by the gravitational field it generates. Mass is calculated by using a modified form of Kepler's third law of planetary motion. Studies using NASA's *Rossi X-ray Timing Explorer* have shown that black holes spin just as stars and planets do. A black hole spins because it retains the angular momentum of the star that formed it. Even though a black hole's electric charge has not been measured, scientists hypothesize that a black hole may become charged when an excess of one type of electric charge falls into it. Super-heated gases in a black hole emit X rays, which can be detected by X-ray telescopes, such as the space-based *Chandra X-ray Observatory*.

Although not everything is known about black holes, there is direct and indirect evidence of their existence. Continued research and special missions will provide a better understanding of black holes.



***Hubble* visible image of galaxy NGC 6240.**



***Chandra* X-ray image of two black holes (blue) in NGC 6240.**

Going Further

Solve The escape velocity of an object leaving the event horizon can be represented by the following equation:

$$v = \sqrt{\frac{2GM}{R_s}}$$

In this equation, G is Newton's universal gravitational constant, M is the mass of the black hole, and R_s is the radius of the black hole. Show that the escape velocity equals the speed of light.

1. Planetary Motion and Gravitation

Vocabulary

- Kepler's first law (p. 172)
- Kepler's second law (p. 173)
- Kepler's third law (p. 173)
- gravitational force (p. 175)
- law of universal gravitation (p. 175)

Key Concepts

- Kepler's first law states that planets move in elliptical orbits, with the Sun at one focus.
- Kepler's second law states that an imaginary line from the Sun to a planet sweeps out equal areas in equal times.
- Kepler's third law states that the square of the ratio of the periods of any two planets is equal to the cube of the ratio of their distances from the Sun.

$$\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_A}{r_B}\right)^3$$

- Newton's law of universal gravitation states that the gravitational force between any two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers. The force is attractive and along a line connecting their centers.

$$F = G\frac{m_1 m_2}{r^2}$$

- Newton's law of universal gravitation can be used to rewrite Kepler's third law to relate the radius and period of a planet to the mass of the Sun.

$$T^2 = \left(\frac{4\pi^2}{Gm_S}\right)r^3$$

7.2 Using the Law of Universal Gravitation

Vocabulary

- gravitational field (p. 183)
- inertial mass (p. 183)
- gravitational mass (p. 184)

Key Concepts

- The speed of an object in circular orbit is given by the following expression.

$$v = \sqrt{\frac{Gm_E}{r}}$$

- The period of a satellite in a circular orbit is given by the following expression.

$$T = 2\pi\sqrt{\frac{r^3}{Gm_E}}$$

- All objects have gravitational fields surrounding them.

$$g = \frac{Gm}{r^2}$$

- Gravitational mass and inertial mass are two essentially different concepts. The gravitational and inertial masses of an object, however, are numerically equal.

$$m_{inertial} = \frac{F_{net}}{a} \qquad m_{grav} = \frac{r^2 F_{grav}}{Gm}$$

- Einstein's general theory of relativity describes gravitational attraction as a property of space itself.

Concept Mapping

22. Create a concept map using these terms: *planets, stars, Newton's law of universal gravitation, Kepler's first law, Kepler's second law, Kepler's third law, Einstein's general theory of relativity.*

Mastering Concepts

23. In 1609, Galileo looked through his telescope at Jupiter and saw four moons. The name of one of the moons that he saw is Io. Restate Kepler's first law for Io and Jupiter. (7.1)

24. Earth moves more slowly in its orbit during summer in the northern hemisphere than it does during winter. Is it closer to the Sun in summer or in winter? (7.1)

25. Is the area swept out per unit of time by Earth moving around the Sun equal to the area swept out per unit of time by Mars moving around the Sun? (7.1)

26. Why did Newton think that a force must act on the Moon? (7.1)

27. How did Cavendish demonstrate that a gravitational force of attraction exists between two small objects? (7.1)

28. What happens to the gravitational force between two masses when the distance between the masses is doubled? (7.1)

29. According to Newton's version of Kepler's third law, how would the ratio T^2/r^3 change if the mass of the Sun were doubled? (7.1)

30. How do you answer the question, "What keeps a satellite up?" (7.2)

31. A satellite is orbiting Earth. On which of the following does its speed depend? (7.2)

a. mass of the satellite
b. distance from Earth
c. mass of Earth

32. What provides the force that causes the centripetal acceleration of a satellite in orbit? (7.2)

33. During space flight, astronauts often refer to forces as multiples of the force of gravity on Earth's surface. What does a force of $5g$ mean to an astronaut? (7.2)

34. Newton assumed that a gravitational force acts directly between Earth and the Moon. How does Einstein's view of the attractive force between the two bodies differ from Newton's view? (7.2)

35. Show that the dimensions of g in the equation $g = F/m$ are in m/s^2. (7.2)

36. If Earth were twice as massive but remained the same size, what would happen to the value of g? (7.2)

Applying Concepts

37. Golf Ball The force of gravity acting on an object near Earth's surface is proportional to the mass of the object. **Figure 7-18** shows a tennis ball and golf ball in free fall. Why does a tennis ball not fall faster than a golf ball?

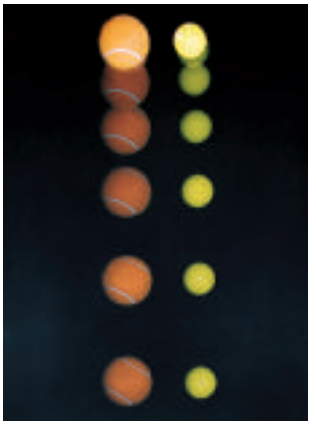


Figure 7-18

38. What information do you need to find the mass of Jupiter using Newton's version of Kepler's third law?

39. The mass of Pluto was not known until a satellite of the planet was discovered. Why?

40. Decide whether each of the orbits shown in **Figure 7-19** is a possible orbit for a planet.

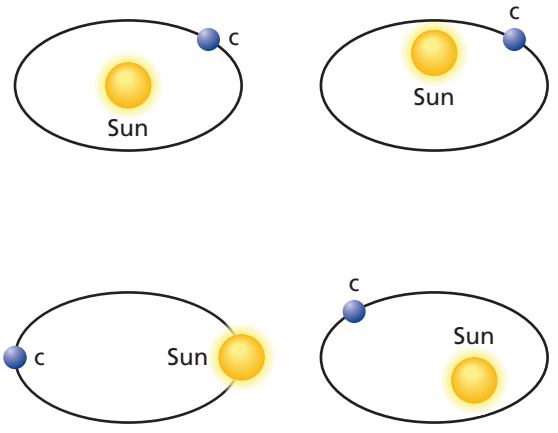


Figure 7-19

41. The Moon and Earth are attracted to each other by gravitational force. Does the more-massive Earth attract the Moon with a greater force than the Moon attracts Earth? Explain.

42. What would happen to the value of G if Earth were twice as massive, but remained the same size?

43. **Figure 7-20** shows a satellite orbiting Earth. Examine the equation $v = \sqrt{\frac{Gm_E}{r}}$, relating the speed of an orbiting satellite and its distance from the center of Earth. Does a satellite with a large or small orbital radius have the greater velocity?

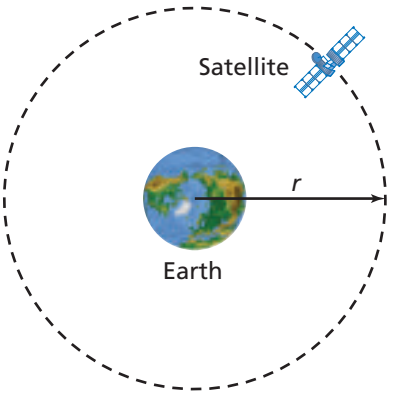


Figure 7-20 (Not to scale)

44. **Space Shuttle** If a space shuttle goes into a higher orbit, what happens to the shuttle's period?

45. Mars has about one-ninth the mass of Earth. **Figure 7-21** shows satellite M, which orbits Mars with the same orbital radius as satellite E, which orbits Earth. Which satellite has a smaller period?

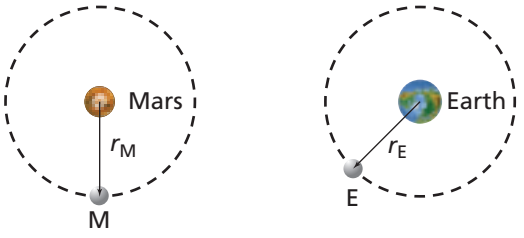


Figure 7-21 (Not to scale)

46. Jupiter has about 300 times the mass of Earth and about ten times Earth's radius. Estimate the size of g on the surface of Jupiter.

47. A satellite is one Earth radius above the surface of Earth. How does the acceleration due to gravity at that location compare to acceleration at the surface of Earth?

48. If a mass in Earth's gravitational field is doubled, what will happen to the force exerted by the field upon the mass?

49. **Weight** Suppose that yesterday your body had a mass of 50.0 kg. This morning you stepped on a scale and found that you had gained weight.

a. What happened, if anything, to your mass?

b. What happened, if anything, to the ratio of your weight to your mass?

50. As an astronaut in an orbiting space shuttle, how would you go about "dropping" an object down to Earth?

51. **Weather Satellites** The weather pictures that you see every day on TV come from a spacecraft in a stationary position relative to the surface of Earth, 35,700 km above Earth's equator. Explain how it can stay in exactly the same position day after day. What would happen if it were closer? Farther out? *Hint: Draw a pictorial model.*

Mastering Problems

7.1 Planetary Motion and Gravitation

52. Jupiter is 5.2 times farther from the Sun than Earth is. Find Jupiter's orbital period in Earth years.

53. **Figure 7-22** shows a Cavendish apparatus like the one used to find G. It has a large lead sphere that is 5.9 kg in mass and a small one with a mass of 0.047 kg. Their centers are separated by 0.055 m. Find the force of attraction between them.

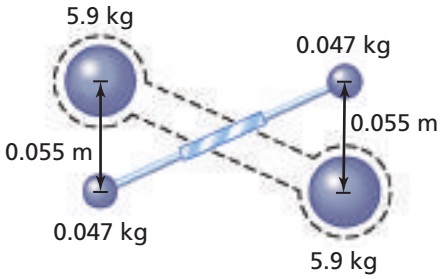


Figure 7-22

54. Use Table 7-1 on p. 173 to compute the gravitational force that the Sun exerts on Jupiter.

55. Tom has a mass of 70.0 kg and Sally has a mass of 50.0 kg. Tom and Sally are standing 20.0 m apart on the dance floor. Sally looks up and sees Tom. She feels an attraction. If the attraction is gravitational, find its size. Assume that both Tom and Sally can be replaced by spherical masses.

56. Two balls have their centers 2.0 m apart, as shown in **Figure 7-23.** One ball has a mass of 8.0 kg. The other has a mass of 6.0 kg. What is the gravitational force between them?

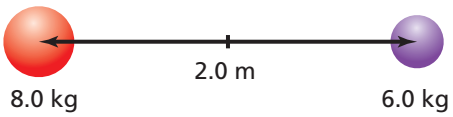


Figure 7-23

57. Two bowling balls each have a mass of 6.8 kg. They are located next to each other with their centers 21.8 cm apart. What gravitational force do they exert on each other?

58. Assume that you have a mass of 50.0 kg. Earth has a mass of 5.97×10^{24} kg and a radius of 6.38×10^{6} m.

a. What is the force of gravitational attraction between you and Earth?
b. What is your weight?

59. The gravitational force between two electrons that are 1.00 m apart is 5.54×10^{-71} N. Find the mass of an electron.

60. A 1.0-kg mass weighs 9.8 N on Earth's surface, and the radius of Earth is roughly 6.4×10^{6} m.

a. Calculate the mass of Earth.
b. Calculate the average density of Earth.

61. Uranus Uranus requires 84 years to circle the Sun. Find Uranus's orbital radius as a multiple of Earth's orbital radius.

62. Venus Venus has a period of revolution of 225 Earth days. Find the distance between the Sun and Venus as a multiple of Earth's orbital radius.

63. If a small planet, D, were located 8.0 times as far from the Sun as Earth is, how many years would it take the planet to orbit the Sun?

64. Two spheres are placed so that their centers are 2.6 m apart. The force between the two spheres is 2.75×10^{-12} N. What is the mass of each sphere if one sphere is twice the mass of the other sphere?

65. The Moon is 3.9×10^{5} km from Earth's center and 1.5×10^{8} km from the Sun's center. If the masses of the Moon, Earth, and the Sun are 7.3×10^{22} kg, 6.0×10^{24} kg, and 2.0×10^{30} kg, respectively, find the ratio of the gravitational forces exerted by Earth and the Sun on the Moon.

66. Toy Boat A force of 40.0 N is required to pull a 10.0-kg wooden toy boat at a constant velocity across a smooth glass surface on Earth. What force would be required to pull the same wooden toy boat across the same glass surface on the planet Jupiter?

67. Mimas, one of Saturn's moons, has an orbital radius of 1.87×10^{8} m and an orbital period of about 23.0 h. Use Newton's version of Kepler's third law to find Saturn's mass.

68. The Moon is 3.9×10^{8} m away from Earth and has a period of 27.33 days. Use Newton's version of Kepler's third law to find the mass of Earth. Compare this mass to the mass found in problem 60.

69. Halley's Comet Every 74 years, comet Halley is visible from Earth. Find the average distance of the comet from the Sun in astronomical units (AU).

70. Area is measured in m^2, so the rate at which area is swept out by a planet or satellite is measured in m^2/s.

a. How quickly is an area swept out by Earth in its orbit about the Sun?
b. How quickly is an area swept out by the Moon in its orbit about Earth? Use 3.9×10^{8} m as the average distance between Earth and the Moon, and 27.33 days as the period of the Moon.

7.2 Using the Law of Universal Gravitation

71. Satellite A geosynchronous satellite is one that appears to remain over one spot on Earth, as shown in **Figure 7-24.** Assume that a geosynchronous satellite has an orbital radius of 4.23×10^{7} m.

a. Calculate its speed in orbit.
b. Calculate its period.

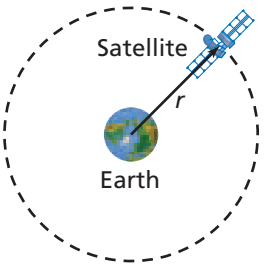


Figure 7-24 (Not to scale)

72. Asteroid The asteroid Ceres has a mass of 7×10^{20} kg and a radius of 500 km.

a. What is g on the surface of Ceres?
b. How much would a 90-kg astronaut weigh on Ceres?

73. Book A 1.25-kg book in space has a weight of 8.35 N. What is the value of the gravitational field at that location?

74. The Moon's mass is 7.34×10^{22} kg, and it is 3.8×10^{8} m away from Earth. Earth's mass is 5.97×10^{24} kg.

a. Calculate the gravitational force of attraction between Earth and the Moon.
b. Find Earth's gravitational field at the Moon.

Chapter 2

Rotational Motion



What You'll Learn

- You will learn how to describe and measure rotational motion.
- You will learn how torque changes rotational velocity.
- You will explore factors that determine the stability of an object.

Why It's Important

You encounter many rotating objects in everyday life, such as CDs, wheels, and amusement-park rides.

Spin Rides Amusement-park rides that spin are designed to thrill riders using the physics of rotational motion. The thrill is produced by a "force" that is present only when the ride spins.

LAUNCH Lab

How do different objects rotate as they roll?

Question

Do different objects of similar size and mass roll at the same rate on an incline?

Procedure

1. You will need a meterstick, a piece of foam board, a ball, a solid can, and a hollow can.
2. Position the foam board on a 20° incline.
3. Place the meterstick horizontally across the foam board, near the top of the incline, and hold it.
4. Place the ball, solid can, and hollow can against the meterstick. The solid can and hollow can should be placed sideways.
5. Simultaneously, release the three objects by lifting the meterstick.
6. As each object accelerates down the incline, due to gravity, observe the order in which each object reaches the bottom.
7. Repeat steps 2–5 two more times.

Analysis

List the objects in order from the greatest to the least acceleration.

Critical Thinking Which of the objects' properties may have contributed to their behavior? List the properties that were similar and those that were different for each object.



1. Describing Rotational Motion

You probably have observed a spinning object many times. How would you measure such an object's rotation? Find a circular object, such as a CD. Mark one point on the edge of the CD so that you can keep track of its position. Rotate the CD to the left (counterclockwise), and as you do so, watch the location of the mark. When the mark returns to its original position, the CD has made one complete revolution. How can you measure a fraction of one revolution? It can be measured in several different ways. A grad is $\frac{1}{400}$ of a revolution, whereas a degree is $\frac{1}{360}$ of a revolution. In mathematics and physics, yet another form of measurement is used to describe fractions of revolutions. In one revolution, a point on the edge travels a distance equal to 2π times the radius of the object. For this reason, the **radian** is defined as $\frac{1}{2}\pi$ of a revolution. In other words, one complete revolution is equal to 2π radians. A radian is abbreviated "rad."

Objectives

- **Describe** angular displacement.
- **Calculate** angular velocity.
- **Calculate** angular acceleration.
- **Solve** problems involving rotational motion.

Vocabulary

radian
angular displacement
angular velocity
angular acceleration

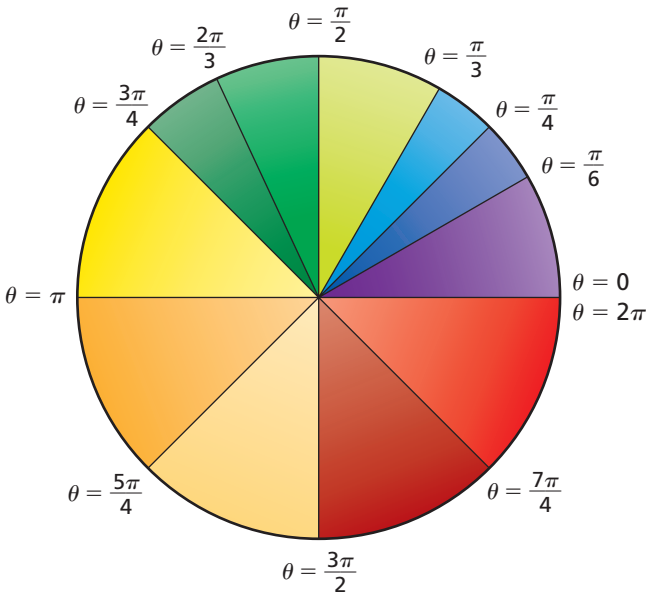


Figure 8-1 The pie chart shows the radian measure of the most common angles, measured in the counterclockwise direction. Each angle is measured from $\theta = 0$.

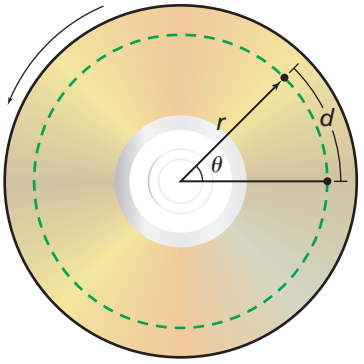
Angular Displacement

The Greek letter theta, θ, is used to represent the angle of revolution. **Figure 8-1** shows the angles in radians for several common fractions of a revolution. Note that counterclockwise rotation is designated as positive, while clockwise is negative. As an object rotates, the change in the angle is called **angular displacement.**

As you know, Earth makes one complete revolution, or 2π rad, in 24 h. In 12 h, its rotation is through π rad. Through what angle does Earth rotate in 6 h? Because 6 h is one-fourth of a day, Earth rotates through an angle of $\frac{\pi}{2}$ rad during that period. Earth's rotation as seen from the north pole is positive. Is it positive or negative when viewed from the south pole?

How far does a point on a rotating object move? You already found that a point on the edge of an object moves 2π times the radius in one revolution. In general, for rotation through an angle, θ, a point at a distance, r, from the center, as shown in **Figure 8-2,** moves a distance given by $d = r\theta$. If r is measured in meters, you might think that multiplying it by θ rad would result in d being measured in m·rad. However, this is not the case. Radians indicate the ratio between d and r. Thus, d is measured in m.

Figure 8-2 The dashed line shows the path of the point on the CD as the CD rotates counterclockwise about its center.



Angular Velocity

How fast does a CD spin? How do you determine its speed of rotation? Recall from Chapter 2 that velocity is displacement divided by the time taken to make the displacement. Likewise, the **angular velocity** of an object is angular displacement divided by the time taken to make the displacement. Thus, the angular velocity of an object is given by the following equation, where angular velocity is represented by the Greek letter omega, ω.

Angular Velocity of an Object $\omega = \frac{\Delta\theta}{\Delta t}$

The angular velocity is equal to the angular displacement divided by the time required to make the rotation.

Recall that if the velocity changes over a time interval, the average velocity is not equal to the instantaneous velocity at any given instant. Similarly, the angular velocity calculated in this way is actually the average angular velocity over a time interval, Δt. Instantaneous angular velocity is equal to the slope of a graph of angular position versus time.

Angular velocity is measured in rad/s. For Earth, $\omega_E = (2\pi \text{ rad})/(24.0 \text{ h})(3600 \text{ s/h}) = 7.27\times10^{-5}$ rad/s. In the same way that counterclockwise rotation produces positive angular displacement, it also results in positive angular velocity.

If an object's angular velocity is ω, then the linear velocity of a point a distance, r, from the axis of rotation is given by $v = r\omega$. The speed at which an object on Earth's equator moves as a result of Earth's rotation is given by $v = r\omega = (6.38\times10^6 \text{ m})(7.27\times10^{-5} \text{ rad/s}) = 464$ m/s. Earth is an example of a rotating, rigid body. Even though different points on Earth rotate different distances in each revolution, all points rotate through the same angle. All parts of a rigid body rotate at the same rate. The Sun, on the other hand, is not a rigid body. Different parts of the Sun rotate at different rates. Most objects that we will consider in this chapter are rigid bodies.

Astronomy Connection

Angular Acceleration

What if angular velocity is changing? For example, if a car were accelerated from 0.0 m/s to 25 m/s in 15 s, then the angular velocity of the wheels also would change from 0.0 rad/s to 78 rad/s in the same 15 s. The wheels would undergo **angular acceleration,** which is defined as the change in angular velocity divided by the time required to make the change. Angular acceleration, α, is represented by the following equation.

Angular Acceleration of an Object $\alpha = \dfrac{\Delta\omega}{\Delta t}$

Angular acceleration is equal to the change in angular velocity divided by the time required to make that change.

Angular acceleration is measured in rad/s^2. If the change in angular velocity is positive, then the angular acceleration also is positive. Angular acceleration defined in this way is also the average angular acceleration over the time interval Δt. One way to find the instantaneous angular acceleration is to find the slope of a graph of angular velocity as a function of time. The linear acceleration of a point at a distance, r, from the axis of an object with angular acceleration, α, is given by $a = r\alpha$. **Table 8-1** is a summary of linear and angular relationships.

Table 8–1

Linear and Angular Measures

Quantity	Linear	Angular	Relationship
Displacement	d (m)	θ (rad)	$d = r\theta$
Velocity	v (m/s)	ω (rad/s)	$v = r\omega$
Acceleration	a (m/s^2)	α (rad/s^2)	$a = r\alpha$

PRACTICE Problems

Additional Problems, Appendix B

1. What is the angular displacement of each of the following hands of a clock in 1 h? State your answer in three significant digits.
   - **a.** the second hand
   - **b.** the minute hand
   - **c.** the hour hand
2. If a truck has a linear acceleration of 1.85 m/s^2 and the wheels have an angular acceleration of 5.23 rad/s^2, what is the diameter of the truck's wheels?
3. The truck in the previous problem is towing a trailer with wheels that have a diameter of 48 cm.
   - **a.** How does the linear acceleration of the trailer compare with that of the truck?
   - **b.** How do the angular accelerations of the wheels of the trailer and the wheels of the truck compare?
4. You want to replace the tires on your car with tires that have a larger diameter. After you change the tires, for trips at the same speed and over the same distance, how will the angular velocity and number of revolutions change?

Angular frequency A rotating object can make many revolutions in a given amount of time. For instance, a spinning wheel can go through several complete revolutions in 1 min. Thus, the number of complete revolutions made by the object in 1 s is called angular frequency. Angular frequency is $f = \omega/2\pi$. In the next section, you will explore the factors that cause the angular frequency to change.

1 Section Review

5. **Angular Displacement** A movie lasts 2 h. During that time, what is the angular displacement of each of the following?
   - **a.** the hour hand
   - **b.** the minute hand
6. **Angular Velocity** The Moon rotates once on its axis in 27.3 days. Its radius is 1.74×10^6 m.
   - **a.** What is the period of the Moon's rotation in seconds?
   - **b.** What is the frequency of the Moon's rotation in rad/s?
   - **c.** What is the linear speed of a rock on the Moon's equator due only to the Moon's rotation?
   - **d.** Compare this speed with the speed of a person on Earth's equator due to Earth's rotation.
7. **Angular Displacement** The ball in a computer mouse is 2.0 cm in diameter. If you move the mouse 12 cm, what is the angular displacement of the ball?
8. **Angular Displacement** Do all parts of the minute hand on a watch have the same angular displacement? Do they move the same linear distance? Explain.
9. **Angular Acceleration** In the spin cycle of a clothes washer, the drum turns at 635 rev/min. If the lid of the washer is opened, the motor is turned off. If the drum requires 8.0 s to slow to a stop, what is the angular acceleration of the drum?
10. **Critical Thinking** A CD-ROM has a spiral track that starts 2.7 cm from the center of the disk and ends 5.5 cm from the center. The disk drive must turn the disk so that the linear velocity of the track is a constant 1.4 m/s. Find the following.
    - **a.** the angular velocity of the disk (in rad/s and rev/min) for the start of the track
    - **b.** the disk's angular velocity at the end of the track
    - **c.** the disk's angular acceleration if the disk is played for 76 min

2. Rotational Dynamics

Objectives

- **Describe** torque and the factors that determine it.
- **Calculate** net torque.
- **Calculate** the moment of inertia.

Vocabulary

lever arm
torque
moment of inertia
Newton's second law for rotational motion

How do you start the rotation of an object? That is, how do you change its angular velocity? Suppose you have a soup can that you want to spin. If you wrap a string around it and pull hard, you could make the can spin rapidly. Later in this chapter, you will learn why gravity, the force of Earth's mass on the can, acts on the center of the can. The force of the string, on the other hand, is exerted at the outer edge of the can, and at right angles to the line from the center of the can, to the point where the string leaves the can's surface.

You have learned that a force changes the velocity of a point object. In the case of a soup can, a force that is exerted in a very specific way changes the angular velocity of an extended object, which is an object that has a definite shape and size. Consider how you open a door: you exert a force. How can you exert the force to open the door most easily? To get the most effect from the least force, you exert the force as far from the axis of rotation as possible, as shown in **Figure 8-3.** In this case, the axis of rotation is an imaginary vertical line through the hinges. The doorknob is near the outer edge of the door. You exert the force on the doorknob at right angles to the door, away from the hinges. Thus, the magnitude of the force, the distance from the axis to the point where the force is exerted, and the direction of the force determine the change in angular velocity.

Lever arm For a given applied force, the change in angular velocity depends on the **lever arm,** which is the perpendicular distance from the axis of rotation to the point where the force is exerted. If the force is perpendicular to the radius of rotation, as it was with the soup can, then the lever arm is the distance from the axis, *r*. For the door, it is the distance from the hinges to the point where you exert the force, as illustrated in **Figure 8-4a,** on the next page. If the force is not perpendicular, the perpendicular component of the force must be found.

The force exerted by the string around the can is perpendicular to the radius. If a force is not exerted perpendicular to the radius, however, the lever arm is reduced. To find the lever arm, extend the line of the force until it forms a right angle with a line from the center of rotation. The distance between the intersection and the axis is the lever arm. Thus, using trigonometry, the lever arm, L, can be calculated by the equation $L = r \sin \theta$, as shown in **Figure 8-4b.** In this equation, r is the distance from the axis of rotation to the point where the force is exerted, and θ is the angle between the force and the radius from the axis of rotation to the point where the force is applied.

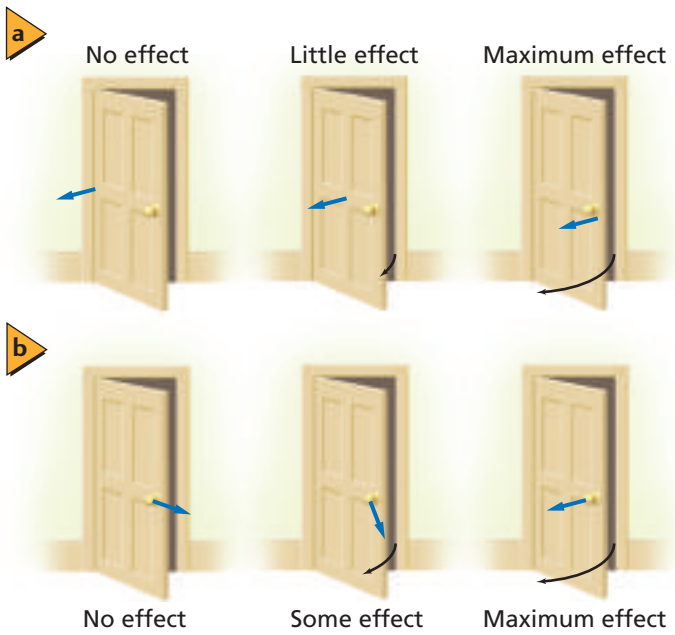


Figure 8-3 When opening a door that is free to rotate about its hinges, the greatest torque is produced when the force is applied farthest from the hinges **(a),** at an angle perpendicular to the door **(b).**

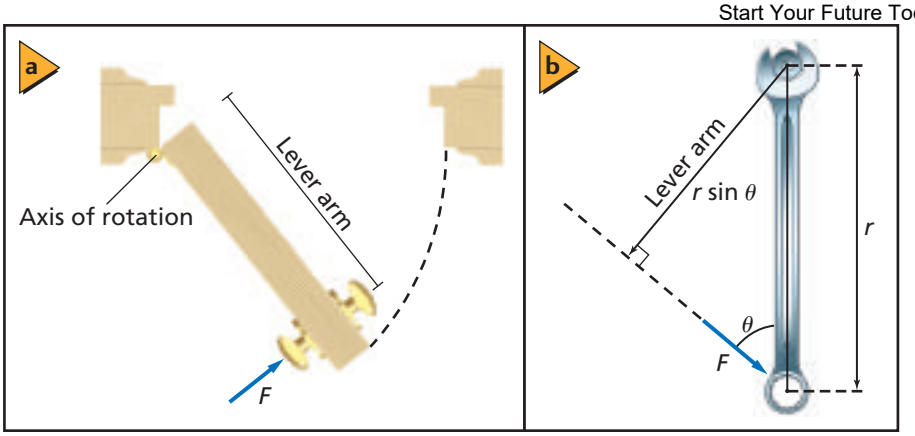


Figure 8-4 The lever arm is along the width of the door, from the hinge to the point where the force is exerted **(a).** The lever arm is equal to $r \sin \theta$, when the angle, θ, between the force and the radius of rotation is not equal to 90° **(b).**

Torque is a measure of how effectively a force causes rotation. The magnitude of torque is the product of the force and the lever arm. Because force is measured in newtons, and distance is measured in meters, torque is measured in newton-meters (N·m). Torque is represented by the Greek letter tau, τ. The equation for torque is shown below.

Torque $\tau = Fr \sin \theta$

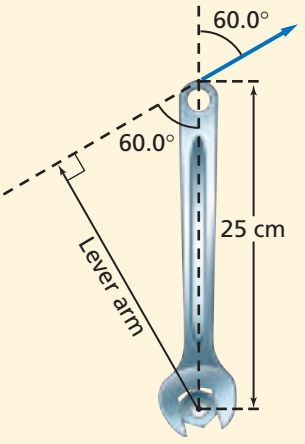
Torque is equal to the force times the lever arm.

EXAMPLE Problem 1

Lever Arm A bolt on a car engine needs to be tightened with a torque of 35 N·m. You use a 25-cm-long wrench and pull on the end of the wrench at an angle of 60.0° from the perpendicular. How long is the lever arm, and how much force do you have to exert?

1 Analyze and Sketch the Problem

- Sketch the situation. Find the lever arm by extending the force vector backwards until a line that is perpendicular to it intersects the axis of rotation.



Known:		Unknown:
$r = 0.25$ m	$\tau = 35$ N·m	$L = ?$
$\theta = 60.0°$		$F = ?$

2 Solve for the Unknown

Solve for the length of the lever arm.

$L = r \sin \theta$

$= (0.25 \text{ m})(\sin 60.0°)$ **Substitute $r = 0.25$ m, $\theta = 60.0°$**

$= 0.22$ m

Solve for the force.

$\tau = Fr \sin \theta$

$F = \dfrac{\tau}{r \sin \theta}$

$= \dfrac{35 \text{ N·m}}{(0.25 \text{ m})(\sin 60.0°)}$ **Substitute $\tau = 35$ N·m, $r = 0.25$ m, $\theta = 60.0°$**

$= 1.6 \times 10^2$ N

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3 Evaluate the Answer

- **Are the units correct?** Force is measured in newtons.
- **Does the sign make sense?** Only the magnitude of the force needed to rotate the wrench clockwise is calculated.

PRACTICE Problems

Additional Problems, Appendix B

11. Consider the wrench in Example Problem 1. What force is needed if it is applied to the wrench at a point perpendicular to the wrench?

12. If a torque of 55.0 N·m is required and the largest force that can be exerted by you is 135 N, what is the length of the lever arm that must be used?

13. You have a 0.234-m-long wrench. A job requires a torque of 32.4 N·m, and you can exert a force of 232 N. What is the smallest angle, with respect to the vertical, at which the force can be exerted?

14. You stand on the pedal of a bicycle. If you have a mass of 65 kg, the pedal makes an angle of 35° above the horizontal, and the pedal is 18 cm from the center of the chain ring, how much torque would you exert?

15. If the pedal in problem 14 is horizontal, how much torque would you exert? How much torque would you exert when the pedal is vertical?

Finding Net Torque

Try the following experiment. Get two pencils, some coins, and some transparent tape. Tape two identical coins to the ends of the pencil and balance it on the second pencil, as shown in **Figure 8-5.** Each coin exerts a torque that is equal to its weight, F_g, times the distance, r, from the balance point to the center of the coin, as follows:

$$\tau = F_g r$$

But the torques are equal and opposite in direction. Thus, the net torque is zero:

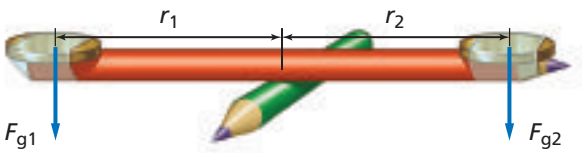
$$\tau_1 - \tau_2 = 0$$

or

$$F_{g1}\, r_1 - F_{g2}\, r_2 = 0$$

How can you make the pencil rotate? You could add a second coin on top of one of the two coins, thereby making the two forces different. You also could slide the balance point toward one end or the other of the pencil, thereby making the two distances different.

Figure 8-5 The torque exerted by the first coin, $F_{g1}r_1$, is equal and opposite in direction to the torque exerted by the second coin, $F_{g2}r_2$, when the pencil is balanced.

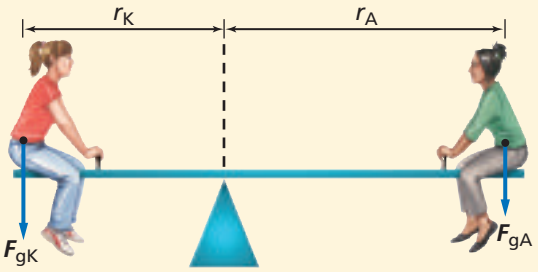


EXAMPLE Problem 2

Balancing Torques Kariann (56 kg) and Aysha (43 kg) want to balance on a 1.75-m-long seesaw. Where should they place the pivot point?

1 Analyze and Sketch the Problem

- Sketch the situation.
- Draw and label the vectors.



Known:

$m_K = 56$ kg

$m_A = 43$ kg

$r_K + r_A = 1.75$ m

Unknown:

$r_K = ?$

$r_A = ?$

2 Solve for the Unknown

Find the two forces.

Kariann:

$$F_{gK} = m_K g$$
$$= (56 \text{ kg})(9.80 \text{ m/s}^2) \quad \text{Substitute } m_K = 56 \text{ kg}, g = 9.80 \text{ m/s}^2$$
$$= 5.5\times10^2 \text{ N}$$

Aysha:

$$F_{gA} = m_A g$$
$$= (43 \text{ kg})(9.80 \text{ m/s}^2) \quad \text{Substitute } m_A = 43 \text{ kg}, g = 9.80 \text{ m/s}^2$$
$$= 4.2\times10^2 \text{ N}$$

Define Kariann's distance in terms of the length of the seesaw and Aysha's distance.

$$r_K = 1.75 \text{ m} - r_A$$

When there is no rotation, the sum of the torques is zero.

$$F_{gK} r_K = F_{gA} r_A$$
$$F_{gK} r_K - F_{gA} r_A = 0.0 \text{ N·m}$$
$$F_{gK}(1.75 \text{ m} - r_A) - F_{gA} r_A = 0.0 \text{ N·m} \quad \text{Substitute } r_K = 1.75 \text{ m} - r_A$$

Solve for r_A.

$$F_{gK}(1.75 \text{ m}) - F_{gK}(r_A) - F_{gA} r_A = 0.0 \text{ N·m}$$
$$F_{gK} r_A + F_{gA} r_A = F_{gK}(1.75 \text{ m})$$
$$(F_{gK} + F_{gA})\, r_A = F_{gK}(1.75 \text{ m})$$
$$r_A = \frac{F_{gK}(1.75 \text{ m})}{(F_{gK} + F_{gA})}$$
$$= \frac{(5.5\times10^2 \text{ N})(1.75 \text{ m})}{(5.5\times10^2 \text{ N} + 4.2\times10^2 \text{ N})} \quad \text{Substitute } F_{gK} = 5.5\times10^2 \text{ N}, F_{gA} = 4.2\times10^2 \text{ N}$$
$$= 0.99 \text{ m}$$

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3 Evaluate the Answer

- **Are the units correct?** Distance is measured in meters.
- **Do the signs make sense?** Distances are positive.
- **Is the magnitude realistic?** Aysha is about 1 m from the center, so Kariann is about 0.75 m away from it. Because Kariann's weight is greater than Aysha's weight, the lever arm on Kariann's side should be shorter. Aysha is farther from the pivot, as expected.

PRACTICE Problems

Additional Problems, Appendix B

16. Ashok, whose mass is 43 kg, sits 1.8 m from the center of a seesaw. Steve, whose mass is 52 kg, wants to balance Ashok. How far from the center of the seesaw should Steve sit?

17. A bicycle-chain wheel has a radius of 7.70 cm. If the chain exerts a 35.0-N force on the wheel in the clockwise direction, what torque is needed to keep the wheel from turning?

18. Two baskets of fruit hang from strings going around pulleys of different diameters, as shown in **Figure 8-6.** What is the mass of basket A?

19. Suppose the radius of the larger pulley in problem 18 was increased to 6.0 cm. What is the mass of basket A now?

20. A bicyclist, of mass 65.0 kg, stands on the pedal of a bicycle. The crank, which is 0.170 m long, makes a 45.0° angle with the vertical, as shown in **Figure 8-7.** The crank is attached to the chain wheel, which has a radius of 9.70 cm. What force must the chain exert to keep the wheel from turning?

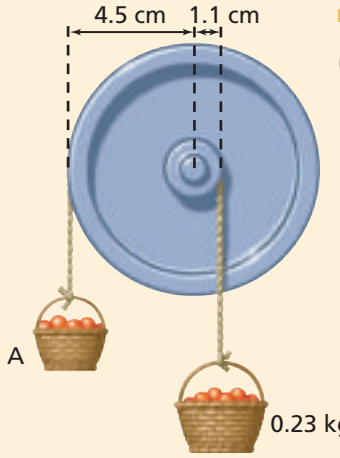


Figure 8-6

(Not to scale)

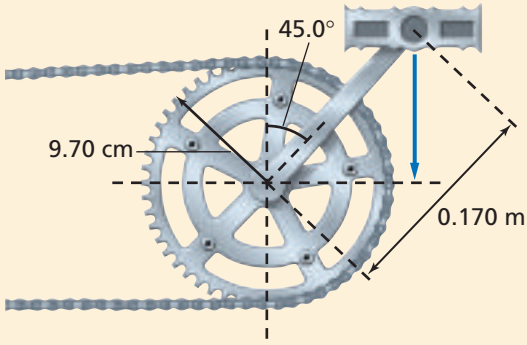


Figure 8-7

The Moment of Inertia

If you exert a force on a point mass, its acceleration will be inversely proportional to its mass. How does an extended object rotate when a torque is exerted on it? To observe firsthand, recover the pencil, the coins, and the transparent tape that you used earlier in this chapter. First, tape the coins at the ends of the pencil. Hold the pencil between your thumb and forefinger, and wiggle it back and forth. Take note of the forces that your thumb and forefinger exert. These forces create torques that change the angular velocity of the pencil and coins.

Now move the coins so that they are only 1 or 2 cm apart. Wiggle the pencil as before. Did the amount of torque and force need to be changed? The torque that was required was much less this time. Thus, the amount of mass is not the only factor that determines how much torque is needed to change angular velocity; the location of that mass also is relevant.

The resistance to rotation is called the **moment of inertia,** which is represented by the symbol I and has units of mass times the square of the distance. For a point object located at a distance, r, from the axis of rotation, the moment of inertia is given by the following equation.

Moment of Inertia of a Point Mass $I = mr^2$

The moment of inertia of a point mass is equal to the mass of the object times the square of the object's distance from the axis of rotation.

Table 8-2

Moments of Inertia for Various Objects

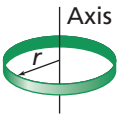
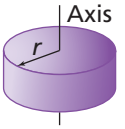
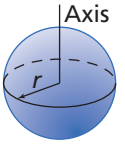
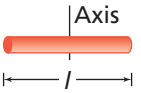
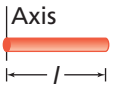
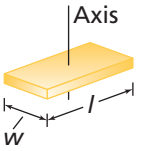
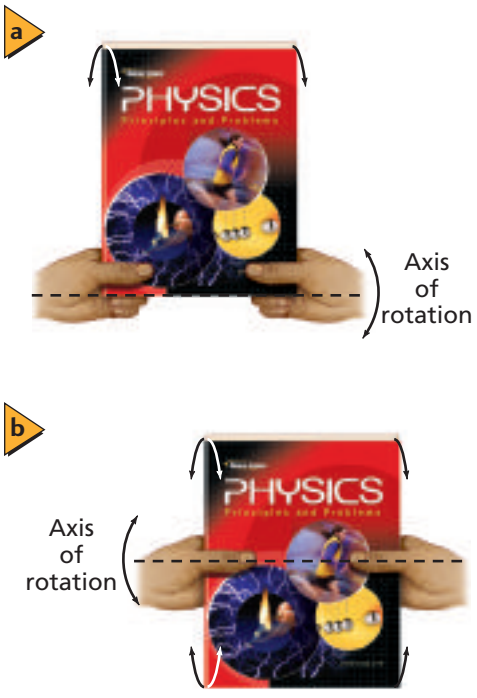
Object	Location of Axis	Diagram	Moment of Inertia
Thin hoop of radius r	Through central diameter		mr^2
Solid, uniform cylinder of radius r	Through center		$\frac{1}{2}mr^2$
Uniform sphere of radius r	Through center		$\frac{2}{5}mr^2$
Long, uniform rod of length l	Through center		$\frac{1}{12}ml^2$
Long, uniform rod of length l	Through end		$\frac{1}{3}ml^2$
Thin, rectangular plate of length l and width w	Through center		$\frac{1}{12}m(l^2 + w^2)$

Figure 8-8 The moment of inertia of a book depends on the axis of rotation. The moment of inertia of the book in **(a)** is larger than the moment of inertia of the book in **(b)** because the average distance of the book's mass from the rotational axis is larger.



As you have seen, the moment of inertia for complex objects, such as the pencil and coins, depends on how far the coins are from the axis of rotation. A bicycle wheel, for example, has almost all of its mass in the rim and tire. Its moment of inertia is almost exactly equal to mr^2, where r is the radius of the wheel. For most objects, however, the mass is distributed continuously and so the moment of inertia is less than mr^2. For example, as shown in **Table 8-2,** for a solid cylinder of radius r, $I = \frac{1}{2}mr^2$, while for a solid sphere, $I = \frac{2}{5}mr^2$.

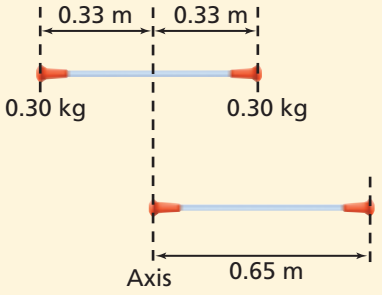
The moment of inertia also depends on the location of the rotational axis, as illustrated in **Figure 8-8.** To observe this first-hand, hold a book in the upright position, by placing your hands at the bottom of the book. Feel the torque needed to rock the book towards you, and then away from you. Now put your hands in the middle of the book and feel the torque needed to rock the book toward you and then away from you. Note that much less torque is needed when your hands are placed in the middle of the book because the average distance of the book's mass from the rotational axis is much less in this case.

EXAMPLE Problem 3

Moment of Inertia A simplified model of a twirling baton is a thin rod with two round objects at each end. The length of the baton is 0.65 m, and the mass of each object is 0.30 kg. Find the moment of inertia of the baton if it is rotated about the midpoint between the round objects. What is the moment of inertia of the baton when it is rotated around one end? Which is greater? Neglect the mass of the rod.

1 Analyze and Sketch the Problem

- Sketch the situation. Show the baton with the two different axes of rotation and the distances from the axes of rotation to the masses.



Known:
$m = 0.30$ kg
$l = 0.65$ m

Unknown:
$I = ?$

2 Solve for the Unknown

Calculate the moment of inertia of each mass separately.

Rotating about the center of the rod:

$$r = \frac{1}{2}l$$
$$= \frac{1}{2}(0.65\text{ m}) \qquad \text{Substitute } l = 0.65\text{ m}$$
$$= 0.33\text{ m}$$
$$I_{\text{single mass}} = mr^2$$
$$= (0.30\text{ kg})(0.33\text{ m})^2 \qquad \text{Substitute } m = 0.30\text{ kg}, r = 0.33\text{ m}$$
$$= 0.033\text{ kg}\cdot\text{m}^2$$

Find the moment of inertia of the baton.

$$I = 2I_{\text{single mass}}$$
$$= 2(0.033\text{ kg}\cdot\text{m}^2) \qquad \text{Substitute } I_{\text{single mass}} = 0.033\text{ kg}\cdot\text{m}^2$$
$$= 0.066\text{ kg}\cdot\text{m}^2$$

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Rotating about one end of the rod:

$$I_{\text{single mass}} = mr^2$$
$$= (0.30\text{ kg})(0.65\text{ m})^2 \qquad \text{Substitute } m = 0.30\text{ kg}, r = 0.65\text{ m}$$
$$= 0.13\text{ kg}\cdot\text{m}^2$$

Find the moment of inertia of the baton.

$$I = I_{\text{single mass}}$$
$$= 0.13\text{ kg}\cdot\text{m}^2$$

The moment of inertia is greater when the baton is swung around one end.

3 Evaluate the Answer

- **Are the units correct?** Moment of inertia is measured in $\text{kg}\cdot\text{m}^2$.
- **Is the magnitude realistic?** Masses and distances are small, and so are the moments of inertia. Doubling the distance increases the moment of inertia by a factor of 4. Thus, doubling the distance overcomes having only one mass contributing.

PRACTICE Problems

Additional Problems, Appendix B

21. Two children of equal masses sit 0.3 m from the center of a seesaw. Assuming that their masses are much greater than that of the seesaw, by how much is the moment of inertia increased when they sit 0.6 m from the center?

22. Suppose there are two balls with equal diameters and masses. One is solid, and the other is hollow, with all its mass distributed at its surface. Are the moments of inertia of the balls equal? If not, which is greater?

23. **Figure 8-9** shows three massive spheres on a rod of very small mass. Consider the moment of inertia of the system, first when it is rotated about sphere A, and then when it is rotated about sphere C. Are the moments of inertia the same or different? Explain. If the moments of inertia are different, in which case is the moment of inertia greater?

24. Each sphere in the previous problem has a mass of 0.10 kg. The distance between spheres A and C is 0.20 m. Find the moment of inertia in the following instances: rotation about sphere A, rotation about sphere C.

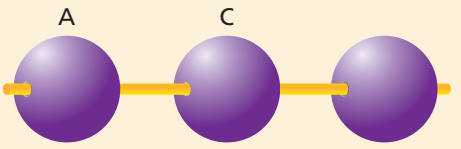


Figure 8-9

Newton's Second Law for Rotational Motion

Newton's second law for linear motion is expressed as $a = F_{net}/m$. If you rewrite this equation to represent rotational motion, acceleration is replaced by angular acceleration, α, force is replaced by net torque, τ_{net}, and mass is replaced by moment of inertia, I. Thus, **Newton's second law for rotational motion** states that angular acceleration is directly proportional to the net torque and inversely proportional to the moment of inertia. This law is expressed by the following equation.

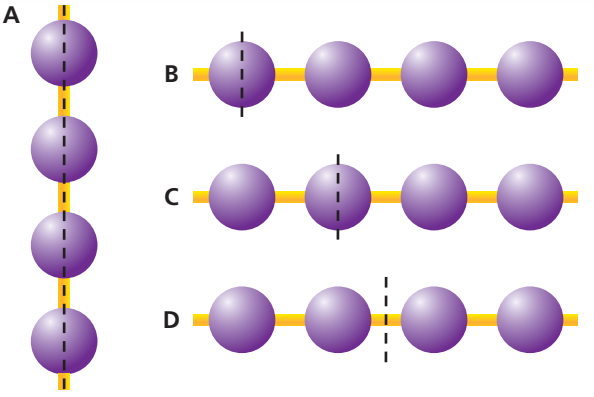
Newton's Second Law for Rotational Motion $\alpha = \frac{\tau_{net}}{I}$

The angular acceleration of an object is equal to the net torque on the object, divided by the moment of inertia.

Recall the coins taped on the pencil. To change the direction of rotation of the pencil—to give it angular acceleration—you had to apply torque to the pencil. The greater the moment of inertia, the more torque needed to produce the same angular acceleration.

CHALLENGE PROBLEM

Rank the objects shown in the diagram according to their moments of inertia about the indicated axes. All spheres have equal masses and all separations are the same.

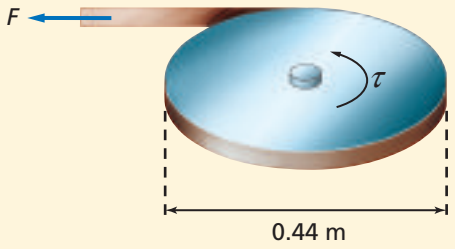


EXAMPLE Problem 4

Torque A solid steel wheel has a mass of 15 kg and a diameter of 0.44 m. It starts at rest. You want to make it rotate at 8.0 rev/s in 15 s.

a. What torque must be applied to the wheel?

b. If you apply the torque by wrapping a strap around the outside of the wheel, how much force should you exert on the strap?



1 Analyze and Sketch the Problem

- Sketch the situation. The torque must be applied in a counterclockwise direction; force must be exerted as shown.

Known:

$m = 15\ \text{kg}$

$r = \frac{1}{2}(0.44\ \text{m}) = 0.22\ \text{m}$

$\omega_i = 0.0\ \text{rad/s}$

$\omega_f = 2\pi(8.0\ \text{rev/s})$

$t = 15\ \text{s}$

Unknown:

$\alpha = ?$

$I = ?$

$\tau = ?$

$F = ?$

2 Solve for the Unknown

a. Solve for angular acceleration.

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

$$= \frac{2\pi(8.0\ \text{rev/s}) - (0.0\ \text{rad/s})}{15\ \text{s}}$$ **Substitute $\omega_f = 2\pi(8.0\ \text{rev/s})$, $\omega_i = 0.0\ \text{rad/s}$**

$$= 3.4\ \text{rad/s}^2$$

Solve for the moment of inertia.

$$I = \frac{1}{2}mr^2$$

$$= \frac{1}{2}(15\ \text{kg})(0.22\ \text{m})^2$$ **Substitute $m = 15\ \text{kg}$, $r = 0.22\ \text{m}$**

$$= 0.36\ \text{kg·m}^2$$

Solve for torque.

$$\tau = I\alpha$$

$$= (0.36\ \text{kg·m}^2)(3.4\ \text{rad/s}^2)$$ **Substitute $I = 0.36\ \text{kg·m}^2$, $\alpha = 3.4\ \text{rad/s}^2$**

$$= 1.2\ \text{kg·m}^2/\text{s}^2$$

$$= 1.2\ \text{N·m}$$

b. Solve for force.

$$\tau = Fr$$

$$F = \frac{\tau}{r}$$

$$= \frac{1.2\ \text{N·m}}{0.22\ \text{m}}$$ **Substitute $\tau = 1.2\ \text{N·m}$, $r = 0.22\ \text{m}$**

$$= 5.5\ \text{N}$$

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3 Evaluate the Answer

- **Are the units correct?** Torque is measured in N·m and force is measured in N.
- **Is the magnitude realistic?** Despite its large mass, the small size of the wheel makes it relatively easy to spin.

PRACTICE Problems

Additional Problems, Appendix B

25. Consider the wheel in Example Problem 4. If the force on the strap were twice as great, what would be the speed of rotation of the wheel after 15 s?

26. A solid wheel accelerates at 3.25 rad/s^2 when a force of 4.5 N exerts a torque on it. If the wheel is replaced by a wheel with all of its mass on the rim, the moment of inertia is given by $I = mr^2$. If the same angular velocity were desired, what force would have to be exerted on the strap?

27. A bicycle wheel can be accelerated either by pulling on the chain that is on the gear or by pulling on a string wrapped around the tire. The wheel's radius is 0.38 m, while the radius of the gear is 0.14 m. If you obtained the needed acceleration with a force of 15 N on the chain, what force would you need to exert on the string?

28. The bicycle wheel in problem 27 is used with a smaller gear whose radius is 0.11 m. The wheel can be accelerated either by pulling on the chain that is on the gear or by pulling string that is wrapped around the tire. If you obtained the needed acceleration with a force of 15 N on the chain, what force would you need to exert on the string?

29. A disk with a moment of inertia of 0.26 kg·m^2 is attached to a smaller disk mounted on the same axle. The smaller disk has a diameter of 0.180 m and a mass of 2.5 kg. A strap is wrapped around the smaller disk, as shown in **Figure 8–10.** Find the force needed to give this system an angular acceleration of 2.57 rad/s^2.

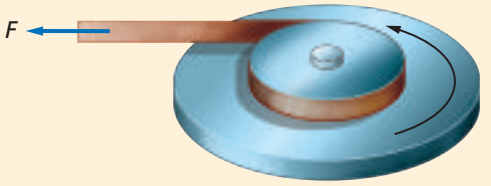


Figure 8-10

In summary, changes in the amount of torque applied to an object, or changes in the moment of inertia, affect the rate of rotation. In this section, you learned how Newton's second law of motion applies to rotational motion. In the next section, you will learn how to keep objects from rotating.

2. Section Review

30. Torque Vijesh enters a revolving door that is not moving. Explain where and how Vijesh should push to produce a torque with the least amount of force.

31. Lever Arm You try to open a door, but you are unable to push at a right angle to the door. So, you push the door at an angle of 55° from the perpendicular. How much harder would you have to push to open the door just as fast as if you were to push it at 90°?

32. Net Torque Two people are pulling on ropes wrapped around the edge of a large wheel. The wheel has a mass of 12 kg and a diameter of 2.4 m. One person pulls in a clockwise direction with a 43-N force, while the other pulls in a counterclockwise direction with a 67-N force. What is the net torque on the wheel?

33. Moment of Inertia Refer to Table 8-2 on page 206 and rank the moments of inertia from least to greatest of the following objects: a sphere, a wheel with almost all of its mass at the rim, and a solid disk. All have equal masses and diameters. Explain the advantage of using the one with the least moment of inertia.

34. Newton's Second Law for Rotational Motion A rope is wrapped around a pulley and pulled with a force of 13.0 N. The pulley's radius is 0.150 m. The pulley's rotational speed goes from 0.0 to 14.0 rev/min in 4.50 s. What is the moment of inertia of the pulley?

35. Critical Thinking A ball on an extremely low-friction, tilted surface, will slide downhill without rotating. If the surface is rough, however, the ball will roll. Explain why, using a free-body diagram.

3. Equilibrium

Why are some vehicles more likely than others to roll over when involved in an accident? What causes a vehicle to roll over? The answer lies in the design of the vehicle. In this section, you will learn some of the factors that cause an object to tip over.

The Center of Mass

How does an object rotate around its center of mass? A wrench may spin about its handle or end-over-end. Does any single point on the wrench follow a straight path? **Figure 8-11a** shows the path of the wrench. You can see that there is a single point whose path traces a straight line, as if the wrench could be replaced by a point particle at that location. The **center of mass** of an object is the point on the object that moves in the same way that a point particle would move.

Locating the center of mass How can you locate the center of mass of an object? First, suspend the object from any point. When the object stops swinging, the center of mass is along the vertical line drawn from the suspension point as shown in **Figure 8-11b.** Draw the line. Then, suspend the object from another point. Again, the center of mass must be below this point. Draw a second vertical line. The center of mass is at the point where the two lines cross, as shown in **Figure 8-11c.** The wrench, racket, and all other freely rotating objects rotate about an axis that goes through their center of mass. Where is the center of mass of a person located?

Objectives

- **Define** center of mass.
- **Explain** how the location of the center of mass affects the stability of an object.
- **Define** the conditions for equilibrium.
- **Describe** how rotating frames of reference give rise to apparent forces.

Vocabulary

center of mass
centrifugal "force"
Coriolis "force"

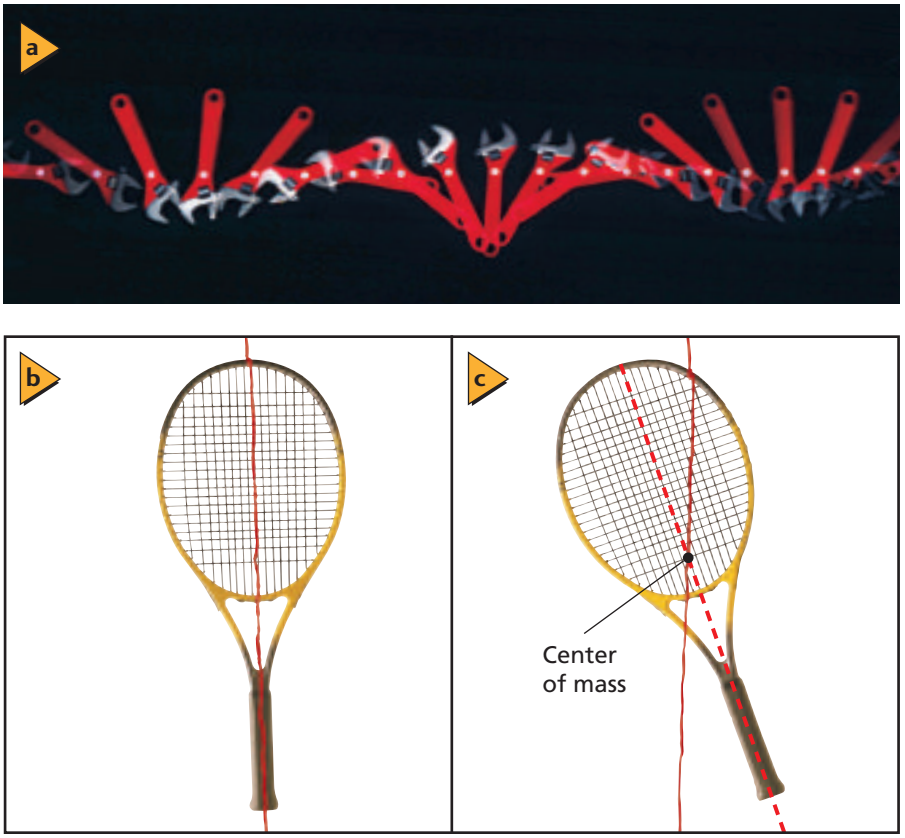
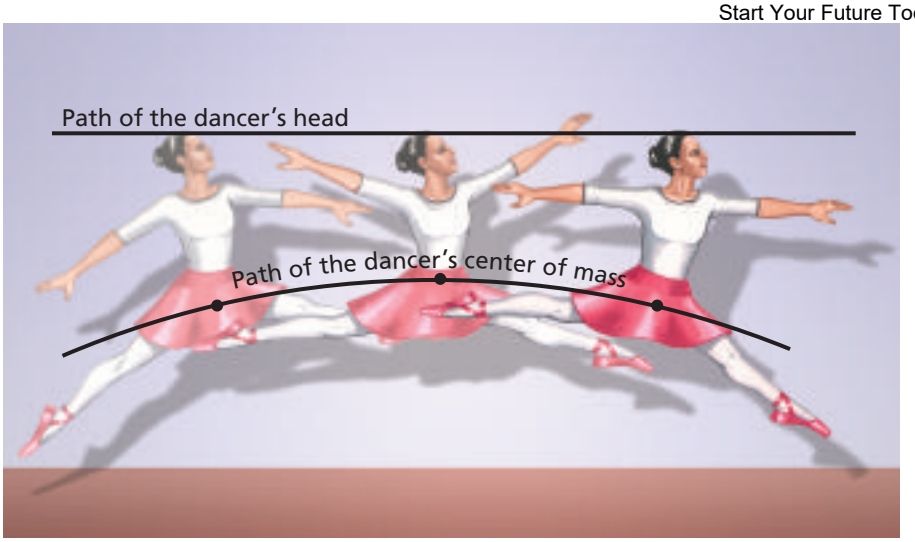


Figure 8-11 The path of the center of mass of a wrench is a straight line **(a).** The center of mass of an object such as a tennis racket can be found by first suspending it from any point **(b).** The point where the strings intersect is the location of the racket's center of mass **(c).**



■ **Figure 8-12** The upward motion of the ballet dancer's head is less than the upward motion of the center of mass. Thus, the head and torso move in a nearly horizontal path. This creates an illusion of floating.

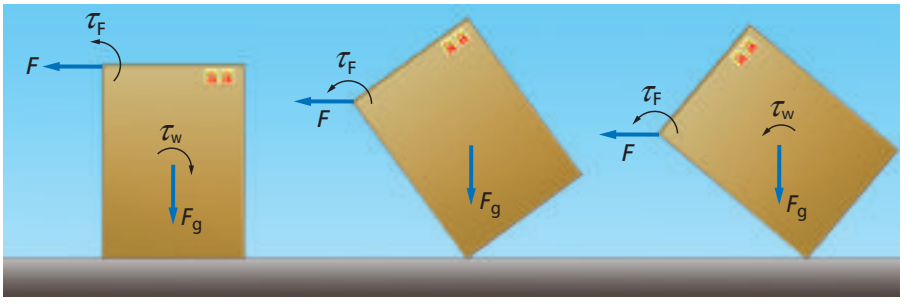
The center of mass of a human body For a person who is standing with his or her arms hanging straight down, the center of mass is a few centimeters below the navel, midway between the front and back of the person's body. It is slightly higher in young children, because of their relatively larger heads. Because the human body is flexible, however, its center of mass is not fixed. If you raise your hands above your head, your center of mass rises 6 to 10 cm. A ballet dancer, for example, can appear to be floating on air by changing her center of mass in a leap. By raising her arms and legs while in the air, as shown in **Figure 8-12,** the dancer moves her center of mass closer to her head. The path of the center of mass is a parabola, so the dancer's head stays at almost the same height for a surprisingly long time.

APPLYING PHYSICS

► **The Fosbury-Flop** In high jumping, a technique called the Fosbury-Flop allows a high jumper to clear the bar when it is placed at the highest position. This is possible because the athlete's center of mass passes below the bar as he or she somersaults over the bar, with his or her back toward it. ◄

Center of Mass and Stability

What factors determine whether a vehicle is stable or prone to roll over in an accident? To understand the problem, think about tipping over a box. A tall, narrow box, standing on end, tips more easily than a low, broad box. Why? To tip a box, as shown in **Figure 8-13,** you must rotate it about a corner. You pull at the top with a force, $\boldsymbol{F}$, applying a torque, τ_F. The weight of the box, acting on the center of mass, $\boldsymbol{F}_g$, applies an opposing torque, τ_w. When the center of mass is directly above the point of support, τ_w is zero. The only torque is the one applied by you. As the box rotates farther, its center of mass is no longer above its base of support, and both torques act in the same direction. At this point, the box tips over rapidly.



■ **Figure 8-13** The bent arrows show the direction of the torque produced by the force exerted to tip over a box.

Stability An object is said to be stable if an external force is required to tip it. The box in Figure 8-13 is stable as long as the direction of the torque due to its weight, τ_w tends to keep it upright. This occurs as long as the box's center of mass lies above its base. To tip the box over, you must rotate its center of mass around the axis of rotation until it is no longer above the base of the box. To rotate the box, you must lift its center of mass. The broader the base, the more stable the object is. For this reason, if you are standing on a bus that is weaving through traffic and you want to avoid falling down, you need to stand with your feet spread apart.

Why do vehicles roll over? **Figure 8-14** shows two vehicles rolling over. Note that the one with the higher center of mass does not have to be tilted very far for its center of mass to be outside its base—its center of mass does not have to be raised as much as the other vehicle's. The lower the location of an object's center of mass, the greater its stability.

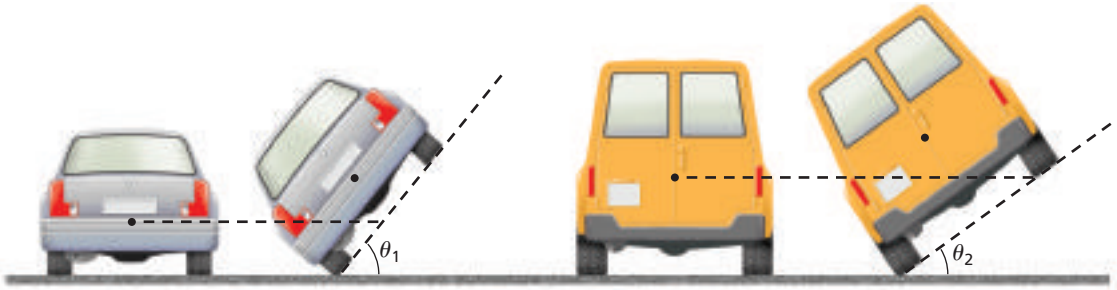
You are stable when you stand flat on your feet. When you stand on tiptoe, however, your center of mass moves forward directly above the balls of your feet, and you have very little stability. A small person can use torque, rather than force, to defend himself or herself against a stronger person. In judo, aikido, and other martial arts, the fighter uses torque to rotate the opponent into an unstable position, where the opponent's center of mass does not lie above his or her feet.

In summary, if the center of mass is outside the base of an object, it is unstable and will roll over without additional torque. If the center of mass is above the base of the object, it is stable. If the base of the object is very narrow and the center of mass is high, then the object is stable, but the slightest force will cause it to tip over.

Conditions for Equilibrium

If your pen is at rest, what is needed to keep it at rest? You could either hold it up or place it on a desk or some other surface. An upward force must be exerted on the pen to balance the downward force of gravity. You must also hold the pen so that it will not rotate. An object is said to be in static equilibrium if both its velocity and angular velocity are zero or constant. Thus, for an object to be in static equilibrium, it must meet two conditions. First, it must be in translational equilibrium; that is, the net force exerted on the object must be zero. Second, it must be in rotational equilibrium; that is, the net torque exerted on the object must be zero.

Figure 8-14 Larger vehicles have a higher center of mass than smaller ones. The higher the center of mass, the smaller the tilt needed to cause the vehicle's center of mass to move outside its base and cause the vehicle to roll over.



MINI LAB

Spinning Tops

1. Cut out two cardboard disks of 10-cm and 15-cm diameter.

2. Use a pencil with an eraser that has rounded edges from use. If it is new, rub it on paper to round it.

3. Spin the pencil and try to make it stand on the eraser. Repeat several times and record your observations.

4. Carefully push the pencil through the center of the 10-cm disk.

5. Spin the pencil with the disk and try to make it stand on the eraser.

6. Move the disk to different points on the pencil. Spin and record your observations.

7. Repeat steps 4-6 with the 15-cm disk.

Analyze and Conclude

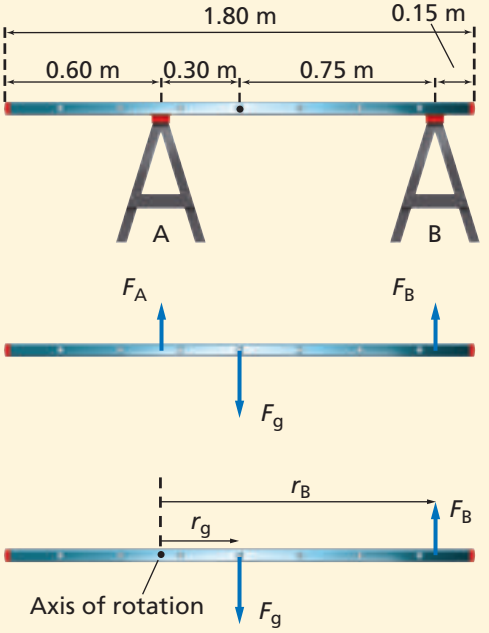
8. Sequence the three trials in order from least to most stable.

9. Describe the location of the pencil's center of mass.

10. Analyze the placement of the disk and its effect on stability.

EXAMPLE Problem 5

Static Equilibrium A 5.8-kg ladder, 1.80 m long, rests on two sawhorses. Sawhorse A is 0.60 m from one end of the ladder, and sawhorse B is 0.15 m from the other end of the ladder. What force does each sawhorse exert on the ladder?



1 Analyze and Sketch the Problem

- Sketch the situation.
- Choose the axis of rotation at the point where F_A acts on the ladder. Thus, the torque due to F_A is zero.

Known:

$m = 5.8$ kg
$l = 1.80$ m
$l_A = 0.60$ m
$l_B = 0.15$ m

Unknown:

$F_A = ?$
$F_B = ?$

2 Solve for the Unknown

For a ladder that has a constant density, the center of mass is at the center rung.

The net force is the sum of all forces on the ladder.

$$F_{net} = F_A + F_B + (-F_g)$$

The ladder is in translational equilibrium, so the net force exerted on it is zero.

$$0.0\ \text{N} = F_A + F_B - F_g$$

Solve for F_A.

$$F_A = F_g - F_B$$

Find the torques due to F_g and F_B.

$$\tau_g = -r_g F_g$$

τ_g is in the clockwise direction.

$$\tau_B = +r_B F_B$$

τ_B is in the counterclockwise direction.

The net torque is the sum of all torques on the object.

$$\tau_{net} = \tau_B + \tau_g$$

$$0.0\ \text{N·m} = \tau_B + \tau_g$$

The ladder is in rotational equilibrium, so $\tau_{net} = 0.0$ N·m.

$$\tau_B = \tau_g$$

$$r_B F_B = r_g F_g$$

Substitute $\tau_B = r_B F_B$, $\tau_g = -r_g F_g$

Solve for F_B.

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$$F_B = \frac{r_g F_g}{r_B}$$

$$= \frac{r_g mg}{r_B}$$

Substitute $F_g = mg$

Using the expression $F_A = F_g - F_B$, substitute in the expressions for F_B and F_g.

$$F_A = F_g - F_B$$

$$= F_g - \frac{r_g mg}{r_B}$$

Substitute $F_B = \frac{r_g mg}{r_B}$

$$= mg - \frac{r_g mg}{r_B}$$

Substitute $F_g = mg$

$$= mg\left(1 - \frac{r_g}{r_B}\right)$$

Solve for r_g.

$$r_g = \frac{1}{2} - l_A \qquad \text{For a ladder, which has a constant density, the center of mass is at the center rung.}$$
$$= 0.90\text{ m} - 0.60\text{ m} \qquad \text{Substitute } \frac{l}{2} = 0.90\text{ m}, l_A = 0.60\text{ m}$$
$$= 0.30\text{ m}$$

Solve for r_B.

$$r_B = (0.90\text{ m} - l_B) + (0.90\text{ m} - l_A)$$
$$= (0.90\text{ m} - 0.15\text{ m}) + (0.90\text{ m} - 0.60\text{ m}) \qquad \text{Substitute } l_B = 0.15\text{ m}, l_A = 0.60\text{ m}$$
$$= 0.75\text{ m} + 0.30\text{ m}$$
$$= 1.05\text{ m}$$

Calculate F_B.

$$F_B = \frac{r_g mg}{r_B}$$
$$= \frac{(0.30\text{ m})(5.8\text{ kg})(9.80\text{ m/s}^2)}{(1.05\text{ m})} \qquad \text{Substitute } r_g = 0.30\text{ m}, m = 5.8\text{ kg}, g = 9.80\text{ m/s}^2, r_B = 1.05\text{ m}$$
$$= 16\text{ N}$$

Calculate F_A.

$$F_A = mg\left(1 - \frac{r_g}{r_B}\right)$$
$$= \left(1 - \frac{(0.30\text{ m})}{(1.05\text{ m})}\right)(5.8\text{ kg})(9.80\text{ m/s}^2) \qquad \text{Substitute } r_g = 0.30\text{ m}, m = 5.8\text{ kg}, g = 9.80\text{ m/s}^2, r_B = 1.05\text{ m}$$
$$= 41\text{ N}$$

3 Evaluate the Answer

- **Are the units correct?** Forces are measured in newtons.
- **Do the signs make sense?** Both forces are upward.
- **Is the magnitude realistic?** The forces add up to the weight of the ladder, and the force exerted by the sawhorse closer to the center of mass is greater, which is correct.

PRACTICE Problems

Additional Problems, Appendix B

36. What would be the forces exerted by the two sawhorses if the ladder in Example Problem 5 had a mass of 11.4 kg?

37. A 7.3-kg ladder, 1.92 m long, rests on two sawhorses, as shown in **Figure 8-15.** Sawhorse A, on the left, is located 0.30 m from the end, and sawhorse B, on the right, is located 0.45 m from the other end. Choose the axis of rotation to be the center of mass of the ladder.

a. What are the torques acting on the ladder?

b. Write the equation for rotational equilibrium.

c. Solve the equation for F_A in terms of F_g.

d. How would the forces exerted by the two sawhorses change if A were moved very close to, but not directly under, the center of mass?

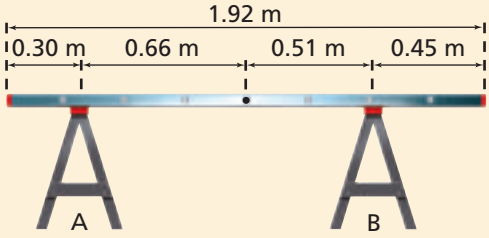


Figure 8-15

38. A 4.5-m-long wooden plank with a 24-kg mass is supported in two places. One support is directly under the center of the board, and the other is at one end. What are the forces exerted by the two supports?

39. A 85-kg diver walks to the end of a diving board. The board, which is 3.5 m long with a mass of 14 kg, is supported at the center of mass of the board and at one end. What are the forces on the two supports?

Rotating Frames of Reference

When you are on a on a rapidly spinning amusement-park ride, it feels like a strong force is pushing you to the outside. A pebble on the floor of the ride would accelerate outward without a horizontal force being exerted on it in the same direction. The pebble would not move in a straight line. In other words, Newton's laws would not apply. This is because rotating frames of reference are accelerated frames. Newton's laws are valid only in inertial or nonaccelerated frames.

Motion in a rotating reference frame is important to us because Earth rotates. The effects of the rotation of Earth are too small to be noticed in the classroom or lab, but they are significant influences on the motion of the atmosphere and therefore, on climate and weather.

Centrifugal "Force"

Suppose you fasten one end of a spring to the center of a rotating platform. An object lies on the platform and is attached to the other end of the spring. As the platform rotates, an observer on the platform sees the object stretch the spring. The observer might think that some force toward the outside of the platform is pulling on the object. This apparent force is called **centrifugal "force."** It is not a real force because there is no physical outward push on the object. Still, this "force" seems real, as anyone who has ever been on an amusement-park ride can attest.

As the platform rotates, an observer on the ground sees things differently. This observer sees the object moving in a circle. The object accelerates toward the center because of the force of the spring. As you know, the acceleration is centripetal acceleration and is given by $a_c = v^2/r$. It also can be written in terms of angular velocity, as $a_c = \omega^2 r$. Centripetal acceleration is proportional to the distance from the axis of rotation and depends on the square of the angular velocity. Thus, if you double the rotational frequency, the acceleration increases by a factor of 4.

The Coriolis "Force"

A second effect of rotation is shown in **Figure 8-16.** Suppose a person standing at the center of a rotating disk throws a ball toward the edge of the disk. Consider the horizontal motion of the ball as seen by two observers and ignore the vertical motion of the ball as it falls.

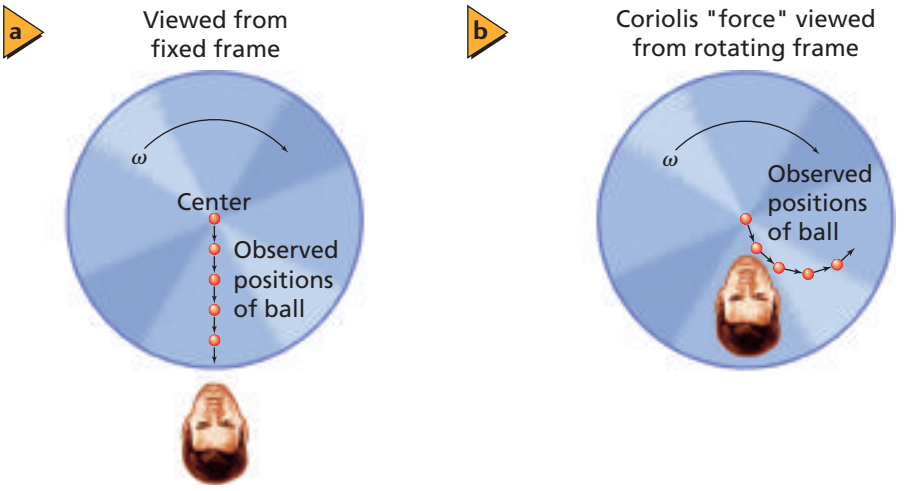


Figure 8-16 The Coriolis "force" exists only in rotating reference frames.

An observer standing outside the disk, as shown in **Figure 8-16a,** sees the ball travel in a straight line at a constant speed toward the edge of the disk. However, the other observer, who is stationed on the disk and rotating with it, as shown in **Figure 8-16b,** sees the ball follow a curved path at a constant speed. A force seems to be acting to deflect the ball. This apparent force is called the **Coriolis "force."** Like the centrifugal "force," the Coriolis "force" is not a real force. It seems to exist because we observe a deflection in horizontal motion when we are in a rotating frame of reference.

Coriolis "force" due to Earth Suppose a cannon is fired from a point on the equator toward a target due north of it. If the projectile were fired directly northward, it would also have an eastward velocity component because of the rotation of Earth. This eastward speed is greater at the equator than at any other latitude. Thus, as the projectile moves northward, it also moves eastward faster than points on Earth below it do. The result is that the projectile lands east of the target as shown in **Figure 8-17.** While an observer in space would see Earth's rotation, an observer on Earth could claim that the projectile missed the target because of the Coriolis "force" on the rocket. Note that for objects moving toward the equator, the direction of the apparent force is westward. A projectile will land west of the target when fired due south.



Figure 8-17 An observer on Earth sees the Coriolis "force" cause a projectile fired due north to deflect to the right of the intended target.

The direction of winds around high- and low-pressure areas results from the Coriolis "force." Winds flow from areas of high to low pressure. Because of the Coriolis "force" in the northern hemisphere, winds from the south go to the east of low-pressure areas. Winds from the north, however, end up west of low-pressure areas. Therefore, winds rotate counterclockwise around low-pressure areas in the northern hemisphere. In the southern hemisphere however, winds rotate clockwise around low-pressure areas.

Meteorology Connection

Most amusement-park rides thrill the riders because they are in accelerated reference frames while on the ride. The "forces" felt by roller-coaster riders at the tops and bottoms of hills, and when moving almost vertically downward, are mostly related to linear acceleration. On Ferris wheels, rotors, other circular rides, and on the curves of roller coasters, centrifugal "forces" provide most of the excitement.

3. Section Review

40. Center of Mass Can the center of mass of an object be located in an area where the object has no mass? Explain.

41. Stability of an Object Why is a modified vehicle with its body raised high on risers less stable than a similar vehicle with its body at normal height?

42. Conditions for Equilibrium Give an example of an object for each of the following conditions.

a. rotational equilibrium, but not translational equilibrium

b. translational equilibrium, but not rotational equilibrium

43. Center of Mass Where is the center of mass of a roll of masking tape?

44. Locating the Center of Mass Describe how you would find the center of mass of this textbook.

45. Rotating Frames of Reference A penny is placed on a rotating, old-fashioned record turntable. At the highest speed, the penny starts sliding outward. What are the forces acting on the penny?

46. Critical Thinking You have learned why the winds around a low-pressure area move in a counterclockwise direction. Would the winds move in the same or opposite direction in the southern hemisphere? Explain.

Horizons Companies

PHYSICS LAB

Translational and Rotational Equilibrium

Alternate CBL instructions can be found on the Web site.
physicspp.com

For maintenance on large buildings, scaffolding can be hung on the outside. In order for the scaffolding to support workers, it must be in translational and rotational equilibrium. If two or more forces act on the scaffolding, each can produce a rotation about either end. Scaffolding with uniform mass distribution acts as though all of the mass is concentrated at its center. In translational equilibrium the object is not accelerating; thus, the upward and downward forces are equal.

In order to achieve rotational equilibrium, the sum of all the clockwise torques must equal the sum of all the counterclockwise torques as measured from a pivot point. That is, the net torque must be zero. In this lab you will model scaffolding hung from two ropes using a meterstick and spring scales, and use numbers to measure the forces on the scaffolding.

QUESTION

What conditions are required for equilibrium when parallel forces act on an object?

Objectives

- **Collect and organize data** about the forces acting on the scaffolding.
- **Describe** clockwise and counterclockwise torque.
- **Compare and contrast** translational and rotational equilibrium.

Safety Precautions



- **Use care to avoid dropping masses.**

Materials

meterstick
two 0-5 N spring scales
two ring stands
two Buret clamps
500-g hooked mass
200-g hooked mass



Procedure

The left spring scale will be considered a pivot point for the purposes of this lab. Therefore, the lever arm will be measured from this point.

1. Place the ring stands 80 cm apart.
2. Attach a Buret clamp to each of the ring stands.
3. Verify that the scales are set to zero before use. If the scales need to be adjusted, ask your teacher for assistance.
4. Hang a spring scale from each Buret clamp attached to a ring stand.
5. Hook the meterstick onto the spring scale in such a manner that the 10-cm mark is supported by one hook and the 90-cm mark is supported by the other hook.
6. Read each spring scale and record the force in Data Table 1.
7. Hang a 500-g mass on the meterstick at the 30-cm mark. This point should be 20-cm from the left scale.
8. Read each spring scale and record the force in Data Table 1.
9. Hang a 200-g mass on the meterstick at the 70-cm mark. This point should be 60 cm from the left scale.
10. Read each spring scale and record the force in Data Table 1.

Data Table 1

Object Added	Distance From Left Scale (m)	Left Scale Reading (N)	Right Scale Reading (N)
Meterstick	0.4		
500-g mass	0.2		
200-g mass	0.6		

Data Table 2

Object Added	τ_c	τ_{cc}	Lever Arm (m)	Force (N)
Meterstick				
500-g mass				
200-g mass				
Right scale				

Data Table 3

Object Added	τ_c (N·m)	τ_{cc} (N·m)
Meterstick		
500-g mass		
200-g mass		
Right scale		
$\Sigma\tau$		

Analyze

1. **Calculate** Find the mass of the meterstick.
2. **Calculate** Find the force, or weight, that results from each object and record it in Data Table 2. For the right scale, read the force it exerts and record it in Data Table 2.
3. Using the point where the left scale is attached as a pivot point, identify the forces located elsewhere that cause the scaffold to rotate clockwise or counterclockwise. Mark these in Data Table 2 with an *x*.
4. Record the lever arm distance of each force from the pivot point in Data Table 2.
5. **Use Numbers** Calculate the torque for each object by multiplying the force and lever arm distance. Record these values in Data Table 3.

Conclude and Apply

1. Is the system in translational equilibrium? How do you know?
2. Draw a free-body diagram of your system, showing all the forces.
3. Compare and contrast the sum of the clockwise torques, $\Sigma\tau_c$, and the counterclockwise torques, $\Sigma\tau_{cc}$.
4. What is the percent difference between $\Sigma\tau_c$ and $\Sigma\tau_{cc}$?

Going Further

Use additional masses at locations of your choice with your teacher's permission and record your data.

Real-World Physics

Research the safety requirements in your area for putting up, using, and dismantling scaffolding.

Physics online

To find out more about rotational motion, visit the Web site: **physicspp.com**

The Stability of Sport-Utility Vehicles

Why are sport-utility vehicles more flippable? Many believe that the large size of the sport-utility vehicle makes it more stable and secure. But, a sport-utility vehicle, as well as other tall vehicles such as vans, is much more likely to roll over than a car.

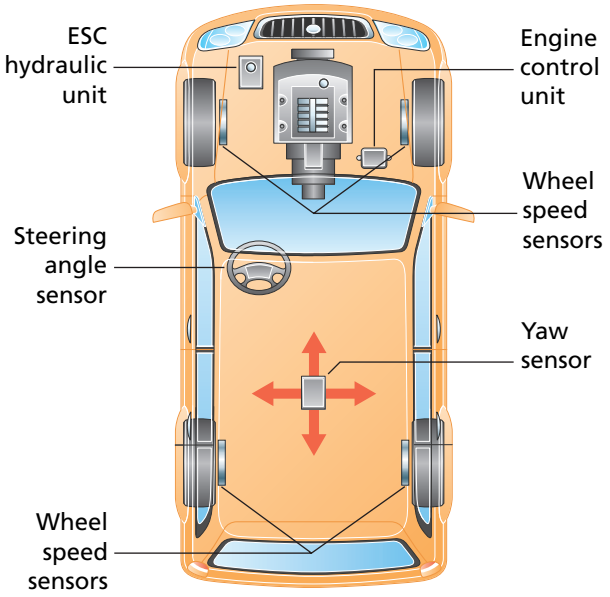
The Problem A sport-utility vehicle has a high center of mass which makes it more likely to topple. Another factor that affects rollover is the static stability factor, which is the ratio of the track width to the center of mass. Track width is defined as half the distance between the two front wheels. The higher the static stability factor, the more likely a vehicle will stay upright.

Many sport-utility vehicles have a center of mass 13 or 15 cm higher than passenger cars. Their track width, however, is about the same as that of passenger cars. Suppose the stability factor for a sport-utility vehicle is 1.06 and 1.43 for a car. Statistics show that in a single-vehicle crash, the sport-utility vehicle has a 37 percent chance of rolling over, while the car has a 10.6 percent chance of rolling over.

However, the static stability factor oversimplifies the issue. Weather and driver behavior are also contributers to rollover crashes. Vehicle factors, such as tires, suspension systems, inertial properties, and advanced handling systems all play a role as well.

It is true that most rollover crashes occur when a vehicle swerves off the road and hits a rut, soft soil, or other surface irregularity. This usually occurs when a driver is not paying proper attention or is speeding. Safe drivers greatly reduce their chances of being involved in a rollover accident by paying attention and driving at the correct speed. Still, weather and driver behavior being equal, the laws of physics indicate that sport-utility vehicles carry an increased risk.

What Is Being Done? Some models are being built with wider track widths or stronger roofs. Optional side-curtain air bags have sensors to keep the bags inflated for up to 6 s, rather than the usual fraction of a second. This will cushion passengers if the vehicle should flip several times.



An ESC system processes information from the sensors and automatically applies the brakes to individual wheels when instability is detected.

A promising new technology called Electronic Stability Control (ESC) can be used to prevent rollover accidents. An ESC system has electronic sensors that detect when a vehicle begins to spin due to oversteering, and also when it begins to slide in a plowlike manner because of understeering. In these instances, an ESC system automatically applies the brakes at one or more wheels, thereby reorienting the vehicle in the right direction.

Safe driving is the key to preventing many automobile accidents. Knowledge of the physics behind rollover accidents and the factors that affect rollover accidents may help make you an informed, safe driver.

Going Further

1. **Hypothesize** In a multi-vehicle accident, sport-utility vehicles generally fare better than the passenger cars involved in the accident. Why is this so?
2. **Debate the Issue** ESC is a life-saving technology. Should it be mandatory in all sport-utility vehicles? Why or why not?

8.1 Describing Rotational Motion

Vocabulary

- radian (p. 197)
- angular displacement (p. 198)
- angular velocity (p. 198)
- angular acceleration (p. 199)

Key Concepts

- Angular position and its changes are measured in radians. One complete revolution is 2π rad.
- Angular velocity is given by the following equation.

$$\omega = \frac{\Delta\theta}{\Delta t}$$

- Angular acceleration is given by the following equation.

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

- For a rotating, rigid object, the angular displacement, velocity, and acceleration can be related to the linear displacement, velocity, and acceleration for any point on the object.

$$d = r\theta \qquad v = r\omega \qquad a = r\alpha$$

8.2 Rotational Dynamics

Vocabulary

- lever arm (p. 201)
- torque (p. 202)
- moment of inertia (p. 205)
- Newton's second law for rotational motion (p. 208)

Key Concepts

- When torque is exerted on an object, its angular velocity changes.
- Torque depends on the magnitude of the force, the distance from the axis of rotation at which it is applied, and the angle between the force and the radius from the axis of rotation to the point where the force is applied.

$$\tau = Fr \sin\theta$$

- The moment of inertia of an object depends on the way the object's mass is distributed about the rotational axis. For a point object:

$$I = mr^2$$

- Newton's second law for rotational motion states that angular acceleration is directly proportional to the net torque and inversely proportional to the moment of inertia.

$$\alpha = \frac{\tau_{\text{net}}}{I}$$

8.3 Equilibrium

Vocabulary

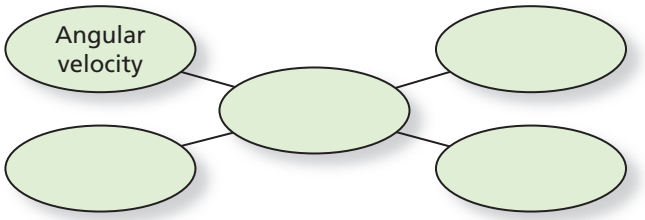
- center of mass (p. 211)
- centrifugal "force" (p. 216)
- Coriolis "force" (p. 217)

Key Concepts

- The center of mass of an object is the point on the object that moves in the same way that a point particle would move.
- An object is stable against rollover if its center of mass is above its base.
- An object is in equilibrium if there are no net forces exerted on it and if there are no net torques acting on it.
- Centrifugal "force" and the Coriolis "force" are two apparent forces that appear when a rotating object is analyzed from a coordinate system that rotates with it.

Concept Mapping

47. Complete the following concept map using the following terms: *angular acceleration, radius, tangential acceleration, centripetal acceleration.*



Mastering Concepts

48. A bicycle wheel rotates at a constant 25 rev/min. Is its angular velocity decreasing, increasing, or constant? (8.1)

49. A toy rotates at a constant 5 rev/min. Is its angular acceleration positive, negative, or zero? (8.1)

50. Do all parts of Earth rotate at the same rate? Explain. (8.1)

51. A unicycle wheel rotates at a constant 14 rev/min. Is the total acceleration of a point on the tire inward, outward, tangential, or zero? (8.1)

52. Think about some possible rotations of your textbook. Are the moments of inertia about these three axes the same or different? Explain. (8.2)

53. Torque is important when tightening bolts. Why is force not important? (8.2)

54. Rank the torques on the five doors shown in **Figure 8-18** from least to greatest. Note that the magnitude of all the forces is the same. (8.2)

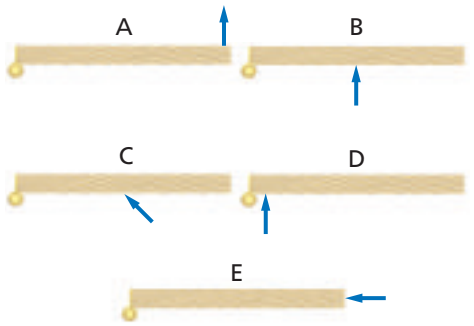


Figure 8-18

55. Explain how you can change an object's angular frequency. (8.2)

56. To balance a car's wheel, it is placed on a vertical shaft and weights are added to make the wheel horizontal. Why is this equivalent to moving the center of mass until it is at the center of the wheel? (8.3)

57. A stunt driver maneuvers a monster truck so that it is traveling on only two wheels. Where is the center of mass of the truck? (8.3)

58. Suppose you stand flat-footed, then you rise and balance on tiptoe. If you stand with your toes touching a wall, you cannot balance on tiptoe. Explain. (8.3)

59. Why does a gymnast appear to be floating on air when she raises her arms above her head in a leap? (8.3)

60. Why is a vehicle with wheels that have a large diameter more likely to roll over than a vehicle with wheels that have a smaller diameter? (8.3)

Applying Concepts

61. Two gears are in contact and rotating. One is larger than the other, as shown in **Figure 8-19.** Compare their angular velocities. Also compare the linear velocities of two teeth that are in contact.



Figure 8-19

62. **Videotape** When a videotape is rewound, why does it wind up fastest towards the end?

63. **Spin Cycle** What does a spin cycle of a washing machine do? Explain in terms of the forces on the clothes and water.

64. How can you experimentally find the moment of inertia of an object?

65. **Bicycle Wheels** Three bicycle wheels have masses that are distributed in three different ways: mostly at the rim, uniformly, and mostly at the hub. The wheels all have the same mass. If equal torques are applied to them, which one will have the greatest angular acceleration? Which one will have the least?

66. **Bowling Ball** When a bowling ball leaves a bowler's hand, it does not spin. After it has gone about half the length of the lane, however, it does spin. Explain how its rotation rate increased and why it does not continue to increase.

67. Flat Tire Suppose your car has a flat tire. You get out your tools and find a lug wrench to remove the nuts off the bolt studs. You find it impossible to turn the nuts. Your friend suggests ways you might produce enough torque to turn them. What three ways might your friend suggest?

68. Tightrope Walkers Tightrope walkers often carry long poles that sag so that the ends are lower than the center as shown in **Figure 8-20.** How does such a pole increase the tightrope walker's stability? *Hint: Consider both center of mass and moment of inertia.*



Figure 8-20

69. Merry-Go-Round While riding a merry-go-round, you toss a key to a friend standing on the ground. For your friend to be able to catch the key, should you toss it a second or two before you reach the spot where your friend is standing or wait until your friend is directly behind you? Explain.

70. Why can you ignore forces that act on the axis of rotation of an object in static equilibrium when determining the net torque?

71. In solving problems about static equilibrium, why is the axis of rotation often placed at a point where one or more forces are acting on the object?

Mastering Problems

8.1 Describing Rotational Motion

72. A wheel is rotated so that a point on the edge moves through 1.50 m. The radius of the wheel is 2.50 m, as shown in **Figure 8-21.** Through what angle (in radians) is the wheel rotated?

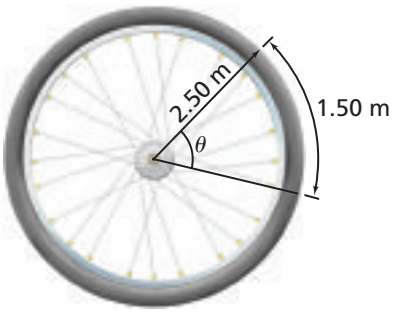


Figure 8-21

73. The outer edge of a truck tire that has a radius of 45 cm has a velocity of 23 m/s. What is the angular velocity of the tire in rad/s?

74. A steering wheel is rotated through 128°, as shown in **Figure 8-22.** Its radius is 22 cm. How far would a point on the steering wheel's edge move?

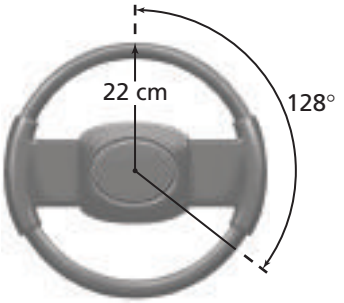


Figure 8-22

75. Propeller A propeller spins at 1880 rev/min.
a. What is its angular velocity in rad/s?
b. What is the angular displacement of the propeller in 2.50 s?

76. The propeller in the previous problem slows from 475 rev/min to 187 rev/min in 4.00 s. What is its angular acceleration?

77. An automobile wheel with a 9.00 cm radius, as shown in **Figure 8-23,** rotates at 2.50 rad/s. How fast does a point 7.00 cm from the center travel?



Figure 8-23

78. Washing Machine A washing machine's two spin cycles are 328 rev/min and 542 rev/min. The diameter of the drum is 0.43 m.
a. What is the ratio of the centripetal accelerations for the fast and slow spin cycles? Recall that $a_c = \frac{v^2}{r}$ and $v = rw$
b. What is the ratio of the linear velocity of an object at the surface of the drum for the fast and slow spin cycles?

79. Find the maximum centripetal acceleration in terms of g for the washing machine in problem 78.

80. A laboratory ultracentrifuge is designed to produce a centripetal acceleration of 0.35×10^6 *g* at a distance of 2.50 cm from the axis. What angular velocity in rev/min is required?

8.2 Rotational Dynamics

81. **Wrench** A bolt is to be tightened with a torque of 8.0 N·m. If you have a wrench that is 0.35 m long, what is the least amount of force you must exert?

82. What is the torque on a bolt produced by a 15-N force exerted perpendicular to a wrench that is 25 cm long, as shown in **Figure 8-24?**

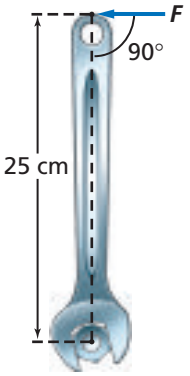


Figure 8-24

83. A toy consisting of two balls, each 0.45 kg, at the ends of a 0.46-m-long, thin, lightweight rod is shown in **Figure 8-25.** Find the moment of inertia of the toy. The moment of inertia is to be found about the center of the rod.

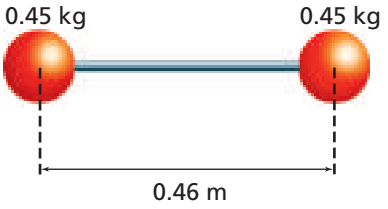


Figure 8-25

84. A bicycle wheel with a radius of 38 cm is given an angular acceleration of 2.67 rad/s^2 by applying a force of 0.35 N on the edge of the wheel. What is the wheel's moment of inertia?

85. **Toy Top** A toy top consists of a rod with a diameter of 8.0-mm and a disk of mass 0.0125 kg and a diameter of 3.5 cm. The moment of inertia of the rod can be neglected. The top is spun by wrapping a string around the rod and pulling it with a velocity that increases from zero to 3.0 m/s over 0.50 s.

a. What is the resulting angular velocity of the top?
b. What force was exerted on the string?

8.3 Equilibrium

86. A 12.5-kg board, 4.00 m long, is being held up on one end by Ahmed. He calls for help, and Judi responds.

a. What is the least force that Judi could exert to lift the board to the horizontal position? What part of the board should she lift to exert this force?
b. What is the greatest force that Judi could exert to lift the board to the horizontal position? What part of the board should she lift to exert this force?

87. Two people are holding up the ends of a 4.25-kg wooden board that is 1.75 m long. A 6.00-kg box sits on the board, 0.50 m from one end, as shown in **Figure 8-26.** What forces do the two people exert?

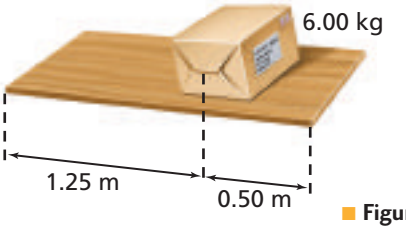


Figure 8-26

88. A car's specifications state that its weight distribution is 53 percent on the front tires and 47 percent on the rear tires. The wheel base is 2.46 m. Where is the car's center of mass?

Mixed Review

89. A wooden door of mass, *m*, and length, *l*, is held horizontally by Dan and Ajit. Dan suddenly drops his end.

a. What is the angular acceleration of the door just after Dan lets go?
b. Is the acceleration constant? Explain.

90. **Topsoil** Ten bags of topsoil, each weighing 175 N, are placed on a 2.43-m-long sheet of wood. They are stacked 0.50 m from one end of the sheet of wood, as shown in **Figure 8-27.** Two people lift the sheet of wood, one at each end. Ignoring the weight of the wood, how much force must each person exert?

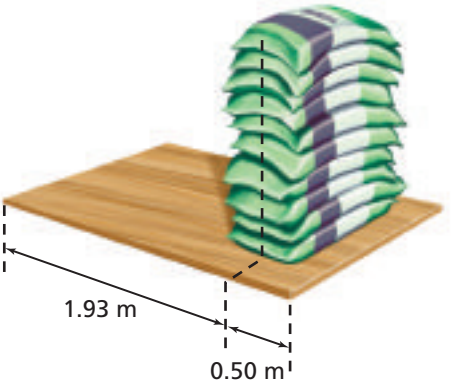


Figure 8-27

91. Basketball A basketball is rolled down the court. A regulation basketball has a diameter of 24.1 cm, a mass of 0.60 kg, and a moment of inertia of 5.8×10^{-3} kg·m^2. The basketball's initial velocity is 2.5 m/s.
- **a.** What is its initial angular velocity?
- **b.** The ball rolls a total of 12 m. How many revolutions does it make?
- **c.** What is its total angular displacement?

92. The basketball in the previous problem stops rolling after traveling 12 m.
- **a.** If its acceleration was constant, what was its angular acceleration?
- **b.** What torque was acting on it as it was slowing down?

93. A cylinder with a 50 m diameter, as shown in **Figure 8-28,** is at rest on a surface. A rope is wrapped around the cylinder and pulled. The cylinder rolls without slipping.
- **a.** After the rope has been pulled a distance of 2.50 m at a constant speed, how far has the center of mass of the cylinder moved?
- **b.** If the rope was pulled a distance of 2.50 m in 1.25 s, how fast was the center of mass of the cylinder moving?
- **c.** What is the angular velocity of the cylinder?

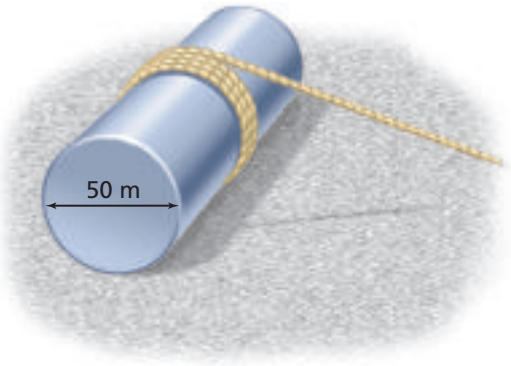


Figure 8-28

94. Hard Drive A hard drive on a modern computer spins at 7200 rpm (revolutions per minute). If the drive is designed to start from rest and reach operating speed in 1.5 s, what is the angular acceleration of the disk?

95. Speedometers Most speedometers in automobiles measure the angular velocity of the transmission and convert it to speed. How will increasing the diameter of the tires affect the reading of the speedometer?

96. A box is dragged across the floor using a rope that is a distance h above the floor. The coefficient of friction is 0.35. The box is 0.50 m high and 0.25 m wide. Find the force that just tips the box.

97. The second hand on a watch is 12 mm long. What is the velocity of its tip?

98. Lumber You buy a 2.44-m-long piece of 10 cm × 10 cm lumber. Your friend buys a piece of the same size and cuts it into two lengths, each 1.22 m long, as shown in **Figure 8-29.** You each carry your lumber on your shoulders.
- **a.** Which load is easier to lift? Why?
- **b.** Both you and your friend apply a torque with your hands to keep the lumber from rotating. Which load is easier to keep from rotating? Why?

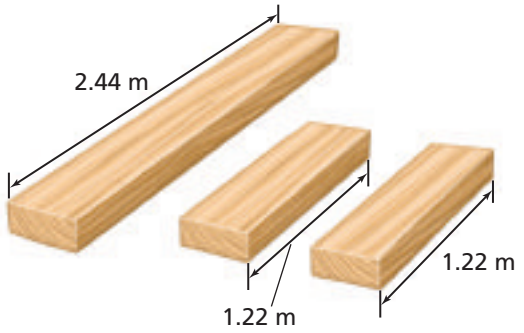


Figure 8-29

99. Surfboard Harris and Paul carry a surfboard that is 2.43 m long and weighs 143 N. Paul lifts one end with a force of 57 N.
- **a.** What force must Harris exert?
- **b.** What part of the board should Harris lift?

100. A steel beam that is 6.50 m long weighs 325 N. It rests on two supports, 3.00 m apart, with equal amounts of the beam extending from each end. Suki, who weighs 575 N, stands on the beam in the center and then walks toward one end. How close to the end can she come before the beam begins to tip?

Thinking Critically

101. Apply Concepts Consider a point on the edge of a rotating wheel.
- **a.** Under what conditions can the centripetal acceleration be zero?
- **b.** Under what conditions can the tangential (linear) acceleration be zero?
- **c.** Can the tangential acceleration be nonzero while the centripetal acceleration is zero? Explain.
- **d.** Can the centripetal acceleration be nonzero while the tangential acceleration is zero? Explain.

102. Apply Concepts When you apply the brakes in a car, the front end dips. Why?

103. **Analyze and Conclude** A banner is suspended from a horizontal, pivoted pole, as shown in **Figure 8-30.** The pole is 2.10 m long and weighs 175 N. The banner, which weighs 105 N, is suspended 1.80 m from the pivot point or axis of rotation. What is the tension in the cable supporting the pole?

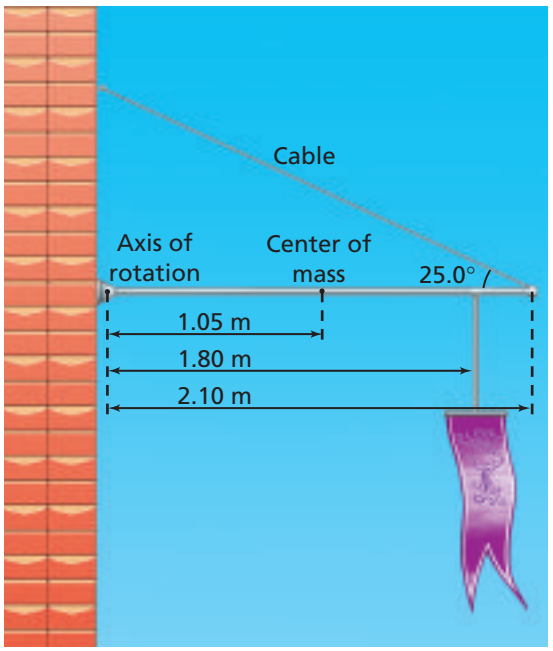


Figure 8-30

104. **Analyze and Conclude** A pivoted lamp pole is shown in **Figure 8-31.** The pole weighs 27 N, and the lamp weighs 64 N.

a. What is the torque caused by each force?
b. Determine the tension in the rope supporting the lamp pole.

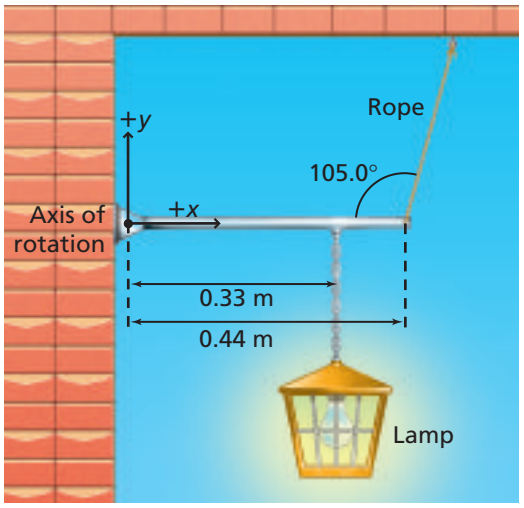


Figure 8-31

105. **Analyze and Conclude** Gerald and Evelyn carry the following objects up a flight of stairs: a large mirror, a dresser, and a television. Evelyn is at the front end, and Gerald is at the bottom end. Assume that both Evelyn and Gerald exert only upward forces.

a. Draw a free-body diagram showing Gerald and Evelyn exerting the same force on the mirror.
b. Draw a free-body diagram showing Gerald exerting more force on the bottom of the dresser.
c. Where would the center of mass of the television have to be so that Gerald carries all the weight?

Writing in Physics

106. Astronomers know that if a satellite is too close to a planet, it will be torn apart by tidal forces. That is, the difference in the gravitational force on the part of the satellite nearest the planet and the part farthest from the planet is stronger than the forces holding the satellite together. Do research on the Roche limit and determine how close the Moon would have to orbit Earth to be at the Roche limit.

107. Automobile engines are rated by the torque that they produce. Research and explain why torque is an important quantity to measure.

Cumulative Review

108. Two blocks, one of mass 2.0 kg and the other of mass 3.0 kg, are tied together with a massless rope. This rope is strung over a massless, resistance-free pulley. The blocks are released from rest. Find the following. (Chapter 4)

a. the tension in the rope
b. the acceleration of the blocks.

109. Eric sits on a see-saw. At what angle, relative to the vertical, will the component of his weight parallel to the plane be equal to one-third the perpendicular component of his weight? (Chapter 5)

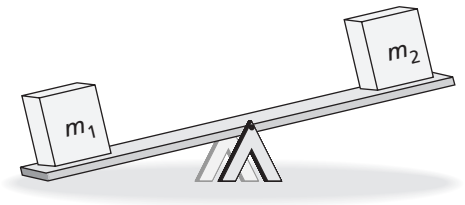
110. The pilot of a plane wants to reach an airport 325 km due north in 2.75 hours. A wind is blowing from the west at 30.0 km/h. What heading and airspeed should be chosen to reach the destination on time? (Chapter 6)

111. A 60.0-kg speed skater with a velocity of 18.0 m/s comes into a curve of 20.0-m radius. How much friction must be exerted between the skates and ice to negotiate the curve? (Chapter 6)

Multiple Choice

1. The illustration below shows two boxes on opposite ends of a board that is 3.0 m long. The board is supported in the middle by a fulcrum. The box on the left has a mass, m_1, of 25 kg, and the box on the right has a mass, m_2, of 15 kg. How far should the fulcrum be positioned from the left side of the board in order to balance the masses horizontally?

Ⓐ 0.38 m
Ⓑ 0.60 m
Ⓒ 1.1 m
Ⓓ 1.9 m



2. A force of 60 N is exerted on one end of a 1.0-m-long lever. The other end of the lever is attached to a rotating rod that is perpendicular to the lever. By pushing down on the end of the lever, you can rotate the rod. If the force on the lever is exerted at an angle of 30°, what torque is exerted on the lever? (sin 30° = 0.5; cos 30° = 0.87; tan 30° = 0.58)

Ⓐ 30 N
Ⓑ 52 N
Ⓒ 60 N
Ⓓ 69 N

3. A child attempts to use a wrench to remove a nut on a bicycle. Removing the nut requires a torque of 10 N·m. The maximum force the child is capable of exerting at a 90° angle is 50 N. What is the length of the wrench the child must use to remove the nut?

Ⓐ 0.1 m
Ⓑ 0.15 m
Ⓒ 0.2 m
Ⓓ 0.25 m

4. A car moves a distance of 420 m. Each tire on the car has a diameter of 42 cm. Which of the following shows how many revolutions each tire makes as they move that distance?

Ⓐ $\frac{5.0\times10^{1}}{\pi}$ rev
Ⓑ $\frac{1.0\times10^{2}}{\pi}$ rev
Ⓒ $\frac{1.5\times10^{2}}{\pi}$ rev
Ⓓ $\frac{1.0\times10^{3}}{\pi}$ rev

5. A thin hoop with a mass of 5.0 kg rotates about a perpendicular axis through its center. A force of 25 N is exerted tangentially to the hoop. If the hoop's radius is 2.0 m, what is its angular acceleration?

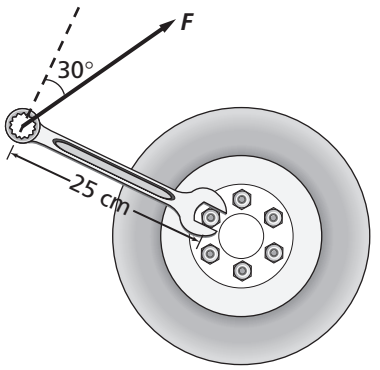
Ⓐ 1.3 rad/s
Ⓑ 2.5 rad/s
Ⓒ 5.0 rad/s
Ⓓ 6.3 rad/s

6. Two of the tires on a farmer's tractor have diameters of 1.5 m. If the farmer drives the tractor at a linear velocity of 3.0 m/s, what is the angular velocity of each tire?

Ⓐ 2.0 rad/s
Ⓑ 2.3 rad/s
Ⓒ 4.0 rad/s
Ⓓ 4.5 rad/s

Extended Answer

7. You use a 25-cm long wrench to remove the lug nuts on a car wheel, as shown in the illustration below. If you pull up on the end of the wrench with a force of 2.0×10^{2} N at an angle of 30°, what is the torque on the wrench? (sin 30° = 0.5, cos 30° = 0.87)



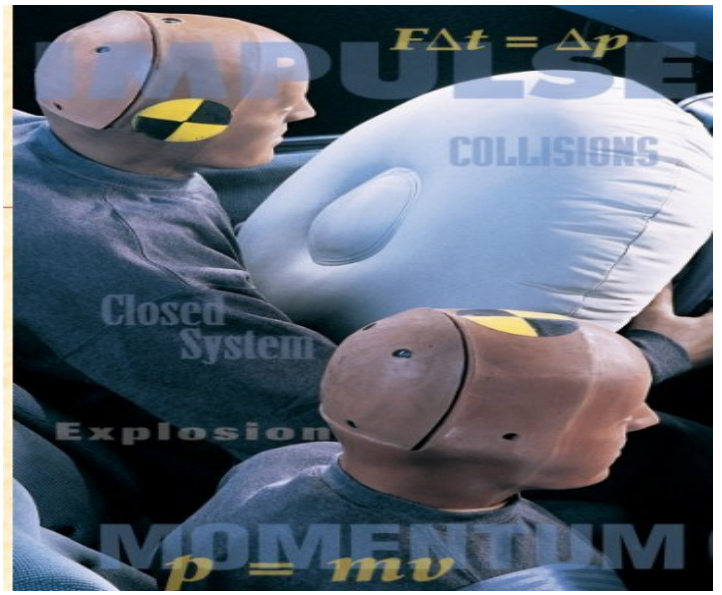
Test-Taking TIP

When Eliminating, Cross It Out

Consider each answer choice individually and cross out the ones you have eliminated. If you cannot write in the test booklet, use the scratch paper to list and cross off the answer choices. You will save time and stop yourself from choosing an answer you have mentally eliminated.

Chapter 3

Momentum and Its Conservation



- You will describe momentum and impulse and apply them to the interactions between objects.
- You will relate Newton's third law of motion to conservation of momentum.
- You will explore the momentum of rotating objects.

Why It's Important

Momentum is the key to success in many sporting events, including baseball, football, ice hockey, and tennis.

Baseball Every baseball player dreams of hitting a home run. When a player hits the ball, at the moment of collision, the ball and the bat are deformed by the collision. The resulting change in momentum determines the batter's success.

LAUNCH Lab

What happens when a hollow plastic ball strikes a bocce ball?

Question

What direction will a hollow plastic ball and a bocce ball move after a head-on collision?

Procedure

1. Roll a bocce ball and a hollow plastic ball toward each other on a smooth surface.
2. Observe the direction each one moves after the collision.
3. Repeat the experiment, this time keeping the bocce ball stationary, while rolling the hollow plastic ball toward it.
4. Observe the direction each one moves after the collision.
5. Repeat the experiment one more time, but keep the hollow plastic ball stationary, while rolling the bocce ball toward it.
6. Observe the direction each one moves after the collision.

Analysis

What factors affect how fast the balls move after the collision? What factors determine the direction each one moves after the collision?

Critical Thinking What factor(s) would cause the bocce ball to move backward after colliding with the hollow plastic ball?



1. Impulse and Momentum

It is always exciting to watch a baseball player hit a home run. The pitcher fires the baseball toward the plate. The batter swings at the baseball and the baseball recoils from the impact of the bat at high speed. Rather than concentrating on the force between the baseball and bat and their resulting accelerations, as in previous chapters, you will approach this collision in a different way in this chapter. The first step in analyzing this type of interaction is to describe what happens before, during, and after the collision between the baseball and bat. You can simplify the collision between the baseball and the bat by making the assumption that all motion is in the horizontal direction. Before the collision, the baseball moves toward the bat. During the collision, the baseball is squashed against the bat. After the collision, however, the baseball moves at a higher velocity away from the bat, and the bat continues in its path, but at a slower velocity.

Objectives

- **Define** the momentum of an object.
- **Determine** the impulse given to an object.
- **Define** the angular momentum of an object.

Vocabulary

impulse
momentum
impulse-momentum theorem
angular momentum
angular impulse-angular momentum theorem

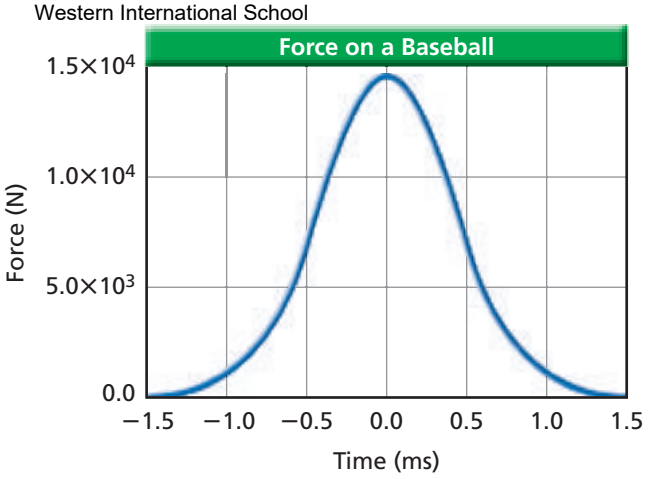


Figure 9-1 The force acting on a baseball increases, then rapidly decreases during a collision, as shown in this force-time graph.

Impulse and Momentum

How are the velocities of the ball, before and after the collision, related to the force acting on it? Newton's second law of motion describes how the velocity of an object is changed by a net force acting on it. The change in velocity of the ball must have been caused by the force exerted by the bat on the ball. The force changes over time, as shown in **Figure 9-1.** Just after contact is made, the ball is squeezed, and the force increases. After the force reaches its maximum, which is more than 10,000 times the weight of the ball, the ball recovers its shape and snaps away from the bat. The force rapidly returns to zero. This whole event takes place within about 3.0 ms. How can you calculate the change in velocity of the baseball?

Impulse Newton's second law of motion, $\boldsymbol{F} = m\boldsymbol{a}$, can be rewritten by using the definition of acceleration as the change in velocity divided by the time needed to make that change. It can be represented by the following equation:

$$\boldsymbol{F} = m\boldsymbol{a} = m\left(\frac{\Delta \boldsymbol{v}}{\Delta t}\right)$$

Multiplying both sides of the equation by the time interval, Δt, results in the following equation:

$$\boldsymbol{F}\Delta t = m\Delta \boldsymbol{v}$$

Impulse, or $\boldsymbol{F}\Delta t$, is the product of the average force on an object and the time interval over which it acts. Impulse is measured in newton-seconds. For instances in which the force varies with time, the magnitude of an impulse is found by determining the area under the curve of a force-time graph, such as the one shown in Figure 9-1.

The right side of the equation, $m\Delta \boldsymbol{v}$, involves the change in velocity: $\Delta \boldsymbol{v} = \boldsymbol{v}_f - \boldsymbol{v}_i$. Therefore, $m\Delta \boldsymbol{v} = m\boldsymbol{v}_f - m\boldsymbol{v}_i$. The product of the object's mass, m, and the object's velocity, $\boldsymbol{v}$, is defined as the **momentum** of the object. Momentum is measured in kg·m/s. An object's momentum, also known as linear momentum, is represented by the following equation.

Momentum $\boldsymbol{p} = m\boldsymbol{v}$

The momentum of an object is equal to the mass of the object times the object's velocity.

Recall the equation $\boldsymbol{F}\Delta t = m\Delta \boldsymbol{v} = m\boldsymbol{v}_f - m\boldsymbol{v}_i$. Because $m\boldsymbol{v}_f = \boldsymbol{p}_f$ and $m\boldsymbol{v}_i = \boldsymbol{p}_i$, this equation can be rewritten as follows: $\boldsymbol{F}\Delta t = m\Delta \boldsymbol{v} = \boldsymbol{p}_f - \boldsymbol{p}_i$. The right side of this equation, $\boldsymbol{p}_f - \boldsymbol{p}_i$, describes the change in momentum of an object. Thus, the impulse on an object is equal to the change in its momentum, which is called the **impulse-momentum theorem.** The impulse-momentum theorem is represented by the following equation.

Impulse-Momentum Theorem $\boldsymbol{F}\Delta t = \boldsymbol{p}_f - \boldsymbol{p}_i$

The impulse on an object is equal to the object's final momentum minus the object's initial momentum.

If the force on an object is constant, the impulse is the product of the force multiplied by the time interval over which it acts. Generally, the force is not constant, however, and the impulse is found by using an average force multiplied by the time interval over which it acts, or by finding the area under a force-time graph.

Because velocity is a vector, momentum also is a vector. Similarly, impulse is a vector because force is a vector. This means that signs will be important for motion in one dimension.

Using the Impulse-Momentum Theorem

What is the change in momentum of a baseball? From the impulse-momentum theorem, you know that the change in momentum is equal to the impulse acting on it. The impulse on a baseball can be calculated by using a force-time graph. In Figure 9-1, the area under the curve is approximately 13.1 N·s. The direction of the impulse is in the direction of the force. Therefore, the change in momentum of the ball also is 13.1 N·s. Because 1 N·s is equal to 1 kg·m/s, the momentum gained by the ball is 13.1 kg·m/s in the direction of the force acting on it.

Assume that a batter hits a fastball. Before the collision of the ball and bat, the ball, with a mass of 0.145 kg, has a velocity of −38 m/s. Assume that the positive direction is toward the pitcher. Therefore, the baseball's momentum is $p_i = (0.145\ \text{kg})(-38\ \text{m/s}) = -5.5\ \text{kg·m/s}$.

What is the momentum of the ball after the collision? Solve the impulse-momentum theorem for the final momentum: $p_f = p_i + F\Delta t$. The ball's final momentum is the sum of the initial momentum and the impulse. Thus, the ball's final momentum is calculated as follows.

$$p_f = p_i + 13.1\ \text{kg·m/s}$$
$$= -5.5\ \text{kg·m/s} + 13.1\ \text{kg·m/s} = +7.6\ \text{kg·m/s}$$

What is the baseball's final velocity? Because $p_f = mv_f$, solving for $\boldsymbol{v}_f$ yields the following:

$$v_f = \frac{p_f}{m} = \frac{+7.6\ \text{kg·m/s}}{+0.145\ \text{kg·m/s}} = +52\ \text{m/s}$$

A speed of 52 m/s is fast enough to clear most outfield fences if the baseball is hit in the correct direction.

Using the Impulse-Momentum Theorem to Save Lives

A large change in momentum occurs only when there is a large impulse. A large impulse can result either from a large force acting over a short period of time or from a smaller force acting over a long period of time.

What happens to the driver when a crash suddenly stops a car? An impulse is needed to bring the driver's momentum to zero. According to the impulse-momentum equation, $\boldsymbol{F}\Delta t = \boldsymbol{p}_f - \boldsymbol{p}_i$. The final momentum, $\boldsymbol{p}_f$, is zero. The initial momentum, $\boldsymbol{p}_i$, is the same with or without an air bag. Thus, the impulse, $\boldsymbol{F}\Delta t$, also is the same. An air bag, such as the one shown in **Figure 9-2,** reduces the force by increasing the time interval during which it acts. It also exerts the force over a larger area of the person's body, thereby reducing the likelihood of injuries.

APPLYING PHYSICS

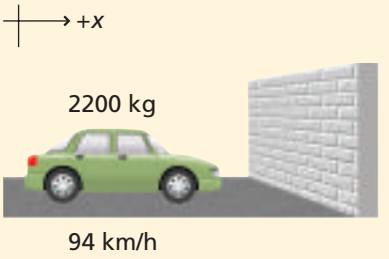
▶ **Running Shoes** Running is hard on the feet. When a runner's foot strikes the ground, the force exerted by the ground on it is as much as four times the runner's weight. The cushioning in an athletic shoe is designed to reduce this force by lengthening the time interval over which the force is exerted. ◀



Figure 9-2 An air bag is inflated during a collision when the force due to the impact triggers the sensor. The chemicals in the air bag's inflation system react and produce a gas that rapidly inflates the air bag.

EXAMPLE Problem 1

Average Force A 2200-kg vehicle traveling at 94 km/h (26 m/s) can be stopped in 21 s by gently applying the brakes. It can be stopped in 3.8 s if the driver slams on the brakes, or in 0.22 s if it hits a concrete wall. What average force is exerted on the vehicle in each of these stops?



1 Analyze and Sketch the Problem

- Sketch the system.
- Include a coordinate axis and select the positive direction to be the direction of the velocity of the car.
- Draw a vector diagram for momentum and impulse.

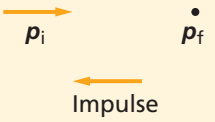
Known:

$m = 2200\ \text{kg}$
$\boldsymbol{v}_i = +26\ \text{m/s}$
$\boldsymbol{v}_f = +0.0\ \text{m/s}$
$\Delta t_{\text{gentle braking}} = 21\ \text{s}$
$\Delta t_{\text{hard braking}} = 3.8\ \text{s}$
$\Delta t_{\text{hitting a wall}} = 0.22\ \text{s}$

Unknown:

$\boldsymbol{F}_{\text{gentle braking}} = ?$
$\boldsymbol{F}_{\text{hard braking}} = ?$
$\boldsymbol{F}_{\text{hitting a wall}} = ?$

Vector diagram



2 Solve for the Unknown

Determine the initial momentum, p_i.

$p_i = mv_i$
$= (2200\ \text{kg})(+26\ \text{m/s})$ **Substitute $m = 2200$ kg, $v_i = +26$ m/s**
$= +5.7\times10^4\ \text{kg·m/s}$

Determine the final momentum, p_f.

$p_f = mv_f$
$= (2200\ \text{kg})(+0.0\ \text{m/s})$ **Substitute $m = 2200$ kg, $v_f = +0.0$ m/s**
$= +0.0\ \text{kg·m/s}$

Apply the impulse-momentum theorem to obtain the force needed to stop the vehicle.

$F\Delta t = p_f - p_i$
$F\Delta t = (+0.0\ \text{kg·m/s}) - (5.7\times10^4\ \text{kg·m/s})$ **Substitute $p_f = 0.0$ kg·m/s, $p_i = 5.7\times10^4$ kg·m/s**
$= -5.7\times10^4\ \text{kg·m/s}$

$$F = \frac{-5.7\times10^4\ \text{kg·m/s}}{\Delta t}$$

$$F_{\text{gentle braking}} = \frac{-5.7\times10^4\ \text{kg·m/s}}{21\ \text{s}}$$ **Substitute $\Delta t_{\text{gentle braking}} = 21$ s**
$= -2.7\times10^3\ \text{N}$

$$F_{\text{hard braking}} = \frac{-5.7\times10^4\ \text{kg·m/s}}{3.8\ \text{s}}$$ **Substitute $\Delta t_{\text{hard braking}} = 3.8$ s**
$= -1.5\times10^4\ \text{N}$

$$F_{\text{hitting a wall}} = \frac{-5.7\times10^4\ \text{kg·m/s}}{0.22\ \text{s}}$$ **Substitute $\Delta t_{\text{hitting a wall}} = 0.22$ s**
$= -2.6\times10^5\ \text{N}$

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3 Evaluate the Answer

- **Are the units correct?** Force is measured in newtons.
- **Does the direction make sense?** Force is exerted in the direction opposite to the velocity of the car and thus, is negative.
- **Is the magnitude realistic?** People weigh hundreds of newtons, so it is reasonable that the force needed to stop a car would be in the thousands of newtons. The impulse is the same for all three stops. Thus, as the stopping time is shortened by more than a factor of 10, the force is increased by more than a factor of 10.

PRACTICE Problems

Additional Problems, Appendix B

1. A compact car, with mass 725 kg, is moving at 115 km/h toward the east. Sketch the moving car.
   - **a.** Find the magnitude and direction of its momentum. Draw an arrow on your sketch showing the momentum.
   - **b.** A second car, with a mass of 2175 kg, has the same momentum. What is its velocity?
2. The driver of the compact car in the previous problem suddenly applies the brakes hard for 2.0 s. As a result, an average force of 5.0×10^3 N is exerted on the car to slow it down.
   - **a.** What is the change in momentum; that is, the magnitude and direction of the impulse, on the car?
   - **b.** Complete the "before" and "after" sketches, and determine the momentum and the velocity of the car now.
3. A 7.0-kg bowling ball is rolling down the alley with a velocity of 2.0 m/s. For each impulse, shown in **Figures 9-3a** and **9-3b,** find the resulting speed and direction of motion of the bowling ball.
4. The driver accelerates a 240.0-kg snowmobile, which results in a force being exerted that speeds up the snowmobile from 6.00 m/s to 28.0 m/s over a time interval of 60.0 s.
   - **a.** Sketch the event, showing the initial and final situations.
   - **b.** What is the snowmobile's change in momentum? What is the impulse on the snowmobile?
   - **c.** What is the magnitude of the average force that is exerted on the snowmobile?
5. Suppose a 60.0-kg person was in the vehicle that hit the concrete wall in Example Problem 1. The velocity of the person equals that of the car both before and after the crash, and the velocity changes in 0.20 s. Sketch the problem.
   - **a.** What is the average force exerted on the person?
   - **b.** Some people think that they can stop their bodies from lurching forward in a vehicle that is suddenly braking by putting their hands on the dashboard. Find the mass of an object that has a weight equal to the force you just calculated. Could you lift such a mass? Are you strong enough to stop your body with your arms?

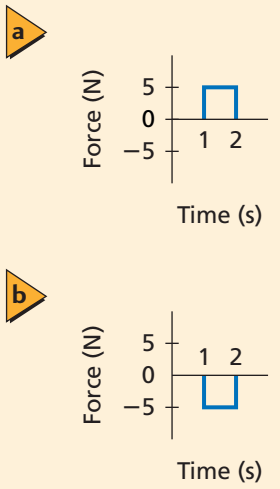


Figure 9-3

2. Angular Momentum

As you learned in Chapter 8, the angular velocity of a rotating object changes only if torque is applied to it. This is a statement of Newton's law for rotational motion, $\tau = I\Delta\omega/\Delta t$. This equation can be rearranged in the same way as Newton's second law of motion was, to produce $\tau\Delta t = I\Delta\omega$.

The left side of this equation, $\tau\Delta t$, is the angular impulse of the rotating object. The right side can be rewritten as $\Delta\omega = \omega_f - \omega_i$. The product of a rotating object's moment of inertia and angular velocity is called **angular momentum,** which is represented by the symbol L. The angular momentum of an object can be represented by the following equation.

Angular Momentum $L = I\omega$

The angular momentum of an object is equal to the product of the object's moment of inertia and the object's angular velocity.

Angular momentum is measured in kg·m^2/s. Just as the linear momentum of an object changes when an impulse acts on it, the angular momentum of an object changes when an angular impulse acts on it. Thus, the angular impulse on the object is equal to the change in the object's angular momentum, which is called the **angular impulse-angular momentum theorem.** The angular impulse-angular momentum theorem is represented by the following equation.

Angular Impulse-Angular Momentum Theorem $\tau\Delta t = L_f - L_i$

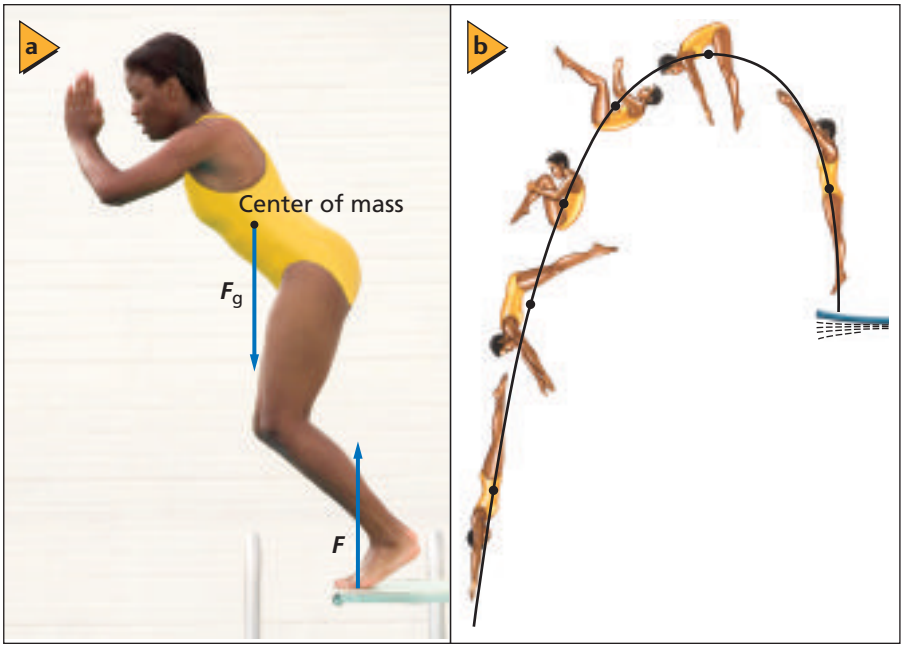
The angular impulse on an object is equal to the object's final angular momentum minus the object's initial angular momentum.

If there are no forces acting on an object, its linear momentum is constant. If there are no torques acting on an object, its angular momentum is also constant. Because an object's mass cannot be changed, if its momentum is constant, then its velocity is also constant. In the case of angular momentum, however, the object's angular velocity does not remain constant. This is because the moment of inertia depends on the object's mass and the way it is distributed about the axis of rotation or revolution. Thus, the angular velocity of an object can change even if no torques are acting on it.

Astronomy Connection

Consider, for example, a planet orbiting the Sun. The torque on the planet is zero because the gravitational force acts directly toward the Sun. Therefore, the planet's angular momentum is constant. When the distance between the planet and the Sun decreases, however, the planet's moment of inertia of revolution in orbit about the Sun also decreases. Thus, the planet's angular velocity increases and it moves faster. This is an explanation of Kepler's second law of planetary motion, based on Newton's laws of motion.

Figure 9-4 The diver's center of mass is in front of her feet as she gets ready to dive **(a).** As the diver changes her moment of inertia by moving her arms and legs to increase her angular momentum, the location of the center of mass changes, but the path of the center of mass remains a parabola **(b).**



Consider the diver in **Figure 9-4.** How does she start rotating her body? She uses the diving board to apply an external torque to her body. Then, she moves her center of mass in front of her feet and uses the board to give a final upward push to her feet. This torque acts over time, Δt, and thus increases the angular momentum of the diver.

Before the diver reaches the water, she can change her angular velocity by changing her moment of inertia. She may go into a tuck position, grabbing her knees with her hands. By moving her mass closer to the axis of rotation, the diver decreases her moment of inertia and increases her angular velocity. When she nears the water, she stretches her body straight, thereby increasing the moment of inertia and reducing the angular velocity. As a result, she goes straight into the water.

An ice-skater uses a similar method to spin. To begin rotating on one foot, the ice-skater applies an external torque to her body by pushing a portion of the other skate into the ice, as shown in **Figure 9-5.** If she pushes on the ice in one direction, the ice will exert a force on her in the opposite direction. The force results in a torque if the force is exerted some distance away from the pivot point, and in a direction that is not toward it. The greatest torque for a given force will result if the push is perpendicular to the lever arm.

The ice-skater then can control his angular velocity by changing her moment of inertia. Both arms and one leg can be extended from the body to slow the rotation, or pulled in close to the axis of rotation to speed it up. To stop spinning, another torque must be exerted by using the second skate to create a way for the ice to exert the needed force.



Figure 9-5 To spin on one foot, an ice-skater extends one leg and pushes on the ice. The ice exerts an equal and opposite force on her body and produces an external torque.

1. Section Review

6. Momentum Is the momentum of a car traveling south different from that of the same car when it travels north at the same speed? Draw the momentum vectors to support your answer.

7. Impulse and Momentum When you jump from a height to the ground, you let your legs bend at the knees as your feet hit the floor. Explain why you do this in terms of the physics concepts introduced in this chapter.

8. Momentum Which has more momentum, a supertanker tied to a dock or a falling raindrop?

9. Impulse and Momentum A 0.174-kg softball is pitched horizontally at 26.0 m/s. The ball moves in the opposite direction at 38.0 m/s after it is hit by the bat.

- **a.** Draw arrows showing the ball's momentum before and after the bat hits it.
- **b.** What is the change in momentum of the ball?
- **c.** What is the impulse delivered by the bat?
- **d.** If the bat and softball are in contact for 0.80 ms, what is the average force that the bat exerts on the ball?

10. Momentum The speed of a basketball as it is dribbled is the same when the ball is going toward the floor as it is when the ball rises from the floor. Is the basketball's change in momentum equal to zero when it hits the floor? If not, in which direction is the change in momentum? Draw the basketball's momentum vectors before and after it hits the floor.

11. Angular Momentum An ice-skater spins with his arms outstretched. When he pulls his arms in and raises them above his head, he spins much faster than before. Did a torque act on the ice-skater? If not, how could his angular velocity have increased?

12. Critical Thinking An archer shoots arrows at a target. Some of the arrows stick in the target, while others bounce off. Assuming that the masses of the arrows and the velocities of the arrows are the same, which arrows produce a bigger impulse on the target? *Hint: Draw a diagram to show the momentum of the arrows before and after hitting the target for the two instances.*

3. Conservation of Momentum

Objectives

- **Relate** Newton's third law to conservation of momentum.
- **Recognize** the conditions under which momentum is conserved.
- **Solve** conservation of momentum problems.

Vocabulary

closed system
isolated system
law of conservation of momentum
law of conservation of angular momentum

In the first section of this chapter, you learned how a force applied during a time interval changes the momentum of a baseball. In the discussion of Newton's third law of motion, you learned that forces are the result of interactions between two objects. The force of a bat on a ball is accompanied by an equal and opposite force of the ball on the bat. Does the momentum of the bat, therefore, also change?

Two-Particle Collisions

The bat, the hand and arm of the batter, and the ground on which the batter is standing are all objects that interact when a batter hits the ball. Thus, the bat cannot be considered a single object. In contrast to this complex system, examine for a moment the much simpler system shown in **Figure 9-6,** the collision of two balls.

During the collision of the two balls, each one briefly exerts a force on the other. Despite the differences in sizes and velocities of the balls, the forces that they exert on each other are equal and opposite, according to Newton's third law of motion. These forces are represented by the following equation: $\boldsymbol{F}_{\text{D on C}} = -\boldsymbol{F}_{\text{C on D}}$

How do the impulses imparted by both balls compare? Because the time intervals over which the forces are exerted are the same, the impulses must be equal in magnitude but opposite in direction. How did the momenta of the balls change as a result of the collision?

According to the impulse-momentum theorem, the change in momentum is equal to the impulse. Compare the changes in the momenta of the two balls.

$$\text{For ball C: } \boldsymbol{p}_{\text{Cf}} - \boldsymbol{p}_{\text{Ci}} = \boldsymbol{F}_{\text{D on C}}\Delta t$$

$$\text{For ball D: } \boldsymbol{p}_{\text{Df}} - \boldsymbol{p}_{\text{Di}} = \boldsymbol{F}_{\text{C on D}}\Delta t$$

Because the time interval over which the forces were exerted is the same, the impulses are equal in magnitude, but opposite in direction. According to Newton's third law of motion, $-\boldsymbol{F}_{\text{C on D}} = \boldsymbol{F}_{\text{D on C}}$. Thus,

$$\boldsymbol{p}_{\text{Cf}} - \boldsymbol{p}_{\text{Ci}} = -(\boldsymbol{p}_{\text{Df}} - \boldsymbol{p}_{\text{Di}}), \text{ or } \boldsymbol{p}_{\text{Cf}} + \boldsymbol{p}_{\text{Df}} = \boldsymbol{p}_{\text{Ci}} + \boldsymbol{p}_{\text{Di}}.$$

This equation states that the sum of the momenta of the balls is the same before and after the collision. That is, the momentum gained by ball D is equal to the momentum lost by ball C. If the system is defined as the two balls, the momentum of the system is constant, and therefore, momentum is conserved for the system.

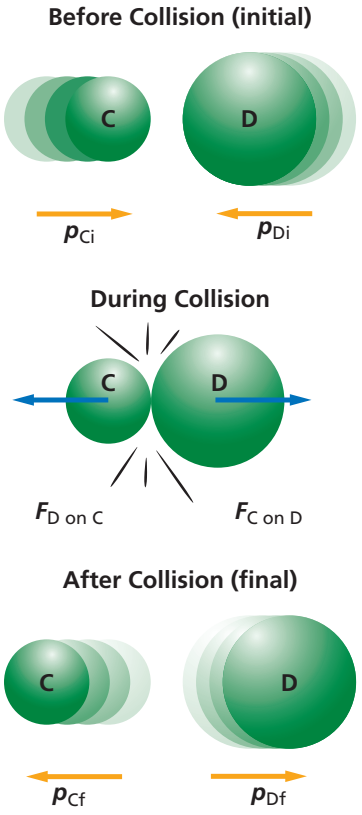


Figure 9-6 When two balls collide, they exert forces on each other that change their momenta.

Momentum in a Closed, Isolated System

Under what conditions is the momentum of the system of two balls conserved? The first and most obvious condition is that no balls are lost and no balls are gained. Such a system, which does not gain or lose mass, is said to be a **closed system.** The second condition required to conserve the momentum of a system is that the forces involved are internal forces; that is, there are no forces acting on the system by objects outside of it.

When the net external force on a closed system is zero, the system is described as an **isolated system.** No system on Earth can be said to be absolutely isolated, however, because there will always be some interactions between a system and its surroundings. Often, these interactions are small enough to be ignored when solving physics problems.

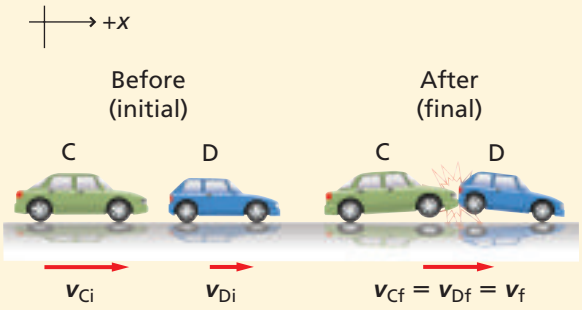
Systems can contain any number of objects, and the objects can stick together or come apart in a collision. Under these conditions, the **law of conservation of momentum** states that the momentum of any closed, isolated system does not change. This law will enable you to make a connection between conditions, before and after an interaction, without knowing any of the details of the interaction.

EXAMPLE Problem 2

Speed A 1875-kg car going 23 m/s rear-ends a 1025-kg compact car going 17 m/s on ice in the same direction. The two cars stick together. How fast do the two cars move together immediately after the collision?

1 Analyze and Sketch the Problem

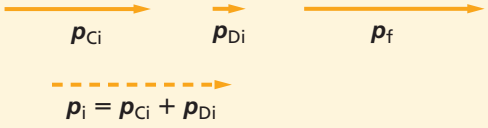
- Define the system.
- Establish a coordinate system.
- Sketch the situation showing the "before" and "after" states.
- Draw a vector diagram for the momentum.



Known:
$m_C = 1875$ kg
$v_{Ci} = +23$ m/s
$m_D = 1025$ kg
$v_{Di} = +17$ m/s

Unknown:
$v_f = ?$

Vector diagram



2 Solve for the Unknown

Momentum is conserved because the ice makes the total external force on the cars nearly zero.

$$p_i = p_f$$
$$p_{Ci} + p_{Di} = p_{Cf} + p_{Df}$$
$$m_C v_{Ci} + m_D v_{Di} = m_C v_{Cf} + m_D v_{Df}$$

Because the two cars stick together, their velocities after the collision, denoted as v_f, are equal.

$$v_{Cf} = v_{Df} = v_f$$
$$m_C v_{Ci} + m_D v_{Di} = (m_C + m_D) v_f$$

Solve for v_f.

$$v_f = \frac{(m_C v_{Ci} + m_D v_{Di})}{(m_C + m_D)}$$
$$= \frac{(1875\text{ kg})(+23\text{ m/s}) + (1025\text{ kg})(+17\text{ m/s})}{(1875\text{ kg} + 1025\text{ kg})}$$
$$= +21\text{ m/s}$$

Substitute $m_C = 1875$ kg, $v_{Ci} = +23$ m/s, $m_D = 1025$ kg, $v_{Di} = +17$ m/s

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3 Evaluate the Answer

- **Are the units correct?** Velocity is measured in m/s.
- **Does the direction make sense?** v_i and v_f are in the positive direction; therefore, v_f should be positive.
- **Is the magnitude realistic?** The magnitude of v_f is between the initial speeds of the two cars, but closer to the speed of the more massive one, so it is reasonable.

PRACTICE Problems

Additional Problems, Appendix B

13. Two freight cars, each with a mass of 3.0×10^5 kg, collide and stick together. One was initially moving at 2.2 m/s, and the other was at rest. What is their final speed?

14. A 0.105-kg hockey puck moving at 24 m/s is caught and held by a 75-kg goalie at rest. With what speed does the goalie slide on the ice?

15. A 35.0-g bullet strikes a 5.0-kg stationary piece of lumber and embeds itself in the wood. The piece of lumber and bullet fly off together at 8.6 m/s. What was the original speed of the bullet?

16. A 35.0-g bullet moving at 475 m/s strikes a 2.5-kg bag of flour that is on ice, at rest. The bullet passes through the bag, as shown in **Figure 9-7,** and exits it at 275 m/s. How fast is the bag moving when the bullet exits?

17. The bullet in the previous problem strikes a 2.5-kg steel ball that is at rest. The bullet bounces backward after its collision at a speed of 5.0 m/s. How fast is the ball moving when the bullet bounces backward?

18. A 0.50-kg ball that is traveling at 6.0 m/s collides head-on with a 1.00-kg ball moving in the opposite direction at a speed of 12.0 m/s. The 0.50-kg ball bounces backward at 14 m/s after the collision. Find the speed of the second ball after the collision.



Figure 9-7

Recoil

It is very important to define a system carefully. The momentum of a baseball changes when the external force of a bat is exerted on it. The baseball, therefore, is not an isolated system. On the other hand, the total momentum of two colliding balls within an isolated system does not change because all forces are between the objects within the system.

Can you find the final velocities of the two in-line skaters in **Figure 9-8?** Assume that they are skating on a smooth surface with no external forces. They both start at rest, one behind the other.



Figure 9-8 The internal forces exerted by Skater C, the boy, and Skater D, the girl, cannot change the total momentum of the system.

Skater C, the boy, gives skater D, the girl, a push. Now, both skaters are moving, making this situation similar to that of an explosion. Because the push was an internal force, you can use the law of conservation of momentum to find the skaters' relative velocities. The total momentum of the system was zero before the push. Therefore, it must be zero after the push.

Before		After
$\boldsymbol{p}_{Ci} + \boldsymbol{p}_{Di}$	=	$\boldsymbol{p}_{Cf} + \boldsymbol{p}_{Df}$
0	=	$\boldsymbol{p}_{Cf} + \boldsymbol{p}_{Df}$
$\boldsymbol{p}_{Cf}$	=	$-\boldsymbol{p}_{Df}$
$m_C\boldsymbol{v}_{Cf}$	=	$-m_D\boldsymbol{v}_{Df}$

The coordinate system was chosen so that the positive direction is to the left. The momenta of the skaters after the push are equal in magnitude but opposite in direction. The backward motion of skater C is an example of recoil. Are the skaters' velocities equal and opposite? The last equation shown above, for the velocity of skater C, can be rewritten as follows:

$$\boldsymbol{v}_{Cf} = \left(\frac{-m_D}{m_C}\right)\boldsymbol{v}_{Df}$$

The velocities depend on the skaters' relative masses. If skater C has a mass of 68.0 kg and skater D's mass is 45.4 kg, then the ratio of their velocities will be 68.0 : 45.4, or 1.50. The less massive skater moves at the greater velocity. Without more information about how hard skater C pushed skater D, however, you cannot find the velocity of each skater.

Propulsion in Space

How does a rocket in space change its velocity? The rocket carries both fuel and oxidizer. When the fuel and oxidizer combine in the rocket motor, the resulting hot gases leave the exhaust nozzle at high speed. If the rocket and chemicals are the system, then the system is a closed system. The forces that expel the gases are internal forces, so the system is also an isolated system. Thus, objects in space can accelerate by using the law of conservation of momentum and Newton's third law of motion.

A NASA space probe, called *Deep Space 1*, performed a flyby of an asteroid a few years ago. The most unusual of the 11 new technologies on board was an ion engine that exerts as much force as a sheet of paper resting on a person's hand. The ion engine shown in **Figure 9-9,** operates differently from a traditional rocket engine. In a traditional rocket engine, the products of the chemical reaction taking place in the combustion chamber are released at high speed from the rear. In the ion engine, however, xenon atoms are expelled at a speed of 30 km/s, producing a force of only 0.092 N. How can such a small force create a significant change in the momentum of the probe? Instead of operating for only a few minutes, as the traditional chemical rockets do, the ion engine can run continuously for days, weeks, or months. Therefore, the impulse delivered by the engine is large enough to increase the momentum of the 490-kg spacecraft until it reaches the speed needed to complete its mission.

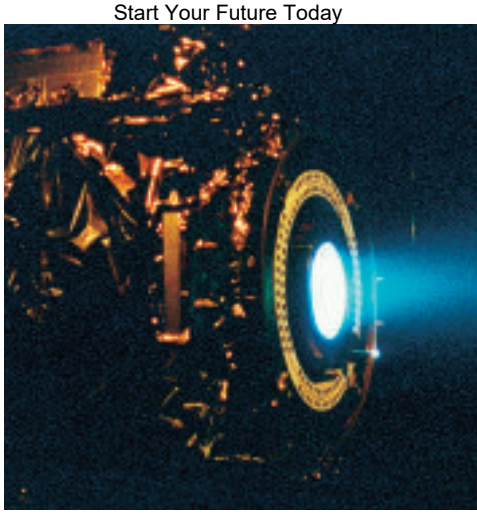


Figure 9-9 The xenon atoms in the ion engine are ionized by bombarding them with electrons. Then, the positively charged xenon ions are accelerated to high speeds.

MINI LAB

Rebound Height

An object's momentum is the product of its mass and velocity.

1. Drop a large rubber ball from about 15 cm above a table.

2. Measure and record the ball's rebound height.

3. Repeat steps 1–2 with a small rubber ball.

4. Hold the small rubber ball on top of, and in contact with, the large rubber ball.

5. Release the two rubber balls from the same height, so that they fall together.

6. Measure the rebound heights of both rubber balls.

Analyze and Conclude

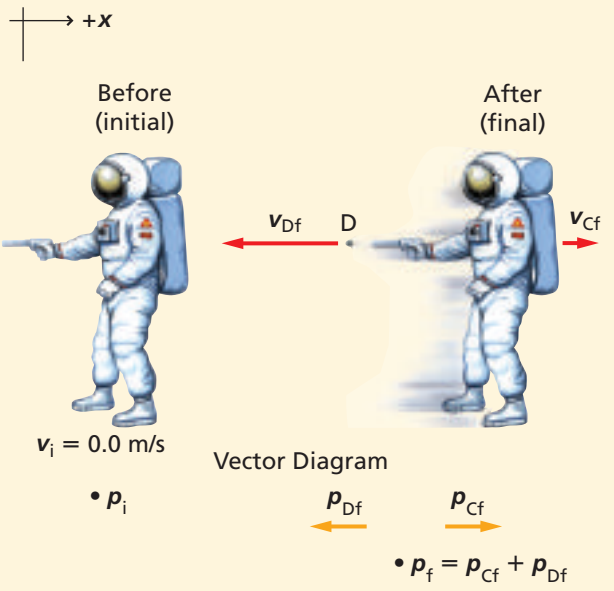
7. Describe the rebound height of each rubber ball dropped by itself.

8. Compare and contrast the rebound heights from number 7 with those from number 6.

9. Explain your observations.

EXAMPLE Problem 3

Speed An astronaut at rest in space fires a thruster pistol that expels 35 g of hot gas at 875 m/s. The combined mass of the astronaut and pistol is 84 kg. How fast and in what direction is the astronaut moving after firing the pistol?



1 Analyze and Sketch the Problem

- Define the system.
- Establish a coordinate axis.
- Sketch the "before" and "after" conditions.
- Draw a vector diagram showing momenta.

Known:

$m_C = 84$ kg
$m_D = 0.035$ kg
$v_{Ci} = v_{Di} = +0.0$ m/s
$v_{Df} = -875$ m/s

Unknown:

$v_{Cf} = ?$

2 Solve for the Unknown

The system is the astronaut, the gun, and the chemicals that produce the gas.

$p_i = p_{Ci} + p_{Di} = +0.0\ \text{kg·m/s}$ **Before the pistol is fired, all parts of the system are at rest; thus, the initial momentum is zero.**

Use the law of conservation of momentum to find p_f.

$$p_i = p_f$$
$$+0.0\ \text{kg·m/s} = p_{Cf} + p_{Df}$$
$$p_{Cf} = -p_{Df}$$

The momentum of the astronaut is equal in magnitude, but opposite in direction to the momentum of the gas leaving the pistol.

Solve for the final velocity of the astronaut, v_{Cf}.

$$m_C v_{Cf} = -m_D v_{Df}$$
$$v_{Cf} = \left(\frac{-m_D v_{Df}}{m_C}\right)$$
$$= \frac{-(0.035\ \text{kg})(-875\ \text{m/s})}{84\ \text{kg}}$$
$$= +0.36\ \text{m/s}$$

Substitute $m_D = 0.035$ kg, $v_{Df} = -875$ m/s, $m_C = 84$ kg

Math Handbook
Isolating a Variable
page 845

3 Evaluate the Answer

- **Are the units correct?** The velocity is measured in m/s.
- **Does the direction make sense?** The velocity of the astronaut is in the opposite direction to that of the expelled gas.
- **Is the magnitude realistic?** The astronaut's mass is much larger than that of the gas, so the velocity of the astronaut is much less than that of the expelled gas.

PRACTICE Problems

Additional Problems, Appendix B

19. A 4.00-kg model rocket is launched, expelling 50.0 g of burned fuel from its exhaust at a speed of 625 m/s. What is the velocity of the rocket after the fuel has burned? *Hint: Ignore the external forces of gravity and air resistance.*

20. A thread holds a 1.5-kg cart and a 4.5-kg cart together. After the thread is burned, a compressed spring pushes the carts apart, giving the 1.5-kg cart a speed of 27 cm/s to the left. What is the velocity of the 4.5-kg cart?

21. Carmen and Judi dock a canoe. 80.0-kg Carmen moves forward at 4.0 m/s as she leaves the canoe. At what speed and in what direction do the canoe and Judi move if their combined mass is 115 kg?

Two-Dimensional Collisions

Up until now, you have looked at momentum in only one dimension. The law of conservation of momentum holds for all closed systems with no external forces. It is valid regardless of the directions of the particles before or after they interact. But what happens in two or three dimensions? **Figure 9-10** shows the result of billiard ball C striking stationary billiard ball D. Consider the two billiard balls to be the system. The original momentum of the moving ball is $\boldsymbol{p}_{Ci}$ and the momentum of the stationary ball is zero. Therefore, the momentum of the system before the collision is equal to $\boldsymbol{p}_{Ci}$.

After the collision, both billiard balls are moving and have momenta. As long as the friction with the tabletop can be ignored, the system is closed and isolated. Thus, the law of conservation of momentum can be used. The initial momentum equals the vector sum of the final momenta, so $\boldsymbol{p}_{Ci} = \boldsymbol{p}_{Cf} + \boldsymbol{p}_{Df}$.

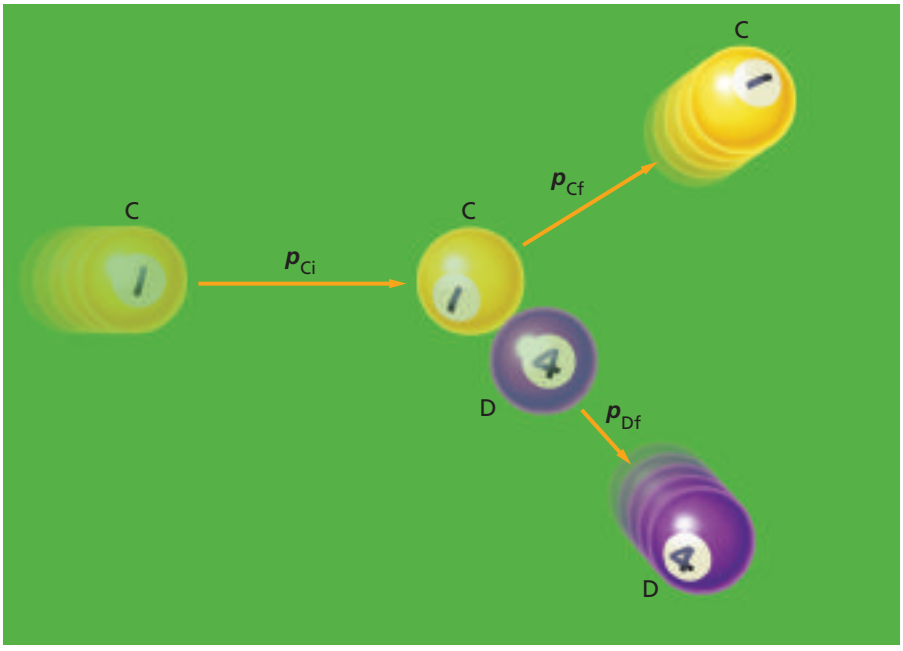
The equality of the momenta before and after the collision also means that the sum of the components of the vectors before and after the collision must be equal. Suppose the x-axis is defined to be in the direction of the initial momentum, then the y-component of the initial momentum is equal to zero. Therefore, the sum of the final y-components also must be zero:

$$\boldsymbol{p}_{Cf,\,y} + \boldsymbol{p}_{Df,\,y} = 0$$

The y-components are equal in magnitude but are in the opposite direction and, thus, have opposite signs. The sum of the horizontal components also is equal:

$$\boldsymbol{p}_{Ci} = \boldsymbol{p}_{Cf,\,x} + \boldsymbol{p}_{Df,\,x}$$

Figure 9-10 The law of conservation of momentum holds for all isolated, closed systems, regardless of the directions of objects before and after a collision.

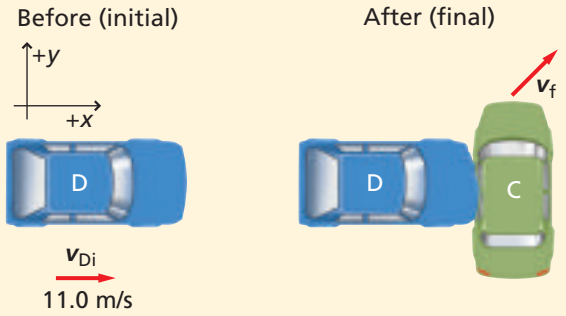


EXAMPLE Problem 4

Speed A 1325-kg car, C, moving north at 27.0 m/s, collides with a 2165-kg car, D, moving east at 11.0 m/s. The two cars are stuck together. In what direction and with what speed do they move after the collision?

1 Analyze and Sketch the Problem

- Define the system.
- Sketch the "before" and "after" states.
- Establish the coordinate axis with the *y*-axis north and the *x*-axis east.
- Draw a momentum-vector diagram.

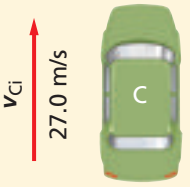


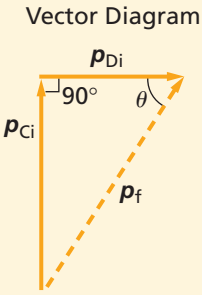
Known:

$m_C = 1325$ kg
$m_D = 2165$ kg
$v_{Ci,\,y} = 27.0$ m/s
$v_{Di,\,x} = 11.0$ m/s

Unknown:

$v_{f,\,x} = ?$
$v_{f,\,y} = ?$
$\theta = ?$





2 Solve for the Unknown

Determine the initial momenta of the cars and the momentum of the system.

$$p_{Ci} = m_C v_{Ci,\,y}$$
$$= (1325\text{ kg})(27.0\text{ m/s})$$ **Substitute $m_C = 1325$ kg, $v_{Ci,\,y} = 27.0$ m/s**
$$= 3.58\times10^4\text{ kg·m/s (north)}$$

$$p_{Di} = m_D v_{Di,\,x}$$
$$= (2165\text{ kg})(11.0\text{ m/s})$$ **Substitute $m_D = 2165$ kg, $v_{Di,\,x} = 11.0$ m/s**
$$= 2.38\times10^4\text{ kg·m/s (east)}$$

Use the law of conservation of momentum to find p_f.

$$p_{f,\,x} = p_{i,\,x} = 2.38\times10^4\text{ kg·m/s}$$ **Substitute $p_{i,\,x} = p_{Di} = 2.38\times10^4$ kg·m/s**
$$p_{f,\,y} = p_{i,\,y} = 3.58\times10^4\text{ kg·m/s}$$ **Substitute $p_{i,\,y} = p_{Ci} = 3.58\times10^4$ kg·m/s**

Use the diagram to set up equations for $p_{f,\,x}$ and $p_{f,\,y}$.

$$p_f = \sqrt{(p_{f,\,x})^2 + (p_{f,\,y})^2}$$
$$= \sqrt{(2.38\times10^4\text{ kg·m/s})^2 + (3.58\times10^4\text{ kg·m/s})^2}$$ **Substitute $p_{f,\,x} = 2.38\times10^4$ kg·m/s, $p_{f,\,y} = 3.58\times10^4$ kg·m/s**
$$= 4.30\times10^4\text{ kg·m/s}$$

Solve for θ.

$$\theta = \tan^{-1}\left(\frac{p_{f,\,y}}{p_{f,\,x}}\right)$$
$$= \tan^{-1}\left(\frac{3.58\times10^4\text{ kg·m/s}}{2.38\times10^4\text{ kg·m/s}}\right)$$ **Substitute $p_{f,\,y} = 3.58\times10^4$ kg·m/s, $p_{f,\,x} = 2.38\times10^4$ kg·m/s**
$$= 56.4°$$

Determine the final speed.

$$v_f = \frac{p_f}{(m_C + m_D)}$$
$$= \frac{4.30\times10^4\text{ kg·m/s}}{(1325\text{ kg} + 2165\text{ kg})}$$ **Substitute $p_f = 4.30\times10^4$ kg·m/s, $m_C = 1325$ kg, $m_D = 2165$ kg**
$$= 12.3\text{ m/s}$$

3 Evaluate the Answer

- **Are the units correct?** The correct unit for speed is m/s.
- **Do the signs make sense?** Answers are both positive and at the appropriate angles.
- **Is the magnitude realistic?** The cars stick together, so v_f must be smaller than v_{Ci}.

PRACTICE Problems

Additional Problems, Appendix B

22. A 925-kg car moving north at 20.1 m/s collides with a 1865-kg car moving west at 13.4 m/s. The two cars are stuck together. In what direction and at what speed do they move after the collision?

23. A 1383-kg car moving south at 11.2 m/s is struck by a 1732-kg car moving east at 31.3 m/s. The cars are stuck together. How fast and in what direction do they move immediately after the collision?

24. A stationary billiard ball, with a mass of 0.17 kg, is struck by an identical ball moving at 4.0 m/s. After the collision, the second ball moves 60.0° to the left of its original direction. The stationary ball moves 30.0° to the right of the moving ball's original direction. What is the velocity of each ball after the collision?

25. A 1345-kg car moving east at 15.7 m/s is struck by a 1923-kg car moving north. They are stuck together and move with an initial velocity of 14.5 m/s at $\theta = 63.5°$. Was the north-moving car exceeding the 20.1 m/s speed limit?

4.Conservation of Angular Momentum

Like linear momentum, angular momentum can be conserved. The **law of conservation of angular momentum** states that if no net external torque acts on an object, then its angular momentum does not change. This is represented by the following equation.

Law of Conservation of Angular Momentum $L_1 = L_2$

An object's initial angular momentum is equal to its final angular momentum.

For example, Earth spins on its axis with no external torques. Its angular momentum is constant. Thus, Earth's angular momentum is conserved. As a result, the length of a day does not change. A spinning ice-skater also demonstrates conservation of angular momentum. **Figure 9-11a** shows an ice-skater spinning with his arms extended. When he pulls in his arms, as shown in **Figure 9-11b,** he begins spinning faster. Without an external torque, his angular momentum does not change; that is, $L = I\omega$ is constant. Thus, the ice-skater's increased angular velocity must be accompanied by a decreased moment of inertia. By pulling his arms close to his body, the ice-skater brings more mass closer to the axis of rotation, thereby decreasing the radius of rotation and decreasing his moment of inertia. You can calculate changes in angular velocity using the law of conservation of angular momentum.

$$L_i = L_f$$

$$\text{thus, } I_i\omega_i = I_f\omega_f$$

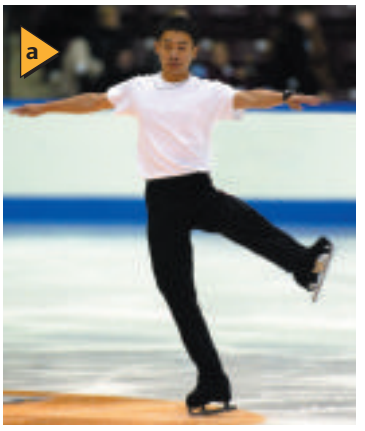
$$\frac{\omega_f}{\omega_i} = \frac{I_i}{I_f}$$

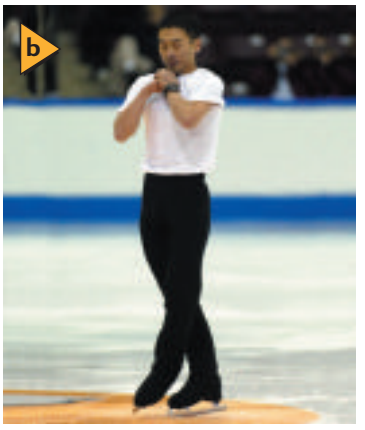
Because frequency is $f = \omega/2\pi$, the above equation can be rewritten as follows:

$$\frac{2\pi(f_f)}{2\pi(f_i)} = \frac{I_i}{I_f}$$

$$\text{thus, } \frac{f_f}{f_i} = \frac{I_i}{I_f}$$

Figure 9-11 When the ice-skater's arms are extended, the moment of inertia increases and his angular velocity decreases **(a).** When his arms are closer to his body the moment of inertia decreases and results in an increased angular velocity **(b).**





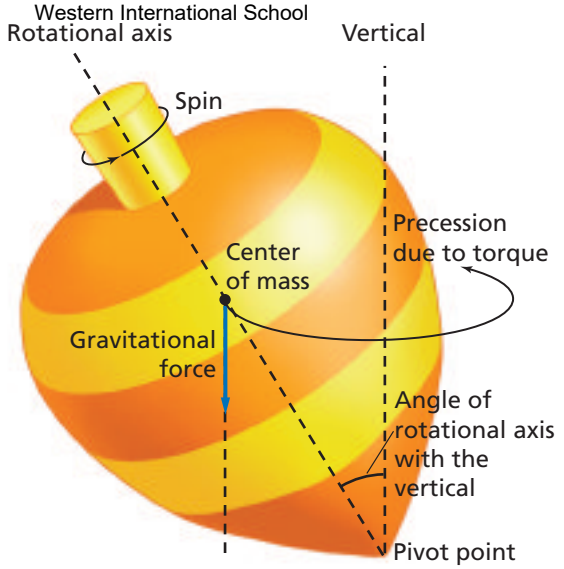


Figure 9-12 The upper end of the top precesses due to the torque acting on the top.

Notice that because f, ω, and I appear as ratios in these equations, any units may be used, as long as the same unit is used for both values of the quantity.

If a torque-free object starts with no angular momentum, it must continue to have no angular momentum. Thus, if part of an object rotates in one direction, another part must rotate in the opposite direction. For example, if you switch on a loosely held electric drill, the drill body will rotate in the direction opposite to the rotation of the motor and bit.

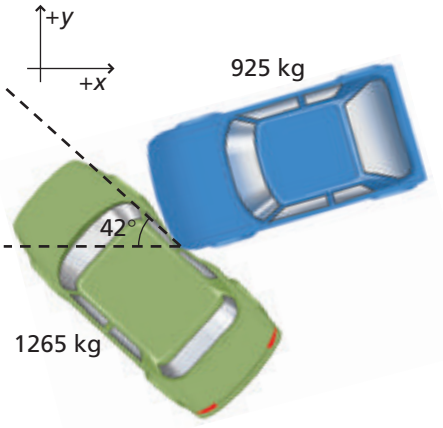
Consider a ball thrown at a weather vane. The ball, moving in a straight line, can start the vane rotating. Consider the ball and vane to be a system. With no external torques, angular momentum is conserved. The vane spins faster if the ball has a large mass, m, a large velocity, v, and hits at right angles as far as possible from the pivot of the vane. The angular momentum of a moving object, such as the ball, is given by $L = mvr$, where r is the perpendicular distance from the axis of rotation.

Tops and Gyroscopes

Because of the conservation of angular momentum, the direction of rotation of a spinning object can be changed only by applying a torque. If you played with a top as a child, you may have spun it by pulling the string wrapped around its axle. When a top is vertical, there is no torque on it, and the direction of its rotation does not change. If the top is tipped, as shown in **Figure 9-12,** a torque tries to rotate it downward. Rather than tipping over, however, the upper end of the top revolves, or precesses slowly about the vertical axis. Because Earth is not a perfect sphere, the Sun exerts a torque on it, causing it to precess. It takes about 26,000 years for Earth's rotational axis to go through one cycle of precession.

CHALLENGE PROBLEM

Your friend was driving her 1265-kg car north on Oak Street when she was hit by a 925-kg compact car going west on Maple Street. The cars stuck together and slid 23.1 m at 42° north of west. The speed limit on both streets is 22 m/s (50 mph). Assume that momentum was conserved during the collision and that acceleration was constant during the skid. The coefficient of kinetic friction between the tires and the pavement is 0.65.



1. Your friend claims that she wasn't speeding, but that the driver of other car was. How fast was your friend driving before the crash?
2. How fast was the other car moving before the crash? Can you support your friend's case in court?

file photo

A gyroscope, such as the one shown in **Figure 9-13,** is a wheel or disk that spins rapidly around one axis while being free to rotate around one or two other axes. The direction of its large angular momentum can be changed only by applying an appropriate torque. Without such a torque, the direction of the axis of rotation does not change. Gyroscopes are used in airplanes, submarines, and spacecraft to keep an unchanging reference direction. Giant gyroscopes are used in cruise ships to reduce their motion in rough water. Gyroscopic compasses, unlike magnetic compasses, maintain direction even when they are not on a level surface.

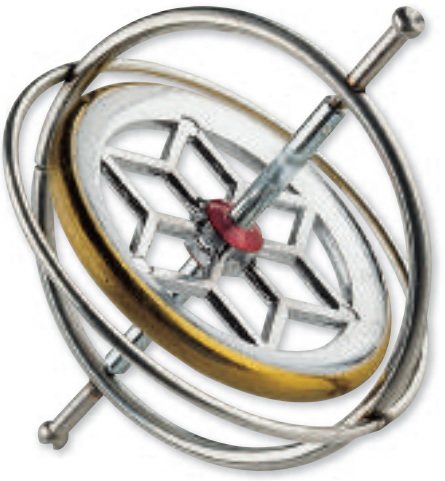


Figure 9-13 Because the orientation of the spin axis of the gyroscope does not change even when it is moved, the gyroscope can be used to fix direction.

A football quarterback uses the gyroscope effect to make an accurate forward pass. As he throws, he spins, or spirals the ball. If the quarterback throws the ball in the direction of its spin axis of rotation, the ball keeps its pointed end forward, thereby reducing air resistance. Thus, the ball can be thrown far and accurately. If its spin direction is slightly off, the ball wobbles. If the ball is not spun, it tumbles end over end.

The flight of a plastic disk also is stabilized by spin. A well-spun plastic disk can fly many meters through the air without wobbling. You are able to perform tricks with a yo-yo because its fast rotational speed keeps it rotating in one plane.

2. Section Review

26. Angular Momentum The outer rim of a plastic disk is thick and heavy. Besides making it easier to catch, how does this affect the rotational properties of the plastic disk?

27. Speed A cart, weighing 24.5 N, is released from rest on a 1.00-m ramp, inclined at an angle of 30.0° as shown in **Figure 9-14.** The cart rolls down the incline and strikes a second cart weighing 36.8 N.

a. Calculate the speed of the first cart at the bottom of the incline.

b. If the two carts stick together, with what initial speed will they move along?

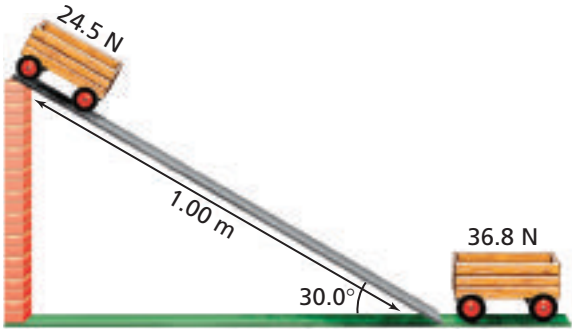


Figure 9-14

28. Conservation of Momentum During a tennis serve, the racket of a tennis player continues forward after it hits the ball. Is momentum conserved in the collision? Explain, making sure that you define the system.

29. Momentum A pole-vaulter runs toward the launch point with horizontal momentum. Where does the vertical momentum come from as the athlete vaults over the crossbar?

30. Initial Momentum During a soccer game, two players come from opposite directions and collide when trying to head the ball. They come to rest in midair and fall to the ground. Describe their initial momenta.

31. Critical Thinking You catch a heavy ball while you are standing on a skateboard, and then you roll backward. If you were standing on the ground, however, you would be able to avoid moving while catching the ball. Explain both situations using the law of conservation of momentum. Explain which system you use in each case.

Alternate CBL instructions can be found on the Web site.
physicspp.com

Sticky Collisions

In this activity, one moving cart will strike a stationary cart. During the collision, the two carts will stick together. You will measure mass and velocity, both before and after the collision. You then will calculate the momentum both before and after the collision.

QUESTION

How is the momentum of a system affected by a sticky collision?

Objectives

- **Describe** how momentum is transferred during a collision.
- **Calculate** the momenta involved.
- **Interpret data** from a collision.
- **Draw conclusions** that support the law of conservation of momentum.

Safety Precautions



Materials

Internet access required

Procedure

1. View Chapter 9 lab video clip 1 at physicspp.com/internet_lab to determine the mass of the carts.
2. Record the mass of each cart.
3. Watch video clip 2: Cart 1 strikes Cart 2.
4. In the video, three frames represent 0.1 s. Record in the data table the distance Cart 1 travels in 0.1 s before the collision.
5. Observe the collision. Record in the data table the distance the Cart 1-Cart 2 system travels in 0.1 s after the collision.
6. Repeat steps 3–5 for video clip 3: Carts 1 and 3 strike Cart 2.
7. Repeat steps 3–5 for video clip 4: Carts 1, 3, and 4 strike Cart 2.
8. Repeat steps 3-5 for video clip 5: Carts 1 and 3 strike Carts 2 and 4.
9. Repeat steps 3–5 for video clip 6: Cart 1 strikes Carts 2, 3, and 4.



Data Tables

Cart	Mass (kg)
1	
2	
3	
4	

Time of Approach (s)	Distance Covered in Approach (cm)	Initial Velocity (cm/s)	Mass of Approaching Cart(s) (g)	Initial Momentum (g·cm/s)	Time of Departure (s)	Distance Covered in Departure (cm)	Final Velocity (cm/s)	Mass of Departing Cart(s) (g)	Final Momentum (g·cm/s)
0.1					0.1				
0.1					0.1				
0.1					0.1				
0.1					0.1				
0.1					0.1				

Analyze

1. Calculate the initial and final velocities for each of the cart systems.
2. Calculate the initial and final momentum for each of the cart systems.
3. **Make and Use Graphs** Make a graph showing final momentum versus initial momentum for all the video clips.

Conclude and Apply

1. What is the relationship between the initial momentum and the final momentum of the cart systems in a sticky collision?
2. In theory, what should be the slope of the line in your graph?
3. The initial and final data numbers may not be the same due to the precision of the instruments, friction, and other variables. Is the initial momentum typically greater or less than the final momentum? Explain.

Going Further

1. Describe what the velocity and momentum data might look like if the carts did not stick together, but rather, bounced off of each other?
2. Design an experiment to test the impact of friction during the collision of the cart systems. Predict how the slope of the line in your graph will change with your experiment, and then try your experiment.

Real-World Physics

1. Suppose a linebacker collides with a stationary quarterback and they become entangled. What will happen to the velocity of the linebacker-quarterback system if momentum is conserved?
2. If a car rear-ends a stationary car so that the two cars become attached, what will happen to the velocity of the first car? The second car?

ShareYourData

Interpret Data Visit physicspp.com/internet_lab to post your findings from the experiment testing the impact of friction during the collisions of the cart systems. Examine the data and graph of the final momentum versus initial momentum on the Web site. Notice how close to or far off the slope is from 1.00.

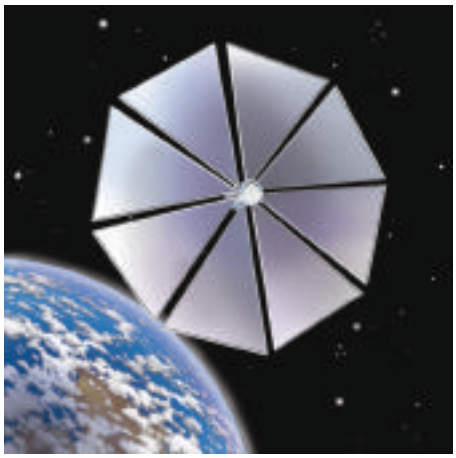
Future Technology

Solar Sailing

Nearly 400 years ago, Johannes Kepler observed that comet tails appeared to be blown by a solar breeze. He suggested that ships would be able to travel in space with sails designed to catch this breeze. Thus, the idea for solar sails was born.

How Does a Solar Sail Work? A solar sail is a spacecraft without an engine. A solar sail works like a giant fabric mirror that is free to move. Solar sails usually are made of 5-micron-thick aluminized polyester film or polyimide film with a 100-nm-thick aluminum layer deposited on one side to form the reflective surface.

Reflected sunlight, rather than rocket fuel, provides the force. Sunlight is made up of individual particles called photons. Photons have momentum, and when a photon bounces off a solar sail, it transfers its momentum to the sail, which propels the spacecraft along.



This artist's rendering shows *Cosmos 1*, the first solar sail scheduled for launch in the near future.

The force of impacting photons is small in comparison to the force rocket fuel can supply. So, small sails experience only a small amount of force from sunlight, while larger sails experience a greater force. Thus, solar sails may be a kilometer or so across.

What speeds can a solar sail achieve? This depends on the momentum transferred to the sail by photons, as well as the sail's mass. To travel quickly through the solar system, a sail and the spacecraft should be lightweight.

Photons supplied by the Sun are constant. They impact the sail every second of every hour of every day during a space flight. The Sun's continuous supply of photons over time allows the sail to build up huge velocities and enables the spacecraft to travel great distances within a convenient time frame. Rockets require enormous amounts of fuel to move large masses, but solar sails only require photons from the Sun. Thus, solar sails may be a superior way to move large masses over great distances in outer space.

Future Journeys The *Cosmos 1*, the first solar-sail prototype, is scheduled for a launch in the very near future. The *Cosmos-1* mission is an international, privately funded venture. The spacecraft looks like a flower with eight huge, solar-sail petals. Being the first solar sail, goals are modest. The mission will be considered successful if the *Cosmos 1* operates for just a few days, accelerating under sunlight pressure.

Solar sails are important, not only for travel, but also for creating new types of space and Earth weather monitoring stations. These stations would be able to provide greater coverage of Earth and more advanced warning of solar storms that cause problems to communication and electric power grids. It is hoped that in the next few decades, solar sails will be used as interplanetary shuttles because of their ability to travel great distances in convenient time frames. Vast distances could someday be traversed by vehicles that do not consume any fuel.

Going Further

1. **Research** how solar sails can help provide advanced warning of solar storms.
2. **Critical Thinking** A certain solar-sail model is predicted to take more time to reach Mars than a rocket-propelled spacecraft would, but less time to go to Pluto than a rocket-propelled spacecraft would. Explain why this is so.

9.1 Impulse and Momentum

Vocabulary

- impulse (p. 230)
- momentum (p. 230)
- impulse-momentum theorem (p. 230)
- angular momentum (p. 233)
- angular impulse-angular momentum theorem (p. 234)

Key Concepts

- When doing a momentum problem, first examine the system before and after the event.
- The momentum of an object is the product of its mass and velocity and is a vector quantity.

$$\boldsymbol{p} = m\boldsymbol{v}$$

- The impulse on an object is the average net force exerted on the object multiplied by the time interval over which the force acts.

$$\text{Impluse} = \boldsymbol{F}\Delta t$$

- The impulse on an object is equal to the change in momentum of the object.

$$\boldsymbol{F}\Delta t = \boldsymbol{p}_\text{f} - \boldsymbol{p}_\text{i}$$

- The angular momentum of a rotating object is the product of its moment of inertia and its angular velocity.

$$L = I\omega$$

- The angular impulse-angular momentum theorem states that the angular impulse on an object is equal to the change in the object's angular momentum.

$$\tau\Delta t = L_\text{f} - L_\text{i}$$

9.2 Conservation of Momentum

Vocabulary

- closed system (p. 236)
- isolated system (p. 237)
- law of conservation of momentum (p. 237)
- law of conservation of angular momentum (p. 243)

Key Concepts

- According to Newton's third law of motion and the law of conservation of momentum, the forces exerted by colliding objects on each other are equal in magnitude and opposite in direction.
- Momentum is conserved in a closed, isolated system.

$$\boldsymbol{p}_\text{f} = \boldsymbol{p}_\text{i}$$

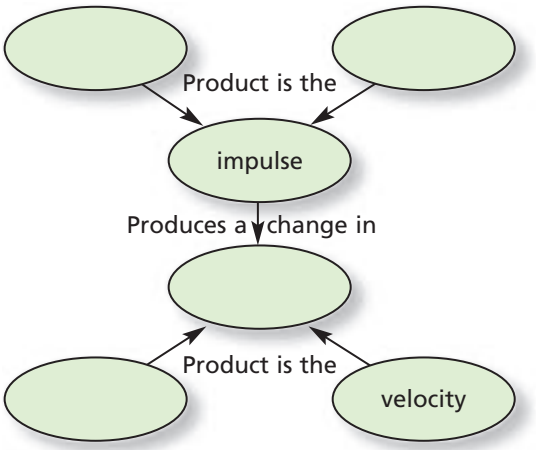
- The law of conservation of momentum can be used to explain the propulsion of rockets.
- Vector analysis is used to solve momentum-conservation problems in two dimensions.
- The law of conservation of angular momentum states that if there are no external torques acting on a system, then the angular momentum is conserved.

$$L_\text{f} = L_\text{i}$$

- Because angular momentum is conserved, the direction of rotation of a spinning object can be changed only by applying a torque.

Concept Mapping

32. Complete the following concept map using the following terms: *mass, momentum, average force, time over which the force is exerted.*



Mastering Concepts

33. Can a bullet have the same momentum as a truck? Explain. (9.1)

34. A pitcher throws a curve ball to the catcher. Assume that the speed of the ball doesn't change in flight. (9.1)
- **a.** Which player exerts the larger impulse on the ball?
- **b.** Which player exerts the larger force on the ball?

35. Newton's second law of motion states that if no net force is exerted on a system, no acceleration is possible. Does it follow that no change in momentum can occur? (9.1)

36. Why are cars made with bumpers that can be pushed in during a crash? (9.1)

37. An ice-skater is doing a spin. (9.1)
- **a.** How can the skater's angular momentum be changed?
- **b.** How can the skater's angular velocity be changed without changing the angular momentum?

38. What is meant by "an isolated system?" (9.2)

39. A spacecraft in outer space increases its velocity by firing its rockets. How can hot gases escaping from its rocket engine change the velocity of the craft when there is nothing in space for the gases to push against? (9.2)

40. A cue ball travels across a pool table and collides with the stationary eight ball. The two balls have equal masses. After the collision, the cue ball is at rest. What must be true regarding the speed of the eight ball? (9.2)

41. Consider a ball falling toward Earth. (9.2)
- **a.** Why is the momentum of the ball not conserved?
- **b.** In what system that includes the falling ball is the momentum conserved?

42. A falling basketball hits the floor. Just before it hits, the momentum is in the downward direction, and after it hits the floor, the momentum is in the upward direction. (9.2)
- **a.** Why isn't the momentum of the basketball conserved even though the bounce is a collision?
- **b.** In what system is the momentum conserved?

43. Only an external force can change the momentum of a system. Explain how the internal force of a car's brakes brings the car to a stop. (9.2)

44. Children's playgrounds often have circular-motion rides. How could a child change the angular momentum of such a ride as it is turning? (9.2)

Applying Concepts

45. Explain the concept of impulse using physical ideas rather than mathematics.

46. Is it possible for an object to obtain a larger impulse from a smaller force than it does from a larger force? Explain.

47. **Foul Ball** You are sitting at a baseball game when a foul ball comes in your direction. You prepare to catch it bare-handed. To catch it safely, should you move your hands toward the ball, hold them still, or move them in the same direction as the moving ball? Explain.

48. A 0.11-g bullet leaves a pistol at 323 m/s, while a similar bullet leaves a rifle at 396 m/s. Explain the difference in exit speeds of the two bullets, assuming that the forces exerted on the bullets by the expanding gases have the same magnitude.

49. An object initially at rest experiences the impulses described by the graph in **Figure 9-15.** Describe the object's motion after impulses A, B, and C.

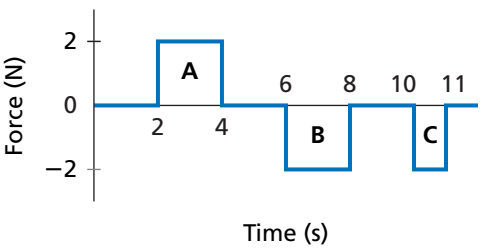


Figure 9-15

Chapter 4

Vibrations and Waves



Periodic motion, from masses on springs to vibrations of atoms, is one of the most important kinds of physical behavior. In this chapter we take a more detailed look at Hooke's law, where the force is proportional to the displacement, tending to restore objects to some equilibrium position. A large number of physical systems can be successfully modeled with this simple idea, including the vibrations of strings, the swinging of a pendulum, and the propagation of waves of all kinds. All these physical phenomena involve periodic motion.

Periodic vibrations can cause disturbances that move through a medium in the form of waves. Many kinds of waves occur in nature, such as sound waves, water waves, seismic waves, and electromagnetic waves. These very different physical phenomena are described by common terms and concepts introduced here.

1. Hooke's Law
2. Elastic Potential Energy
3. Comparing Simple Harmonic Motion with Uniform Circular Motion
4. Position, Velocity, and Acceleration as a Function of Time
5. Motion of a Pendulum
6. Damped Oscillations
7. Waves
8. Frequency, Amplitude, and Wavelength
9 The Speed of Waves on Strings
10. Interference of Waves
11. Reflection of Waves

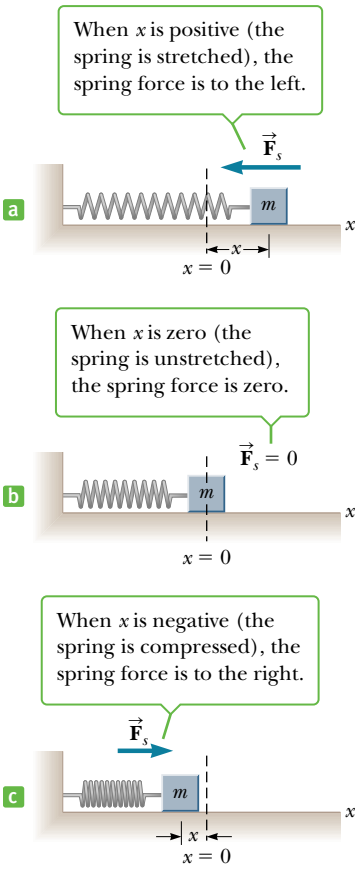
◄ Hooke's law

1. Hooke's Law

One of the simplest types of vibrational motion is that of an object attached to a spring, previously discussed in the context of energy in Chapter 5. We assume the object moves on a frictionless horizontal surface. If the spring is stretched or compressed a small distance x from its unstretched or equilibrium position and then released, it exerts a force on the object as shown in Active Figure 13.1 (page 438). From experiment the spring force F_s is found to obey the equation

$$F_s = -kx \qquad [13.1]$$

where x is the displacement of the object from its equilibrium position ($x = 0$) and k is a positive constant called the **spring constant.** This force law for springs was



Active Figure 13.1
The force exerted by a spring on an object varies with the displacement of the object from the equilibrium position, $x = 0$.

discovered by Robert Hooke in 1678 and is known as **Hooke's law.** The value of k is a measure of the stiffness of the spring. Stiff springs have large k values, and soft springs have small k values.

The negative sign in Equation 13.1 means that the force exerted by the spring is always directed *opposite* the displacement of the object. When the object is to the right of the equilibrium position, as in Active Figure 13.1a, x is positive and F_s is negative. This means that the force is in the negative direction, to the left. When the object is to the left of the equilibrium position, as in Active Figure 13.1c, x is negative and F_s is positive, indicating that the direction of the force is to the right. Of course, when $x = 0$, as in Active Figure 13.1b, the spring is unstretched and $F_s = 0$. Because the spring force always acts toward the equilibrium position, it is sometimes called a restoring force. **A restoring force always pushes or pulls the object toward the equilibrium position.**

Suppose the object is initially pulled a distance A to the right and released from rest. The force exerted by the spring on the object pulls it back toward the equilibrium position. As the object moves toward $x = 0$, the magnitude of the force decreases (because x decreases) and reaches zero at $x = 0$. The object gains speed as it moves toward the equilibrium position, however, reaching its maximum speed when $x = 0$. The momentum gained by the object causes it to overshoot the equilibrium position and compress the spring. As the object moves to the left of the equilibrium position (negative x-values), the spring force acts on it to the right, steadily increasing in strength, and the speed of the object decreases. The object finally comes briefly to rest at $x = -A$ before accelerating back towards $x = 0$ and ultimately returning to the original position at $x = A$. The process is then repeated, and the object continues to oscillate back and forth over the same path. This type of motion is called **simple harmonic motion. Simple harmonic motion occurs when the net force along the direction of motion obeys Hooke's law—when the net force is proportional to the displacement from the equilibrium point and is always directed toward the equilibrium point.**

Not all periodic motions over the same path can be classified as simple harmonic motion. A ball being tossed back and forth between a parent and a child moves repetitively, but the motion isn't simple harmonic motion because the force acting on the ball doesn't take the form of Hooke's law, Equation 13.1.

The motion of an object suspended from a vertical spring is also simple harmonic. In this case the force of gravity acting on the attached object stretches the spring until equilibrium is reached and the object is suspended at rest. By definition, the equilibrium position of the object is $x = 0$. When the object is moved away from equilibrium by a distance x and released, a net force acts toward the equilibrium position. Because the net force is proportional to x, the motion is simple harmonic.

The following three concepts are important in discussing any kind of periodic motion:

- The **amplitude** A is the maximum distance of the object from its equilibrium position. In the absence of friction, an object in simple harmonic motion oscillates between the positions $x = -A$ and $x = +A$.
- The **period** T is the time it takes the object to move through one complete cycle of motion, from $x = A$ to $x = -A$ and back to $x = A$.
- The **frequency** f is the number of complete cycles or vibrations per unit of time, and is the reciprocal of the period ($f = 1/T$).

The acceleration of an object moving with simple harmonic motion can be found by using Hooke's law in the equation for Newton's second law, $F = ma$. This gives

$$ma = F = -kx$$

$$a = -\frac{k}{m}x$$

Equation 13.2, an example of a *harmonic oscillator equation,* gives the acceleration as a function of position. Because the maximum value of x is defined to be the amplitude A, the acceleration ranges over the values $-kA/m$ to $+kA/m$. In the next section we will find equations for velocity as a function of position and for position as a function of time. Springs satisfying Hooke's law are also called *ideal* springs. In real springs, spring mass, internal friction, and varying elasticity affect the force law and motion.

Tip 13.1 Constant-Acceleration Equations Don't Apply

The acceleration a of a particle in simple harmonic motion is *not* constant; it changes, varying with x, so we can't apply the constant acceleration kinematic equations of Chapter 2.

Quick Quiz

13.1 A block on the end of a horizontal spring is pulled from equilibrium at $x = 0$ to $x = A$ and released. Through what total distance does it travel in one full cycle of its motion? (a) $A/2$ (b) A (c) $2A$ (d) $4A$

13.2 For a simple harmonic oscillator, which of the following pairs of vector quantities can't both point in the same direction? (The position vector is the displacement from equilibrium.) (a) position and velocity (b) velocity and acceleration (c) position and acceleration

EXAMPLE 13.1 Simple Harmonic Motion on a Frictionless Surface

GOAL Calculate forces and accelerations for a horizontal spring system.

PROBLEM A 0.350-kg object attached to a spring of force constant 1.30×10^2 N/m is free to move on a frictionless horizontal surface, as in Active Figure 13.1. If the object is released from rest at $x = 0.100$ m, find the force on it and its acceleration at $x = 0.100$ m, $x = 0.050\ 0$ m, $x = 0$ m, $x = -0.050\ 0$ m, and $x = -0.100$ m.

STRATEGY Substitute given quantities into Hooke's law to find the forces, then calculate the accelerations with Newton's second law. The amplitude A is the same as the point of release from rest, $x = 0.100$ m.

SOLUTION

Write Hooke's force law:

$$F_s = -kx$$

Substitute the value for k and take $x = A = 0.100$ m, finding the spring force at that point:

$$F_{max} = -kA = -(1.30 \times 10^2\ \text{N/m})(0.100\ \text{m}) = \boxed{-13.0\ \text{N}}$$

Solve Newton's second law for a and substitute to find the acceleration at $x = A$:

$$ma = F_{max}$$

$$a = \frac{F_{max}}{m} = \frac{-13.0\ \text{N}}{0.350\ \text{kg}} = \boxed{-37.1\ \text{m/s}^2}$$

Repeat the same process for the other four points, assembling a table:

Position (m)	Force (N)	Acceleration (m/s^2)
0.100	−13.0	−37.1
0.050	−6.50	−18.6
0	0	0
−0.050	+6.50	+18.6
−0.100	+13.0	+37.1

REMARKS The table above shows that when the initial position is halved, the force and acceleration are also halved. Further, positive values of x give negative values of the force and acceleration, whereas negative values of x give positive values of the force and acceleration. As the object moves to the left and passes the equilibrium point, the spring force becomes positive (for negative values of x), slowing the object down.

QUESTION 13.1 Will doubling a given displacement always result in doubling the magnitude of the spring force? Explain.

EXERCISE 13.1 For the same spring and mass system, find the force exerted by the spring and the position x when the object's acceleration is $+9.00$ m/s^2.

ANSWERS 3.15 N, −2.42 cm

EXAMPLE 13.2 Mass on a Vertical Spring

GOAL Apply Newton's second law together with the force of gravity and Hooke's law.

PROBLEM A spring is hung vertically (Fig. 13.2a), and an object of mass m attached to the lower end is then slowly lowered a distance d to the equilibrium point (Fig. 13.2b). **(a)** Find the value of the spring constant if the spring is displaced by 2.0 cm and the mass is 0.55 kg. **(b)** If a second identical spring is attached to the object in parallel with the first spring (Fig. 13.2d), where is the new equilibrium point of the system? **(c)** What is the effective spring constant of the two springs acting as one?

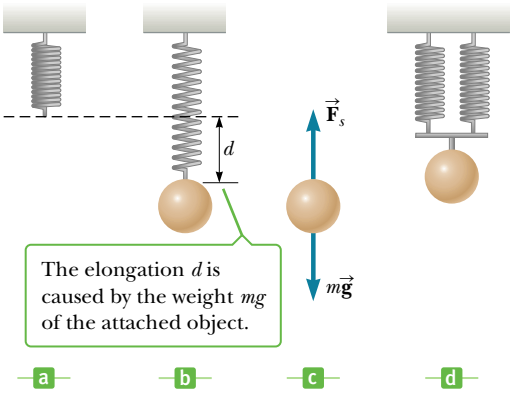


Figure 13.2 (Example 13.2) (a)–(c) Determining the spring constant. Because the upward spring force balances the weight when the system is in equilibrium, it follows that $k = mg/d$. (d) A system involving two springs in parallel.

STRATEGY This example is an application of Newton's second law. The spring force is upward, balancing the downward force of gravity mg when the system is in equilibrium. (See Fig. 13.2c.) Because the suspended object is in equilibrium, the forces on the object sum to zero, and it's possible to solve for the spring constant k. Part (b) is solved the same way, but has two spring forces balancing the force of gravity. The spring constants are known, so the second law for equilibrium can be solved for the displacement of the spring. Part (c) involves using the displacement found in part (b). Treating the two springs as one equivalent spring, the second law then leads to the effective spring constant of the two-spring system.

SOLUTION

(a) Find the value of the spring constant if the spring is displaced by 2.0 cm and the mass of the object is 0.55 kg.

Apply Newton's second law to the object (with $a = 0$) and solve for the spring constant k:

$$\sum F = F_g + F_s = -mg + kd = 0$$

$$k = \frac{mg}{d} = \frac{(0.55\ \text{kg})(9.80\ \text{m/s}^2)}{2.0 \times 10^{-2}\ \text{m}} = 2.7 \times 10^2\ \text{N/m}$$

(b) If a second identical spring is attached to the object in parallel with the first spring (Fig. 13.2d), find the new equilibrium point of the system.

Apply Newton's second law, but with two springs acting on the object:

$$\sum F = F_g + F_{s1} + F_{s2} = -mg + kd_2 + kd_2 = 0$$

Solve for d_2:

$$d_2 = \frac{mg}{2k} = \frac{(0.55\ \text{kg})(9.80\ \text{m/s}^2)}{2(2.7 \times 10^2\ \text{N/m})} = 1.0 \times 10^{-2}\ \text{m}$$

(c) What is the effective spring constant of the two springs acting as one?

Write the second law for the system, with an effective spring constant k_{eff}:

$$\sum F = F_g + F_s = -mg + k_{\text{eff}} d_2 = 0$$

Solve for k_{eff}:

$$k_{\text{eff}} = \frac{mg}{d_2} = \frac{(0.55\ \text{kg})(9.80\ \text{m/s}^2)}{1.0 \times 10^{-2}\ \text{m}} = 5.4 \times 10^2\ \text{N/m}$$

REMARKS In this example, the spring force is positive because it's directed upward. Once the object is displaced from the equilibrium position and released, it oscillates around the equilibrium position, just like the horizontal spring. Notice that attaching an extra identical spring in parallel is equivalent to having a single spring with twice the force constant. When the springs are attached end to end in series, however, the exercise illustrates that, all other things being equal, longer springs have smaller force constants than shorter springs.

QUESTION 13.2 Generalize: When two springs with force constants k_1 and k_2 act in parallel on an object, what is the spring constant k_{eq} of the single spring that would be equivalent to the two springs, in terms of k_1 and k_2?

EXERCISE 13.2 When a 75.0-kg man slowly adds his weight to a vertical spring attached to the ceiling, he reaches equilibrium when the spring is stretched by 6.50 cm. (a) Find the spring constant. (b) If a second, identical spring is hung on the first, and the man again adds his weight to the system, by how much does the system of springs stretch? (c) What would be the spring constant of a single, equivalent spring?

ANSWERS (a) 1.13×10^4 N/kg (b) 13.0 cm (c) 5.65×10^3 N/m

2. Elastic Potential Energy

In this section we review the material covered in Section 4 of Chapter 5.

A system of interacting objects has potential energy associated with the configuration of the system. A compressed spring has potential energy that, when allowed to expand, can do work on an object, transforming spring potential energy into the object's kinetic energy. As an example, Figure 13.3 shows a ball being projected from a spring-loaded toy gun, where the spring is compressed a distance x. As the gun is fired, the compressed spring does work on the ball and imparts kinetic energy to it.

Recall that the energy stored in a stretched or compressed spring or some other elastic material is called elastic potential energy, PE_s, given by

$$PE_s \equiv \tfrac{1}{2}kx^2 \qquad [13.3]$$

◀ Elastic potential energy

Recall also that the law of conservation of energy, including both gravitational and spring potential energy, is given by

$$(KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f \qquad [13.4]$$

If nonconservative forces such as friction are present, then the change in mechanical energy must equal the work done by the nonconservative forces:

$$W_{nc} = (KE + PE_g + PE_s)_f - (KE + PE_g + PE_s)_i \qquad [13.5]$$

Rotational kinetic energy must be included in both Equation 13.4 and Equation 13.5 for systems involving torques.

As an example of the energy conversions that take place when a spring is included in a system, consider Figure 13.4 (page 442). A block of mass m slides on a frictionless horizontal surface with constant velocity $\vec{\mathbf{v}}_i$ and collides with a coiled spring. The description that follows is greatly simplified by assuming the spring is very light (an ideal spring) and therefore has negligible kinetic energy. As the spring is compressed, it exerts a force to the left on the block. At maximum compression, the block comes to rest for just an instant (Fig. 13.4c). The initial total energy in the system (block plus spring) before the collision is the kinetic energy of the block. After the block collides with the spring and the spring is partially compressed, as in Figure 13.4b, the block has kinetic energy $\frac{1}{2}mv^2$ (where $v < v_i$) and the spring has potential energy $\frac{1}{2}kx^2$. When the block stops for an instant at the point of maximum compression, the kinetic energy is zero. Because the spring force is conservative and because there are no external forces that can do work on the system, **the total mechanical energy of the system consisting of the block**

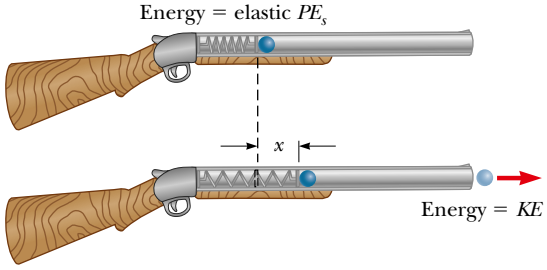


Figure 13.3 A ball projected from a spring-loaded gun. The elastic potential energy stored in the spring is transformed into the kinetic energy of the ball.

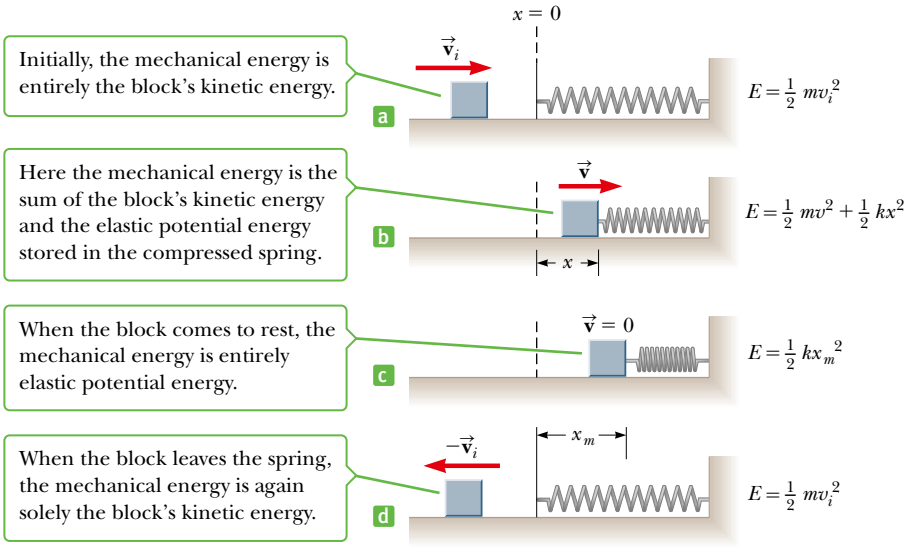


Figure 13.4 A block sliding on a frictionless horizontal surface collides with a light spring. In the absence of friction, the mechanical energy in this process remains constant.

and spring remains constant. Energy is transformed from the kinetic energy of the block to the potential energy stored in the spring. As the spring expands, the block moves in the opposite direction and regains all its initial kinetic energy, as in Figure 13.4d.

When an archer pulls back on a bowstring, elastic potential energy is stored in both the bent bow and stretched bowstring (Fig. 13.5). When the arrow is released, the potential energy stored in the system is transformed into the kinetic energy of the arrow. Devices such as crossbows and slingshots work the same way.

APPLICATION
Archery



Volodymyr Vyshnivetskyy/istockphoto.com

Figure 13.5 Elastic potential energy is stored in this drawn bow.

Quick Quiz

13.3 When an object moving in simple harmonic motion is at its maximum displacement from equilibrium, which of the following is at a maximum? (a) velocity, (b) acceleration, or (c) kinetic energy

EXAMPLE 13.3 Stop That Car!

GOAL Apply conservation of energy and the work–energy theorem with spring and gravitational potential energy.

PROBLEM A 13 000-N car starts at rest and rolls down a hill from a height of 10.0 m (Fig. 13.6). It then moves across a level surface and collides with a light spring-loaded guardrail. **(a)** Neglecting any losses due to friction, and ignoring the rotational kinetic energy of the wheels, find the maximum distance the spring is compressed. Assume a spring constant of 1.0×10^6 N/m. **(b)** Calculate the maximum acceleration of the car after contact with the spring, assuming no frictional losses. **(c)** If the spring is compressed by only 0.30 m, find the change in the mechanical energy due to friction.

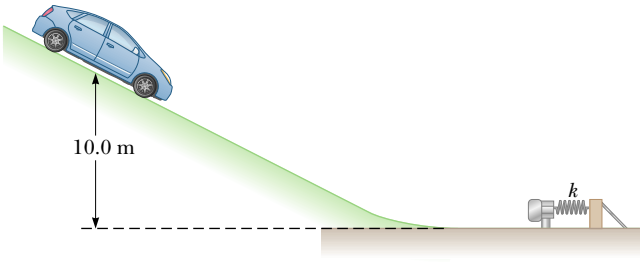


Figure 13.6 (Example 13.3) A car starts from rest on a hill at the position shown. When the car reaches the bottom of the hill, it collides with a spring-loaded guardrail.

STRATEGY Because friction losses are neglected, use conservation of energy in the form of Equation 13.4 to solve for the spring displacement in part (a). The initial and final values of the car's kinetic energy are zero, so the initial potential energy of the car–spring–Earth system is completely converted to elastic potential energy in the spring at the end of the ride. In part (b) apply Newton's second law, substituting the answer to part (a) for x because the maximum compression will give the maximum acceleration. In part (c) friction is no longer neglected, so use the work–energy theorem, Equation 13.5. The change in mechanical energy must equal the mechanical energy lost due to friction.

SOLUTION

(a) Find the maximum spring compression, assuming no energy losses due to friction.

Apply conservation of mechanical energy. Initially, there is only gravitational potential energy, and at maximum compression of the guardrail, there is only spring potential energy.

$$(KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f$$
$$0 + mgh + 0 = 0 + 0 + \tfrac{1}{2}kx^2$$

Solve for x:

$$x = \sqrt{\frac{2mgh}{k}} = \sqrt{\frac{2(13\,000\text{ N})(10.0\text{ m})}{1.0 \times 10^6\text{ N/m}}} = \boxed{0.51\text{ m}}$$

(b) Calculate the maximum acceleration of the car by the spring, neglecting friction.

Apply Newton's second law to the car:

$$ma = -kx \quad \rightarrow \quad a = -\frac{kx}{m} = -\frac{kxg}{mg} = -\frac{kxg}{w}$$

Substitute values:

$$a = -\frac{(1.0 \times 10^6\text{ N/m})(0.51\text{ m})(9.8\text{ m/s}^2)}{13\,000\text{ N}}$$
$$= \boxed{-380\text{ m/s}^2}$$

(c) If the compression of the guardrail is only 0.30 m, find the change in the mechanical energy due to friction.

Use the work–energy theorem:

$$W_{nc} = (KE + PE_g + PE_s)_f - (KE + PE_g + PE_s)_i$$
$$= (0 + 0 + \tfrac{1}{2}kx^2) - (0 + mgh + 0)$$
$$= \tfrac{1}{2}(1.0 \times 10^6\text{ N/m})(0.30)^2 - (13\,000\text{ N})(10.0\text{ m})$$
$$W_{nc} = \boxed{-8.5 \times 10^4\text{ J}}$$

REMARKS The answer to part (b) is about 40 times greater than the acceleration of gravity, so we'd better be wearing our seat belts. Note that the solution didn't require calculation of the velocity of the car.

QUESTION 13.3 True or False: In the absence of energy losses due to friction, doubling the height of the hill doubles the maximum acceleration delivered by the spring.

EXERCISE 13.3 A spring-loaded gun fires a 0.100-kg puck along a tabletop. The puck slides up a curved ramp and flies straight up into the air. If the spring is displaced 12.0 cm from equilibrium and the spring constant is 875 N/m, how high does the puck rise, neglecting friction? (b) If instead it only rises to a height of 5.00 m because of friction, what is the change in mechanical energy?

ANSWERS (a) 6.43 m (b) −1.40 J

In addition to studying the preceding example, it's a good idea to review those given in Section 5.4.

Velocity as a Function of Position

Conservation of energy provides a simple method of deriving an expression for the velocity of an object undergoing periodic motion as a function of position. The object in question is initially at its maximum extension A (Fig. 13.7a on page 444) and is then released from rest. The initial energy of the system is entirely elastic potential energy stored in the spring, $\frac{1}{2}kA^2$. As the object moves toward the origin to some new position x (Fig. 13.7b), part of this energy is transformed into kinetic energy, and the potential energy stored in the spring is reduced to $\frac{1}{2}kx^2$. Because the total energy of the system is equal to $\frac{1}{2}kA^2$ (the initial energy stored

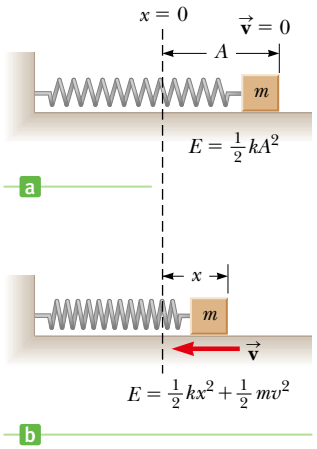


Figure 13.7 (a) An object attached to a spring on a frictionless surface is released from rest with the spring extended a distance A. Just before the object is released, the total energy is the elastic potential energy $\frac{1}{2}kA^2$. (b) When the object reaches position x, it has kinetic energy $\frac{1}{2}mv^2$ and the elastic potential energy has decreased to $\frac{1}{2}kx^2$.

in the spring), we can equate this quantity to the sum of the kinetic and potential energies at the position x:

$$\tfrac{1}{2}kA^2 = \tfrac{1}{2}mv^2 + \tfrac{1}{2}kx^2$$

Solving for v, we get

$$v = \pm\sqrt{\frac{k}{m}(A^2 - x^2)} \qquad [13.6]$$

This expression shows that the object's speed is a maximum at $x = 0$ and is zero at the extreme positions $x = \pm A$.

The right side of Equation 13.6 is preceded by the ±sign because the square root of a number can be either positive or negative. If the object in Figure 13.7 is moving to the right, v is positive; if the object is moving to the left, v is negative.

EXAMPLE 13.4 The Object–Spring System Revisited

GOAL Apply the time-independent velocity expression, Equation 13.6, to an object-spring system.

PROBLEM A 0.500-kg object connected to a light spring with a spring constant of 20.0 N/m oscillates on a frictionless horizontal surface. **(a)** Calculate the total energy of the system and the maximum speed of the object if the amplitude of the motion is 3.00 cm. **(b)** What is the velocity of the object when the displacement is 2.00 cm? **(c)** Compute the kinetic and potential energies of the system when the displacement is 2.00 cm.

STRATEGY The total energy of the system can be found most easily at $x = A$, where the kinetic energy is zero. There, the potential energy alone is equal to the total energy. Conservation of energy then yields the speed at $x = 0$. For part (b), obtain the velocity by substituting the given value of x into the time-independent velocity equation. Using this result, the kinetic energy asked for in part (c) can be found by substitution, and the potential energy can be found by substitution into Equation 13.3.

SOLUTION

(a) Calculate the total energy and maximum speed if the amplitude is 3.00 cm.

Substitute $x = A = 3.00$ cm and $k = 20.0$ N/m into the equation for the total mechanical energy E:

$$\begin{aligned} E &= KE + PE_g + PE_s \\ &= 0 + 0 + \tfrac{1}{2}kA^2 = \tfrac{1}{2}(20.0\ \text{N/m})(3.00 \times 10^{-2}\ \text{m})^2 \\ &= 9.00 \times 10^{-3}\ \text{J} \end{aligned}$$

Use conservation of energy with $x_i = A$ and $x_f = 0$ to compute the speed of the object at the origin:

$$(KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f$$
$$0 + 0 + \tfrac{1}{2}kA^2 = \tfrac{1}{2}mv_{max}^2 + 0 + 0$$
$$\tfrac{1}{2}mv_{max}^2 = 9.00 \times 10^{-3}\,\text{J}$$
$$v_{max} = \sqrt{\frac{18.0 \times 10^{-3}\,\text{J}}{0.500\,\text{kg}}} = 0.190\ \text{m/s}$$

(b) Compute the velocity of the object when the displacement is 2.00 cm.

Substitute known values directly into Equation 13.6:

$$v = \pm\sqrt{\frac{k}{m}(A^2 - x^2)}$$
$$= \pm\sqrt{\frac{20.0\ \text{N/m}}{0.500\ \text{kg}}[(0.030\,0\ \text{m})^2 - (0.020\,0\ \text{m})^2]}$$
$$= \pm 0.141\ \text{m/s}$$

(c) Compute the kinetic and potential energies when the displacement is 2.00 cm.

Substitute into the equation for kinetic energy:

$$KE = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(0.500\ \text{kg})(0.141\ \text{m/s})^2 = 4.97 \times 10^{-3}\,\text{J}$$

Substitute into the equation for spring potential energy:

$$PE_s = \tfrac{1}{2}kx^2 = \tfrac{1}{2}(20.0\ \text{N/m})(2.00 \times 10^{-2}\ \text{m})^2$$
$$= 4.00 \times 10^{-3}\,\text{J}$$

REMARKS With the given information, it is impossible to choose between the positive and negative solutions in part (b). Notice that the sum $KE + PE_s$ in part (c) equals the total energy E found in part (a), as it should (except for a small discrepancy due to rounding).

QUESTION 13.4 True or False: Doubling the initial displacement doubles the speed of the object at the equilibrium point.

EXERCISE 13.4 For what values of x is the speed of the object 0.10 m/s? **ANSWER** ±2.55 cm

3. Comparing Simple Harmonic Motion with Uniform Circular Motion

We can better understand and visualize many aspects of simple harmonic motion along a straight line by looking at its relationship to uniform circular motion. Active Figure 13.8 is a top view of an experimental arrangement that is useful for this purpose. A ball is attached to the rim of a turntable of radius A, illuminated from the side by a lamp. We find that **as the turntable rotates with constant angular speed, the shadow of the ball moves back and forth with simple harmonic motion.**

This fact can be understood from Equation 13.6, which says that the velocity of an object moving with simple harmonic motion is related to the displacement by

$$v = C\sqrt{A^2 - x^2}$$

where C is a constant. To see that the shadow also obeys this relation, consider Figure 13.9 (page 446), which shows the ball moving with a constant speed v_0 in a direction tangent to the circular path. At this instant, the velocity of the ball in the x-direction is given by $v = v_0 \sin\theta$, or

$$\sin\theta = \frac{v}{v_0}$$

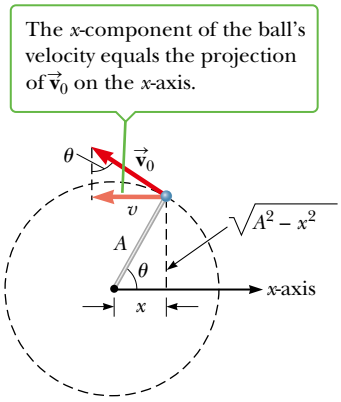


Figure 13.9 The ball rotates with constant speed v_0.

From the larger triangle in Figure 13.9 we can obtain a second expression for sin θ:

$$\sin\theta = \frac{\sqrt{A^2 - x^2}}{A}$$

Equating the right-hand sides of the two expressions for sin θ, we find the following relationship between the velocity v and the displacement x:

$$\frac{v}{v_0} = \frac{\sqrt{A^2 - x^2}}{A}$$

or

$$v = \frac{v_0}{A}\sqrt{A^2 - x^2} = C\sqrt{A^2 - x^2}$$

The velocity of the ball in the x-direction is related to the displacement x in exactly the same way as the velocity of an object undergoing simple harmonic motion. The shadow therefore moves with simple harmonic motion.

A valuable example of the relationship between simple harmonic motion and circular motion can be seen in vehicles and machines that use the back-and-forth motion of a piston to create rotational motion in a wheel. Consider the drive wheel of a locomotive. In Figure 13.10, the rods are connected to a piston that moves back and forth in simple harmonic motion. The rods transform the back-and-forth motion of the piston into rotational motion of the wheels. A similar mechanism in an automobile engine transforms the back-and-forth motion of the pistons to rotational motion of the crankshaft.

Period and Frequency

The period T of the shadow in Active Figure 13.8, which represents the time required for one complete trip back and forth, is also the time it takes the ball to make one complete circular trip on the turntable. Because the ball moves through the distance $2\pi A$ (the circumference of the circle) in the time T, the speed v_0 of the ball around the circular path is

APPLICATION
Pistons and Drive Wheels



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Figure 13.10 The drive wheel mechanism of an old locomotive.

$$v_0 = \frac{2\pi A}{T}$$

and the period is

$$T = \frac{2\pi A}{v_0} \qquad [13.7]$$

Imagine that the ball moves from P to Q, a quarter of a revolution, in Active Figure 13.8. The motion of the shadow is equivalent to the horizontal motion of an object on the end of a spring. For this reason, the radius A of the circular motion is the same as the amplitude A of the simple harmonic motion of the shadow. During the quarter of a cycle shown, the shadow moves from a point where the energy of the system (ball and spring) is solely elastic potential energy to a point where the energy is solely kinetic energy. By conservation of energy, we have

$$\tfrac{1}{2}kA^2 = \tfrac{1}{2}mv_0^2$$

which can be solved for A/v_0:

$$\frac{A}{v_0} = \sqrt{\frac{m}{k}}$$

Substituting this expression for A/v_0 in Equation 13.7, we find that the period is

The period of an object–spring system moving with simple harmonic motion ▶

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad [13.8]$$

Equation 13.8 represents the time required for an object of mass m attached to a spring with spring constant k to complete one cycle of its motion. The square root of the mass is in the numerator, so a large mass will mean a large period, in agreement with intuition. The square root of the spring constant k is in the denominator, so a large spring constant will yield a small period, again agreeing with intuition. It's also interesting that the period doesn't depend on the amplitude A.

The inverse of the period is the frequency of the motion:

$$f = \frac{1}{T} \qquad [13.9]$$

Therefore, the frequency of the periodic motion of a mass on a spring is

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad [13.10]$$

◀ Frequency of an object–spring system

The units of frequency are cycles per second (s^{-1}), or **hertz** (Hz). The **angular frequency** ω is

$$\omega = 2\pi f = \sqrt{\frac{k}{m}} \qquad [13.11]$$

◀ Angular frequency of an object–spring system

The frequency and angular frequency are actually closely related concepts. The unit of frequency is cycles per second, where a cycle may be thought of as a unit of angular measure corresponding to 2π radians, or 360°. Viewed in this way, angular frequency is just a unit conversion of frequency. Radian measure is used for angles mainly because it provides a convenient and natural link between linear and angular quantities.

Tip 13.2 Twin Frequencies

The *frequency* gives the number of cycles per second, whereas the *angular frequency* gives the number of radians per second. These two physical concepts are nearly identical and are linked by the conversion factor 2π rad/cycle.

Although an ideal mass–spring system has a period proportional to the square root of the object's mass m, experiments show that a graph of T^2 versus m doesn't pass through the origin. This is because the spring itself has a mass. The coils of the spring oscillate just like the object, except the amplitudes are smaller for all coils but the last. For a cylindrical spring, energy arguments can be used to show that the *effective* additional mass of a light spring is one-third the mass of the spring. The square of the period is proportional to the total oscillating mass, so a graph of T^2 versus total mass (the mass hung on the spring plus the effective oscillating mass of the spring) would pass through the origin.

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Quick Quiz

13.4 An object of mass m is attached to a horizontal spring, stretched to a displacement A from equilibrium and released, undergoing harmonic oscillations on a frictionless surface with period T_0. The experiment is then repeated with a mass of $4m$. What's the new period of oscillation? (a) $2T_0$ (b) T_0 (c) $T_0/2$ (d) $T_0/4$

13.5 Consider the situation in Quick Quiz 13.4. Is the subsequent total mechanical energy of the object with mass $4m$ (a) greater than, (b) less than, or (c) equal to the original total mechanical energy?

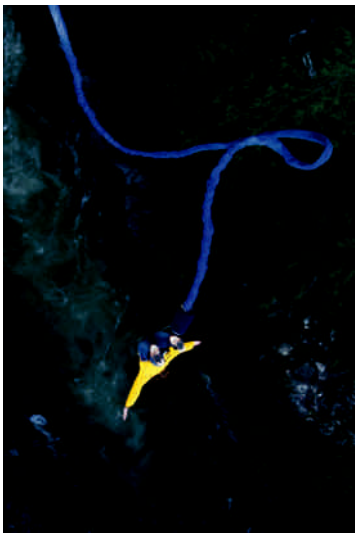
APPLYING PHYSICS 13.1 Bungee Jumping

A bungee cord can be modeled as a spring. If you go bungee jumping, you will bounce up and down at the end of the elastic cord after your dive off a bridge (Fig. 13.11 on page 448). Suppose you perform a dive and measure the frequency of your bouncing. You then move to another bridge, but find that the bungee cord is too long for dives off this bridge. What possible solutions might be applied? In terms of the original frequency, what is the frequency of vibration associated with the solution?

EXPLANATION There are two possible solutions: Make the bungee cord smaller or fold it in half. The latter would be the safer of the two choices, as we'll see. The force exerted by the bungee cord, modeled as a spring, is proportional to the separation of the coils as the spring is extended. First, we extend the spring by a given distance and measure the distance between coils. We then cut the spring in half. If one of the half-springs is now extended by the same distance, the coils will be twice as far apart as

they were for the complete spring. Therefore, it takes twice as much force to stretch the half-spring through the same displacement, so the half-spring has a spring constant twice that of the complete spring. The folded bungee cord can then be modeled as two half-springs in parallel. Each half has a spring constant that is twice the original spring constant of the bungee cord. In addition, an object hanging on the folded bungee cord will experience two forces, one from each half-spring. As a result, the required force for a given extension will be four times as much as for the original bungee cord. The effective spring constant of the folded bungee cord is therefore four times as large as the original spring constant. Because the frequency of oscillation is proportional to the square root of the spring constant, your bouncing frequency on the folded cord will be twice what it was on the original cord.

This discussion neglects the fact that the coils of a spring have an initial separation. It's also important to remember that a shorter coil may lose elasticity more readily, possibly even going beyond the elastic limit for the material, with disastrous results. Bungee jumping is dangerous; discretion is advised! ■



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Figure 13.11 (Applying Physics 13.1) A bungee jumper relies on elastic forces to pull him up short of a deadly impact.

EXAMPLE 13.5 That Car Needs Shock Absorbers!

GOAL Understand the relationships between period, frequency, and angular frequency.

PROBLEM A 1.30×10^3-kg car is constructed on a frame supported by four springs. Each spring has a spring constant of 2.00×10^4 N/m. If two people riding in the car have a combined mass of 1.60×10^2 kg, find the frequency of vibration of the car when it is driven over a pothole in the road. Find also the period and the angular frequency. Assume the weight is evenly distributed.

STRATEGY Because the weight is evenly distributed, each spring supports one-fourth of the mass. Substitute this value and the spring constant into Equation 13.10 to get the frequency. The reciprocal is the period, and multiplying the frequency by 2π gives the angular frequency.

SOLUTION

Compute one-quarter of the total mass:

$$m = \tfrac{1}{4}(m_{\text{car}} + m_{\text{pass}}) = \tfrac{1}{4}(1.30 \times 10^3 \text{ kg} + 1.60 \times 10^2 \text{ kg}) = 365 \text{ kg}$$

Substitute into Equation 13.10 to find the frequency:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{2.00 \times 10^4 \text{ N/m}}{365 \text{ kg}}} = 1.18 \text{ Hz}$$

Invert the frequency to get the period:

$$T = \frac{1}{f} = \frac{1}{1.18 \text{ Hz}} = 0.847 \text{ s}$$

Multiply the frequency by 2π to get the angular frequency:

$$\omega = 2\pi f = 2\pi(1.18 \text{ Hz}) = 7.41 \text{ rad/s}$$

REMARKS Solving this problem didn't require any knowledge of the size of the pothole because the frequency doesn't depend on the amplitude of the motion.

QUESTION 13.5 True or False: The frequency of vibration of a heavy vehicle is greater than that of a lighter vehicle, assuming the two vehicles are supported by the same set of springs.

EXERCISE 13.5 A 45.0-kg boy jumps on a 5.00-kg pogo stick with spring constant 3 650 N/m. Find (a) the angular frequency, (b) the frequency, and (c) the period of the boy's motion.

ANSWERS (a) 8.54 rad/s (b) 1.36 Hz (c) 0.735 s

4. Position, Velocity, and Acceleration as a Function of Time

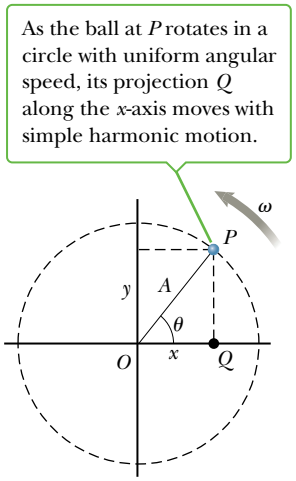


Figure 13.12 A reference circle.

We can obtain an expression for the position of an object moving with simple harmonic motion as a function of time by returning to the relationship between simple harmonic motion and uniform circular motion. Again, consider a ball on the rim of a rotating turntable of radius A, as in Figure 13.12. We refer to the circle made by the ball as the *reference circle* for the motion. We assume the turntable revolves at a *constant* angular speed ω. As the ball rotates on the reference circle, the angle θ made by the line OP with the x-axis changes with time. Meanwhile, the projection of P on the x-axis, labeled point Q, moves back and forth along the axis with simple harmonic motion.

From the right triangle OPQ, we see that $\cos\theta = x/A$. Therefore, the x-coordinate of the ball is

$$x = A\cos\theta$$

Because the ball rotates with constant angular speed, it follows that $\theta = \omega t$ (see Chapter 7), so we have

$$x = A\cos(\omega t) \qquad [13.12]$$

In one complete revolution, the ball rotates through an angle of 2π rad in a time equal to the period T. In other words, the motion repeats itself every T seconds. Therefore,

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T} = 2\pi f \qquad [13.13]$$

where f is the frequency of the motion. The angular speed of the ball as it moves around the reference circle is the same as the angular frequency of the projected simple harmonic motion. Consequently, Equation 13.12 can be written

$$x = A\cos(2\pi f t) \qquad [13.14a]$$

This cosine function represents the position of an object moving with simple harmonic motion as a function of time, and is graphed in Figure 13.13a. Because the cosine function varies between 1 and −1, x varies between A and $-A$. The shape of the graph is called *sinusoidal*.

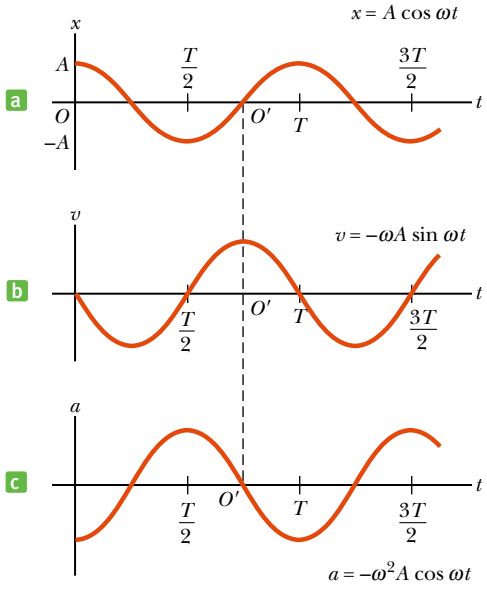


Figure 13.13 (a) Displacement, (b) velocity, and (c) acceleration versus time for an object moving with simple harmonic motion under the initial conditions $x_0 = A$ and $v_0 = 0$ at $t = 0$.

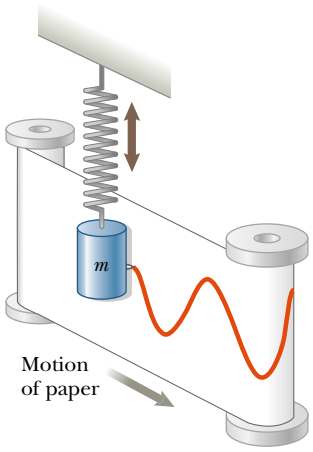


Figure 13.14 An experimental apparatus for demonstrating simple harmonic motion. A pen attached to the oscillating object traces out a sinusoidal wave on the moving chart paper.

Figures 13.13b and 13.13c represent curves for velocity and acceleration as a function of time. To find the equation for the velocity, use Equations 13.6 and 13.14a together with the identity $\cos^2\theta + \sin^2\theta = 1$, obtaining

$$v = -A\omega \sin(2\pi ft) \qquad \textbf{[13.14b]}$$

where we have used the fact that $\omega = \sqrt{k/m}$. The $\pm$ sign is no longer needed, because sine can take both positive and negative values. Deriving an expression for the acceleration involves substituting Equation 13.14a into Equation 13.2, Newton's second law for springs:

$$a = -A\omega^2 \cos(2\pi ft) \qquad \textbf{[13.14c]}$$

The detailed steps of these derivations are left as an exercise for the student. Notice that when the displacement x is at a maximum, at $x = A$ or $x = -A$, the velocity is zero, and when x is zero, the magnitude of the velocity is a maximum. Further, when $x = +A$, its most positive value, the acceleration is a maximum but in the negative x-direction, and when x is at its most negative position, $x = -A$, the acceleration has its maximum value in the positive x-direction. These facts are consistent with our earlier discussion of the points at which v and a reach their maximum, minimum, and zero values.

The maximum values of the position, velocity, and acceleration are always equal to the magnitude of the expression in front of the trigonometric function in each equation because the largest value of either cosine or sine is 1.

Figure 13.14 illustrates one experimental arrangement that demonstrates the sinusoidal nature of simple harmonic motion. An object connected to a spring has a marking pen attached to it. While the object vibrates vertically, a sheet of paper is moved horizontally with constant speed. The pen traces out a sinusoidal pattern.

Quick Quiz

13.6 If the amplitude of a system moving in simple harmonic motion is doubled, which of the following quantities *doesn't* change? (a) total energy (b) maximum speed (c) maximum acceleration (d) period

EXAMPLE 13.6 The Vibrating Object–Spring System

GOAL Identify the physical parameters of a harmonic oscillator from its mathematical description.

PROBLEM **(a)** Find the amplitude, frequency, and period of motion for an object vibrating at the end of a horizontal spring if the equation for its position as a function of time is

$$x = (0.250\ \text{m}) \cos\left(\frac{\pi}{8.00}\, t\right)$$

(b) Find the maximum magnitude of the velocity and acceleration. **(c)** What are the position, velocity, and acceleration of the object after 1.00 s has elapsed?

STRATEGY In part (a) the amplitude and frequency can be found by comparing the given equation with the standard form in Equation 13.14a, matching up the numerical values with the corresponding terms in the standard form. In part (b) the maximum speed will occur when the sine function in Equation 13.14b equals 1 or −1, the extreme values of the sine function (and similarly for the acceleration and the cosine function). In each case, find the magnitude of the expression in front of the trigonometric function. Part (c) is just a matter of substituting values into Equations 13.14a–13.14c.

SOLUTION

(a) Find the amplitude, frequency, and period.

Write the standard form given by Equation 13.14a and underneath it write the given equation:

(1) $x = A\cos(2\pi ft)$

(2) $x = (0.250\ \text{m}) \cos\left(\frac{\pi}{8.00}\, t\right)$

Equate the factors in front of the cosine functions to find the amplitude:

$A = 0.250\ \text{m}$

The angular frequency ω is the factor in front of t in Equations (1) and (2). Equate these factors:

$$\omega = 2\pi f = \frac{\pi}{8.00}\ \text{rad/s} = 0.393\ \text{rad/s}$$

Divide ω by 2π to get the frequency f:

$$f = \frac{\omega}{2\pi} = \boxed{0.062\ 5\ \text{Hz}}$$

The period T is the reciprocal of the frequency:

$$T = \frac{1}{f} = \boxed{16.0\ \text{s}}$$

(b) Find the maximum magnitudes of the velocity and the acceleration.

Calculate the maximum speed from the factor in front of the sine function in Equation 13.14b:

$$v_{max} = A\omega = (0.250\ \text{m})(0.393\ \text{rad/s}) = \boxed{0.098\ 3\ \text{m/s}}$$

Calculate the maximum acceleration from the factor in front of the cosine function in Equation 13.14c:

$$a_{max} = A\omega^2 = (0.250\ \text{m})(0.393\ \text{rad/s})^2 = \boxed{0.038\ 6\ \text{m/s}^2}$$

(c) Find the position, velocity, and acceleration of the object after 1.00 s.

Substitute $t = 1.00$ s in the given equation:

$$x = (0.250\ \text{m})\cos(0.393\ \text{rad}) = \boxed{0.231\ \text{m}}$$

Substitute values into the velocity equation:

$$v = -A\omega \sin(\omega t)$$
$$= -(0.250\ \text{m})(0.393\ \text{rad/s}) \sin(0.393\ \text{rad/s} \cdot 1.00\ \text{s})$$
$$v = \boxed{-0.037\ 6\ \text{m/s}}$$

Substitute values into the acceleration equation:

$$a = -A\omega^2 \cos(\omega t)$$
$$= -(0.250\ \text{m})(0.393\ \text{rad/s}^2)^2 \cos(0.393\ \text{rad/s} \cdot 1.00\ \text{s})$$
$$a = \boxed{-0.035\ 7\ \text{m/s}^2}$$

REMARKS In evaluating the sine or cosine function, the angle is in radians, so you should either set your calculator to evaluate trigonometric functions based on radian measure or convert from radians to degrees.

QUESTION 13.6 If the mass is doubled, is the magnitude of the acceleration of the system at any position (a) doubled, (b) halved, or (c) unchanged?

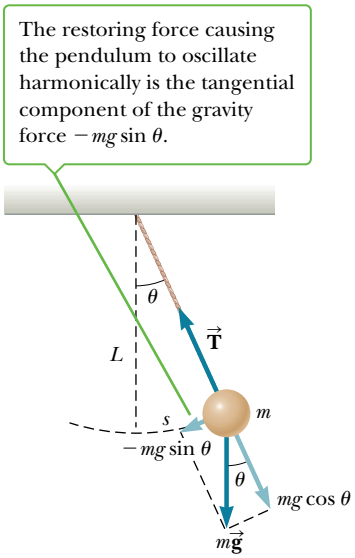
EXERCISE 13.6 If the object–spring system is described by $x = (0.330\ \text{m})\cos(1.50t)$, find (a) the amplitude, the angular frequency, the frequency, and the period, (b) the maximum magnitudes of the velocity and acceleration, and (c) the position, velocity, and acceleration when $t = 0.250$ s.

ANSWERS (a) $A = 0.330$ m, $\omega = 1.50$ rad/s, $f = 0.239$ Hz, $T = 4.18$ s (b) $v_{max} = 0.495$ m/s, $a_{max} = 0.743\ \text{m/s}^2$ (c) $x = 0.307$ m, $v = -0.181$ m/s, $a = -0.691\ \text{m/s}^2$

5. Motion of a Pendulum

A simple pendulum is another mechanical system that exhibits periodic motion. It consists of a small bob of mass m suspended by a light string of length L fixed at its upper end, as in Active Figure 13.15 (page 452). (By a light string, we mean that the string's mass is assumed to be very small compared with the mass of the bob and hence can be ignored.) When released, the bob swings to and fro over the same path, but is its motion simple harmonic?

Answering this question requires examining the restoring force—the force of gravity—that acts on the pendulum. The pendulum bob moves along a circular arc, rather than back and forth in a straight line. When the oscillations are small, however, the motion of the bob is nearly straight, so Hooke's law may apply approximately.



Active Figure 13.15
A simple pendulum consists of a bob of mass m suspended by a light string of length L. (L is the distance from the pivot to the center of mass of the bob.)

Tip 13.3 Pendulum Motion Is Not Harmonic

Remember that the pendulum *does not* exhibit true simple harmonic motion for *any* angle. If the angle is less than about 15°, the motion can be *modeled* as approximately simple harmonic.

In Active Figure 13.15, s is the displacement of the bob from equilibrium along the arc. Hooke's law is $F = -kx$, so we are looking for a similar expression involving s, $F_t = -ks$, where F_t is the force acting in a direction tangent to the circular arc. From the figure, the restoring force is

$$F_t = -mg\sin\theta$$

Since $s = L\theta$, the equation for F_t can be written as

$$F_t = -mg\sin\left(\frac{s}{L}\right)$$

This expression isn't of the form $F_t = -ks$, so in general, the motion of a pendulum is *not* simple harmonic. For small angles less than about 15 degrees, however, the angle θ measured in radians and the sine of the angle are approximately equal. For example, $\theta = 10.0° = 0.175$ rad, and $\sin(10.0°) = 0.174$. Therefore, if we restrict the motion to *small* angles, the approximation $\sin\theta \approx \theta$ is valid, and the restoring force can be written

$$F_t = -mg\sin\theta \approx -mg\theta$$

Substituting $\theta = s/L$, we obtain

$$F_t = -\left(\frac{mg}{L}\right)s$$

This equation follows the general form of Hooke's force law $F_t = -ks$, with $k = mg/L$. We are justified in saying that a pendulum undergoes simple harmonic motion only when it swings back and forth at small amplitudes (or, in this case, small values of θ, so that $\sin\theta \cong \theta$).

Recall that for the object–spring system, the angular frequency is given by Equation 13.11:

$$\omega = 2\pi f = \sqrt{\frac{k}{m}}$$

Substituting the expression of k for a pendulum, we obtain

$$\omega = \sqrt{\frac{mg/L}{m}} = \sqrt{\frac{g}{L}}$$

This angular frequency can be substituted into Equation 13.12, which then mathematically describes the motion of a pendulum. The frequency is just the angular frequency divided by 2π, while the period is the reciprocal of the frequency, or

The period of a simple pendulum depends only on L and g ▶

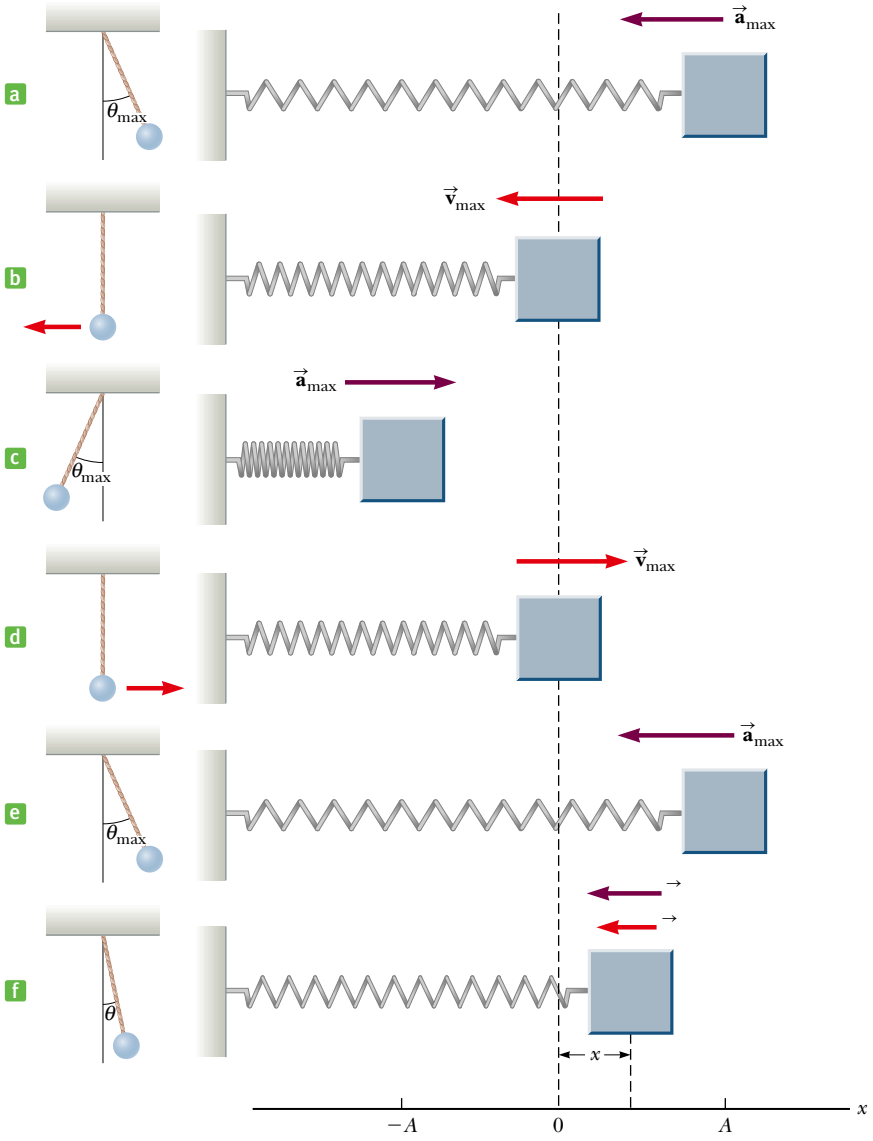
$$T = 2\pi\sqrt{\frac{L}{g}} \qquad [13.15]$$

This equation reveals the somewhat surprising result that the period of a simple pendulum doesn't depend on the mass, but only on the pendulum's length and on the free-fall acceleration. Further, the amplitude of the motion isn't a factor as long as it's relatively small. The analogy between the motion of a simple pendulum and the object–spring system is illustrated in Active Figure 13.16.

Galileo first noted that the period of a pendulum was independent of its amplitude. He supposedly observed this while attending church services at the cathedral in Pisa. The pendulum he studied was a swinging chandelier that was set in motion when someone bumped it while lighting candles. Galileo was able to measure its period by timing the swings with his pulse.

APPLICATION
Pendulum Clocks

The dependence of the period of a pendulum on its length and on the free-fall acceleration allows us to use a pendulum as a timekeeper for a clock. A number of clock designs employ a pendulum, with the length adjusted so that its period serves as the basis for the rate at which the clock's hands turn. Of course, these



Active Figure 13.16
Simple harmonic motion for an object–spring system, and its analogy, the motion of a simple pendulum.

clocks are used at different locations on the Earth, so there will be some variation of the free-fall acceleration. To compensate for this variation, the pendulum of a clock should have some movable mass so that the effective length can be adjusted.

Geologists often make use of the simple pendulum and Equation 13.15 when prospecting for oil or minerals. Deposits beneath the Earth's surface can produce irregularities in the free-fall acceleration over the region being studied. A specially designed pendulum of known length is used to measure the period, which in turn is used to calculate *g*. Although such a measurement in itself is inconclusive, it's an important tool for geological surveys.

APPLICATION
Use of Pendulum in Prospecting

Quick Quiz

13.7 A simple pendulum is suspended from the ceiling of a stationary elevator, and the period is measured. If the elevator moves with constant velocity, does the period (a) increase, (b) decrease, or (c) remain the same? If the elevator accelerates upward, does the period (a) increase, (b) decrease, or (c) remain the same?

13.8 A pendulum clock depends on the period of a pendulum to keep correct time. Suppose a pendulum clock is keeping correct time and then Dennis the Menace slides the bob of the pendulum downward on the oscillating rod. Does the clock run (a) slow, (b) fast, or (c) correctly?

13.9 The period of a simple pendulum is measured to be T on the Earth. If the same pendulum were set in motion on the Moon, would its period be (a) less than T, (b) greater than T, or (c) equal to T?

EXAMPLE 13.7 Measuring the Value of g

GOAL Determine g from pendulum motion.

PROBLEM Using a small pendulum of length 0.171 m, a geophysicist counts 72.0 complete swings in a time of 60.0 s. What is the value of g in this location?

STRATEGY First calculate the period of the pendulum by dividing the total time by the number of complete swings. Solve Equation 13.15 for g and substitute values.

SOLUTION

Calculate the period by dividing the total elapsed time by the number of complete oscillations:

$$T = \frac{\text{time}}{\#\text{ of oscillations}} = \frac{60.0\text{ s}}{72.0} = 0.833\text{ s}$$

Solve Equation 13.15 for g and substitute values:

$$T = 2\pi\sqrt{\frac{L}{g}} \quad \rightarrow \quad T^2 = 4\pi^2\frac{L}{g}$$

$$g = \frac{4\pi^2 L}{T^2} = \frac{(39.5)(0.171\text{ m})}{(0.833\text{ s})^2} = 9.73\text{ m/s}^2$$

REMARKS Measuring such a vibration is a good way of determining the local value of the acceleration of gravity.

QUESTION 13.7 True or False: A simple pendulum of length 0.50 m has a larger frequency of vibration than a simple pendulum of length 1.0 m.

EXERCISE 13.7 What would be the period of the 0.171-m pendulum on the Moon, where the acceleration of gravity is 1.62 m/s^2?

ANSWER 2.04 s

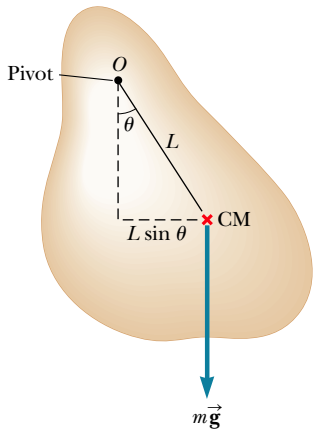


Figure 13.17 A physical pendulum pivoted at O.

The Physical Pendulum

The simple pendulum discussed thus far consists of a mass attached to a string. A pendulum, however, can be made from an object of any shape. The general case is called the *physical pendulum*.

In Figure 13.17 a rigid object is pivoted at point O, which is a distance L from the object's center of mass. The center of mass oscillates along a circular arc, just like the simple pendulum. The period of a physical pendulum is given by

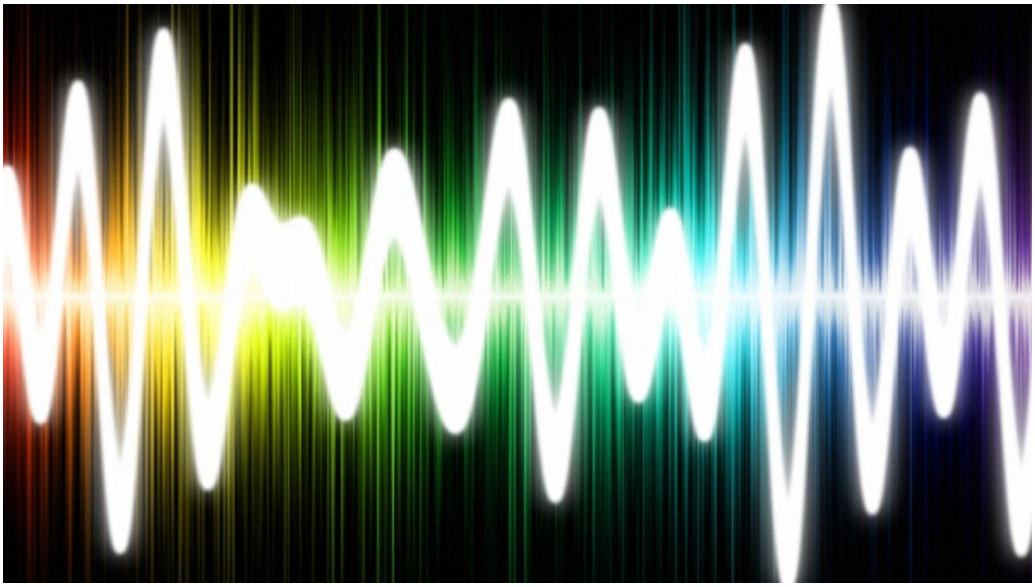
$$T = 2\pi\sqrt{\frac{I}{mgL}} \qquad [13.16]$$

where I is the object's moment of inertia and m is the object's mass. As a check, notice that in the special case of a simple pendulum with an arm of length L and negligible mass, the moment of inertia is $I = mL^2$. Substituting into Equation 13.16 results in

$$T = 2\pi\sqrt{\frac{mL^2}{mgL}} = 2\pi\sqrt{\frac{L}{g}}$$

which is the correct period for a simple pendulum.

6. Waves



The world is full of waves: sound waves, waves on a string, seismic waves, and electromagnetic waves, such as visible light, radio waves, television signals, and x-rays. All these waves have as their source a vibrating object, so we can apply the concepts of simple harmonic motion in describing them.

In the case of sound waves, the vibrations that produce waves arise from sources such as a person's vocal chords or a plucked guitar string. The vibrations of elec-trons in an antenna produce radio or television waves, and the simple up-and-down motion of a hand can produce a wave on a string. Certain concepts are com-mon to all waves, regardless of their nature. In the remainder of this chapter, we focus our attention on the general properties of waves. In later chapters we will study specific types of waves, such as sound waves and electromagnetic waves.

What Is a Wave?

When you drop a pebble into a pool of water, the disturbance produces water waves, which move away from the point where the pebble entered the water. A leaf floating near the disturbance moves up and down and back and forth about its original position, but doesn't undergo any net displacement attributable to the disturbance. This means that the water wave (or disturbance) moves from one place to another, *but the water isn't carried with it.*

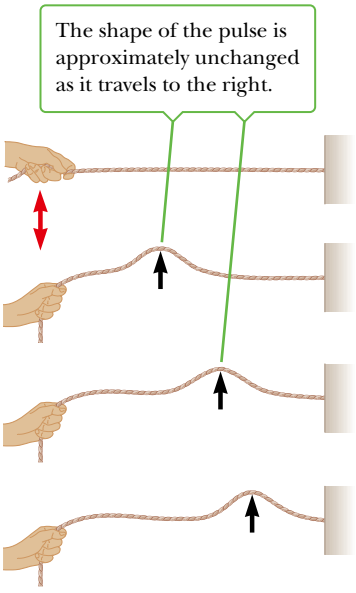
When we observe a water wave, we see a rearrangement of the water's surface. Without the water, there wouldn't be a wave. Similarly, a wave traveling on a string wouldn't exist without the string. Sound waves travel through air as a result of pressure variations from point to point. Therefore, we can consider a wave to be *the motion of a disturbance.* In Chapter 21 we discuss electromagnetic waves, which don't require a medium.

The mechanical waves discussed in this chapter require (1) some source of disturbance, (2) a medium that can be disturbed, and (3) some physical connection or mechanism through which adjacent portions of the medium can influence each other. All waves carry energy and momentum. The amount of energy transmitted through a medium and the mechanism responsible for the transport of energy differ from case to case. The energy carried by ocean waves during a storm, for example, is much greater than the energy carried by a sound wave generated by a single human voice.

APPLYING PHYSICS 11.2 Burying Bond

At one point in *On Her Majesty's Secret Service,* a James Bond film from the 1960s, Bond was escaping on skis. He had a good lead and was a hard-to-hit moving target. There was no point in wasting bullets shooting at him, so why did the bad guys open fire?

EXPLANATION These misguided gentlemen had a good understanding of the physics of waves. An impulsive sound, like a gunshot, can cause an acoustical disturbance that propagates through the air. If it impacts a ledge of snow that is ready to break free, an avalanche can result. Such a disaster occurred in 1916 during World War I when Austrian soldiers in the Alps were smothered by an avalanche caused by cannon fire. So the bad guys, who have never been able to hit Bond with a bullet, decided to use the sound of gunfire to start an avalanche. ■



Active Figure 13.21 A hand moves the end of a stretched string up and down once (red arrow), causing a pulse to travel along the string.

Types of Waves

One of the simplest ways to demonstrate wave motion is to flip one end of a long string that is under tension and has its opposite end fixed, as in Active Figure 13.21. The bump (called a pulse) travels to the right with a definite speed. A disturbance of this type is called a **traveling wave.** The figure shows the shape of the string at three closely spaced times.

As such a wave pulse travels along the string, **each segment of the string that is disturbed moves in a direction perpendicular to the wave motion.** Figure 13.22 illustrates this point for a particular tiny segment *P.* The string never moves in the direction of the wave. A traveling wave in which the particles of the disturbed medium move in a direction perpendicular to the wave velocity is called a **transverse wave.** Figure 13.23a illustrates the formation of transverse waves on a long spring.

In another class of waves, called **longitudinal waves, the elements of the medium undergo displacements parallel to the direction of wave motion.** Sound waves in air are longitudinal. Their disturbance corresponds to a series of high- and low-pressure regions that may travel through air or through any material medium with a certain speed. A longitudinal pulse can easily be produced in a stretched spring, as in Figure 13.23b. The free end is pumped back and forth along the length of the spring. This action produces compressed and stretched regions of the coil that travel along the spring, parallel to the wave motion.

Waves need not be purely transverse or purely longitudinal: ocean waves exhibit a superposition of both types. When an ocean wave encounters a cork, the cork executes a circular motion, going up and down while going forward and back.

Another type of wave, called a **soliton,** consists of a solitary wave front that propagates in isolation. Ordinary water waves generally spread out and dissipate, but solitons tend to maintain their form. The study of solitons began in 1849, when Scottish engineer John Scott Russell noticed a solitary wave leaving the turbulence in front of a barge and propagating forward all on its own. The wave maintained its shape and traveled down a canal at about 10 mi/h. Russell chased the wave two miles on horseback before losing it. Only in the 1960s did scientists take solitons seriously; they are now widely used to model physical phenomena, from elementary particles to the Giant Red Spot of Jupiter.

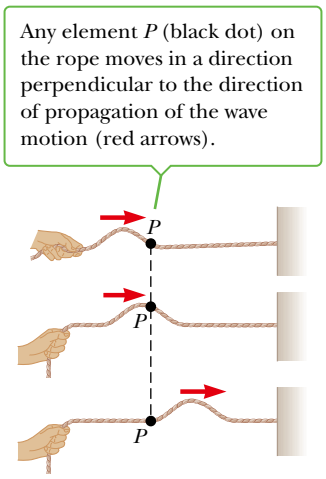


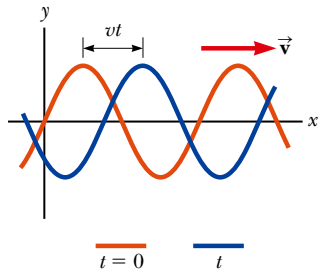
Figure 13.22 A pulse traveling on a stretched string is a transverse wave.

Picture of a Wave

Active Figure 13.24 shows the curved shape of a vibrating string. This pattern is a sinusoidal curve, the same as in simple harmonic motion. The brown curve can be thought of as a snapshot of a traveling wave taken at some instant of time, say, $t = 0$; the blue curve is a snapshot of the same traveling wave at a later time. This picture can also be used to represent a wave on water. In such a case, a high point would correspond to the *crest* of the wave and a low point to the *trough* of the wave.

The same waveform can be used to describe a longitudinal wave, even though no up-and-down motion is taking place. Consider a longitudinal wave traveling on a spring. Figure 13.25a is a snapshot of this wave at some instant, and Figure 13.25b shows the sinusoidal curve that represents the wave. Points where the coils of the spring are compressed correspond to the crests of the waveform, and stretched regions correspond to troughs.

The type of wave represented by the curve in Figure 13.25b is often called a *density wave* or *pressure wave,* because the crests, where the spring coils are compressed, are regions of high density, and the troughs, where the coils are stretched, are regions of low density. Sound waves are longitudinal waves, propagating as a series of high- and low-density regions.



Active Figure 13.24
A one-dimensional sinusoidal wave traveling to the right with a speed v. The brown curve is a snapshot of the wave at $t = 0$, and the blue curve is another snapshot at some later time t.

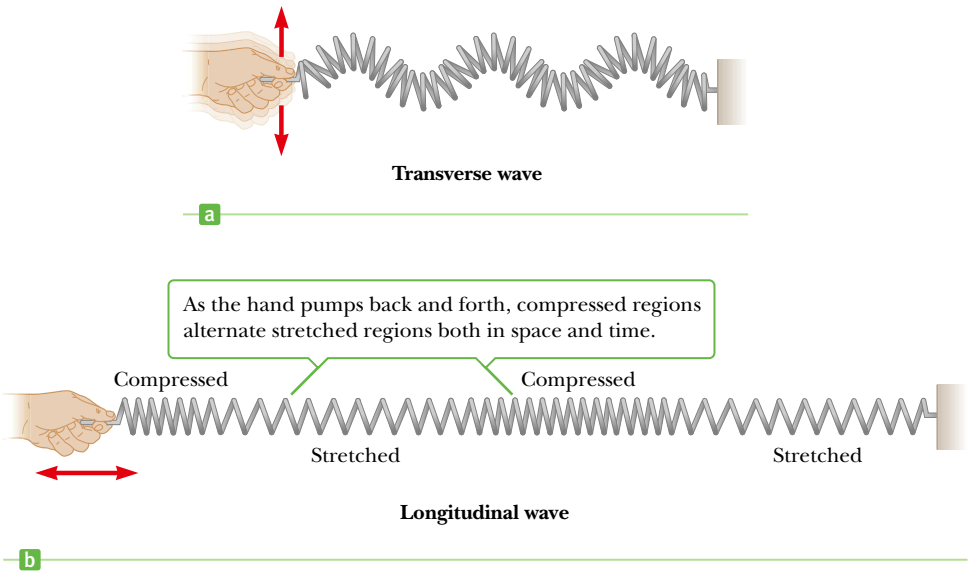


Figure 13.23 (a) A transverse wave is set up in a spring by moving one end of the spring perpendicular to its length. (b) A longitudinal wave along a stretched spring.

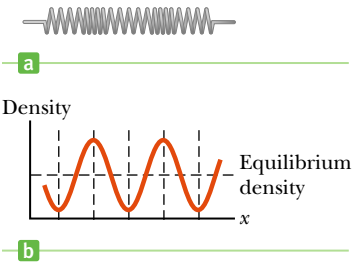
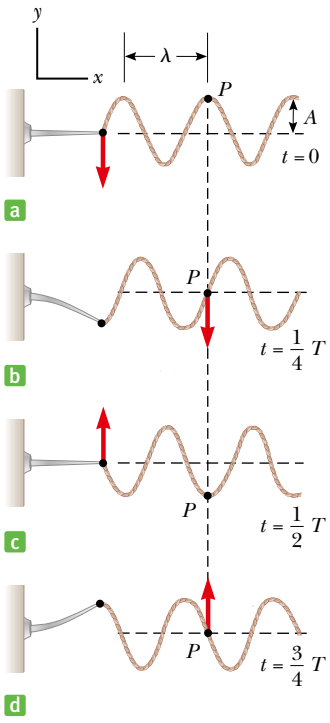


Figure 13.25 (a) A longitudinal wave on a spring. (b) The crests of the waveform correspond to compressed regions of the spring, and the troughs correspond to stretched regions of the spring.



Active Figure 13.26
One method for producing traveling waves on a continuous string. The left end of the string is connected to a blade that is set vibrating. Every part of the string, such as point *P*, oscillates vertically with simple harmonic motion.

7. Frequency, Amplitude, and Wavelength

Active Figure 13.26 illustrates a method of producing a continuous wave or a steady stream of pulses on a very long string. One end of the string is connected to a blade that is set vibrating. As the blade oscillates vertically with simple harmonic motion, a traveling wave moving to the right is set up in the string. Active Figure 13.26 consists of views of the wave at intervals of one-quarter of a period. Note that **each small segment of the string, such as *P*, oscillates vertically in the *y*-direction with simple harmonic motion.** That must be the case because each segment follows the simple harmonic motion of the blade. Every segment of the string can therefore be treated as a simple harmonic oscillator vibrating with the same frequency as the blade that drives the string.

The frequencies of the waves studied in this course will range from rather low values for waves on strings and waves on water, to values for sound waves between 20 Hz and 20 000 Hz (recall that 1 Hz = 1 s^{-1}), to much higher frequencies for electromagnetic waves. These waves have different physical sources, but can be described with the same concepts.

The horizontal dashed line in Active Figure 13.26 represents the position of the string when no wave is present. The maximum distance the string moves above or below this equilibrium value is called the **amplitude** A of the wave. For the waves we work with, the amplitudes at the crest and the trough will be identical.

Active Figure 13.26a illustrates another characteristic of a wave. The horizontal arrows show the distance between two successive points that behave identically. This distance is called the **wavelength** λ (the Greek letter lambda).

We can use these definitions to derive an expression for the speed of a wave. We start with the defining equation for the **wave speed** v:

$$v = \frac{\Delta x}{\Delta t}$$

The wave speed is the speed at which a particular part of the wave—say, a crest—moves through the medium.

A wave advances a distance of one wavelength in a time interval equal to one period of the vibration. Taking $\Delta x = \lambda$ and $\Delta t = T$, we see that

$$v = \frac{\lambda}{T}$$

Because the frequency is the reciprocal of the period, we have

Wave speed ▶

$$v = f\lambda \qquad [13.17]$$

This important general equation applies to many different types of waves, such as sound waves and electromagnetic waves.

EXAMPLE 13.8 A Traveling Wave

GOAL Obtain information about a wave directly from its graph.

PROBLEM A wave traveling in the positive *x*-direction is pictured in Figure 13.27a. Find the amplitude, wavelength, speed, and period of the wave if it has a frequency of 8.00 Hz. In Figure 13.27a, $\Delta x = 40.0$ cm and $\Delta y = 15.0$ cm.

STRATEGY The amplitude and wavelength can be read directly from the figure: The maximum vertical displacement is the amplitude, and the distance from one crest to the next is the wavelength. Multiplying the wavelength by the frequency gives the speed, whereas the period is the reciprocal of the frequency.

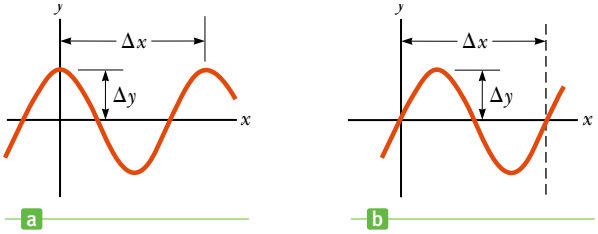


Figure 13.27 (a) (Example 13.8) (b) (Exercise 13.8)

SOLUTION

The maximum wave displacement is the amplitude A: $A = \Delta y = 15.0 \text{ cm} = 0.150 \text{ m}$

The distance from crest to crest is the wavelength: $\lambda = \Delta x = 40.0 \text{ cm} = 0.400 \text{ m}$

Multiply the wavelength by the frequency to get the speed: $v = f\lambda = (8.00 \text{ Hz})(0.400 \text{ m}) = 3.20 \text{ m/s}$

Take the reciprocal of the frequency to get the period: $T = \frac{1}{f} = \frac{1}{8.00 \text{ Hz}} = 0.125 \text{ s}$

REMARKS It's important not to confuse the wave with the medium it travels in. A wave is energy transmitted through a medium; some waves, such as light waves, don't require a medium.

QUESTION 13.8 Is the frequency of a wave affected by the wave's amplitude?

EXERCISE 13.8 A wave traveling in the positive x-direction is pictured in Figure 13.27b. Find the amplitude, wavelength, speed, and period of the wave if it has a frequency of 15.0 Hz. In the figure, $\Delta x = 72.0$ cm and $\Delta y = 25.0$ cm.

ANSWERS $A = 0.250$ m, $\lambda = 0.720$ m, $v = 10.8$ m/s, $T = 0.0667$ s

EXAMPLE 13.9 Sound and Light

GOAL Perform elementary calculations using speed, wavelength, and frequency.

PROBLEM A wave has a wavelength of 3.00 m. Calculate the frequency of the wave if it is **(a)** a sound wave and **(b)** a light wave. Take the speed of sound as 343 m/s and the speed of light as 3.00×10^8 m/s.

SOLUTION

(a) Find the frequency of a sound wave with $\lambda = 3.00$ m.

Solve Equation 3.17 for the frequency and substitute: **(1)** $f = \frac{v}{\lambda} = \frac{343 \text{ m/s}}{3.00 \text{ m}} = 114 \text{ Hz}$

(b) Find the frequency of a light wave with $\lambda = 3.00$ m.

Substitute into Equation (1), using the speed of light for c: $f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{3.00 \text{ m}} = 1.00 \times 10^8 \text{ Hz}$

REMARKS The same equation can be used to find the frequency in each case, despite the great difference between the physical phenomena. Notice how much larger frequencies of light waves are than frequencies of sound waves.

QUESTION 13.9 A wave in one medium encounters a new medium and enters it. Which of the following wave properties will be affected in this process? (a) wavelength (b) frequency (c) speed

EXERCISE 13.9 (a) Find the wavelength of an electromagnetic wave with frequency 9.00 GHz = 9.00×10^9 Hz (G = giga = 10^9), which is in the microwave range. (b) Find the speed of a sound wave in an unknown fluid medium if a frequency of 567 Hz has a wavelength of 2.50 m.

ANSWERS (a) 0.033 3 m (b) 1.42×10^3 m/s

8. The Speed of Waves on Strings

In this section we focus our attention on the speed of a transverse wave on a stretched string.

For a vibrating string, there are two speeds to consider. One is the speed of the physical string that vibrates up and down, transverse to the string, in the y-direction. The other is the *wave speed,* which is the rate at which the disturbance propagates along the length of the string in the x-direction. We wish to find an expression for the wave speed.

If a horizontal string under tension is pulled vertically and released, it starts at its maximum displacement, $y = A$, and takes a certain amount of time to go to $y = -A$ and back to A again. This amount of time is the period of the wave, and is the same as the time needed for the wave to advance *horizontally* by one wavelength. Dividing the wavelength by the period of one transverse oscillation gives the wave speed.

For a fixed wavelength, a string under greater tension F has a greater wave speed because the period of vibration is shorter, and the wave advances one wavelength during one period. It also makes sense that a string with greater mass per unit length, μ, vibrates more slowly, leading to a longer period and a slower wave speed. The wave speed is given by

$$v = \sqrt{\frac{F}{\mu}} \qquad [13.18]$$

where F is the tension in the string and μ is the mass of the string per unit length, called the *linear density*. From Equation 13.18, it's clear that a larger tension F results in a larger wave speed, whereas a larger linear density μ gives a slower wave speed, as expected.

According to Equation 13.18, the propagation speed of a mechanical wave, such as a wave on a string, depends only on the properties of the string through which the disturbance travels. It doesn't depend on the amplitude of the vibration. This turns out to be generally true of waves in various media.

A proof of Equation 13.18 requires calculus, but dimensional analysis can easily verify that the expression is dimensionally correct. Note that the dimensions of F are ML/T^2, and the dimensions of μ are M/L. The dimensions of F/μ are therefore L^2/T^2, so those of $\sqrt{F/\mu}$ are L/T, giving the dimensions of speed. No other combination of F and μ is dimensionally correct, so in the case in which the tension and mass density are the only relevant physical factors, we have verified Equation 13.18 up to an overall constant.

APPLICATION
Bass Guitar Strings

According to Equation 13.18, we can increase the speed of a wave on a stretched string by increasing the tension in the string. Increasing the mass per unit length, on the other hand, decreases the wave speed. These physical facts lie behind the metallic windings on the bass strings of pianos and guitars. The windings increase the mass per unit length, μ, decreasing the wave speed and hence the frequency, resulting in a lower tone. Tuning a string to a desired frequency is a simple matter of changing the tension in the string.

EXAMPLE 13.10 A Pulse Traveling on a String

GOAL Calculate the speed of a wave on a string.

PROBLEM A uniform string has a mass M of 0.030 0 kg and a length L of 6.00 m. Tension is maintained in the string by suspending a block of mass $m = 2.00$ kg from one end (Fig. 13.28). **(a)** Find the speed of a transverse wave pulse on this string. **(b)** Find the time it takes the pulse to travel from the wall to the pulley. Neglect the mass of the hanging part of the string.

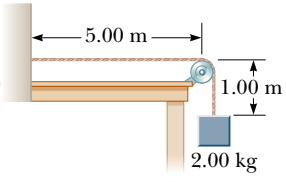


Figure 13.28 (Example 13.10) The tension F in the string is maintained by the suspended block. The wave speed is given by the expression $v = \sqrt{F/\mu}$.

STRATEGY The tension F can be obtained from Newton's second law for equilibrium applied to the block, and the mass per unit length of the string is $\mu = M/L$. With these quantities, the speed of the transverse pulse can be found by substitution into Equation 13.18. Part (b) requires the formula $d = vt$.

SOLUTION

(a) Find the speed of the wave pulse.

Apply the second law to the block: the tension F is equal and opposite to the force of gravity.

$$\sum F = F - mg = 0 \quad \rightarrow \quad F = mg$$

Substitute expressions for F and μ into Equation 13.18:

$$v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{mg}{M/L}}$$

$$= \sqrt{\frac{(2.00\ \text{kg})(9.80\ \text{m/s}^2)}{(0.030\ 0\ \text{kg})/(6.00\ \text{m})}} = \sqrt{\frac{19.6\ \text{N}}{0.005\ 00\ \text{kg/m}}}$$

$$= 62.6\ \text{m/s}$$

(b) Find the time it takes the pulse to travel from the wall to the pulley.

Solve the distance formula for time:

$$t = \frac{d}{v} = \frac{5.00\ \text{m}}{62.6\ \text{m/s}} = 0.079\ 9\ \text{s}$$

REMARKS Don't confuse the speed of the wave on the string with the speed of the sound wave produced by the vibrating string. (See Chapter 14.)

QUESTION 13.10 If the mass of the block is quadrupled, what happens to the speed of the wave?

EXERCISE 13.10 To what tension must a string with mass 0.010 0 kg and length 2.50 m be tightened so that waves will travel on it at a speed of 125 m/s?

ANSWER 62.5 N

9. Interference of Waves

Many interesting wave phenomena in nature require two or more waves passing through the same region of space at the same time. **Two traveling waves can meet and pass through each other without being destroyed or even altered.** For instance, when two pebbles are thrown into a pond, the expanding circular waves don't destroy each other. In fact, the ripples pass through each other. Likewise, when sound waves from two sources move through air, they pass through each other. In the region of overlap, the resultant wave is found by adding the displacements of the individual waves. For such analyses, the **superposition principle** applies:

> When two or more traveling waves encounter each other while moving through a medium, the resultant wave is found by adding together the displacements of the individual waves point by point.

Experiments show that the superposition principle is valid only when the individual waves have small amplitudes of displacement, which is an assumption we make in all our examples.

Figures 13.29a and 13.29b show two waves of the same amplitude and frequency. If at some instant of time these two waves were traveling through the same region of space, the resultant wave at that instant would have a shape like that shown in Figure 13.29c. For example, suppose the waves are water waves of amplitude 1 m. At the instant they overlap so that crest meets crest and trough meets trough, the resultant wave has an amplitude of 2 m. Waves coming together like that are said to be *in phase* and to exhibit **constructive interference.**

Figures 13.30a (page 462) and 13.30b show two similar waves. In this case, however, the crest of one coincides with the trough of the other, so one wave is *inverted* relative to the other. The resultant wave, shown in Figure 13.30c, is seen to be a

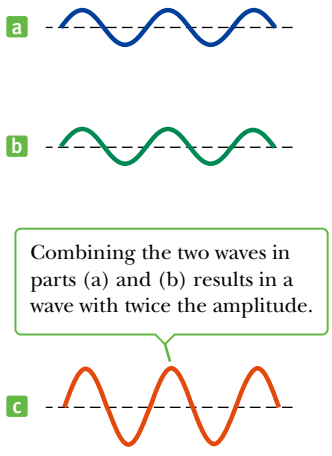


Figure 13.29 Constructive interference. If two waves having the same frequency and amplitude are in phase, as in (a) and (b), the resultant wave when they combine (c) has the same frequency as the individual waves, but twice their amplitude.

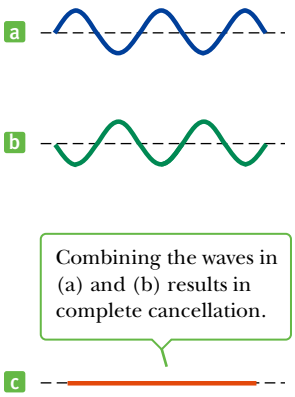
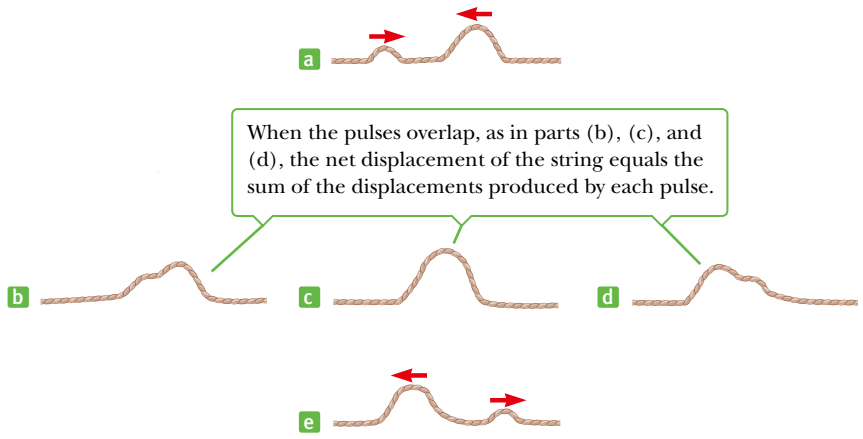


Figure 13.30 Destructive interference. The two waves in (a) and (b) have the same frequency and amplitude but are 180° out of phase.



Active Figure 13.32
Two wave pulses traveling on a stretched string in opposite directions pass through each other.

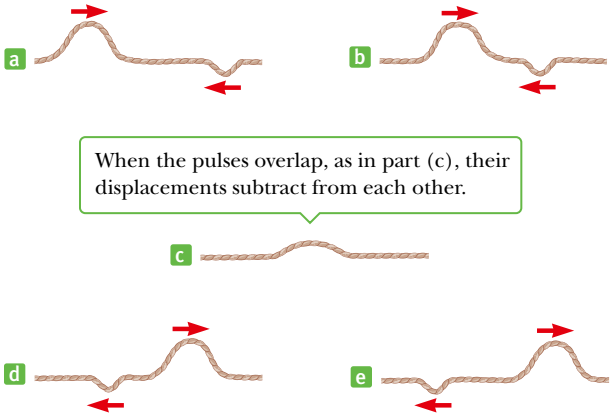


Martin Dohrn/SPL/Photo Researchers, Inc.

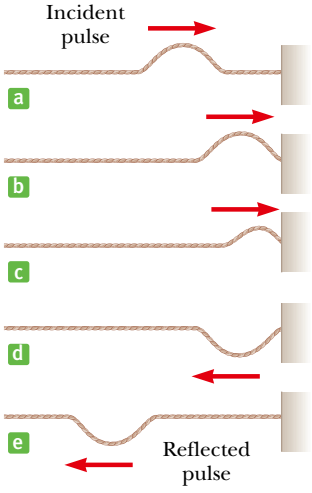
Figure 13.31 Interference patterns produced by outward-spreading waves from many drops of liquid falling into a body of water.

state of complete cancellation. If these were water waves coming together, one of the waves would exert an upward force on an individual drop of water at the same instant the other wave was exerting a downward force. The result would be no motion of the water at all. In such a situation the two waves are said to be 180° out of phase and to exhibit **destructive interference.** Figure 13.31 illustrates the interference of water waves produced by drops of water falling into a pond.

Active Figure 13.32 shows constructive interference in two pulses moving toward each other along a stretched string; Active Figure 13.33 shows destructive interference in two pulses. Notice in both figures that when the two pulses separate, their shapes are unchanged, as if they had never met!



Active Figure 13.33
Two wave pulses traveling in opposite directions with displacements that are inverted relative to each other.

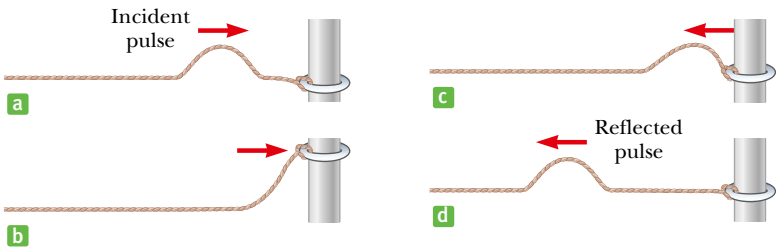


Active Figure 13.34
The reflection of a traveling wave at the fixed end of a stretched string. Note that the reflected pulse is inverted, but its shape remains the same.

10. Reflection of Waves

In our discussion so far, we have assumed waves could travel indefinitely without striking anything. Often, such conditions are not realized in practice. Whenever a traveling wave reaches a boundary, part or all of the wave is reflected. For example, consider a pulse traveling on a string that is fixed at one end (Active Fig. 13.34). When the pulse reaches the wall, it is reflected.

Note that the reflected pulse is inverted. This can be explained as follows: When the pulse meets the wall, the string exerts an upward force on the wall. According to Newton's third law, the wall must exert an equal and opposite (downward) reaction force on the string. This downward force causes the pulse to invert on reflection.



Active Figure 13.35
The reflection of a traveling wave at the free end of a stretched string. In this case the reflected pulse is not inverted.

Now suppose the pulse arrives at the string's end, and the end is attached to a ring of negligible mass that is free to slide along the post without friction (Active Fig. 13.35). Again the pulse is reflected, but this time it is not inverted. On reaching the post, the pulse exerts a force on the ring, causing it to accelerate upward. The ring is then returned to its original position by the downward component of the tension in the string.

An alternate method of showing that a pulse is reflected without inversion when it strikes a free end of a string is to send the pulse down a string hanging vertically. When the pulse hits the free end, it's reflected without inversion, just as is the pulse in Active Figure 13.35.

Finally, when a pulse reaches a boundary, it's partly reflected and partly transmitted past the boundary into the new medium. This effect is easy to observe in the case of two ropes of different density joined at some boundary.

SUMMARY

1. Hooke's Law

Simple harmonic motion occurs when the net force on an object along the direction of motion is proportional to the object's displacement and in the opposite direction:

$$F_s = -kx \qquad [13.1]$$

This equation is called Hooke's law. The time required for one complete vibration is called the **period** of the motion. The reciprocal of the period is the **frequency** of the motion, which is the number of oscillations per second.

When an object moves with simple harmonic motion, its **acceleration** as a function of position is

$$a = -\frac{k}{m}x \qquad [13.2]$$

2. Elastic Potential Energy

The energy stored in a stretched or compressed spring or in some other elastic material is called **elastic potential energy:**

$$PE_s \equiv \tfrac{1}{2}kx^2 \qquad [13.3]$$

The **velocity** of an object as a function of position, when the object is moving with simple harmonic motion, is

$$v = \pm\sqrt{\frac{k}{m}(A^2 - x^2)} \qquad [13.6]$$

3. Comparing Simple Harmonic Motion with Uniform Circular Motion

The **period** of an object of mass m moving with simple harmonic motion while attached to a spring of spring constant k is

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad [13.8]$$

where T is independent of the amplitude A.

The **frequency** of an object–spring system is $f = 1/T$. The **angular frequency** ω of the system in rad/s is

$$\omega = 2\pi f = \sqrt{\frac{k}{m}} \qquad [13.11]$$

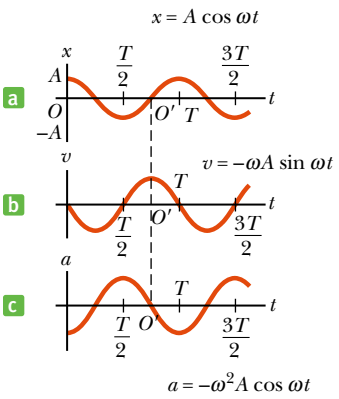
4. Position, Velocity, and Acceleration as a Function of Time

When an object is moving with simple harmonic motion, the **position, velocity,** and **acceleration** of the object as a function of time are given by

$$x = A\cos(2\pi ft) \qquad [13.14a]$$

$$v = -A\omega\sin(2\pi ft) \qquad [13.14b]$$

$$a = -A\omega^2\cos(2\pi ft) \qquad [13.14c]$$



(a) Displacement, (b) velocity, and (c) acceleration versus time for an object moving with simple harmonic motion under the initial conditions $x_0 = A$ and $v_0 = 0$ at $t = 0$.

5. Motion of a Pendulum

A **simple pendulum** of length L moves with simple harmonic motion for small angular displacements from the vertical, with a period of

$$T = 2\pi\sqrt{\frac{L}{g}} \qquad [13.15]$$

7 Waves

In a **transverse wave** the elements of the medium move in a direction perpendicular to the direction of the wave. An example is a wave on a stretched string.

In a **longitudinal wave** the elements of the medium move parallel to the direction of the wave velocity. An example is a sound wave.

8 Frequency, Amplitude, and Wavelength

The relationship of the speed, wavelength, and frequency of a wave is

$$v = f\lambda \qquad [13.17]$$

This relationship holds for a wide variety of different waves.

9. The Speed of Waves on Strings

The speed of a wave traveling on a stretched string of mass per unit length μ and under tension F is

$$v = \sqrt{\frac{F}{\mu}} \qquad [13.18]$$

10. Interference of Waves

The **superposition principle** states that if two or more traveling waves are moving through a medium, the resultant wave is found by adding the individual waves together point by point. When waves meet crest to crest and trough to trough, they undergo **constructive interference.** When crest meets trough, the waves undergo **destructive interference.**

11. Reflection of Waves

When a wave pulse reflects from a rigid boundary, the pulse is inverted. When the boundary is free, the reflected pulse is not inverted.

MULTIPLE-CHOICE QUESTIONS

ENHANCED WebAssign The multiple-choice questions in this chapter may be assigned online in Enhanced WebAssign.

1. The distance between the crest of a water wave and the next trough is 2 m. If the frequency of a particular wave is 2 Hz, what is the speed of the wave? (a) 4 m/s (b) 1 m/s (c) 8 m/s (d) 2 m/s (e) impossible to determine from the information given

2. The position of an object moving with simple harmonic motion is given by $x = 4\cos(6\pi t)$, where x is in meters and t is in seconds. What is the period of the oscillating system? (a) 4 s (b) $\frac{1}{6}$ s (c) $\frac{1}{3}$ s (d) 6π s (e) impossible to determine from the information given

3. A block-spring system vibrating on a frictionless, horizontal surface with an amplitude of 6.0 cm has a total energy of 12 J. If the block is replaced by one having twice the mass of the original block and the amplitude of the motion is again 6.0 cm, what is the energy of the more massive system? (a) 12 J (b) 24 J (c) 6 J (d) 48 J (e) none of those answers

4. A mass of 0.40 kg, hanging from a spring with a spring constant of 80.0 N/m, is set into an up-and-down simple harmonic motion. If the mass is displaced from equilibrium by 0.10 m and released from rest, what is its speed when moving through the equilibrium point? (a) 0 (b) 1.4 m/s (c) 2.0 m/s (d) 3.4 m/s (e) 4.2 m/s

5. If an object of mass m attached to a light spring is replaced by one of mass $9m$, the frequency of the vibrating system changes by what multiplicative factor? (a) $\frac{1}{9}$ (b) $\frac{1}{3}$ (c) 3.0 (d) 9.0 (e) 6.0

6. An object of mass 0.40 kg, hanging from a spring with a spring constant of 8.0 N/m, is set into an up-and-down simple harmonic motion. What is the magnitude of the acceleration of the object when it is at its maximum displacement of 0.10 m? (a) 0 (b) 0.45 m/s^2 (c) 1.0 m/s^2 (d) 2.0 m/s^2 (e) 2.40 m/s^2

7. A runaway railroad car with mass 3.0×10^5 kg coasts across a level track at 2.0 m/s when it collides elastically with a spring-loaded bumper at the end of the track. If the spring constant of the bumper is 2.0×10^6 N/m, what is the maximum compression of the spring during the collision? (a) 0.77 m (b) 0.58 m (c) 0.34 m (d) 1.07 m (e) 1.24 m

8. If a simple pendulum oscillates with a small amplitude and its length is doubled, what happens to the

frequency of its motion? (a) It doubles. (b) It becomes $\sqrt{2}$ times as large. (c) It becomes half as large. (d) It becomes $1/\sqrt{2}$ as large. (e) It remains the same.

9. A simple pendulum has a period of 2.5 s. What is its period if its length is made four times as large? (a) 0.625 s (b) 1.25 s (c) 2.5 s (d) 3.54 s (e) 5.0 s

10. A particle on a spring moves in simple harmonic motion along the *x*-axis between turning points at x_1 = 100 cm and x_2 = 140 cm. At which of the following positions does the particle have its maximum kinetic energy? (a) 100 cm (b) 110 cm (c) 120 cm (d) 130 cm (e) 140 cm

11. Which of the following statements is *not* true regarding a mass–spring system that moves with simple harmonic motion in the absence of friction? (a) The total energy of the system remains constant. (b) The energy of the system is continually transformed between kinetic and potential energy. (c) The total energy of the system is proportional to the square of the amplitude. (d) The potential energy stored in the system is greatest when the mass passes through the equilibrium position. (e) The velocity of the oscillating mass has its maximum value when the mass passes through the equilibrium position.

12. A block is attached to a spring hanging vertically. After being slowly lowered, it hangs at rest with the spring stretched by 15.0 cm. If the block is raised back up and released from rest with the spring unstretched, what maximum distance does it fall? (a) 7.5 cm (b) 15.0 cm (c) 30.0 cm (d) 60.0 cm (e) impossible to determine without knowing the mass and spring constant

13. An object–spring system moving with simple harmonic motion has an amplitude *A*. When the kinetic energy of the object equals twice the potential energy stored in the spring, what is the position *x* of the object? (a) *A* (b) $\frac{1}{3}A$ (c) $A/\sqrt{3}$ (d) 0 (e) none of those answers

14. You stand on the end of a diving board and bounce to set it into oscillation. You find a maximum response in terms of the amplitude of oscillation of the end of the board when you bounce at a frequency *f*. You now move to the middle of the board and repeat the experiment. Is the resonance frequency of the forced oscillations at this point (a) higher, (b) lower, or (c) the same as *f*?

CONCEPTUAL QUESTIONS

ENHANCED WebAssign The conceptual questions in this chapter may be assigned online in Enhanced WebAssign.

1. An object–spring system undergoes simple harmonic motion with an amplitude *A*. Does the total energy change if the mass is doubled but the amplitude isn't changed? Are the kinetic and potential energies at a given point in its motion affected by the change in mass? Explain.

2. If an object–spring system is hung vertically and set into oscillation, why does the motion eventually stop?

3. An object is hung on a spring, and the frequency of oscillation of the system, *f*, is measured. The object, a second identical object, and the spring are carried to space in the space shuttle. The two objects are attached to the ends of the spring, and the system is taken out into space on a space walk. The spring is extended, and the system is released to oscillate while floating in space. The coils of the spring don't bump into one another. What is the frequency of oscillation for this system in terms of *f*?

4. If a spring is cut in half, what happens to its spring constant?

5. A pendulum bob is made from a sphere filled with water. What would happen to the frequency of vibration of this pendulum if the sphere had a hole in it that allowed the water to leak out slowly?

6. If a pendulum clock keeps perfect time at the base of a mountain, will it also keep perfect time when it is moved to the top of the mountain? Explain.

7. (a) Is a bouncing ball an example of simple harmonic motion? (b) Is the daily movement of a student from home to school and back simple harmonic motion?

8. If a grandfather clock were running slow, how could we adjust the length of the pendulum to correct the time?

9. What happens to the speed of a wave on a string when the frequency is doubled? Assume the tension in the string remains the same.

10. If you stretch a rubber hose and pluck it, you can observe a pulse traveling up and down the hose. What happens to the speed of the pulse if you stretch the hose more tightly? What happens to the speed if you fill the hose with water?

11. Explain why the kinetic and potential energies of an object–spring system can never be negative.

12. A grandfather clock depends on the period of a pendulum to keep correct time. Suppose such a clock is calibrated correctly and then the temperature of the room in which it resides increases. Does the clock run slow, fast, or correctly? *Hint:* A material expands when its temperature increases.

PROBLEMS

ENHANCED WebAssign The problems in this chapter may be assigned online in Enhanced WebAssign. Selected problems also have Watch It video solutions.

1. denotes straightforward problem; 2. denotes intermediate problem;
3. denotes challenging problem
1. denotes full solution available in *Student Solutions Manual/ Study Guide*

1. denotes problems most often assigned in Enhanced WebAssign
BIO denotes biomedical problems
GP denotes guided problems
M denotes Master It tutorial available in Enhanced WebAssign
Q|C denotes asking for quantitative and conceptual reasoning
S denotes symbolic reasoning problem

13.1 Hooke's Law

1. A block of mass m = 0.60 kg attached to a spring with force constant 130 N/m is free to move on a frictionless, horizontal surface as in Figure P13.1. The block is released from rest after the spring is stretched a distance A = 0.13 m. At that instant, find (a) the force on the block and (b) its acceleration.

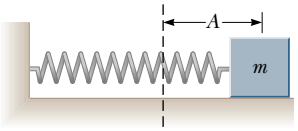


Figure P13.1

2. A spring oriented vertically is attached to a hard horizontal surface as in Figure P13.2. The spring has a force constant of 1.46 kN/m. How much is the spring compressed when a object of mass m = 2.30 kg is placed on top of the spring and the system is at rest?

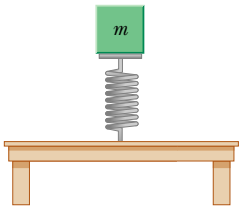


Figure P13.2

3. The force constant of a spring is 137 N/m. Find the magnitude of the force required to (a) compress the spring by 4.80 cm from its unstretched length and (b) stretch the spring by 7.36 cm from its unstretched length.

4. A load of 50 N attached to a spring hanging vertically stretches the spring 5.0 cm. The spring is now placed horizontally on a table and stretched 11 cm. (a) What force is required to stretch the spring by that amount? (b) Plot a graph of force (on the y-axis) versus spring displacement from the equilibrium position along the x-axis.

5. A spring is hung from a ceiling, and an object attached to its lower end stretches the spring by a distance d = 5.00 cm from its unstretched position when the system is in equilibrium as in Figure P13.5. If the spring constant is 47.5 N/m, determine the mass of the object.

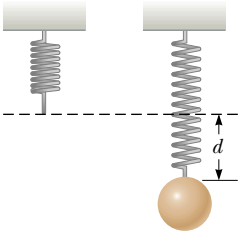


Figure P13.5

6. An archer must exert a force of 375 N on the bowstring shown in Figure P13.6a such that the string makes an angle of θ = 35.0° with the vertical. (a) Determine the tension in the bowstring. (b) If the applied force is replaced by a stretched spring as in Figure P13.6b and the spring is stretched 30.0 cm from its unstretched length, what is the spring constant?

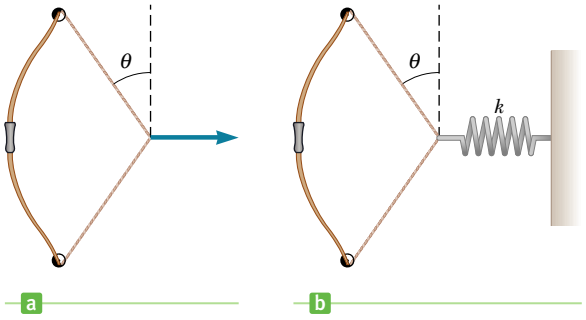


Figure P13.6

7. Q|C A spring 1.50 m long with force constant 475 N/m is hung from the ceiling of an elevator, and a block of mass 10.0 kg is attached to the bottom of the spring. (a) By how much is the spring stretched when the block is slowly lowered to its equilibrium point? (b) If the elevator subsequently accelerates upward at 2.00 m/s^2, what is the position of the block, taking the equilibrium position found in part (a) as y = 0 and upwards as the positive y-direction. (c) If the elevator cable snaps during the acceleration, describe the subsequent motion of the block relative to the freely falling elevator. What is the amplitude of its motion?

13.2 Elastic Potential Energy

8. A block of mass m = 2.00 kg is attached to a spring of force constant k = 5.00 × 10^2 N/m that lies on a horizontal frictionless surface as shown in Figure P13.8. The block is pulled to a position x_i = 5.00 cm to the right of equilibrium and released from rest. Find (a) the work required to stretch the spring and (b) the speed the block has as it passes through equilibrium.

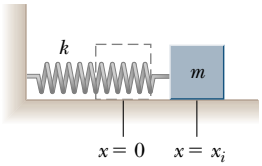


Figure P13.8

9. A slingshot consists of a light leather cup containing a stone. The cup is pulled back against two parallel rubber bands. It takes a force of 15 N to stretch either one of these bands 1.0 cm. (a) What is the potential energy

stored in the two bands together when a 50-g stone is placed in the cup and pulled back 0.20 m from the equilibrium position? (b) With what speed does the stone leave the slingshot?

10. An archer pulls her bowstring back 0.400 m by exerting a force that increases uniformly from zero to 230 N. (a) What is the equivalent spring constant of the bow? (b) How much work is done in pulling the bow?

11. A child's toy consists of a piece of plastic attached to a spring (Fig. P13.11). The spring is compressed against the floor a distance of 2.00 cm, and the toy is released. If the toy has a mass of 100 g and rises to a maximum height of 60.0 cm, estimate the force constant of the spring.

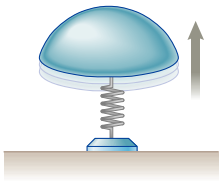


Figure P13.11

12. An automobile having a mass of 1 000 kg is driven into a brick wall in a safety test. The bumper behaves like a spring with constant 5.00×10^6 N/m and is compressed 3.16 cm as the car is brought to rest. What was the speed of the car before impact, assuming no energy is lost in the collision with the wall?

13. A 10.0-g bullet is fired into, and embeds itself in, a 2.00-kg block attached to a spring with a force constant of 19.6 N/m and whose mass is negligible. How far is the spring compressed if the bullet has a speed of 300 m/s just before it strikes the block and the block slides on a frictionless surface? *Note:* You must use conservation of momentum in this problem because of the inelastic collision between the bullet and block.

14. S An object–spring system moving with simple harmonic motion has an amplitude A. (a) What is the total energy of the system in terms of k and A only? (b) Suppose at a certain instant the kinetic energy is twice the elastic potential energy. Write an equation describing this situation, using only the variables for the mass m, velocity v, spring constant k, and position x. (c) Using the results of parts (a) and (b) and the conservation of energy equation, find the positions x of the object when its kinetic energy equals twice the potential energy stored in the spring. (The answer should in terms of A only.)

15. GP A horizontal block–spring system with the block on a frictionless surface has total mechanical energy $E = 47.0$ J and a maximum displacement from equilibrium of 0.240 m. (a) What is the spring constant? (b) What is the kinetic energy of the system at the equilibrium point? (c) If the maximum speed of the block is 3.45 m/s, what is its mass? (d) What is the speed of the block when its displacement is 0.160 m? (e) Find the kinetic energy of the block at $x = 0.160$ m. (f) Find the potential energy stored in the spring when $x = 0.160$ m. (g) Suppose the same system is released from rest at $x = 0.240$ m on a rough surface so that it loses 14.0 J by the time it reaches its first turning point (after passing equilibrium at $x = 0$). What is its position at that instant?

16. Q|C A 0.250-kg block resting on a frictionless, horizontal surface is attached to a spring having force constant 83.8 N/m as in Figure P13.16. A horizontal force $\vec{\mathbf{F}}$ causes the spring to stretch a distance of 5.46 cm from its equilibrium position. (a) Find the value of F. (b) What is the total energy stored in the system when the spring is stretched? (c) Find the magnitude of the acceleration of the block immediately after the applied force is removed. (d) Find the speed of the block when it first reaches the equilibrium position. (e) If the surface is not frictionless but the block still reaches the equilibrium position, how would your answer to part (d) change? (f) What other information would you need to know to find the actual answer to part (d) in this case?

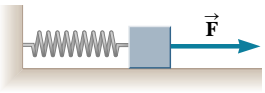


Figure P13.16

13.3 Comparing Simple Harmonic Motion with Uniform Circular Motion

13.4 Position, Velocity, and Acceleration as a Function of Time

17. A 0.40-kg object connected to a light spring with a force constant of 19.6 N/m oscillates on a frictionless horizontal surface. If the spring is compressed 4.0 cm and released from rest, determine (a) the maximum speed of the object, (b) the speed of the object when the spring is compressed 1.5 cm, and (c) the speed of the object as it passes the point 1.5 cm from the equilibrium position. (d) For what value of x does the speed equal one-half the maximum speed?

18. An object–spring system oscillates with an amplitude of 3.5 cm. If the spring constant is 250 N/m and the object has a mass of 0.50 kg, determine (a) the mechanical energy of the system, (b) the maximum speed of the object, and (c) the maximum acceleration of the object.

19. M At an outdoor market, a bunch of bananas attached to the bottom of a vertical spring of force constant 16.0 N/m is set into oscillatory motion with an amplitude of 20.0 cm. It is observed that the maximum speed of the bunch of bananas is 40.0 cm/s. What is the weight of the bananas in newtons?

20. A 50.0-g object is attached to a horizontal spring with a force constant of 10.0 N/m and released from rest with an amplitude of 25.0 cm. What is the velocity of the object when it is halfway to the equilibrium position if the surface is frictionless?

21. A horizontal spring attached to a wall has a force constant of $k = 850$ N/m. A block of mass $m = 1.00$ kg is attached to the spring and rests on a frictionless, horizontal surface as in Figure P13.21 on page 468. (a) The

block is pulled to a position $x_i = 6.00$ cm from equilibrium and released. Find the potential energy stored in the spring when the block is 6.00 cm from equilibrium. (b) Find the speed of the block as it passes through the equilibrium position. (c) What is the speed of the block when it is at a position $x_i/2 = 3.00$ cm?

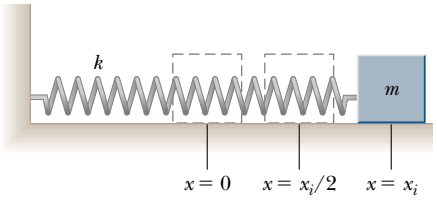


Figure P13.21

22. An object moves uniformly around a circular path of radius 20.0 cm, making one complete revolution every 2.00 s. What are (a) the translational speed of the object, (b) the frequency of motion in hertz, and (c) the angular speed of the object?

23. Consider the simplified single-piston engine in Figure P13.23. If the wheel rotates at a constant angular speed ω, explain why the piston oscillates in simple harmonic motion.

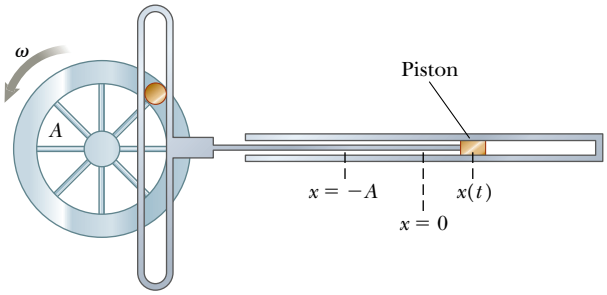


Figure P13.23

24. The period of motion of an object–spring system is $T = 0.528$ s when an object of mass $m = 238$ g is attached to the spring. Find (a) the frequency of motion in hertz and (b) the force constant of the spring. (c) If the total energy of the oscillating motion is 0.234 J, find the amplitude of the oscillations.

25. M A vertical spring stretches 3.9 cm when a 10-g object is hung from it. The object is replaced with a block of mass 25 g that oscillates up and down in simple harmonic motion. Calculate the period of motion.

26. When four people with a combined mass of 320 kg sit down in a 2.0×10^3-kg car, they find that their weight compresses the springs an additional 0.80 cm. (a) What is the effective force constant of the springs? (b) The four people get out of the car and bounce it up and down. What is the frequency of the car's vibration?

27. A cart of mass 250 g is placed on a frictionless horizontal air track. A spring having a spring constant of 9.5 N/m is attached between the cart and the left end of the track. If the cart is displaced 4.5 cm from its equilibrium position, find (a) the period at which it oscillates, (b) its maximum speed, and (c) its speed when it is located 2.0 cm from its equilibrium position.

28. The position of an object connected to a spring varies with time according to the expression $x = (5.2 \text{ cm}) \sin(8.0\pi t)$. Find (a) the period of this motion, (b) the frequency of the motion, (c) the amplitude of the motion, and (d) the first time after $t = 0$ that the object reaches the position $x = 2.6$ cm.

29. A 326-g object is attached to a spring and executes simple harmonic motion with a period of 0.250 s. If the total energy of the system is 5.83 J, find (a) the maximum speed of the object, (b) the force constant of the spring, and (c) the amplitude of the motion.

30. S An object executes simple harmonic motion with an amplitude A. (a) At what values of its position does its speed equal half its maximum speed? (b) At what values of its position does its potential energy equal half the total energy?

31. A 2.00-kg object on a frictionless horizontal track is attached to the end of a horizontal spring whose force constant is 5.00 N/m. The object is displaced 3.00 m to the right from its equilibrium position and then released, initiating simple harmonic motion. (a) What is the force (magnitude and direction) acting on the object 3.50 s after it is released? (b) How many times does the object oscillate in 3.50 s?

32. GP A spring of negligible mass stretches 3.00 cm from its relaxed length when a force of 7.50 N is applied. A 0.500-kg particle rests on a frictionless horizontal surface and is attached to the free end of the spring. The particle is displaced from the origin to $x = 5.00$ cm and released from rest at $t = 0$. (a) What is the force constant of the spring? (b) What are the angular frequency ω, the frequency, and the period of the motion? (c) What is the total energy of the system? (d) What is the amplitude of the motion? (e) What are the maximum velocity and the maximum acceleration of the particle? (f) Determine the displacement x of the particle from the equilibrium position at $t = 0.500$ s. (g) Determine the velocity and acceleration of the particle when $t = 0.500$ s.

33. Given that $x = A \cos(\omega t)$ is a sinusoidal function of time, show that v (velocity) and a (acceleration) are also sinusoidal functions of time. *Hint:* Use Equations 13.6 and 13.2.

13.5 Motion of a Pendulum

34. A man enters a tall tower, needing to know its height. He notes that a long pendulum extends from the ceiling almost to the floor and that its period is 15.5 s. (a) How tall is the tower? (b) If this pendulum is taken to the Moon, where the free-fall acceleration is 1.67 m/s^2, what is the period there?

35. A simple pendulum has a length of 52.0 cm and makes 82.0 complete oscillations in 2.00 min. Find (a) the period of the pendulum and (b) the value of g at the location of the pendulum.

36. A "seconds" pendulum is one that moves through its equilibrium position once each second. (The period of the pendulum is 2.000 s.) The length of a seconds pendulum is 0.992 7 m at Tokyo and 0.994 2 m at Cambridge, England. What is the ratio of the free-fall accelerations at these two locations?

37. A pendulum clock that works perfectly on the Earth is taken to the Moon. (a) Does it run fast or slow there? (b) If the clock is started at 12:00 midnight, what will it read after one Earth day (24.0 h)? Assume the free-fall acceleration on the Moon is 1.63 m/s^2.

38. A coat hanger of mass $m = 0.238$ kg oscillates on a peg as a physical pendulum as shown in Figure P13.38. The distance from the pivot to the center of mass of the coat hanger is $d = 18.0$ cm and the period of the motion is $T = 1.25$ s. Find the moment of inertia of the coat hanger about the pivot.

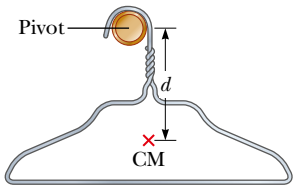


Figure P13.38

39. M The free-fall acceleration on Mars is 3.7 m/s^2. (a) What length of pendulum has a period of 1 s on Earth? (b) What length of pendulum would have a 1-s period on Mars? An object is suspended from a spring with force constant 10 N/m. Find the mass suspended from this spring that would result in a period of 1 s (c) on Earth and (d) on Mars.

40. A simple pendulum is 5.00 m long. (a) What is the period of simple harmonic motion for this pendulum if it is located in an elevator accelerating upward at 5.00 m/s^2? (b) What is its period if the elevator is accelerating downward at 5.00 m/s^2? (c) What is the period of simple harmonic motion for the pendulum if it is placed in a truck that is accelerating horizontally at 5.00 m/s^2?

13.6 Damped Oscillations

13.7 Waves

13.8 Frequency, Amplitude, and Wavelength

41. The sinusoidal wave shown in Figure P13.41 is traveling in the positive x-direction and has a frequency of 18.0 Hz. Find the (a) amplitude, (b) wavelength, (c) period, and (d) speed of the wave.

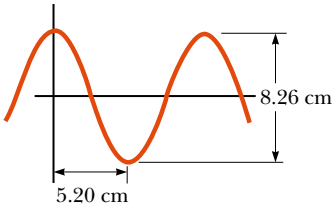


Figure P13.41

42. An object attached to a spring vibrates with simple harmonic motion as described by Figure P13.42. For this motion, find (a) the amplitude, (b) the period, (c) the angular frequency, (d) the maximum speed, (e) the maximum acceleration, and (f) an equation for its position x in terms of a sine function.

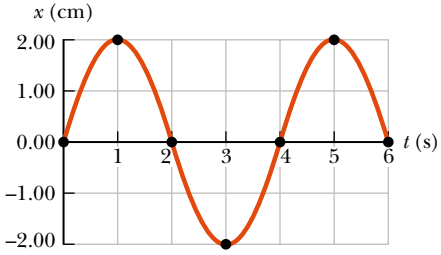


Figure P13.42

43. Light waves are electromagnetic waves that travel at 3.00×10^8 m/s. The eye is most sensitive to light having a wavelength of 5.50×10^{-7} m. Find (a) the frequency of this light wave and (b) its period.

44. The distance between two successive minima of a transverse wave is 2.76 m. Five crests of the wave pass a given point along the direction of travel every 14.0 s. Find (a) the frequency of the wave and (b) the wave speed.

45. A harmonic wave is traveling along a rope. It is observed that the oscillator that generates the wave completes 40.0 vibrations in 30.0 s. Also, a given maximum travels 425 cm along the rope in 10.0 s. What is the wavelength?

46. BIO A bat can detect small objects, such as an insect, whose size is approximately equal to one wavelength of the sound the bat makes. If bats emit a chirp at a frequency of 60.0×10^3 Hz and the speed of sound in air is 343 m/s, what is the smallest insect a bat can detect?

47. A cork on the surface of a pond bobs up and down two times per second on ripples having a wavelength of 8.50 cm. If the cork is 10.0 m from shore, how long does it take a ripple passing the cork to reach the shore?

48. Ocean waves are traveling to the east at 4.0 m/s with a distance of 20 m between crests. With what frequency do the waves hit the front of a boat (a) when the boat is at anchor and (b) when the boat is moving westward at 1.0 m/s?

13.9 The Speed of Waves on Strings

49. An ethernet cable is 4.00 m long and has a mass of 0.200 kg. A transverse wave pulse is produced by plucking one end of the taut cable. The pulse makes four trips down and back along the cable in 0.800 s. What is the tension in the cable?

50. A circus performer stretches a tightrope between two towers. He strikes one end of the rope and sends a wave along it toward the other tower. He notes that it takes the wave 0.800 s to reach the opposite tower, 20.0 m away. If a 1.00-m length of the rope has a mass of 0.350 kg, find the tension in the tightrope.

51. A piano string of mass per unit length 5.00×10^{-3} kg/m is under a tension of 1 350 N. Find the speed with which a wave travels on this string.

52. Q|C S A student taking a quiz finds on a reference sheet the two equations

$$f = \frac{1}{T} \quad \text{and} \quad v = \sqrt{\frac{T}{\mu}}$$

She has forgotten what T represents in each equation. (a) Use dimensional analysis to determine the units required for T in each equation. (b) Explain how you can identify the physical quantity each T represents from the units.

53. Transverse waves with a speed of 50.0 m/s are to be produced on a stretched string. A 5.00-m length of string with a total mass of 0.060 0 kg is used. (a) What is the required tension in the string? (b) Calculate the wave speed in the string if the tension is 8.00 N.

54. An astronaut on the Moon wishes to measure the local value of g by timing pulses traveling down a wire that has a large object suspended from it. Assume a wire of mass 4.00 g is 1.60 m long and has a 3.00-kg object suspended from it. A pulse requires 36.1 ms to traverse the length of the wire. Calculate g_{Moon} from these data. (You may neglect the mass of the wire when calculating the tension in it.)

55. A simple pendulum consists of a ball of mass 5.00 kg hanging from a uniform string of mass 0.060 0 kg and length L. If the period of oscillation of the pendulum is 2.00 s, determine the speed of a transverse wave in the string when the pendulum hangs vertically.

56. A string is 50.0 cm long and has a mass of 3.00 g. A wave travels at 5.00 m/s along this string. A second string has the same length, but half the mass of the first. If the two strings are under the same tension, what is the speed of a wave along the second string?

57. M Tension is maintained in a string as in Figure P13.57. The observed wave speed is $v = 24.0$ m/s when the suspended mass is $m = 3.00$ kg. (a) What is the mass per unit length of the string? (b) What is the wave speed when the suspended mass is $m = 2.00$ kg?

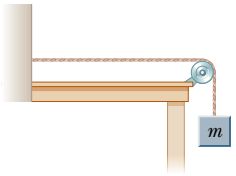


Figure P13.57

58. The elastic limit of a piece of steel wire is 2.70×10^9 Pa. What is the maximum speed at which transverse wave pulses can propagate along the wire without exceeding its elastic limit? (The density of steel is 7.86×10^3 kg/m^3.)

59. Q|C A 2.65-kg power line running between two towers has a length of 38.0 m and is under a tension of 12.5 N. (a) What is the speed of a transverse pulse set up on the line? (b) If the tension in the line was unknown, describe a procedure a worker on the ground might use to estimate the tension.

60. S A taut clothesline has length L and a mass M. A transverse pulse is produced by plucking one end of the clothesline. If the pulse makes n round trips along the clothesline in t seconds, find expressions for (a) the speed of the pulse in terms of n, L, and t and (b) the tension F in the clothesline in terms of the same variables and mass M.

13.10 Interference of Waves

13.11 Reflection of Waves

61. A wave of amplitude 0.30 m interferes with a second wave of amplitude 0.20 m traveling in the same direction. What are (a) the largest and (b) the smallest resultant amplitudes that can occur, and under what conditions will these maxima and minima arise?

Additional Problems

62. The position of a 0.30-kg object attached to a spring is described by

$$x = (0.25 \text{ m}) \cos (0.4\pi t)$$

Find (a) the amplitude of the motion, (b) the spring constant, (c) the position of the object at $t = 0.30$ s, and (d) the object's speed at $t = 0.30$ s.

63. An object of mass 2.00 kg is oscillating freely on a vertical spring with a period of 0.600 s. Another object of unknown mass on the same spring oscillates with a period of 1.05 s. Find (a) the spring constant k and (b) the unknown mass.

64. A certain tuning fork vibrates at a frequency of 196 Hz while each tip of its two prongs has an amplitude of 0.850 mm. (a) What is the period of this motion? (b) Find the wavelength of the sound produced by the vibrating fork, taking the speed of sound in air to be 343 m/s.

65. A simple pendulum has mass 1.20 kg and length 0.700 m. (a) What is the period of the pendulum near the surface of Earth? (b) If the same mass is attached to a spring, what spring constant would result in the period of motion found in part (a)?

66. A 500-g block is released from rest and slides down a frictionless track that begins 2.00 m above the horizontal, as shown in Figure P13.66. At the bottom of the track, where the surface is horizontal, the block strikes and sticks to a light spring with a spring constant of 20.0 N/m. Find the maximum distance the spring is compressed.

Chapter 5

Static Electricity



Why It's Important

Static electricity enables the operation of devices such as printers and copiers, but it has harmful effects on electronic components and in the form of lightning.

Lightning The tiny spark that you experience when you touch a doorknob and the dazzling display of lightning in a storm are both examples of the discharge of static electricity. The charging processes and the means of discharging are vastly different in scale, but they are similar in their fundamental nature.

LAUNCH Lab

Which forces act over a distance?

Question

What happens when a plastic ruler is rubbed with wool and then brought near a pile of paper scraps?

Procedure

1. Place 15-20 scraps of paper from a hole punch on the table.
2. Take a plastic ruler and rub it with a piece of wool.
3. Bring the ruler close to the pieces of paper. Observe the effect the ruler has on the scraps of paper.

Analysis

What happens to the pieces of paper when the ruler is brought close to them? What happens to the pieces of paper that come in contact with the ruler? Did you observe any unexpected results when the ruler was brought close to the paper scraps? If so, describe these results.

Critical Thinking

What forces are acting on the pieces of paper before the ruler is brought close to them? What can you infer about the forces on the paper after the ruler is brought near?

Based on your answers to the previous questions, form a hypothesis that explains the effect the ruler has on the scraps of paper.



1. Electric Charge

You may have had the experience of rubbing your shoes on a carpet to create a spark when you touched someone. In 1752, Benjamin Franklin set off a flurry of research in the field of electricity when his famous kite experiment showed that lightning is similar to the sparks caused by friction. In his experiment, Franklin flew a kite with a key attached to the string. As a thunderstorm approached, the loose threads of the kite string began to stand up and repel one another, and when Franklin brought his knuckle close to the key, he experienced a spark. Electric effects produced in this way are called static electricity.

In this chapter, you will investigate **electrostatics,** the study of electric charges that can be collected and held in one place. The effects of electrostatics are observable over a vast scale, from huge displays of lightning to the submicroscopic world of atoms and molecules. Current electricity, which is produced by batteries and generators, will be explored in later chapters.

Objectives

- **Demonstrate** that charged objects exert forces, both attractive and repulsive.
- **Recognize** that charging is the separation, not the creation, of electric charges.
- **Describe** the differences between conductors and insulators.

Vocabulary

electrostatics
neutral
insulator
conductor

Charged Objects

Have you ever noticed the way that your hair is attracted to the comb when you comb your hair on a dry day or the way that your hair stands on end after it is rubbed with a balloon? Perhaps you also have found that socks sometimes stick together when you take them out of a clothes dryer. If so, you will recognize the attraction of the bits of paper to a plastic ruler demonstrated by the Launch Lab and shown in **Figure 20-1.** You might have noticed the way the paper pieces jumped up to the ruler as you worked through the Launch Lab. There must be a new, relatively strong force causing this upward acceleration because it is larger than the downward acceleration caused by the gravitational force of Earth.

There are other differences between this new force and gravity. Paper is attracted to a plastic ruler only after the ruler has been rubbed; if you wait a while, the attractive property of the ruler disappears. Gravity, on the other hand, does not require rubbing and does not disappear. The ancient Greeks noticed effects similar to that of the ruler when they rubbed amber. The Greek word for amber is *elektron,* and today this attractive property is called electric. An object that exhibits electric interaction after rubbing is said to be charged.

Like charges You can explore electric interactions with simple objects, such as transparent tape. Fold over about 5 mm of the end of a strip of tape for a handle, and then stick the remaining 8- to 12-cm-long part of the tape strip on a dry, smooth surface, such as your desktop. Stick a second, similar piece of tape next to the first. Quickly pull both strips off the desk and bring them near each other. A new property causes the strips to repel each other: they are electrically charged. Because they were prepared in the same way, they must have the same type of charge. Thus, you have demonstrated that two objects with the same type of charge repel each other.



Figure 20-1 Rubbing a plastic ruler with wool produces a new force of attraction between the ruler and bits of paper. When the ruler is brought close to bits of paper, the attractive electric force accelerates the paper bits upward against the force of gravity.

You can learn more about this charge by doing some simple experiments. You may have found that the tape is attracted to your hand. Are both sides attracted, or just one? If you wait a while, especially in humid weather, you will find that the electric charge disappears. You can restore it by again sticking the tape to the desk and pulling it off. You also can remove its charge by gently rubbing your fingers down both sides of the tape.

Opposite charges Now, stick one strip of tape on the desk and place the second strip on top of the first. As shown in **Figure 20-2a,** use the handle of the bottom strip of tape to pull the two off the desk together. Rub them with your fingers until they are no longer attracted to your hand. You now have removed all the electric charge. With one hand on the handle of one strip and the other on the handle of the second strip, quickly pull the two strips apart. You will find that they are now both charged. They once again are attracted to your hands. Do they still repel each other? No, they now attract each other. They are charged, but they are no longer charged alike. They have opposite charges and therefore attract each other.

Is tape the only object that you can charge? Once again, stick one strip of tape to the desk and the second strip on top. Label the bottom strip *B* and the top strip *T*. Pull the pair off together. Discharge them, then pull them apart. Stick the handle end of each strip to the edge of a table, the bottom of a lamp shade, or some similar object. The two should hang down a short distance apart. Finally, rub a comb or pen on your clothing and bring it near one strip of tape and then the other. You will find that one strip will be attracted to the comb, while the other will be repelled by it, as shown in **Figure 20-2b.** You now can explore the interactions of charged objects with the strips of tape.





Figure 20-2 Strips of tape can be given opposite charges **(a)** and then be used to demonstrate the interactions of like and opposite charges **(b).**

Experimenting with charge Try to charge other objects, such as glasses and plastic bags. Rub them with different materials, such as silk, wool, and plastic wrap. If the air is dry, scuff your shoes on carpet and bring your finger near the strips of tape. To test silk or wool, slip a plastic bag over your hand before holding the cloth. After rubbing, take your hand out of the bag and bring both the bag and cloth near the strips of tape.

Most charged objects will attract one strip and repel the other. You will never find an object that repels both strips of tape, although you might find some that attract both. For example, your finger will attract both strips. You will explore this effect later in this chapter.

Types of charge From your experiments, you can make a list of objects labeled *B*, for bottom, which have the same charge as the tape stuck on the desk. Another list can be made of objects labeled *T*, which have the same charge as the top strip of tape. There are only two lists, because there are only two types of charge. Benjamin Franklin called them positive and negative charges. Using Franklin's convention, when hard rubber and plastic are rubbed, they become negatively charged. When materials such as glass and wool are rubbed, they become positively charged.

Color Convention

- Positive charges are shown in **red.**
- Negative charges are shown in **blue.**

Just as you showed that an uncharged pair of tape strips became oppositely charged, you probably were able to show that if you rubbed plastic with wool, the plastic became negatively charged and the wool positively charged. The two kinds of charges were not created alone, but in pairs. These experiments suggest that matter normally contains both charges, positive and negative. Contact in some way separates the two. To explore this further, you must consider the microscopic picture of matter.

A Microscopic View of Charge

Electric charges exist within atoms. In 1897, J.J. Thomson discovered that all materials contain light, negatively charged particles that he called electrons. Between 1909 and 1911, Ernest Rutherford, a student of Thomson from New Zealand, discovered that the atom has a massive, positively charged nucleus. When the positive charge of the nucleus equals the negative charge of the surrounding electrons, then the atom is **neutral.**

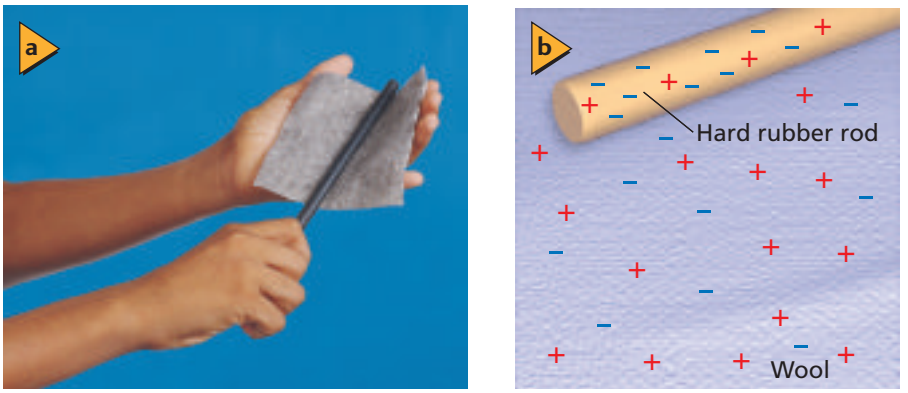


Figure 20-3 When wool is used to charge a rubber rod, electrons are removed from the wool atoms and cling to the rubber atoms. In this way, both objects become charged.

With the addition of energy, the outer electrons can be removed from atoms. An atom missing electrons has an overall positive charge, and consequently, any matter made of these electron-deficient atoms is positively charged. The freed electrons can remain unattached or become attached to other atoms, resulting in negatively charged particles. From a microscopic viewpoint, acquiring charge is a process of transferring electrons.

Separation of charge If two neutral objects are rubbed together, each can become charged. For instance, when rubber and wool are rubbed together, electrons from atoms on the wool are transferred to the rubber, as shown in **Figure 20-3.** The extra electrons on the rubber result in a net negative charge. The electrons missing from the wool result in a net positive charge. The combined total charge of the two objects remains the same. Charge is conserved, which is one way of saying that individual charges never are created or destroyed. All that happens is that the positive and negative charges are separated through a transfer of electrons.

Complex processes that affect the tires of a moving car or truck can cause the tires to become charged. Processes inside a thundercloud can cause the cloud bottom to become negatively charged and the cloud top to become positively charged. In both these cases, charge is not created, but separated.

Conductors and Insulators

Hold a plastic rod or comb at its midpoint and rub only one end. You will find that only the rubbed end becomes charged. In other words, the charges that you transferred to the plastic stayed where they were put; they did not move. A material through which a charge will not move easily is called an electric **insulator.** The strips of tape that you charged earlier in this chapter acted in this way. Glass, dry wood, most plastics, cloth, and dry air are all good insulators.

Suppose that you support a metal rod on an insulator so that it is isolated, or completely surrounded by insulators. If you then touch the charged comb to one end of the metal rod, you will find that the charge spreads very quickly over the entire rod. A material that allows charges to move about easily is called an electric **conductor.** Electrons carry, or conduct, electric charge through the metal. Metals are good conductors because at least one electron on each atom of the metal can be removed easily. These electrons act as if they no longer belong to any one atom, but to the metal as a whole; consequently, they move freely throughout the piece of metal. **Figure 20-4** contrasts how charges behave when they are

APPLYING PHYSICS

► Conductor or Insulator?

It might be tempting to classify an element as solely a conductor or solely an insulator, but the classification can change depending on the form the element takes. For example, carbon in the form of diamond is an insulator, but carbon in the form of graphite can conduct charge. This is because the carbon atoms in diamonds are tightly bonded to four other carbons, while the carbon atoms in graphite form three stronger bonds and a fourth, weaker bond that allows electrons a limited amount of movement. As a result, graphite is a much better conductor than diamond, even though both are simply carbon atoms. ◄

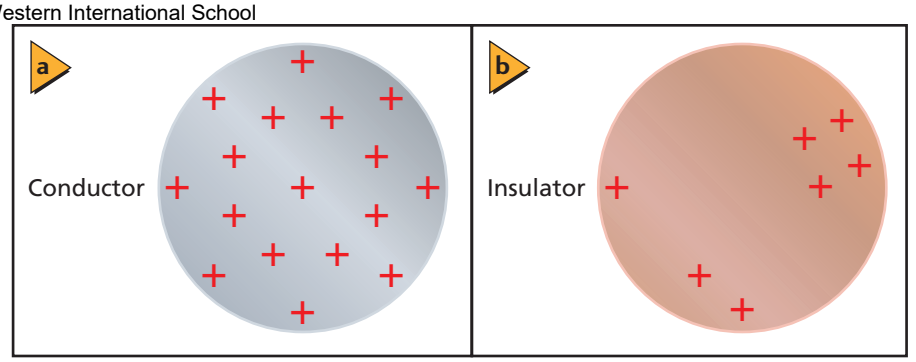


Figure 20-4 Charges placed on a conductor will spread over the entire surface **(a).** Charges placed on an insulator will remain where they are placed **(b).**

placed on a conductor with how they behave on an insulator. Copper and aluminum are both excellent conductors and are used commercially to carry electricity. Plasma, a highly ionized gas, and graphite also are good conductors of electric charge.

When air becomes a conductor Air is an insulator; however, under certain conditions, charges move through air as if it were a conductor. The spark that jumps between your finger and a doorknob after you have rubbed your feet on a carpet discharges you. In other words, you have become neutral because the excess charges have left you. Similarly, lightning discharges a thundercloud. In both of these cases, air became a conductor for a brief moment. Recall that conductors must have charges that are free to move. For a spark or lightning to occur, freely moving charged particles must be formed in the normally neutral air. In the case of lightning, excess charges in the cloud and on the ground are great enough to remove electrons from the molecules in the air. The electrons and positively or negatively charged atoms form a plasma, which is a conductor. The discharge of Earth and the thundercloud by means of this conductor forms a luminous arc called lightning. In the case of your finger and the doorknob, the discharge is called a spark.

1. Section Review

1. **Charged Objects** After a comb is rubbed on a wool sweater, it is able to pick up small pieces of paper. Why does the comb lose that ability after a few minutes?
2. **Types of Charge** In the experiments described earlier in this section, how could you find out which strip of tape, B or T, is positively charged?
3. **Types of Charge** A pith ball is a small sphere made of a light material, such as plastic foam, often coated with a layer of graphite or aluminum paint. How could you determine whether a pith ball that is suspended from an insulating thread is neutral, is charged positively, or is charged negatively?
4. **Charge Separation** A rubber rod can be charged negatively when it is rubbed with wool. What happens to the charge of the wool? Why?
5. **Conservation of Charge** An apple contains trillions of charged particles. Why don't two apples repel each other when they are brought together?
6. **Charging a Conductor** Suppose you hang a long metal rod from silk threads so that the rod is isolated. You then touch a charged glass rod to one end of the metal rod. Describe the charges on the metal rod.
7. **Charging by Friction** You can charge a rubber rod negatively by rubbing it with wool. What happens when you rub a copper rod with wool?
8. **Critical Thinking** It once was proposed that electric charge is a type of fluid that flows from objects with an excess of the fluid to objects with a deficit. Why is the current two-charge model better than the single-fluid model?

2. Electric Force

Objectives

- **Summarize** the relationships between electric forces, charges, and distance.
- **Explain** how to charge objects by conduction and induction.
- **Develop** a model of how charged objects can attract a neutral object.
- **Apply** Coulomb's law to problems in one and two dimensions.

Vocabulary

electroscope
charging by conduction
charging by induction
grounding
Coulomb's law
coulomb
elementary charge

Electric forces must be strong because they can easily produce accelerations larger than the acceleration caused by gravity. You also have learned that they can be either repulsive or attractive, while gravitational forces always are attractive. Over the years, many scientists made attempts to measure electric forces. Daniel Bernoulli, best known for his work with fluids, made some crude measurements in 1760. In the 1770s, Henry Cavendish showed that electric forces must obey an inverse square force law, but being extremely shy, he did not publish his work. His manuscripts were discovered over a century later, after all his work had been duplicated by others.

Forces on Charged Bodies

The forces that you observed on tape strips also can be demonstrated by suspending a negatively charged, hard rubber rod so that it turns easily, as shown in **Figure 20-5.** If you bring another negatively charged rod near the suspended rod, the suspended rod will turn away. The negative charges on the rods repel each other. It is not necessary for the rods to make contact. The force, called the electric force, acts at a distance. If a positively charged glass rod is suspended and a similarly charged glass rod is brought close, the two positively charged rods also will repel each other. If a negatively charged rod is brought near a positively charged rod, however, the two will attract each other, and the suspended rod will turn toward the oppositely charged rod. The results of your tape experiments and these actions of charged rods can be summarized in the following way:

- There are two kinds of electric charges: positive and negative.
- Charges exert forces on other charges at a distance.
- The force is stronger when the charges are closer together.
- Like charges repel; opposite charges attract.

Neither a strip of tape nor a large rod that is hanging in open air is a very sensitive or convenient way of determining charge. Instead, a device called an electroscope is used. An **electroscope** consists of a metal knob connected by a metal stem to two thin, lightweight pieces of metal foil, called leaves. **Figure 20-6** shows a neutral electroscope. Note that the leaves hang loosely and are enclosed to eliminate stray air currents.

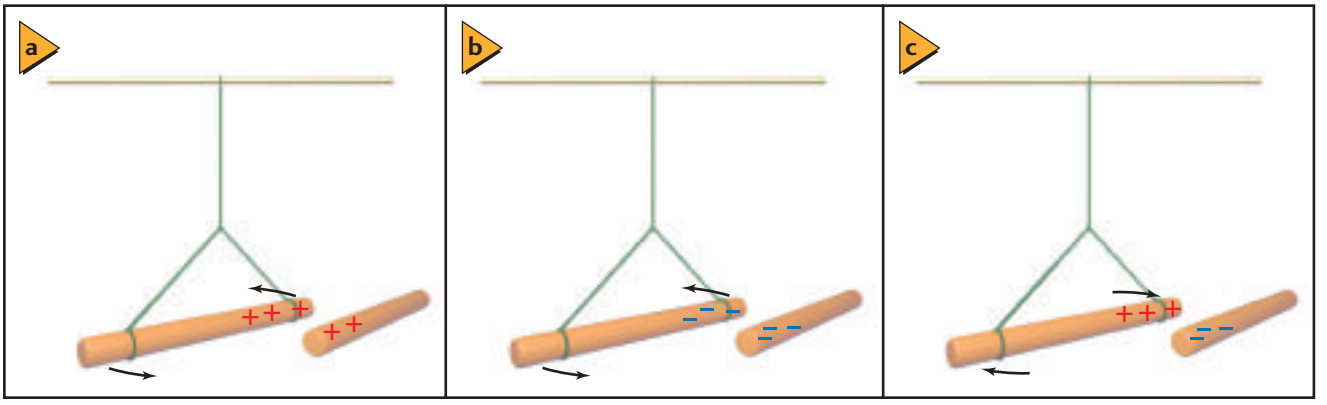


Figure 20-5 A charged rod, when brought close to another charged and suspended rod, will attract or repel the suspended rod.

Charging by conduction When a negatively charged rod is touched to the knob of an electroscope, electrons are added to the knob. These charges spread over all the metal surfaces. As shown in **Figure 20-7a,** the two leaves are charged negatively and repel each other; therefore, they spread apart. The electroscope has been given a net charge. Charging a neutral body by touching it with a charged body is called **charging by conduction.** The leaves also will spread apart if the electroscope is charged positively. How, then, can you find out whether the electroscope is charged positively or negatively? The type of charge can be determined by observing the leaves when a rod of known charge is brought close to the knob. The leaves will spread farther apart if the rod and the electroscope have the same charge, as shown in **Figure 20-7b.** The leaves will fall slightly if the electroscope's charge is opposite that of the rod, as in **Figure 20-7c.**

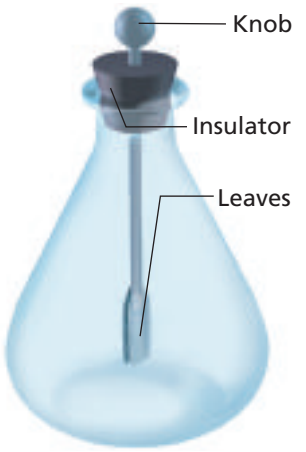


Figure 20-6 An electroscope is a device used for detecting charges. In a neutral electroscope, the leaves hang loosely, almost touching one another.

Separation of charge on neutral objects Earlier in this chapter, when you brought your finger near either charged strip of tape, the tape was attracted to your finger. Your finger, however, was neutral—it had equal amounts of positive and negative charge. You know that in conductors, charges can move easily, and that in the case of sparks, electric forces can change insulators into conductors. Given this information, you can develop a plausible model for the force that your finger exerted on the strips of tape.

Suppose you move your finger, or any uncharged object, close to a positively charged object. The negative charges in your finger will be attracted to the positively charged object, and the positive charges in your finger will be repelled. Your finger will remain neutral, but the positive and negative charges will be separated. The electric force is stronger for charges that are closer together; therefore, the separation results in an attractive force between your finger and the charged object. The force that a charged ruler exerts on neutral pieces of paper is the result of the same process, the separation of charges.

The negative charges at the bottom of thunderclouds also can cause charge separation in Earth. Positive charges in the ground are attracted to Earth's surface under the cloud. The forces of the charges in the cloud and those on Earth's surface can break molecules into positively and negatively charged particles. These charged particles are free to move, and they establish a conducting path from the ground to the cloud. The lightning that you observe occurs when a bolt travels at speeds on the order of 500,000 km/h along the conducting path and discharges the cloud.

Figure 20-7 A negatively charged electroscope will have its leaves spread apart **(a).** A negatively charged rod pushes electrons down to the leaves, causing them to spread farther apart **(b).** A positively charged rod attracts some of the electrons, causing the leaves to spread apart less **(c).**



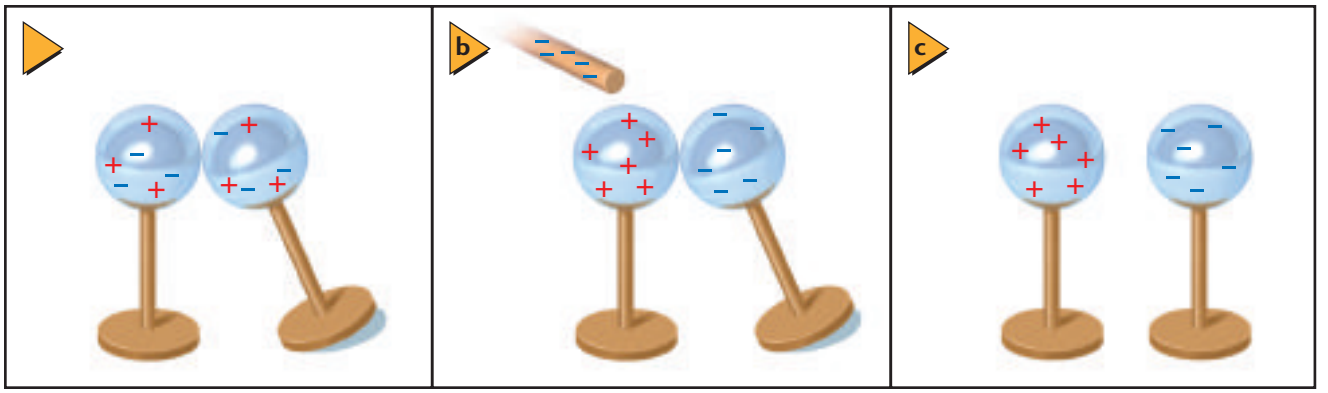


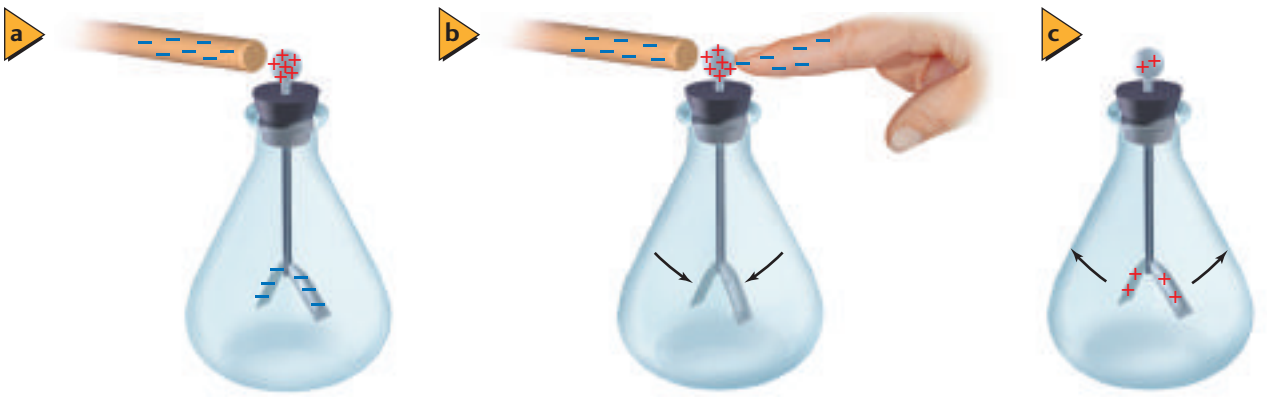
Figure 20-8 One method of charging by induction begins with neutral spheres that are touching **(a).** A charged rod is brought near them **(b),** then the spheres are separated and the charged rod is removed **(c).** The charges on the separated spheres are equal in magnitude, but opposite in sign.

Charging by induction Suppose that two identical, insulated metal spheres are touching, as shown in **Figure 20-8a.** When a negatively charged rod is brought close to one, as in **Figure 20-8b,** electrons from the first sphere will be forced onto the sphere farther from the rod and will make it negatively charged. The closer sphere is now positively charged. If the spheres are separated while the rod is nearby, each sphere will have a charge, and the charges will be equal but opposite, as shown in **Figure 20-8c.** This process of charging an object without touching it is called **charging by induction.**

A single object can be charged by induction through **grounding,** which is the process of connecting a body to Earth to eliminate excess charge. Earth is a very large sphere, and it can absorb great amounts of charge without becoming noticeably charged itself. If a charged body is touched to Earth, almost any amount of charge can flow to Earth.

If a negatively charged rod is brought close to the knob of an electroscope, as in **Figure 20-9a,** electrons are repelled onto the leaves. If the knob is then grounded on the side opposite the charged rod, electrons will be pushed from the electroscope into the ground until the leaves are neutral, as in **Figure 20-9b.** Removing the ground before the rod leaves the electroscope with a deficit of electrons, and it will be positively charged, as in **Figure 20-9c.** Grounding also can be used as a source of electrons. If a positive rod is brought near the knob of a grounded electroscope, electrons will be attracted from the ground, and the electroscope will obtain a negative charge. When this process is employed, the charge induced on the electroscope is opposite that of the object used to charge it. Because the rod never touches the electroscope, its charge is not transferred, and it can be used many times to charge objects by induction.

Figure 20-9 A negatively charged rod induces a separation of charges in an electroscope **(a).** The electroscope is grounded, and negative charges are pushed from the electroscope to the ground **(b).** The ground is removed before the rod, and the electroscope is left with a positive charge **(c).**



Coulomb's Law

You have seen that a force acts between two or more charged objects. In your experiments with tape, you found that the force depends on distance. The closer you brought the charged comb to the tape, the stronger the force was. You also found that the more you charged the comb, the stronger the force was. How can you vary the quantity of charge in a controlled way? This problem was solved in 1785 by French physicist Charles Coulomb. The type of apparatus used by Coulomb is shown in **Figure 20-10.** An insulating rod with small conducting spheres, A and A′, at each end was suspended by a thin wire. A similar sphere, B, was placed in contact with sphere A. When they were touched with a charged object, the charge spread evenly over the two spheres. Because they were the same size, they received equal amounts of charge. The symbol for charge is q. Therefore, the amount of charge on the spheres can be represented by the notation q_A and q_B.

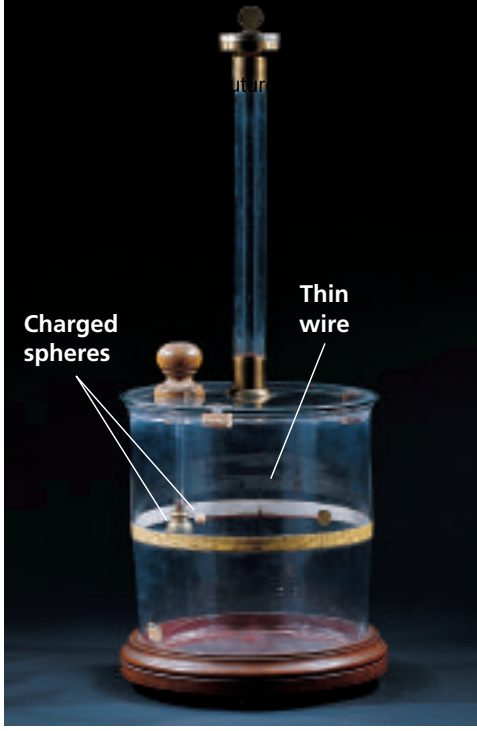


Figure 20-10 Coulomb used a similar type of apparatus to measure the force between two spheres, A and B. He observed the deflection of A while varying the distance between A and B.

Force depends on distance Coulomb found how the force between the two charged spheres depended on the distance. First, he carefully measured the amount of force needed to twist the suspending wire through a given angle. He then placed equal charges on spheres A and B and varied the distance, r, between them. The force moved A, which twisted the suspending wire. By measuring the deflection of A, Coulomb could calculate the force of repulsion. He showed that the force, F, varied inversely with the square of the distance between the centers of the spheres.

$$F \propto \frac{1}{r^2}$$

Force depends on charge To investigate the way in which the force depended on the amount of charge, Coulomb had to change the charges on the spheres in a measured way. He first charged spheres A and B equally, as before. Then he selected an uncharged sphere, C, of the same size as sphere B. When C was placed in contact with B, the spheres shared the charge that had been on B alone. Because the two were the same size, B then had only half of its original charge. Therefore, the charge on B was only one-half the charge on A. After Coulomb adjusted the position of B so that the distance, r, between A and B was the same as before, he found that the force between A and B was half of its former value. That is, he found that the force varied directly with the charge of the bodies.

$$F \propto q_A q_B$$

After many similar measurements, Coulomb summarized the results in a law now known as **Coulomb's law:** the magnitude of the force between charge q_A and charge q_B, separated by a distance r, is proportional to the magnitude of the charges and inversely proportional to the square of the distance between them.

$$F \propto \frac{q_A q_B}{r^2}$$

The unit of charge: the coulomb The amount of charge that an object has is difficult to measure directly. Coulomb's experiments, however, showed that the quantity of charge could be related to force. Thus, Coulomb could define a standard quantity of charge in terms of the amount of force that it produces. The SI standard unit of charge is called the **coulomb** (C).

MINI LAB

Investigating Induction and Conduction

Use a balloon and an electroscope to investigate charging by induction and charging by conduction.

1. Predict what will happen if you charge a balloon by rubbing it with wool and bring it near a neutral electroscope.

2. Predict what will happen if you touch the balloon to the electroscope.

3. Test your predictions.

Analyze and Conclude

4. Describe your results.

5. Explain the movements of the leaves in each step of the experiment. Include diagrams.

6. Describe the results if the wool had been used to charge the electroscope.

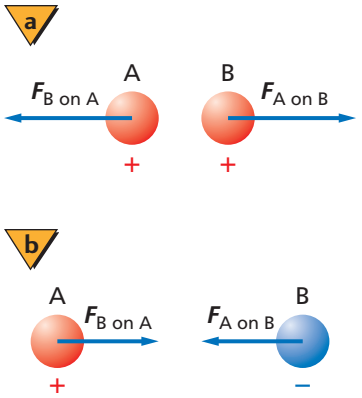


Figure 20-11 The rule for determining the direction of force is: like charges repel; unlike charges attract.

One coulomb is the charge of 6.24×10^{18} electrons or protons. A typical lightning bolt can carry 5 C to 25 C of charge. The charge on a single electron is 1.60×10^{-19} C. The magnitude of the charge of an electron is called the **elementary charge.** Even small pieces of matter, such as coins, contain up to 10^{6} C of negative charge. This enormous amount of negative charge produces almost no external effects because it is balanced by an equal amount of positive charge. If the charge is unbalanced, even as small a charge as 10^{-9} C can result in large forces.

According to Coulomb's law, the magnitude of the force on charge q_A caused by charge q_B a distance r away can be written as follows.

Coulomb's Law $F = K\frac{q_A q_B}{r^2}$

The force between two charges is equal to Coulomb's constant, times the product of the two charges, divided by the square of the distance between them.

When the charges are measured in coulombs, the distance in meters, and the force in newtons, the constant, K, is 9.0×10^{9} N·m^2/C^2.

The Coulomb's law equation gives the magnitude of the force that charge q_A exerts on q_B and also the force that q_B exerts on q_A. These two forces are equal in magnitude but opposite in direction. You can observe this example of Newton's third law of motion in action when you bring two strips of tape with like charges together. Each exerts forces on the other. If you bring a charged comb near either strip of tape, the strip, with its small mass, moves readily. The acceleration of the comb and you is, of course, much less because of the much greater mass.

The electric force, like all other forces, is a vector quantity. Force vectors need both a magnitude and a direction. However, the Coulomb's law equation above gives only the magnitude of the force. To determine the direction, you need to draw a diagram and interpret charge relations carefully. If two positively charged objects, A and B, are brought near, the forces they exert on each other are repulsive, as shown in **Figure 20-11a.** If, instead, B is negatively charged, the forces are attractive, as shown in **Figure 20-11b.**

PROBLEM-SOLVING **Strategies**

Electric Force Problems

Use these steps to find the magnitude and direction of the force between charges.

1. Sketch the system showing all distances and angles to scale.
2. Diagram the vectors of the system.
3. Use Coulomb's law to find the magnitude of the force.
4. Use your diagram along with trigonometric relations to find the direction of the force.
5. Perform all algebraic operations on both the numbers and the units. Make sure that the units match the variables in question.
6. Consider the magnitude of your answer. Is it reasonable?

EXAMPLE Problem 1

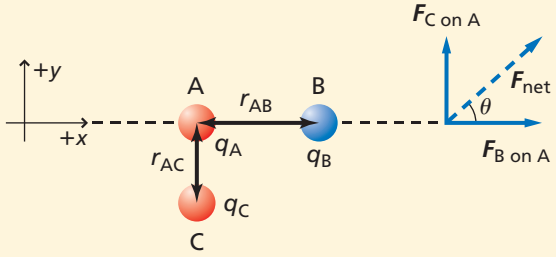
Coulomb's Law in Two Dimensions Sphere A, with a charge of +6.0 μC, is located near another charged sphere, B. Sphere B has a charge of −3.0 μC and is located 4.0 cm to the right of A.

a. What is the force of sphere B on sphere A?

b. A third sphere, C, with a +1.5-μC charge, is added to the configuration. If it is located 3.0 cm directly beneath A, what is the new net force on sphere A?

1 Analyze and Sketch the Problem

- Establish coordinate axes and sketch the spheres.
- Show and label the distances between the spheres.
- Diagram and label the force vectors.



Known:

$q_A = +6.0\ \mu C$ $\quad r_{AB} = 4.0\ cm$

$q_B = -3.0\ \mu C$ $\quad r_{AC} = 3.0\ cm$

$q_C = +1.5\ \mu C$

Unknown:

$\boldsymbol{F}_{B\text{ on }A} = ?$

$\boldsymbol{F}_{C\text{ on }A} = ?$

$\boldsymbol{F}_{net} = ?$

2 Solve for the Unknown

a. Find the force of sphere B on sphere A.

$$F_{B\text{ on }A} = K\frac{q_A q_B}{r_{AB}^2}$$

$$= (9.0\times10^9\ N\cdot m^2/C^2)\frac{(6.0\times10^{-6}\ C)(3.0\times10^{-6}\ C)}{(4.0\times10^{-2}\ m)^2}$$

Substitute $q_A = 6.0\ \mu C$, $q_B = 3.0\ \mu C$, $r_{AB} = 4.0\ cm$

$$= 1.0\times10^2\ N$$

Because spheres A and B have unlike charges, the force of B on A is to the right.

b. Find the force of sphere C on sphere A.

$$F_{C\text{ on }A} = K\frac{q_A q_C}{r_{AC}^2}$$

$$= (9.0\times10^9\ N\cdot m^2/C^2)\frac{(6.0\times10^{-6}\ C)(1.5\times10^{-6}\ C)}{(3.0\times10^{-2}\ m)^2}$$

Substitute $q_A = 6.0\ \mu C$, $q_C = 1.5\ \mu C$, $r_{AC} = 3.0\ cm$

$$= 9.0\times10^1\ N$$

Spheres A and C have like charges, which repel. The force of C on A is upward.

Find the vector sum of $\boldsymbol{F}_{B\text{ on }A}$ and $\boldsymbol{F}_{C\text{ on }A}$ to find $\boldsymbol{F}_{net}$ on sphere A.

$$F_{net} = \sqrt{F_{B\text{ on }A}^2 + F_{C\text{ on }A}^2}$$

$$= \sqrt{(1.0\times10^2\ N)^2 + (9.0\times10^1\ N)^2}$$

Substitute $F_{B\text{ on }A} = 1.0\times10^2\ N$, $F_{C\text{ on }A} = 9.0\times10^1\ N$

$$= 130\ N$$

$$\tan\theta = \frac{F_{C\text{ on }A}}{F_{B\text{ on }A}}$$

$$\theta = \tan^{-1}\left(\frac{F_{C\text{ on }A}}{F_{B\text{ on }A}}\right)$$

$$= \tan^{-1}\left(\frac{9.0\times10^1\ N}{1.0\times10^2\ N}\right)$$

Substitute $F_{C\text{ on }A} = 9.0\times10^1\ N$, $F_{B\text{ on }A} = 1.0\times10^2\ N$

$$= 42°$$

$\boldsymbol{F}_{net} = 130\ N, 42°$ above the x-axis

Math Handbook

3 Evaluate the Answer

- **Are the units correct?** $(N\cdot m^2/C^2)(C)(C)/m^2 = N$. The units work out to be newtons.
- **Does the direction make sense?** Like charges repel; unlike charges attract.
- **Is the magnitude realistic?** The magnitude of the net force is in agreement with the magnitudes of the component forces.

PRACTICE Problems

Additional Problems, Appendix B

9. A negative charge of -2.0×10^{-4} C and a positive charge of 8.0×10^{-4} C are separated by 0.30 m. What is the force between the two charges?

10. A negative charge of -6.0×10^{-6} C exerts an attractive force of 65 N on a second charge that is 0.050 m away. What is the magnitude of the second charge?

11. The charge on B in Example Problem 1 is replaced by a charge of $+3.00\ \mu$C. Diagram the new situation and find the net force on A.

12. Sphere A is located at the origin and has a charge of $+2.0\times10^{-6}$ C. Sphere B is located at +0.60 m on the *x*-axis and has a charge of -3.6×10^{-6} C. Sphere C is located at +0.80 m on the *x*-axis and has a charge of $+4.0\times10^{-6}$ C. Determine the net force on sphere A.

13. Determine the net force on sphere B in the previous problem.

As you use the Coulomb's law equation, keep in mind that Coulomb's law is valid only for point charges or uniform spherical charge distributions. That is, a charged sphere may be treated as if all the charge were located at its center if the charge is spread evenly across its entire surface or throughout its volume. If a sphere is a conductor and another charge is brought near it, the charges on the sphere will be attracted or repelled, and the charge no longer will act as if it were at the sphere's center. Therefore, it is important to consider how large and how far apart two charged spheres are before applying Coulomb's law. The problems in this textbook assume that charged spheres are small enough and far enough apart to be considered point charges unless otherwise noted. When shapes such as long wires or flat plates are considered, Coulomb's law must be modified to account for the nonpoint charge distributions.

Application of Electrostatic Forces

There are many applications of electric forces on particles. For example, these forces can collect soot in smokestacks, thereby reducing air pollution, as shown in **Figure 20-12.** Tiny paint droplets, charged by

CHALLENGE PROBLEM

As shown in the figure on the right, two spheres of equal mass, *m*, and equal positive charge, *q*, are a distance, *r*, apart.

1. Derive an expression for the charge, *q*, that must be on each sphere so that the spheres are in equilibrium; that is, so that the attractive and repulsive forces between them are balanced.

2. If the distance between the spheres is doubled, how will that affect the expression for the value of *q* that you determined in the previous problem? Explain.

3. If the mass of each sphere is 1.50 kg, determine the charge on each sphere needed to maintain the equilibrium.

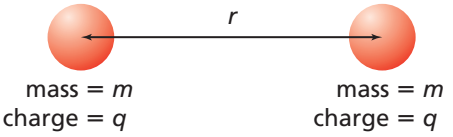




Figure 20-12 The fly ash being released by these smokestacks is a by-product of burning coal. Static-electricity precipitators can be used to reduce fly ash emissions.

induction, can be used to paint automobiles and other objects very uniformly. Photocopy machines use static electricity to place black toner on a page so that a precise reproduction of the original document is made. In other instances, applications are concerned with the control of static charge. For example, static charge can ruin film if it attracts dust, and electronic equipment can be damaged by the discharge of static charge. In these cases, applications are designed to avoid the buildup of static charge and to safely eliminate any charge that does build up.

2. Section Review

14. Force and Charge How are electric force and charge related? Describe the force when the charges are like charges and the force when the charges are opposite charges.

15. Force and Distance How are electric force and distance related? How would the force change if the distance between two charges were tripled?

16. Electroscopes When an electroscope is charged, the leaves rise to a certain angle and remain at that angle. Why do they not rise farther?

17. Charging an Electroscope Explain how to charge an electroscope positively using

a. a positive rod.

b. a negative rod.

18. Attraction of Neutral Objects What two properties explain why a neutral object is attracted to both positively and negatively charged objects?

19. Charging by Induction In an electroscope being charged by induction, what happens when the charging rod is moved away before the ground is removed from the knob?

20. Electric Forces Two charged spheres are held a distance, r, apart. One sphere has a charge of $+3\mu C$, and the other sphere has a charge of $+9\mu C$. Compare the force of the $+3\mu C$ sphere on the $+9\mu C$ sphere with the force of the $+9\mu C$ sphere on the $+3\mu C$ sphere.

21. Critical Thinking Suppose that you are testing Coulomb's law using a small, positively charged plastic sphere and a large, positively charged metal sphere. According to Coulomb's law, the force depends on $1/r^2$, where r is the distance between the centers of the spheres. As the spheres get close together, the force is smaller than expected from Coulomb's law. Explain.

Charged Objects

In this chapter, you observed and studied phenomena that result from the separation of electric charges. You learned that hard rubber and plastic tend to become negatively charged when they are rubbed, while glass and wool tend to become positively charged. But what happens if two objects that tend to become negatively charged are rubbed together? Will electrons be transferred? If so, which material will gain electrons, and which will lose them? In this physics lab, you will design a procedure to further your investigations of positive and negative charges.

QUESTION

How can you test materials for their ability to hold positive and negative charges?

Objectives

- **Observe** that different materials tend to become positively or negatively charged.
- **Compare and contrast** the ability of materials to acquire and hold positive and negative charges.
- **Interpret data** to order a list of materials from strongest tendency to be negatively charged to strongest tendency to be positively charged.

Safety Precautions





Materials

15-cm plastic ruler
thread
ring stand with ring
masking tape
materials to be charged, such as rubber rods, plastic rods, glass rods, PVC pipe, copper pipe, steel pipe, pencils, pens, wool, silk, plastic wrap, plastic sandwich bags, waxed paper, and aluminum foil

Procedure

1. Use the lab photo as a guide to suspend a 15-cm plastic ruler. It is advisable to wash the ruler in soapy water, then rinse and dry it thoroughly before each use, especially if it is a humid day. The thread should be attached at the midpoint of the ruler with two or three wraps of masking tape between the thread and ruler.
2. Use the following situations as a reference for types of charges a material can have: 1) a plastic ruler rubbed with wool gives the plastic ruler an excess negative charge and the wool an excess positive charge, and 2) a plastic ruler rubbed with plastic wrap gives the plastic ruler an excess positive charge and the plastic wrap an excess negative charge.

Data Table

Material 1	Material 2	Charge on Ruler (+, −, 0)	Observation of Ruler's Movements	Charge on Material 1 (+, −, 0)	Charge on Material 2 (+, −, 0)

3. Design a procedure to test which objects tend to become negatively charged and which tend to become positively charged. Try various combinations of materials and record your observations in the data table.
4. Develop a test to see if an object is neutral. Remember that a charged ruler may be attracted to a neutral object if it induces a separation of charge in the neutral object.
5. Be sure to check with your teacher and have your procedure approved before you proceed with your lab.

Analyze

1. **Observe and Infer** As you brought charged materials together, could you detect a force between the charged materials? Describe this force.
2. **Formulate Models** Make a drawing of the charge distribution on the two materials for one of your trials. Use this drawing to explain why the materials acted the way they did during your experiments with them.
3. **Draw Conclusions** Which materials hold an excess charge? Which materials do not hold a charge very well?
4. **Draw Conclusions** Which materials tend to become negatively charged? Which tend to become positively charged?
5. **Interpret Data** Use your data table to list the relative tendencies of materials to be positively or negatively charged.

Conclude and Apply

1. Explain what is meant by the phrases *excess charge* and *charge imbalance* when referring to static electricity.
2. Does excess charge remain on a material or does it dissipate over time?
3. Could you complete this physics lab using a metal rod in place of the suspended plastic ruler? Explain.
4. Clear plastic wrap seals containers of food. Why does plastic wrap cling to itself after it is pulled from its container?

Going Further

Review the information in your textbook about electroscopes. Redesign the lab using an electroscope, rather than a suspended ruler, to test for the type of charge on an object.

Real-World Physics

Trucks often have a rubber strap or a chain that drags along the road. Why are they used?

Physics online

To find out more about static charge, visit the Web site: physicspp.com

Future Technology

Spacecraft and Static Electricity

Most objects on Earth do not build up substantial static-electric charges because a layer of moisture clings to surfaces, allowing charges to migrate to or from the ground. As you learned in this chapter, Earth can absorb almost any amount of charge. However, there is no moisture in space, and Earth is far away. Charged particles ejected from the Sun, or in the ionosphere, strike and cling to spacecraft, charging their surfaces to thousands of volts.

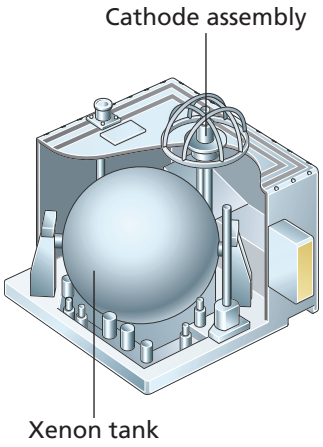
Plasma and Charging

In Chapter 13, you learned that plasma consists of free electrons and positive ions. Orbiting spacecraft are surrounded by a thin cloud of this plasma. The electrons in plasma can move far more easily than more massive positive ions. Thus, spacecraft surfaces tend to attract electrons and develop a negative charge. This negative charge eventually attracts some heavy positive ions, which strike the spacecraft and can damage its surface.

On the *International Space Station,* an additional difficulty stems from the array of solar panels that convert energy from the Sun into electricity. When the arrays are powering the space station, the voltage on the surface of the craft tends to be close to the voltage of the solar array. As a result, it is possible that an electric arc could form between the space station and the plasma that surrounds it.



Plasma Contactor Unit (PCU)



This is a cutaway drawing of the PCU.

Consequences of an Arc Arcs are extremely hot and carry a great deal of current. They can prematurely ignite retro-rockets or explosive bolts and interfere with the operation of the spacecraft's electronic equipment. The solar panels are particularly susceptible to arc damage. In addition to damage to the spacecraft's components, there is a remote chance that the buildup of charge might endanger astronauts on space walks.

To discharge the potential difference and protect craft and crew, the space station's skin must be connected by a conductor, called a plasma contactor, to the plasma cloud surrounding it. The connection begins on board the station, where a stream of xenon gas from a tank in the Plasma Contactor Unit (PCU) is ionized by an electric current. This ionization takes place in the cathode assembly. The ionized xenon, now in the plasma state, passes out of the craft through the cathode assembly. It is this stream of conductive plasma that connects the craft to the surrounding plasma cloud, thereby reducing the potential difference to safe levels.

Future Applications

Future spacecraft might integrate the plasma contactor into the propulsion system. For example, the Variable Specific Impulse Magnetoplasma Rocket (VASIMR) could use the plasma exhaust that it produces to provide an electric connection between the spacecraft and the surrounding plasma. Scientists think that this type of rocket could be used in the future to travel between planets.

Going Further

1. **Apply** What is the purpose of a plasma contactor? How is it similar to using your finger to ground an electroscope?
2. **Research** How could scientists assess the charge on the surface of the *International Space Station?*

1. Electric Charge

Vocabulary

- electrostatics (p. 541)
- neutral (p. 543)
- insulator (p. 544)
- conductor (p. 544)

Key Concepts

- There are two kinds of electric charge, positive and negative. Interactions of these charges explain the attraction and repulsion that you observed in the strips of tape.
- Electric charge is not created or destroyed; it is conserved. Charging is the separation, not creation, of electric charges.
- Objects can be charged by the transfer of electrons. An area with excess electrons has a net negative charge; an area with a deficit of electrons has a net positive charge.
- Charges added to one part of an insulator remain on that part. Insulators include glass, dry wood, plastics, and dry air.
- Charges added to a conductor quickly spread over the surface of the object. In general, examples of conductors include graphite, metals, and matter in the plasma state.
- Under certain conditions, charges can move through a substance that is ordinarily an insulator. Lightning moving through air is one example.

20.2 Electric Force

Vocabulary

- electroscope (p. 546)
- charging by conduction (p. 547)
- charging by induction (p. 548)
- grounding (p. 548)
- Coulomb's law (p. 549)
- coulomb (p. 549)
- elementary charge (p. 550)

Key Concepts

- When an electroscope is charged, electric forces cause its thin metal leaves to spread apart.
- An object can be charged by conduction by touching it with a charged object.
- A charged object will induce a separation of charges within a neutral conductor. This process will result in an attractive force between the charged object and the neutral conductor.
- To charge a conductor by induction, a charged object is first brought near it, causing a separation of charges. Then, the conductor to be charged is separated, trapping opposite charges on the two halves.
- Grounding is the removal of excess charge by touching an object to Earth. Grounding can be used in the process of charging an electroscope by induction.
- Coulomb's law states that the force between two charged particles varies directly with the product of their charges and inversely with the square of the distance between them.

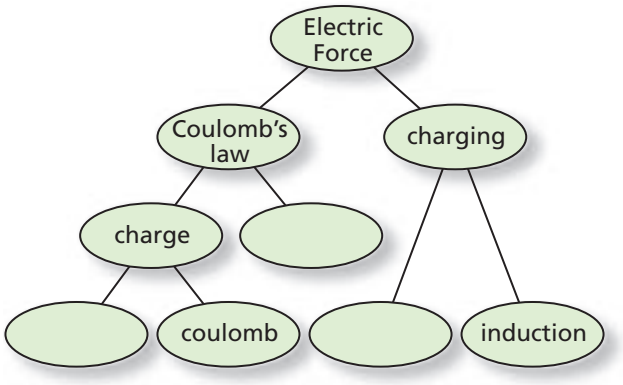
$$F = K\frac{q_A q_B}{r^2}$$

To determine the direction of the force, remember the following rule: like charges repel; unlike charges attract.

- The SI unit of charge is the coulomb. One coulomb (C) is the magnitude of the charge of 6.24×10^{18} electrons or protons. The elementary charge, the charge of a proton or electron, is 1.60×10^{-19} C.

Concept Mapping

22. Complete the concept map below using the following terms: *conduction, distance, elementary charge.*



Mastering Concepts

23. If you comb your hair on a dry day, the comb can become positively charged. Can your hair remain neutral? Explain. (20.1)

24. List some insulators and conductors. (20.1)

25. What property makes metal a good conductor and rubber a good insulator? (20.1)

26. **Laundry** Why do socks taken from a clothes dryer sometimes cling to other clothes? (20.2)

27. **Compact Discs** If you wipe a compact disc with a clean cloth, why does the CD then attract dust? (20.2)

28. **Coins** The combined charge of all electrons in a nickel is hundreds of thousands of coulombs. Does this imply anything about the net charge on the coin? Explain. (20.2)

29. How does the distance between two charges impact the force between them? If the distance is decreased while the charges remain the same, what happens to the force? (20.2)

30. Explain how to charge a conductor negatively if you have only a positively charged rod. (20.2)

Applying Concepts

31. How does the charge of an electron differ from the charge of a proton? How are they similar?

32. Using a charged rod and an electroscope, how can you find whether or not an object is a conductor?

33. A charged rod is brought near a pile of tiny plastic spheres. Some of the spheres are attracted to the rod, but as soon as they touch the rod, they are flung off in different directions. Explain.

34. **Lightning** Lightning usually occurs when a negative charge in a cloud is transported to Earth. If Earth is neutral, what provides the attractive force that pulls the electrons toward Earth?

35. Explain what happens to the leaves of a positively charged electroscope when rods with the following charges are brought close to, but not touching, the electroscope.

a. positive
b. negative

36. As shown in **Figure 20-13,** Coulomb's law and Newton's law of universal gravitation appear to be similar. In what ways are the electric and gravitational forces similar? How are they different?

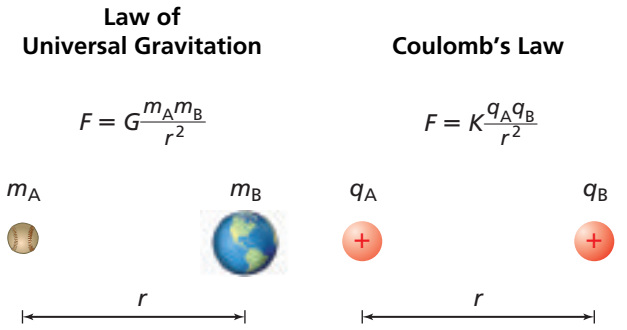


Figure 20-13 (Not to scale)

37. The constant, K, in Coulomb's equation is much larger than the constant, G, in the universal gravitation equation. Of what significance is this?

38. The text describes Coulomb's method for charging two spheres, A and B, so that the charge on B was exactly half the charge on A. Suggest a way that Coulomb could have placed a charge on sphere B that was exactly one-third the charge on sphere A.

39. Coulomb measured the deflection of sphere A when spheres A and B had equal charges and were a distance, r, apart. He then made the charge on B one-third the charge on A. How far apart would the two spheres then have had to be for A to have had the same deflection that it had before?

40. Two charged bodies exert a force of 0.145 N on each other. If they are moved so that they are one-fourth as far apart, what force is exerted?

41. Electric forces between charges are enormous in comparison to gravitational forces. Yet, we normally do not sense electric forces between us and our surroundings, while we do sense gravitational interactions with Earth. Explain.

Mastering Problems

20.2 Electric Force

42. Two charges, q_A and q_B, are separated by a distance, r, and exert a force, F, on each other. Analyze Coulomb's law and identify what new force would exist under the following conditions.
- **a.** q_A is doubled
- **b.** q_A and q_B are cut in half
- **c.** r is tripled
- **d.** r is cut in half
- **e.** q_A is tripled and r is doubled

43. Lightning A strong lightning bolt transfers about 25 C to Earth. How many electrons are transferred?

44. Atoms Two electrons in an atom are separated by 1.5×10^{-10} m, the typical size of an atom. What is the electric force between them?

45. A positive and a negative charge, each of magnitude 2.5×10^{-5} C, are separated by a distance of 15 cm. Find the force on each of the particles.

46. A force of 2.4×10^{2} N exists between a positive charge of 8.0×10^{-5} C and a positive charge of 3.0×10^{-5} C. What distance separates the charges?

47. Two identical positive charges exert a repulsive force of 6.4×10^{-9} N when separated by a distance of 3.8×10^{-10} m. Calculate the charge of each.

48. A positive charge of 3.0 μC is pulled on by two negative charges. As shown in **Figure 20-14,** one negative charge, -2.0 μC, is 0.050 m to the west, and the other, -4.0 μC, is 0.030 m to the east. What total force is exerted on the positive charge?

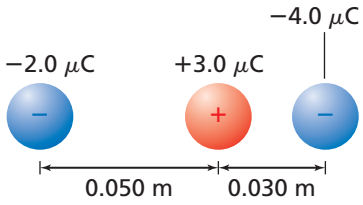


Figure 20-14

49. **Figure 20-15** shows two positively charged spheres, one with three times the charge of the other. The spheres are 16 cm apart, and the force between them is 0.28 N. What are the charges on the two spheres?

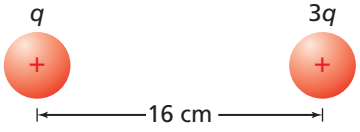


Figure 20-15

50. Charge in a Coin How many coulombs of charge are on the electrons in a nickel? Use the following method to find the answer.
- **a.** Find the number of atoms in a nickel. A nickel has a mass of about 5 g. A nickel is 75 percent Cu and 25 percent Ni, so each mole of the coin's atoms will have a mass of about 62 g.
- **b.** Find the number of electrons in the coin. On average, each atom has 28.75 electrons.
- **c.** Find the coulombs on the electrons.

51. Three particles are placed in a line. The left particle has a charge of -55 μC, the middle one has a charge of $+45$ μC, and the right one has a charge of -78 μC. The middle particle is 72 cm from each of the others, as shown in **Figure 20-16.**
- **a.** Find the net force on the middle particle.
- **b.** Find the net force on the right particle.

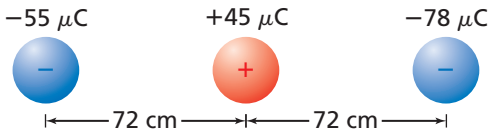


Figure 20-16

Mixed Review

52. A small metal sphere with charge 1.2×10^{-5} C is touched to an identical neutral sphere and then placed 0.15 m from the second sphere. What is the electric force between the two spheres?

53. Atoms What is the electric force between an electron and a proton placed 5.3×10^{-11} m apart, the approximate radius of a hydrogen atom?

54. A small sphere of charge 2.4 μC experiences a force of 0.36 N when a second sphere of unknown charge is placed 5.5 cm from it. What is the charge of the second sphere?

55. Two identically charged spheres placed 12 cm apart have an electric force of 0.28 N between them. What is the charge of each sphere?

56. In an experiment using Coulomb's apparatus, a sphere with a charge of 3.6×10^{-8} C is 1.4 cm from a second sphere of unknown charge. The force between the spheres is 2.7×10^{-2} N. What is the charge of the second sphere?

57. The force between a proton and an electron is 3.5×10^{-10} N. What is the distance between these two particles?

Thinking Critically

58. **Apply Concepts** Calculate the ratio of the electric force to the gravitational force between the electron and the proton in a hydrogen atom.

59. **Analyze and Conclude** Sphere A, with a charge of $+64\ \mu C$, is positioned at the origin. A second sphere, B, with a charge of $-16\ \mu C$, is placed at $+1.00$ m on the x-axis.
- **a.** Where must a third sphere, C, of charge $+12\ \mu C$ be placed so there is no net force on it?
- **b.** If the third sphere had a charge of $+6\ \mu C$, where should it be placed?
- **c.** If the third sphere had a charge of $-12\ \mu C$, where should it be placed?

60. Three charged spheres are located at the positions shown in **Figure 20-17.** Find the total force on sphere B.

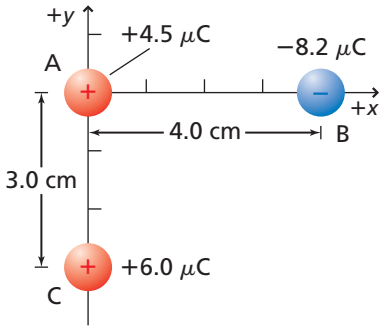


Figure 20-17

61. The two pith balls in **Figure 20-18** each have a mass of 1.0 g and an equal charge. One pith ball is suspended by an insulating thread. The other is brought to 3.0 cm from the suspended ball. The suspended ball is now hanging with the thread forming an angle of 30.0° with the vertical. The ball is in equilibrium with F_E, F_g, and F_T. Calculate each of the following.
- **a.** F_g on the suspended ball
- **b.** F_E
- **c.** the charge on the balls

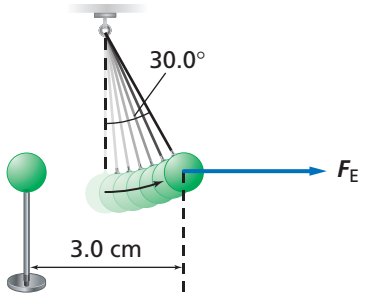


Figure 20-18

62. Two charges, q_A and q_B, are at rest near a positive test charge, q_T, of $7.2\ \mu C$. The first charge, q_A, is a positive charge of $3.6\ \mu C$ located 2.5 cm away from q_T at 35°; q_B is a negative charge of $-6.6\ \mu C$ located 6.8 cm away at 125°.
- **a.** Determine the magnitude of each of the forces acting on q_T.
- **b.** Sketch a force diagram.
- **c.** Graphically determine the resultant force acting on q_T.

Writing in Physics

63. **History of Science** Research several devices that were used in the seventeenth and eighteenth centuries to study static electricity. Examples that you might consider include the Leyden jar and the Wimshurst machine. Discuss how they were constructed and how they worked.

64. In Chapter 13, you learned that forces exist between water molecules that cause water to be denser as a liquid between 0°C and 4°C than as a solid at 0°C. These forces are electrostatic in nature. Research electrostatic intermolecular forces, such as van der Waals forces and dipole-dipole forces, and describe their effects on matter.

Cumulative Review

65. Explain how a pendulum can be used to determine the acceleration of gravity. (Chapter 14)

66. A submarine that is moving 12.0 m/s sends a sonar ping of frequency 1.50×10^3 Hz toward a seamount that is directly in front of the submarine. It receives the echo 1.800 s later. (Chapter 15)
- **a.** How far is the submarine from the seamount?
- **b.** What is the frequency of the sonar wave that strikes the seamount?
- **c.** What is the frequency of the echo received by the submarine?

67. **Security Mirror** A security mirror is used to produce an image that is three-fourths the size of an object and is located 12.0 cm behind the mirror. What is the focal length of the mirror? (Chapter 17)

68. A 2.00-cm-tall object is located 20.0 cm away from a diverging lens with a focal length of 24.0 cm. What are the image position, height, and orientation? Is this a real or a virtual image? (Chapter 18)

69. **Spectrometer** A spectrometer contains a grating of 11,500 slits/cm. Find the angle at which light of wavelength 527 nm has a first-order bright band. (Chapter 19)

Multiple Choice

1. How many electrons have been removed from a positively charged electroscope if it has a net charge of 7.5×10^{-11} C?

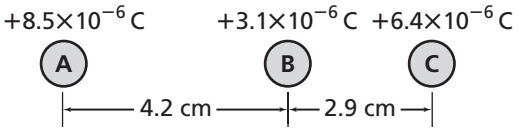
Ⓐ 7.5×10^{-11} electrons
Ⓑ 2.1×10^{-9} electrons
Ⓒ 1.2×10^{8} electrons
Ⓓ 4.7×10^{8} electrons

2. The force exerted on a particle with a charge of 5.0×10^{-9} C by a second particle that is 4 cm away is 8.4×10^{-5} N. What is the charge of the second particle?

Ⓐ 4.2×10^{-13} C Ⓒ 3.0×10^{-9} C
Ⓑ 2.0×10^{-9} C Ⓓ 6.0×10^{-5} C

3. Three charges, A, B, and C, are located in a line, as shown below. What is the net force on charge B?

Ⓐ 78 N toward A Ⓒ 130 N toward A
Ⓑ 78 N toward C Ⓓ 210 N toward C



4. What is the charge on an electroscope that has an excess of 4.8×10^{10} electrons?

Ⓐ 3.3×10^{-30} C Ⓒ 7.7×10^{-9} C
Ⓑ 4.8×10^{-10} C Ⓓ 4.8×10^{10} C

5. Two charged bodies exert a force of 86 N on each other. If they are moved so that they are six times farther apart, what is the new force that they will exert on each other?

Ⓐ 2.4 N Ⓒ 86 N
Ⓑ 14 N Ⓓ 5.2×10^{2} N

6. Two equally charged bodies exert a force of 90 N on each other. If one of the bodies is exchanged for a body of the same size, but three times as much charge, what is the new force that they will exert on each other?

Ⓐ 10 N Ⓒ 2.7×10^{2} N
Ⓑ 30 N Ⓓ 8.1×10^{2} N

7. An alpha particle has a mass of 6.68×10^{-27} kg and a charge of 3.2×10^{-19} C. What is the ratio of the electrostatic force to the gravitational force between two alpha particles?

Ⓐ 1 Ⓒ 2.3×10^{15}
Ⓑ 4.8×10^{7} Ⓓ 3.1×10^{35}

8. Charging a neutral body by touching it with a charged body is called charging by _____ .

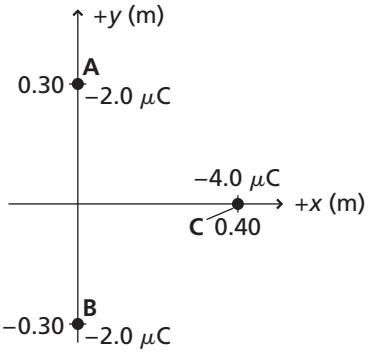
Ⓐ conduction Ⓒ grounding
Ⓑ induction Ⓓ discharging

9. Macy rubs a balloon with wool, giving the balloon a charge of -8.9×10^{-14} C. What is the force between the balloon and a metal sphere that is charged to 25 C and is 2 km away?

Ⓐ 8.9×10^{-15} N Ⓒ 2.2×10^{-12} N
Ⓑ 5.0×10^{-9} N Ⓓ 5.6×10^{4} N

Extended Answer

10. According to the diagram, what is the net force exerted by charges A and B on charge C? In your answer, include a diagram showing the force vectors $\boldsymbol{F}_{\text{A on C}}$, $\boldsymbol{F}_{\text{B on C}}$, and $\boldsymbol{F}_{\text{net}}$.



Test-Taking TIP

Slow Down

Check to make sure you are answering the question that each problem is posing. Read the questions and answer choices very carefully. Remember that doing most of the problems and getting them right is always preferable to doing all of the problems and getting a lot of them wrong.

Chapter 6

Electric Field



- You will relate electric fields to electric forces and distinguish between them.
- You will relate electric potential difference to work and energy.
- You will describe how charges are distributed on conductors.
- You will explain how capacitors store electric charges.

Why It's Important

Electricity is an essential form of energy for modern societies.

High-Energy Discharge
A high-voltage generator produces the glow you see inside these discharge spheres.

LAUNCH Lab

How do charged objects interact at a distance?

Question

How is a charged object affected by interaction with other charged objects at a distance?

Procedure

1. Inflate and tie off two balloons. Attach a $\frac{1}{2}$-m length of string to each balloon.
2. Rub one balloon back and forth on your shirt six to eight times, causing it to become charged. Hang it from a cabinet, table, or other support by the string with a piece of tape.
3. Rub the second balloon the same way and then suspend it from its string.
4. **Observe** Slowly bring the second balloon toward the suspended one. How do the balloons behave? Tape the second balloon so it hangs by its string next to the first balloon.
5. **Observe** Bring your hand toward the charged balloons. What happens?

Analysis

What did you observe as the two balloons were brought near each other? What happened as your hand was brought near the balloons?

Critical Thinking

With what two objects have you previously observed similar behaviors of action at a distance?



1. Creating and Measuring Electric Fields

Electric force, like gravitational force, which you studied in Chapter 8, varies inversely as the square of the distance between two point objects. Both forces can act from great distances. How can a force be exerted across what seems to be empty space? Michael Faraday suggested that because an electrically charged object, A, creates a force on another charged object, B, anywhere in space, object A must somehow change the properties of space. Object B somehow senses the change in space and experiences a force due to the properties of the space at its location. We call the changed property of space an **electric field.** An electric field means that the interaction is not between two distant objects, but between an object and the field at its location.

The forces exerted by electric fields can do work, transferring energy from the field to another charged object. This energy is something you use on a daily basis, whether you plug an appliance into an electric outlet or use a battery-powered, portable device. In this chapter, you will learn more about electric fields, forces, and electric energy.

Objectives

- **Define** an electric field.
- **Solve** problems relating to charge, electric fields, and forces.
- **Diagram** electric field lines.

Vocabulary

electric field
electric field line

The Electric Field

How can you measure an electric field? Place a small charged object at some location. If there is an electric force on it, then there is an electric field at that point. The charge on the object that is used to test the field, called the test charge, must be small enough that it doesn't affect other charges.

Consider **Figure 21-1,** which illustrates a charged object with a charge of q. Suppose you place the positive test charge at some point, A, and measure a force, $\boldsymbol{F}$. According to Coulomb's law, the force is directly proportional to the strength of the test charge, q'. That is, if the charge is doubled, so is the force. Therefore, the ratio of the force to the charge is a constant. If you divide the force, $\boldsymbol{F}$, by the test charge, q', you obtain a vector quantity, $\boldsymbol{F}/q'$. This quantity does not depend on the test charge, only on the force, $\boldsymbol{F}$, and the location of point A. The electric field at point A, the location of q', is represented by the following equation.

Color Convention

- Electric field lines are **indigo.**
- Positive charges are **red.**
- Negative charges are **blue.**

Electric Field Strengh $\boldsymbol{E} = \dfrac{\boldsymbol{F}_{\text{on } q'}}{q'}$

The strength of an electric field is equal to the force on a positive test charge divided by the strength of the test charge.

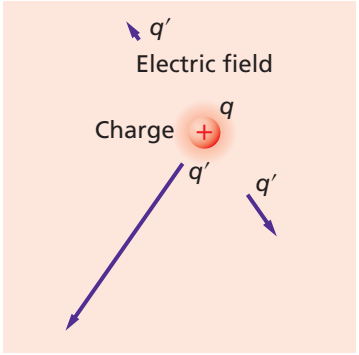


Figure 21-1 Arrows can be used to represent the magnitude and direction of the electric field about an electric charge at various locations.

The direction of an electric field is the direction of the force on a positive test charge. The magnitude of the electric field strength is measured in newtons per coulomb, N/C.

A picture of an electric field can be made by using arrows to represent the field vectors at various locations, as shown in Figure 21-1. The length of the arrow is used to show the strength of the field. The direction of the arrow shows the field direction. To find the field from two charges, the fields from the individual charges are added vectorially. A test charge can be used to map out the field resulting from any collection of charges. Typical electric field strengths produced by charge collections are shown in **Table 21-1.**

An electric field should be measured only by a very small test charge. This is because the test charge also exerts a force on q. It is important that the force exerted by the test charge does not cause charge to be redistributed on a conductor, thereby causing q to move to another location and thus, changing the force on q' as well as the electric field strength being measured. A test charge always should be small enough so that its effect on q is negligible.

Table 21-1

Approximate Values of Typical Electric Fields

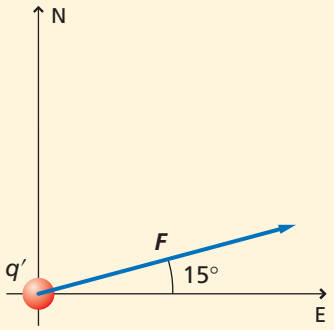
Field	Value (N/C)
Near a charged, hard-rubber rod	1×10^{3}
In a television picture tube	1×10^{5}
Needed to create a spark in air	3×10^{6}
At an electron's orbit in a hydrogen atom	5×10^{11}

EXAMPLE Problem 1

Electric Field Strength An electric field is measured using a positive test charge of 3.0×10^{-6} C. This test charge experiences a force of 0.12 N at an angle of 15° north of east. What are the magnitude and direction of the electric field strength at the location of the test charge?

1 Analyze and Sketch the Problem

- Draw and label the test charge, q'.
- Show and label the coordinate system centered on the test charge.
- Diagram and label the force vector at 15° north of east.



Known:

$q' = +3.0\times10^{-6}$ C

$\boldsymbol{F}$ = 0.12 N at 15° N of E

Unknown:

$\boldsymbol{E}$ = ?

2 Solve for the Unknown

$$E = \frac{F}{q'}$$

$$= \frac{0.12\text{ N}}{3.0\times10^{-6}\text{ N/C}} \qquad \text{Substitute } \boldsymbol{F} = 0.12\text{ N}, q' = 3.0\times10^{-6}\text{ C}$$

$$= 4.0\times10^{4}\text{ N/C}$$

The force on the test charge and the electric field are in the same direction.

$\boldsymbol{E}$ = 4.0×10^{4} N/C at 15° N of E

Math Handbook
Operations with Significant Digits
pages 835–836

3 Evaluate the Answer

- **Are the units correct?** Electric field strength is correctly measured in N/C.
- **Does the direction make sense?** The field direction is in the direction of the force because the test charge is positive.
- **Is the magnitude realistic?** This field strength is consistent with the values listed in Table 21-1.

PRACTICE Problems

Additional Problems, Appendix B

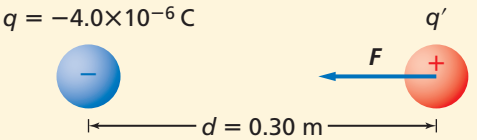
1. A positive test charge of 5.0×10^{-6} C is in an electric field that exerts a force of 2.0×10^{-4} N on it. What is the magnitude of the electric field at the location of the test charge?
2. A negative charge of 2.0×10^{-8} C experiences a force of 0.060 N to the right in an electric field. What are the field's magnitude and direction at that location?
3. A positive charge of 3.0×10^{-7} C is located in a field of 27 N/C directed toward the south. What is the force acting on the charge?
4. A pith ball weighing 2.1×10^{-3} N is placed in a downward electric field of 6.5×10^{4} N/C. What charge (magnitude and sign) must be placed on the pith ball so that the electric force acting on it will suspend it against the force of gravity?
5. You are probing the electric field of a charge of unknown magnitude and sign. You first map the field with a 1.0×10^{-6}-C test charge, then repeat your work with a 2.0×10^{-6}-C test charge.
   a. Would you measure the same forces at the same place with the two test charges? Explain.
   b. Would you find the same field strengths? Explain.

EXAMPLE Problem 2

Electric Field Strength What is the electric field strength at a point that is 0.30 m to the right of a small sphere with a charge of -4.0×10^{-6} C?

1 Analyze and Sketch the Problem

- Draw and label the sphere and its charge, q, and the test charge, q'.
- Show and label the distance between the charges.
- Diagram and label the force vector acting on q'.



Known:
$q = -4.0\times10^{-6}$ C
$d = 0.30$ m

Unknown:
$\boldsymbol{E} = ?$

2 Solve for the Unknown

The force and the magnitude of the test charge are unknown, so use Coulomb's law in combination with the electric field strength.

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$$E = \frac{F}{q'}$$

$$= K\frac{qq'}{d^2q'} \qquad \text{Substitute } F = K\frac{qq'}{d^2}$$

$$= K\frac{q}{d^2}$$

$$= \left(9.0\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\frac{(-4.0\times10^{-6}\ \text{C})}{(0.30\ \text{m})^2} \qquad \text{Substitute } K = 9.0\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2,\ q = -4.0\times10^{-6}\ \text{C},\ d = 0.30\ \text{m}$$

$$= -4.0\times10^{5}\ \text{N/C}$$

$\boldsymbol{E} = 4.0\times10^{5}$ N/C toward the sphere, or to the left

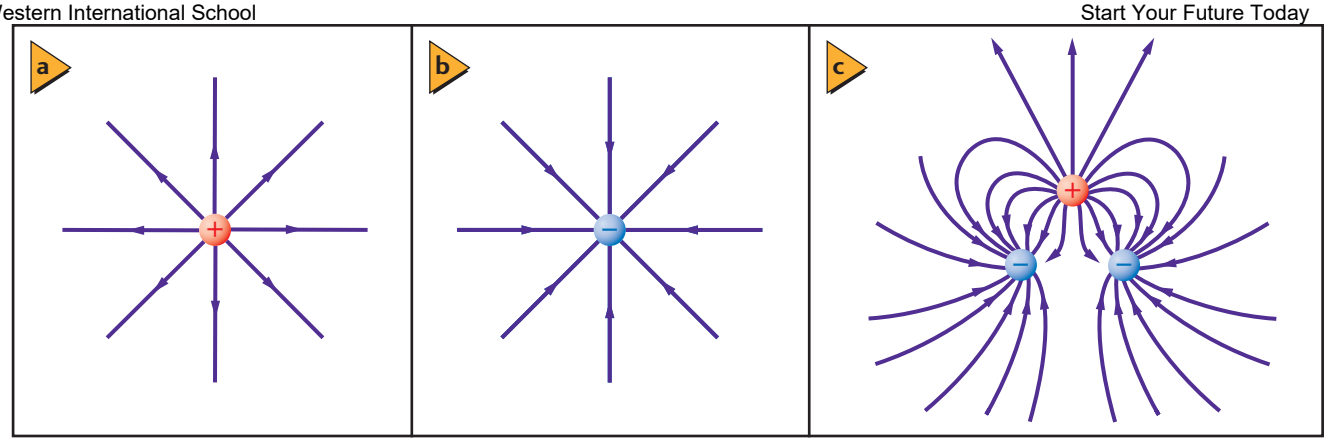
3 Evaluate the Answer

- **Are the units correct?** $(\text{N}\cdot\text{m}^2/\text{C}^2)(\text{C})/\text{m}^2 = \text{N/C}$. The units work out to be N/C, which is correct for electric field strength.
- **Does the direction make sense?** The negative sign indicates that the positive test charge is attracted toward the negative point charge.
- **Is the magnitude realistic?** This field strength is consistent with the values listed in Table 21-1.

PRACTICE Problems

Additional Problems, Appendix B

6. What is the magnitude of the electric field strength at a position that is 1.2 m from a point charge of 4.2×10^{-6} C?
7. What is the magnitude of the electric field strength at a distance twice as far from the point charge in problem 6?
8. What is the electric field at a position that is 1.6 m east of a point charge of $+7.2\times10^{-6}$ C?
9. The electric field that is 0.25 m from a small sphere is 450 N/C toward the sphere. What is the charge on the sphere?
10. How far from a point charge of $+2.4\times10^{-6}$ C must a test charge be placed to measure a field of 360 N/C?



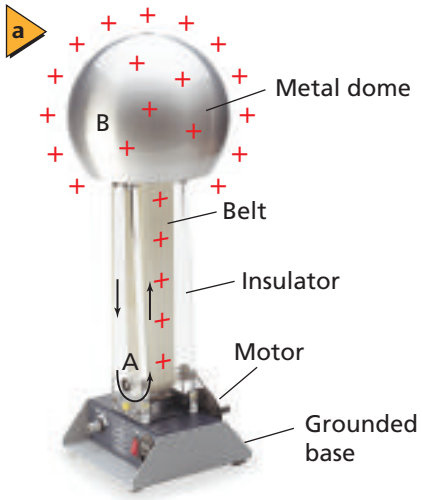
■ **Figure 21-2** Lines of force are drawn perpendicularly away from a positively charged object **(a)** and perpendicularly into a negatively charged object **(b).** Electric field lines are shown between like charged and oppositely charged objects **(c).**

So far, you have measured an electric field at a single point. Now, imagine moving the test charge to another location. Measure the force on it again and calculate the electric field. Repeat this process again and again until you assign every location in space a measurement of the vector quantity of the electric field strength associated with it. The field is present even if there is no test charge to measure it. Any charge placed in an electric field experiences a force on it resulting from the electric field at that location. The strength of the force depends on the magnitude of the field, $\boldsymbol{E}$, and the magnitude of the charge, q. Thus, $\boldsymbol{F} = \boldsymbol{E}q$. The direction of the force depends on the direction of the field and the sign of the charge.

Picturing the Electric Field

A picture of an electric field is shown in **Figure 21-2.** Each of the lines used to represent the actual field in the space around a charge is called an **electric field line.** The direction of the field at any point is the tangent drawn to a field line at that point. The strength of the electric field is indicated by the spacing between the lines. The field is strong where the lines are close together. It is weaker where the lines are spaced farther apart. Although only two-dimensional models can be shown here, remember that electric fields exist in three dimensions.

The direction of the force on a positive test charge near another positive charge is away from the other charge. Thus, the field lines extend radially outward like the spokes of a wheel, as shown in **Figure 21-2a.** Near a negative charge, the direction of the force on the positive test charge is toward the negative charge, so the field lines point radially inward, as shown in **Figure 21-2b.** When there are two or more charges, the field is the vector sum of the fields resulting from the individual charges. The field lines become curved and the pattern is more complex, as shown in **Figure 21-2c.** Note that field lines always leave a positive charge and enter a negative charge, and that they never cross each other.





■ **Figure 21-3** In the Van de Graaff generator **(a),** charge is transferred onto a moving belt at A, and from the belt to the metal dome at B. An electric motor does the work needed to increase the electric potential energy. When a person touches a Van de Graaff generator, the results can be dramatic **(b).**

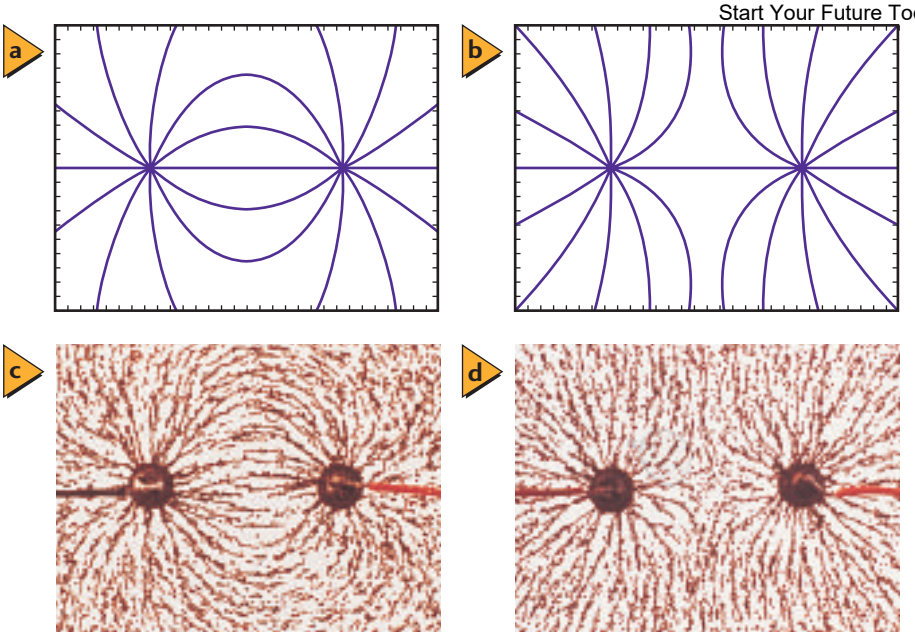


Figure 21-4 Lines of force between unlike charges **(a, c)** and between like charges **(b, d)** describe the behavior of a positively charged object in a field. The top figures are computer tracings of electric field lines.

Robert Van de Graaff devised the high-voltage electrostatic generator in the 1930s. Van de Graaff's machine, shown in **Figure 21-3a** on the previous page, is a device that transfers large amounts of charge from one part of the machine to a metal terminal at the top of the device. Charge is transferred onto a moving belt at the base of the generator, position A, and is transferred off the belt at the metal dome at the top, position B. An electric motor does the work needed to increase the electric potential energy. A person touching the terminal of a Van de Graaff machine is charged electrically. The charges on the person's hairs repel each other, causing the hairs to follow the field lines, as shown in **Figure 21-3b.**

Another method of visualizing field lines is to use grass seed in an insulating liquid, such as mineral oil. The electric forces cause a separation of charge in each long, thin grass seed. The seeds then turn so that they line up along the direction of the electric field. The seeds thus form a pattern of the electric field lines, as in **Figure 21-4.** Field lines do not really exist. They are simply a means of providing a model of an electric field. Electric fields, on the other hand, do exist. Although they provide a method of calculating the force on a charged body, they do not explain why charged bodies exert forces on each other.

1. Section Review

11. Measuring Electric Fields Suppose you are asked to measure the electric field in space. How do you detect the field at a point? How do you determine the magnitude of the field? How do you choose the magnitude of the test charge? What do you do next?

12. Field Strength and Direction A positive test charge of magnitude 2.40×10^{-8} C experiences a force of 1.50×10^{-3} N toward the east. What is the electric field at the position of the test charge?

13. Field Lines In Figure 21-4, can you tell which charges are positive and which are negative? What would you add to complete the field lines?

14. Field Versus Force How does the electric field, $\boldsymbol{E}$, at the test charge differ from the force, $\boldsymbol{F}$, on it?

15. Critical Thinking Suppose the top charge in Figure 21-2c is a test charge measuring the field resulting from the two negative charges. Is it small enough to produce an accurate measure? Explain.

2. Applications of Electric Fields

As you have learned, the concept of energy is extremely useful in mechanics. The law of conservation of energy allows us to solve motion problems without knowing the forces in detail. The same is true in the study of electrical interactions. The work performed moving a charged particle in an electric field can result in the particle's gaining potential, or kinetic energy, or both. Because this chapter investigates charges at rest, only changes in potential energy will be discussed.

Objectives

- **Define** electric potential difference.
- **Calculate** potential difference from the work required to move a charge.
- **Describe** how charges are distributed on solid and hollow conductors.
- **Solve** problems pertaining to capacitance.

Vocabulary

electric potential difference
volt
equipotential
capacitor
capacitance

Energy and Electric Potential

Recall the change in gravitational potential energy of a ball when it is lifted, as shown in **Figure 21-5.** Both the gravitational force, $\boldsymbol{F}$, and the gravitational field, $\boldsymbol{g} = \boldsymbol{F}/m$, point toward Earth. If you lift a ball against the force of gravity, you do work on it, thereby increasing its potential energy.

The situation is similar with two unlike charges: they attract each other, and so you must do work to pull one charge away from the other. When you do the work, you transfer energy to the charge where that energy is stored as potential energy. The larger the test charge, the greater the increase in its potential energy, ΔPE.

Although the force on the test charge depends on its magnitude, q', the electric field it experiences does not. The electric field, $\boldsymbol{E} = \boldsymbol{F}/q'$, is the force per unit charge. In a similar way, the **electric potential difference,** ΔV, is defined as the work done moving a positive test charge between two points in an electric field divided by the magnitude of the test charge.

Electric Potential Difference $\Delta V = \dfrac{W_{\text{on } q'}}{q'}$

The difference in electrical potential is the ratio of the work needed to move a charge to the strength of that charge.

Electric potential difference is measured in joules per coulomb. One joule per coulomb is called a **volt** (J/C = V).

Consider the situation shown in **Figure 21-6** on the next page. The negative charge creates an electric field toward itself. Suppose you place a small positive test charge in the field at position A. It will experience a force in the direction of the field. If you now move the test charge away from the negative charge to position B, as in **Figure 21-6a,** you will have to exert a force, $\boldsymbol{F}$, on the charge. Because the force that you exert is in the same direction as the displacement, the work that you do on the test charge is positive. Therefore, there also will be a positive change in the electric potential difference. The change in potential difference does not depend on the magnitude of the test charge. It depends only on the field and the displacement.

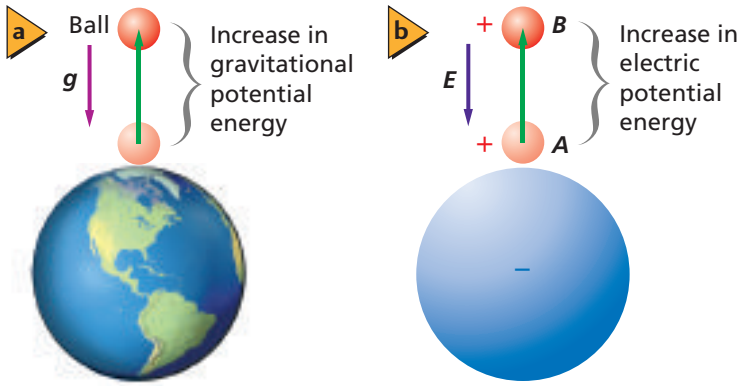


Figure 21-5 Work is needed to move an object against the force of gravity **(a)** and against the electric force **(b).** In both cases, the potential energy of the object is increased.

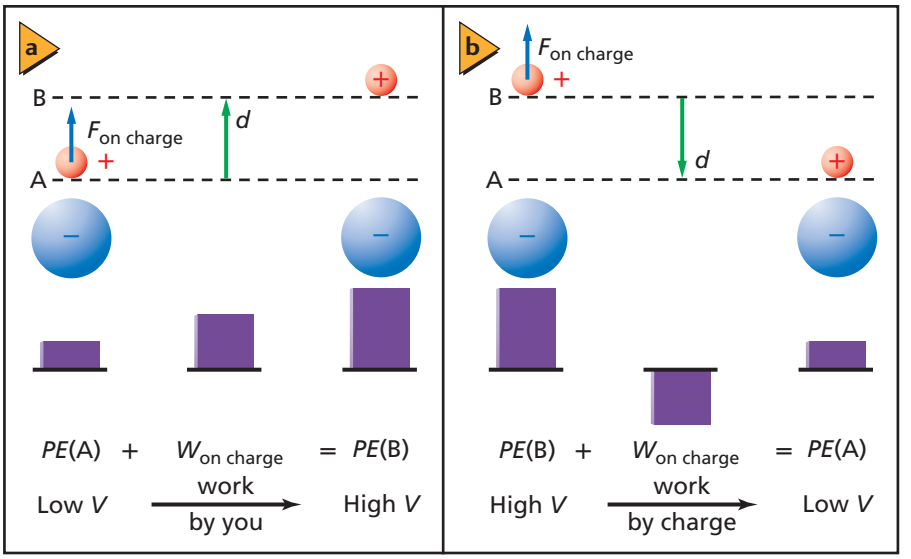


Figure 21-6 Electric potential difference is determined by measuring the work per unit charge. If you move unlike charges apart, you increase the electric potential difference **(a).** If you move unlike charges closer together, you reduce the electric potential difference **(b).**

Suppose you now move the test charge back to position A from position B, as in **Figure 21-6b.** The force that you exert is now in the direction opposite the displacement, so the work that you do is negative. The electric potential difference is also negative. In fact, it is equal and opposite to the potential difference for the move from position A to position B. The electric potential difference does not depend on the path used to go from one position to another. It does depend on the two positions.

Is there always an electric potential difference between the two positions? Suppose you move the test charge in a circle around the negative charge. The force that the electric field exerts on the test charge is always perpendicular to the direction in which you moved it, so you do no work. Therefore, the electric potential difference is zero. Whenever the electric potential difference between two or more positions is zero, those positions are said to be at **equipotential.**

Only differences in potential energy can be measured. The same is true of electric potential; thus, only differences in electric potential are important. The electric potential difference from point A to point B is defined as $\Delta V = V_B - V_A$. Electric potential differences are measured with a voltmeter. Sometimes, the electric potential difference is simply called the voltage. Do not confuse electric potential difference, ΔV, with the unit for volts, V.

APPLYING PHYSICS

▶ **Static Electricity** Modern electronic devices, such as personal computers, contain components that are easily damaged by static electric discharges. To prevent damage to these sensitive components during repair, a technician will wear a conductive strap around his or her wrist. The other end of this strap is clipped to a grounded piece of metal. The strap conducts charge away from the technician and eliminates any possible potential difference with the grounded equipment. ◀

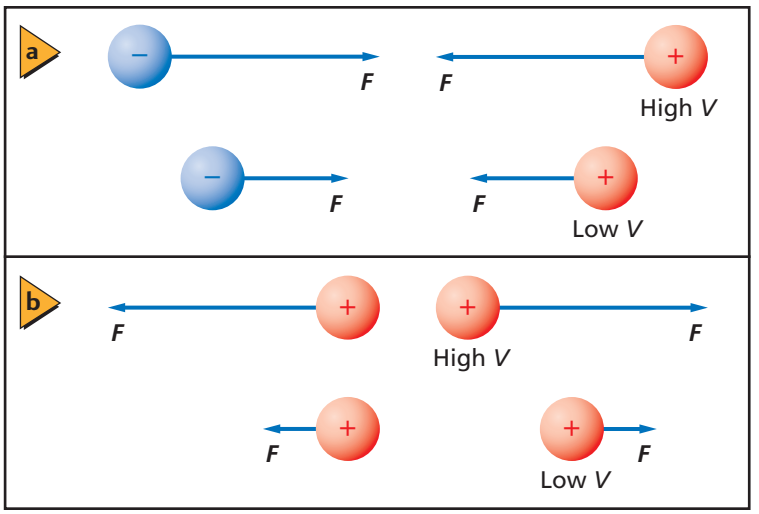


Figure 21-7 Electric potential is smaller when two unlike charges are closer together **(a)** and larger when two like charges are closer together **(b).**

You have seen that electric potential difference increases as a positive test charge is separated from a negative charge. What happens when a positive test charge is separated from a positive charge? There is a repulsive force between these two charges. Potential energy decreases as the two charges are moved farther apart. Therefore, the electric potential is smaller at points farther from the positive charge, as shown in **Figure 21-7.**

As you learned in Chapter 11, the potential energy of a system can be defined as zero at any reference point. In the same way, the electric potential of any point can be defined as zero. No matter what reference point is chosen, the value of the electric potential difference from point A to point B always will be the same.

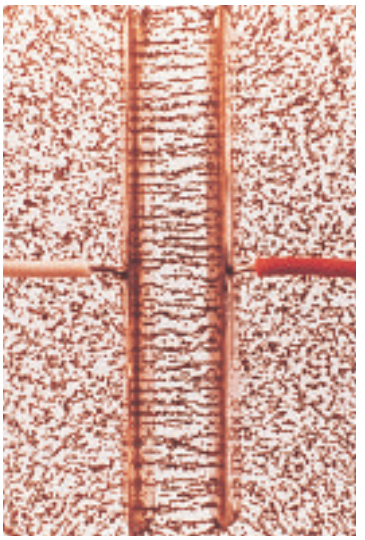


Figure 21-8 A representation of an electric field between parallel plates is shown.

The Electric Potential in a Uniform Field

A uniform electric force and field can be made by placing two large, flat, conducting plates parallel to each other. One is charged positively and the other is charged negatively. The electric field between the plates is constant, except at the edges of the plates, and its direction is from the positive to the negative plate. The pattern formed by the grass seeds pictured in **Figure 21-8** represents the electric field between parallel plates.

If a positive test charge, q', is moved a distance, d, in the direction opposite the electric field direction, the work done is found by the relationship $W_{\text{on } q'} = Fd$. Thus, the electric potential difference, the work done per unit charge, is $\Delta V = Fd/q' = (F/q')d$. Now, the electric field intensity is the force per unit charge, $E = F/q'$. Therefore, the electric potential difference, ΔV, between two points a distance, d, apart in a uniform field, E, is represented by the following equation.

Electric Potential Difference in a Uniform Field $\Delta V = Ed$

The electrical potential difference in a uniform field is equal to the product of electric field intensity and the distance moved by a charge.

The electric potential increases in the direction opposite the electric field direction. That is, the electric potential is higher near the positively charged plate. By dimensional analysis, the product of the units of E and d is (N/C)(m). This is equivalent to one J/C, which is the definition of 1 V.

PRACTICE Problems

Additional Problems, Appendix B

16. The electric field intensity between two large, charged, parallel metal plates is 6000 N/C. The plates are 0.05 m apart. What is the electric potential difference between them?

17. A voltmeter reads 400 V across two charged, parallel plates that are 0.020 m apart. What is the electric field between them?

18. What electric potential difference is applied to two metal plates that are 0.200 m apart if the electric field between them is 2.50×10^3 N/C?

19. When a potential difference of 125 V is applied to two parallel plates, the field between them is 4.25×10^3 N/C. How far apart are the plates?

20. A potential difference of 275 V is applied to two parallel plates that are 0.35 cm apart. What is the electric field between the plates?

EXAMPLE Problem 3

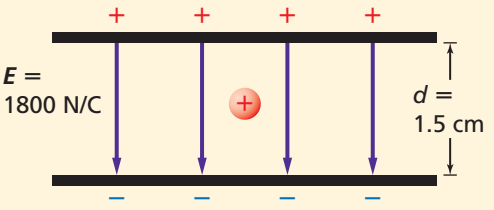
Work Required to Move a Proton Between Charged Parallel Plates Two charged parallel plates are 1.5 cm apart. The magnitude of the electric field between the plates is 1800 N/C.

a. What is the electric potential difference between the plates?

b. What work is required to move a proton from the negative plate to the positive plate?

1 Analyze and Sketch the Problem

- Draw the plates separated by 1.5 cm.
- Label one plate with positive charges and the other with negative charges.
- Draw uniformly spaced electric field lines from the positive plate to the negative plate.
- Indicate the electric field strength between the plates.
- Place a proton in the electric field.



Known:

$E = 1800$ N/C

$d = 1.5$ cm

$q = 1.60\times10^{-19}$ C

Unknown:

$\Delta V = ?$

$W = ?$

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2 Solve for the Unknown

$$\Delta V = Ed$$
$$= (1800 \text{ N/C})(0.015 \text{ m})$$ **Substitute $E = 1800$ N/C, $d = 0.015$ m**
$$= 27 \text{ V}$$

$$\Delta V = \frac{W}{q}$$
$$W = q\Delta V$$
$$= (1.60\times10^{-19} \text{ C})(27 \text{ V})$$ **Substitute $q = 1.60\times10^{-19}$ C, $\Delta V = 27$ V**
$$= 4.3\times10^{-18} \text{ J}$$

3 Evaluate the Answer

- **Are the units correct?** (N/C)(m) = N·m/C = J/C = V. The units work out to be volts. C·V = C(J/C) = J, the unit for work.
- **Does the sign make sense?** Positive work must be done to move a positive charge toward a positive plate.
- **Is the magnitude realistic?** With such a small charge moved through a potential difference of a few volts, the work performed will be small.

PRACTICE Problems

Additional Problems, Appendix B

21. What work is done when 3.0 C is moved through an electric potential difference of 1.5 V?

22. A 12-V car battery can store 1.44×10^{6} C when it is fully charged. How much work can be done by this battery before it needs recharging?

23. An electron in a television picture tube passes through a potential difference of 18,000 V. How much work is done on the electron as it passes through that potential difference?

24. If the potential difference in problem 18 is between two parallel plates that are 2.4 cm apart, what is the magnitude of the electric field between them?

25. The electric field in a particle-accelerator machine is 4.5×10^{5} N/C. How much work is done to move a proton 25 cm through that field?

Millikan's Oil-Drop Experiment

One important application of the uniform electric field between two parallel plates is the measurement of the charge of an electron. This first was determined by American physicist Robert A. Millikan in 1909. **Figure 21-9** shows the method used by Millikan to measure the charge carried by a single electron. First, fine oil drops were sprayed from an atomizer into the air. These drops were charged by friction with the atomizer as they were sprayed. Gravity acting on the drops caused them to fall, and a few of them entered the hole in the top plate of the apparatus. An electric potential difference then was placed across the two plates. The resulting electric field between the plates exerted a force on the charged drops. When the top plate was made positive enough, the electric force caused negatively charged drops to rise. The electric potential difference between the plates was adjusted to suspend a charged drop between the plates. At this point, the downward force of Earth's gravitational field and the upward force of the electric field were equal in magnitude.

The magnitude of the electric field, E, was determined from the electric potential difference between the plates. A second measurement had to be made to find the weight of the drop using the relationship mg, which was too tiny to measure by ordinary methods. To make this measurement, a drop first was suspended. Then, the electric field was turned off, and the rate of the fall of the drop was measured. Because of friction with the air molecules, the oil drop quickly reached terminal velocity, which was related to the mass of the drop by a complex equation. Using the measured terminal velocity to calculate mg and knowing E, the charge, q, could be calculated.

Charge on an electron Millikan found that there was a great deal of variation in the charges of the drops. When he used X rays to ionize the air and add or remove electrons from the drops, he noted, however, that the changes in the charge on the drops were always a multiple of 1.60×10^{-19} C. The changes were caused by one or more electrons being added to or removed from the drops. Millikan concluded that the smallest change in charge that could occur was the amount of charge of one electron. Therefore, Millikan proposed that each electron always has the same charge, 1.60×10^{-19} C. Millikan's experiment showed that charge is quantized. This means that an object can have only a charge with a magnitude that is some integral multiple of the charge of an electron.

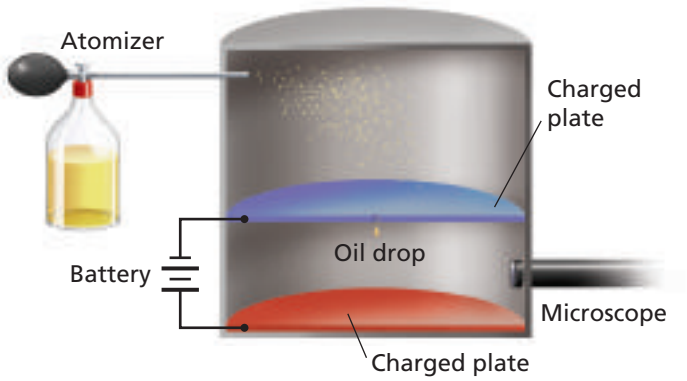


Figure 21-9 This illustration shows a cross-sectional view of the apparatus that Millikan used to determine the charge on an electron.

MINI LAB

Electric Fields

Tie a pith ball on the end of a 20-cm nylon thread and tie the other end to a plastic straw. Holding the straw horizontally, notice that the ball hangs straight down. Now rub a piece of wool on a 30 cm × 30 cm square of plastic foam to charge both objects. Stand the foam vertically. Hold the straw and touch the pith ball to the wool.

1. **Predict** what will happen when the ball is close to the foam.
2. **Test** your prediction by slowly bringing the hanging ball toward the charged plastic foam.
3. **Predict** the ball's behavior at different locations around the foam, and test your prediction.
4. **Observe** the angle of the thread as you move the pith ball to different regions around the foam.

Analyze and Conclude

5. **Explain,** in terms of the electric field, why the ball swings toward the charged plastic.
6. **Compare** the angle of the thread at various points around the foam. Why did it change?
7. **Infer** what the angle of the thread indicates about the strength and the direction of the electric field.

EXAMPLE Problem 4

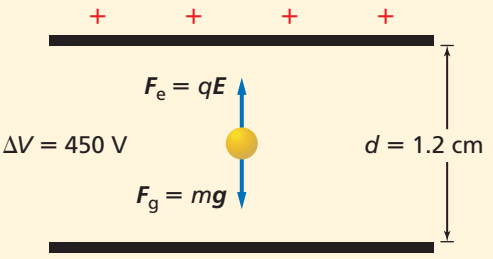
Finding the Charge on an Oil Drop In a Millikan oil-drop experiment, a drop has been found to weigh 2.4×10^{-14} N. The parallel plates are separated by a distance of 1.2 cm. When the potential difference between the plates is 450 V, the drop is suspended, motionless.

a. What is the charge on the oil drop?

b. If the upper plate is positive, how many excess electrons are on the oil drop?

1 Analyze and Sketch the Problem

- Draw the plates with the oil drop suspended between them.
- Draw and label vectors representing the forces.
- Indicate the potential difference and the distance between the plates.



Known:

$\Delta V = 450$ V

$F_g = 2.4\times10^{-14}$ N

$d = 1.2$ cm

Unknown:

charge on drop, q = ?

number of electrons, n = ?

2 Solve for the Unknown

To be suspended, the electric force and gravitational force must be balanced.

$$F_e = F_g$$

$$qE = F_g \qquad \text{Substitute } F_e = qE$$

$$\frac{q\Delta V}{d} = F_g \qquad \text{Substitute } E = \frac{\Delta V}{d}$$

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Solve for q.

$$q = \frac{F_g d}{\Delta V}$$

$$= \frac{(2.4\times10^{-14}\text{ N})(0.012\text{ m})}{450\text{ V}} \qquad \text{Substitute } F_g = 2.4\times10^{-14}\text{ N},\ d = 0.012\text{ m},\ \Delta V = 450\text{ V}$$

$$= 6.4\times10^{-19}\text{ C}$$

Solve for the number of electrons on the drop.

$$n = \frac{q}{e}$$

$$= \frac{6.4\times10^{-19}\text{ C}}{1.6\times10^{-19}\text{ C}} \qquad \text{Substitute } q = 6.4\times10^{-19}\text{ C},\ e = 1.6\times10^{-19}\text{ C}$$

$$= 4$$

3 Evaluate the Answer

- **Are the units correct?** N·m/V = J/(J/C) = C, the unit for charge.
- **Is the magnitude realistic?** This is a small whole number of elementary charges.

PRACTICE Problems

Additional Problems, Appendix B

26. A drop is falling in a Millikan oil-drop apparatus with no electric field. What forces are acting on the oil drop, regardless of its acceleration? If the drop is falling at a constant velocity, describe the forces acting on it.

27. An oil drop weighs 1.9×10^{-15} N. It is suspended in an electric field of 6.0×10^{3} N/C. What is the charge on the drop? How many excess electrons does it carry?

28. An oil drop carries one excess electron and weighs 6.4×10^{-15} N. What electric field strength is required to suspend the drop so it is motionless?

29. A positively charged oil drop weighing 1.2×10^{-14} N is suspended between parallel plates separated by 0.64 cm. The potential difference between the plates is 240 V. What is the charge on the drop? How many electrons is the drop missing?

Sharing of Charge

All systems come to equilibrium when the energy of the system is at a minimum. For example, if a ball is placed on a hill, it finally will come to rest in a valley where its gravitational potential energy is smallest. This also would be the location where its gravitational potential has been reduced by the largest amount. This same principle explains what happens when an insulated, positively charged metal sphere, such as the one shown in **Figure 21-10,** touches a second, uncharged sphere.

The excess charges on sphere A repel each other, so when the neutral sphere, B, touches sphere A, there is a net force on the charges on A toward B. Suppose that you were to physically move the charges, individually, from A to B. When you move the first charge, the other charges on A would push it toward B, so, to control its speed, you would have to exert a force in the opposite direction. Therefore, you do negative work on it, and the electric potential difference from A to B is negative. When the next few charges are moved, they feel a small repulsive force from the charges already on B, but there is still a net positive force in that direction. At some point, the force pushing a charge off A will equal the repulsive force from the charges on B, and the electric potential difference is zero. After this point of equilibrium, work would have to be done to move the next charge to B, so this would not happen by itself and would require an increase in the energy of the system. However, if you did continue to move charges, the electric potential difference from A to B would then be positive. Thus, you can see that charges would move from A to B without external forces until there is no electric potential difference between the two spheres.

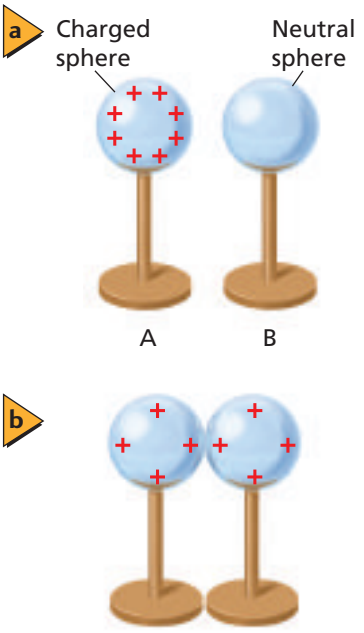


Figure 21-10 A charged sphere shares charge equally with a neutral sphere of equal size when they are placed in contact with each other.

Different sizes of spheres Suppose that the two spheres have different sizes, as in **Figure 21-11.** Although the total numbers of charges on the two spheres are the same, the larger sphere has a larger surface area, so the charges can spread farther apart, and the repulsive force between them is reduced. Thus, if the two spheres now are touched together, there will be a net force that will move charges from the smaller to the larger sphere. Again, the charges will move to the sphere with the lower electric potential until there is no electric potential difference between the two spheres. In this case, the larger sphere will have a larger charge when equilibrium is reached.

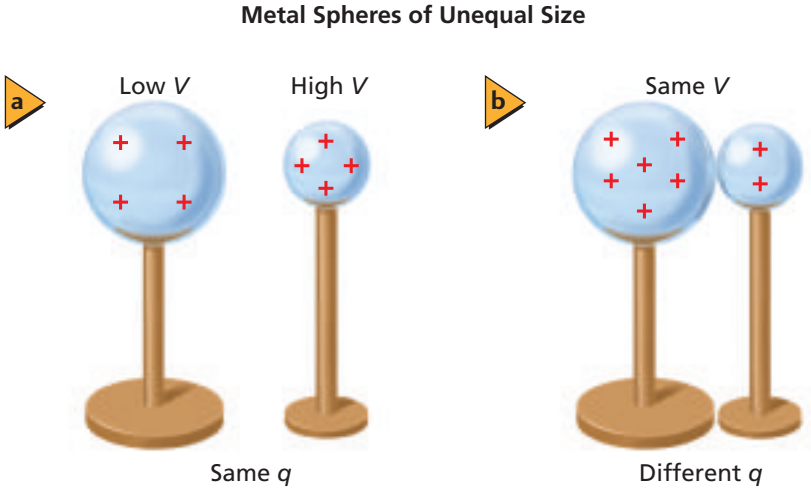


Figure 21-11 Charges are transferred from a sphere with high potential to a sphere with lower potential when they touch. The charges move to create no potential difference.



Figure 21-12 The ground wire on a fuel truck prevents ignition of the gasoline vapors.

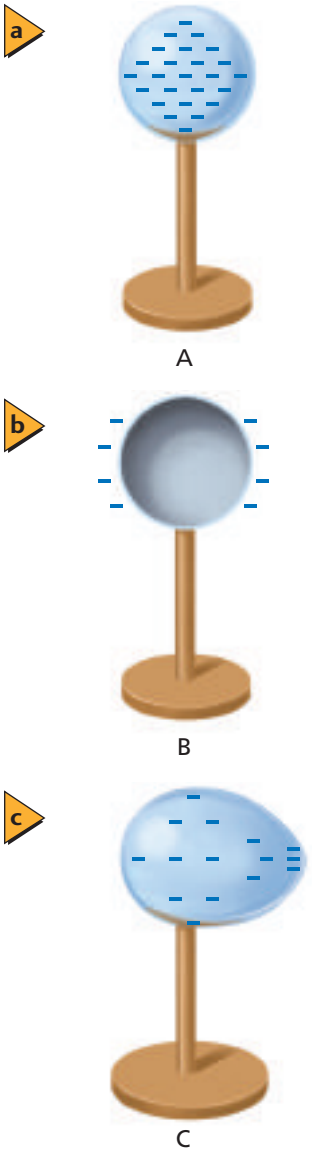


Figure 21-13 On a conducting sphere, **(a),** the charge is evenly distributed around the surface. The charges on the hollow sphere, **(b),** are entirely on the outer surface. In irregular shapes, **(c),** the charges will be closest together at sharp points.

The same principle explains how charges move on the individual spheres, or on any conductor. They distribute themselves so that the net force on each charge is zero. With no force, there is no electric field along the surface of the conductor. Thus, there is no electric potential difference anywhere on the surface. The surface of a conductor is, therefore, an equipotential surface.

If a charged body is grounded by touching Earth, almost any amount of charge can flow to Earth until the electric potential difference between that body and Earth is reduced to zero. Gasoline trucks, for example, can become charged by friction. If the charge on a gasoline truck were to jump to Earth through gasoline vapor, it could cause an explosion. To prevent this, a metal wire on the truck safely conducts the charge to the ground, as shown in **Figure 21-12.** Similarly, if a computer is not grounded, an electric potential difference between the computer and Earth can occur. If a person then touches the computer, charges could flow through the computer to the person and damage the equipment or hurt the person.

Electric Fields Near Conductors

The charges on a conductor are spread as far apart as they can be to make the energy of the system as low as possible. The result is that all charges are on the surface of a solid conductor. If the conductor is hollow, excess charges will move to the outer surface. If a closed metal container is charged, there will be no charges on the inside surfaces of the container. In this way, a closed metal container shields the inside from electric fields. For example, people inside a car are protected from the electric fields generated by lightning. Likewise, on an open coffee can, there will be very few charges inside and none near the bottom. Even if the inner surface of an object is pitted or bumpy, giving it a larger surface area than the outer surface, the charge still will be entirely on the outside.

On the outside of a conductor, however, the electric field often is not zero. Even though the surface of a conductor is at an equipotential, the electric field around the outside of it depends on the shape of the conductor, as well as on the electric potential difference between it and Earth. The charges are closer together at sharp points of a conductor, as indicated in **Figure 21-13.** Therefore, the field lines are closer together and the field is stronger. This field can become so strong that when electrons are knocked off of atoms by passing cosmic rays, the electrons and resulting ions are accelerated by the field, causing them to strike other atoms, resulting in more ionization of atoms. This chain reaction is what results in the pink glow,

such as that seen inside a gas-discharge sphere. If the field is strong enough, when the particles hit other molecules they will produce a stream of ions and electrons that form a plasma, which is a conductor. The result is a spark, or, in extreme cases, lightning. To reduce discharges and sparking, conductors that are highly charged or that operate at high potentials are made smooth in shape to reduce the electric fields.

In contrast, a lightning rod is pointed so that the electric field will be strong near the end of the rod. As the field accelerates electrons and ions, they form the start of a conducting path from the rod to the clouds. As a result of the rod's sharply pointed shape, charges in the clouds spark to the rod, rather than to a chimney or other high point on a house or other building. From the rod, a conductor takes the charges safely to the ground.

Lightning usually requires a potential difference of millions of volts between Earth and the clouds. Even a small gas-discharge tube operates at several thousand volts. Household wiring, on the other hand, does not normally carry a high enough potential difference to cause such discharges.

Storing Charges: The Capacitor

When you lift a book, you increase its gravitational potential energy. This can be interpreted as storing energy in a gravitational field. In a similar way, you can store energy in an electric field. In 1746, Dutch physician and physicist Pieter Van Musschenbroek invented a small device that could store a large electric charge. In honor of the city in which he worked, it was called a Leyden jar. Benjamin Franklin used a Leyden jar to store the charge from lightning and in many other experiments. A version of the Leyden jar is still in use today in electric equipment. This new device for storing a charge has a new form, is much smaller in size, and is called a **capacitor.**

As charge is added to an object, the electric potential difference between that object and Earth increases. For a given shape and size of an object, the ratio of charge stored to electric potential difference, $q/\Delta V$, is a constant called the **capacitance,** C. For a small sphere far from the ground, even a small amount of added charge will increase the electric potential difference. Thus, C is small. A larger sphere can hold more charge for the same increase in electric potential difference, and its capacitance is larger.

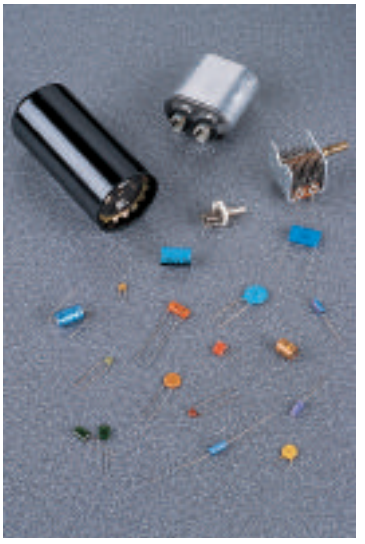
Capacitors are designed to have specific capacitances. All capacitors are made up of two conductors that are separated by an insulator. The two conductors have equal and opposite charges. Capacitors are used today in electric circuits to store charge. Commercial capacitors, such as those shown in **Figure 21-14,** typically contain strips of aluminum foil separated by thin plastic that are tightly rolled up to conserve space.

The capacitance of a capacitor is independent of the charge on it, and can be measured by first placing charge q on one plate and charge $-q$ on the other, and then measuring the electric potential difference, ΔV, that results. The capacitance is found by using the following equation, and is measured in farads, F.

Capacitance $C = \frac{q}{\Delta V}$

Capacitance is the ratio of charge on one plate to potential difference.

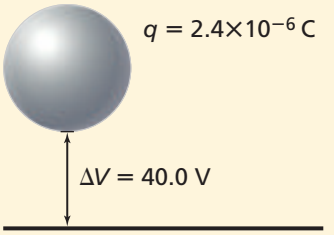
■ **Figure 21-14** Various types of capacitors are pictured below.



The farad as a unit of measure One farad, F, named after Michael Faraday, is one coulomb per volt, C/V. Just as 1 C is a large amount of charge, 1 F is also a fairly large capacitance. Most capacitors used in modern electronics have capacitances between 10 picofarads (10×10^{-12} F) and 500 microfarads (500×10^{-6} F). However, memory capacitors that are used to prevent loss of memory in some computers can have capacitance from 0.5 F to 1.0 F. Note that if the charge is increased, the electric potential difference also increases. The capacitance depends only on the construction of the capacitor, not on the charge, q.

EXAMPLE Problem 5

Finding Capacitance A sphere has an electric potential difference between it and Earth of 40.0 V when it has been charged to 2.4×10^{-6} C. What is its capacitance?



1 Analyze and Sketch the Problem

- Draw a sphere above Earth and label the charge and potential difference.

Known:
$\Delta V = 40.0$ V
$q = 2.4\times10^{-6}$ C

Unknown:
$C = ?$

2 Solve for the Unknown

$$C = \frac{q}{\Delta V}$$

$$= \frac{2.4\times10^{-6}\text{ C}}{40.0\text{ V}}$$ Substitute $\Delta V = 40.0$ V, $q = 2.4\times10^{-6}$ C

$$= 6.0\times10^{-8}\text{ F}$$

$$= 0.060\ \mu\text{F}$$

Math Handbook
Operations with Significant Digits
pages 835–836

3 Evaluate the Answer

- **Are the units correct?** C/V = F. The units are farads.
- **Is the magnitude realistic?** A small capacitance would store a small charge at a low voltage.

PRACTICE Problems

Additional Problems, Appendix B

30. A 27-μF capacitor has an electric potential difference of 45 V across it. What is the charge on the capacitor?

31. Both a 3.3-μF and a 6.8-μF capacitor are connected across a 24-V electric potential difference. Which capacitor has a greater charge? What is it?

32. The same two capacitors as in problem 31 are each charged to 3.5×10^{-4} C. Which has the larger electric potential difference across it? What is it?

33. A 2.2-μF capacitor first is charged so that the electric potential difference is 6.0 V. How much additional charge is needed to increase the electric potential difference to 15.0 V?

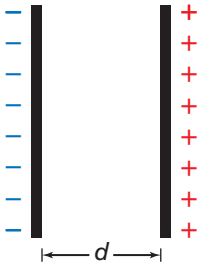
34. When a charge of 2.5×10^{-5} C is added to a capacitor, the potential difference increases from 12.0 V to 14.5 V. What is the capacitance of the capacitor?

CHALLENGE PROBLEM

The plates of a capacitor attract each other because they carry opposite charges. A capacitor consisting of two parallel plates that are separated by a distance, d, has capacitance, C.

1. Derive an expression for the force between the two plates when the capacitor has charge, q.

2. What charge must be stored on a 22-μF capacitor to have a force of 2.0 N between the plates if they are separated by 1.5 mm?



Varieties of capacitors Capacitors have many shapes and sizes, as shown in Figure 21-14. Some are large enough to fill whole rooms and can store enough charge to create artificial lightning or power giant lasers that release thousands of joules of energy in a few billionths of a second. Capacitors in television sets can store enough charge at several hundred volts to be very dangerous if they are touched. These capacitors can remain charged for hours after the televisions have been turned off. This is why you should not open the case of a television or a computer monitor even if it is unplugged.

The capacitance of a capacitor is controlled by varying the surface area of the two conductors, or plates, within a capacitor, by the distance between the plates, and by the nature of the insulating material. Capacitors are named for the type of insulator, or dielectric, used to separate the plates, and include ceramic, mica, polyester, paper, and air. Higher capacitance is obtained by increasing the surface area and decreasing the separation of the plates. Certain dielectrics have the ability to effectively offset some of the charge on the plates and allow more charge to be stored.

2. Section Review

35. Potential Difference What is the difference between electric potential energy and electric potential difference?

36. Electric Field and Potential Difference Show that a volt per meter is the same as a newton per coulomb.

37. Millikan Experiment When the charge on an oil drop suspended in a Millikan apparatus is changed, the drop begins to fall. How should the potential difference on the plates be changed to bring the drop back into balance?

38. Charge and Potential Difference In problem 37, if changing the potential difference has no effect on the falling drop, what does this tell you about the new charge on the drop?

39. Capacitance How much charge is stored on a 0.47-μF capacitor when a potential difference of 12 V is applied to it?

40. Charge Sharing If a large, positively charged, conducting sphere is touched by a small, negatively charged, conducting sphere, what can be said about the following?

a. the potentials of the two spheres

b. the charges on the two spheres

41. Critical Thinking Referring back to Figure 21-3a, explain how charge continues to build up on the metal dome of a Van de Graaff generator. In particular, why isn't charge repelled back onto the belt at point B?

Alternate CBL instructions can be found on the Web site.
physicspp.com

Charging of Capacitors

A capacitor is an electric device that is made from two conductors, or plates, that are separated by an insulator. It is designed to have a specific capacitance. The capacitance depends on the physical characteristics and geometric arrangement of the conductors and the insulator. In the circuit schematic, the capacitor appears to create an open circuit, even when the switch is in the closed position. However, because capacitors store charge, when the switch is closed, charge from the battery will move to the capacitor. The equal, but opposite charges on the two plates within the capacitor establishes a potential difference, or voltage. As charge is added to the capacitor, the electric potential difference increases. In this laboratory activity you will examine the charging of several different capacitors.

QUESTION

How do the charging times of different capacitors vary with capacitance?

Objectives

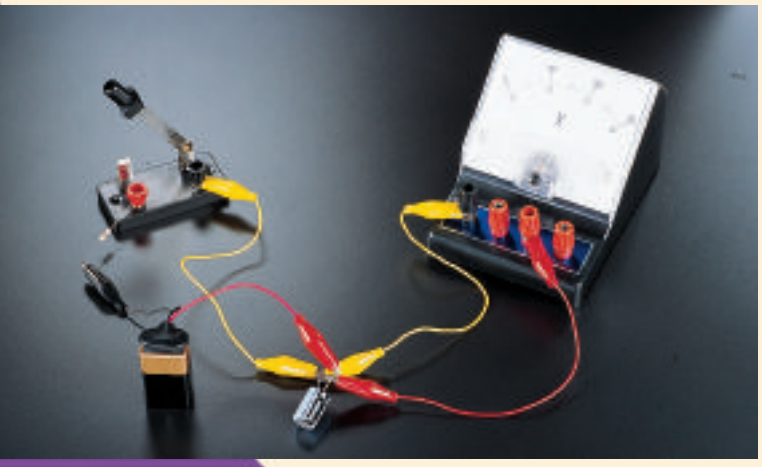
- **Collect and organize** data on the rate of charge of different capacitors.
- **Compare and contrast** the rate of charging for different capacitances.
- **Make and use graphs** of potential difference versus time for several capacitors.

Safety Precautions



Materials

9-V battery
9-V battery clip
hook-up wires
switch
voltmeter
47-kΩ resistor
stopwatch
capacitors: 1000 μF, 500 μF, 240 μF



Procedure

1. Before you begin, leave the switch open (off). Do not attach the battery at this time. ***CAUTION: Be careful to avoid a short circuit, especially by permitting the leads from the battery clip to touch each other.*** Connect the circuit, as illustrated. Do this by connecting either end of the resistor to one side of the switch. The resistor is used to reduce the charging of the capacitor to a measurable rate. Connect the other end of the resistor to the negative side of the 9-V battery clip. Inspect your 1000-μF capacitor to determine whether either end is marked with a negative sign, or an arrow with negative signs on it, that points to the lead that is to be connected to the negative side of the battery. Connect this negative lead to the other side of the switch. Attach the unconnected (positive) lead of the capacitor to the positive lead from the battery clip.
2. Connect the positive terminal of the voltmeter to the positive side of the capacitor and the negative terminal to the negative side of the capacitor. Compare your circuit to the photo to verify your connections. Attach the battery after your teacher has inspected the circuit.
3. Prepare a data table having columns for time and potential difference on each of the three different capacitors.
4. One person should watch the time and another should record potential difference at the designated times. Close the switch and measure the voltage at 5-s intervals. Open the switch after you have collected data.

Data Table

Time (s)	Voltage (V) across 1000 μF	Voltage (V) across 500 μF	Voltage (V) across 240 μF	Time (s)	Voltage (V) across 1000 μF	Voltage (V) across 500 μF	Voltage (V) across 240 μF
0				55			
5				60			
10				65			
15				70			
20				75			
25				80			
30				85			
35				90			
40				95			
45				100			
50				105			

5. When you have completed the trial, take a short piece of wire and place it across both ends of the capacitor. This will cause the capacitor to discharge.
6. Replace the 1000-μF capacitor with a 500-μF capacitor. Repeat steps 4–5 and enter data into the appropriate columns of your data table for the 500-μF capacitor.
7. Replace the 500-μF capacitor with a 240-μF capacitor. Repeat steps 4–5 and enter data into the appropriate column of your data table for this last capacitor.

Analyze

1. **Observe and Infer** Does each capacitor charge to 9 V? Propose an explanation for the observed behavior.
2. **Make and Use Graphs** Prepare a graph that plots the time horizontally and the potential difference vertically. Make a separate labeled line for each capacitor.

Conclude and Apply

1. **Interpret Data** Does the voltage on the capacitor immediately jump to the battery's potential difference (9-V)? Explain the reason for the observed behavior.
2. **Infer** Does the larger capacitor require a longer time to become fully charged? Explain why or why not.

Going Further

1. The time for a capacitor to charge to the voltage of the battery depends upon its capacitance and the opposition to the flow of charge in the circuit. In this lab, the opposition to the flow of charge was controlled by the 47-kΩ resistor that was placed in the circuit. In circuits with a capacitor and resistance, such as in this activity, the time in seconds to charge the capacitor to 63.3 percent of the applied voltage is equal to the product of the capacitor and resistance. This is called the time constant. Therefore, $T = RC$, where T is in seconds, R is in ohms, and C is in microfarads. Calculate the time constant for each of the capacitors with the 47-kΩ resistor.
2. Compare your time constants to the values from your graph.

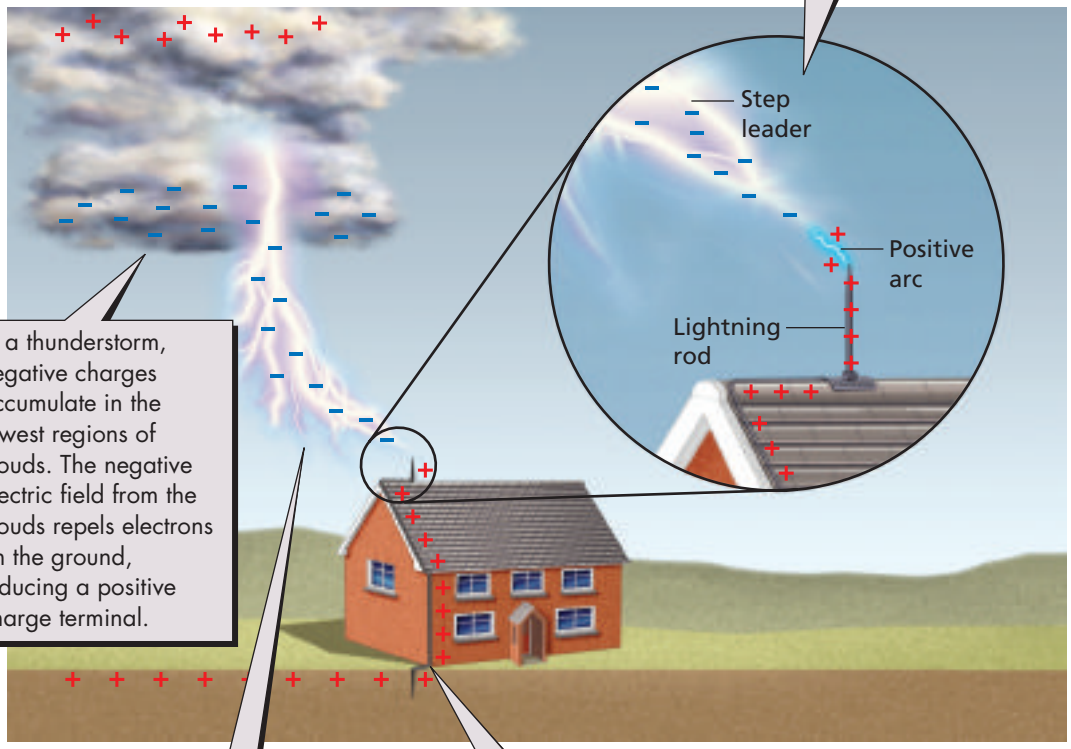
Real-World Physics

Explain Small, disposable, flash cameras, as well as regular electronic flash units, require time before the flash is ready to be used. A capacitor stores the energy for the flash. Explain what might be going on during the time you must wait to take your next picture.

How it Works

Lightning Rods

Lightning can be very destructive because it creates huge currents in materials that are poor conductors and generates a great deal of heat. In addition to protecting a structure by dissipating some of the charge before lightning strikes, lightning rods are excellent conductors that provide a safe path for the current. Benjamin Franklin is credited with inventing the lightning rod in the 1750s.



1 In a thunderstorm, negative charges accumulate in the lowest regions of clouds. The negative electric field from the clouds repels electrons on the ground, inducing a positive charge terminal.

2 The strong electric field accelerates electrons and ions, causing a chain reaction in the air, forming plasma. The ionized air is a conductor, and it branches out from the cloud forming what are called step leaders.

3 Positive charges spark out from the lightning rod, meeting the step leader. The conducting path is complete and current neutralizes the separation of charges. Even if the strike does not hit the lightning rod directly, the massive current still can leap to the rod, which is the path of least resistance to the ground.

4 The current travels safely through the conductor to the ground terminal.

Thinking Critically

1. **Hypothesize** Along what path would the current travel if a house without a lightning rod were struck by lightning?
2. **Evaluate** Should the resistance between the ground terminal and Earth be high or low?
3. **Infer** What are the dangers of an incorrectly installed lightning rod system?

1. Creating and Measuring Electric Fields

Vocabulary

- electric field (p. 563)
- electric field line (p. 567)

Key Concepts

- An electric field exists around any charged object. The field produces forces on other charged objects.
- The electric field is the force per unit charge.

$$\boldsymbol{E} = \frac{\boldsymbol{F}}{q'}$$

- The direction of the electric field is the direction of the force on a tiny, positive test charge.
- Electric field lines provide a picture of the electric field. They are directed away from positive charges and toward negative charges. They never cross, and their density is related to the strength of the field.

21.2 Applications of Electric Fields

Vocabulary

- electric potential difference (p. 569)
- volt (p. 569)
- equipotential (p. 570)
- capacitor (p. 577)
- capacitance (p. 577)

Key Concepts

- Electric potential difference is the change in potential energy per unit charge in an electric field.

$$\Delta V = \frac{W}{q'}$$

- Electric potential differences are measured in volts.
- The electric field between two parallel plates is uniform between the plates, except near the edges. In a uniform field, the potential difference is related to the field strength by the following.

$$\Delta V = Ed$$

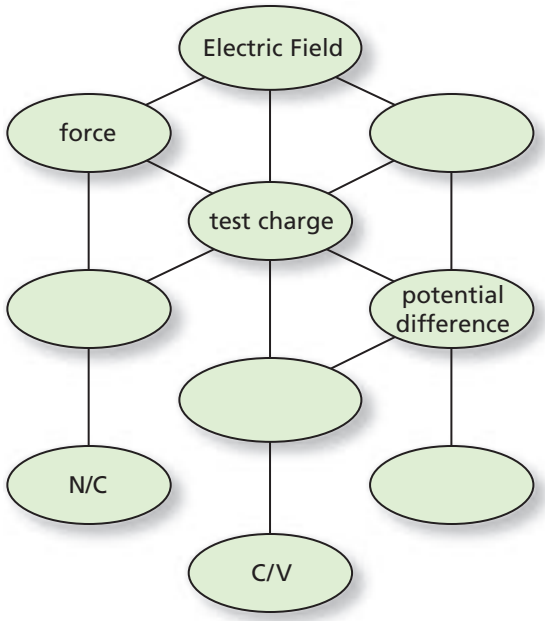
- Robert Millikan's experiments showed that electric charge is quantized.
- Robert Millikan also showed that the negative charge carried by an electron is 1.60×10^{-19} C.
- Charges will move in a conductor until the electric potential is the same everywhere on the conductor.
- Grounding makes the potential difference between an object and Earth equal to zero.
- Grounding can prevent sparks resulting from a neutral object making contact with objects that have built-up charge on them.
- Electric fields are strongest near sharply pointed conductors.
- Capacitance is the ratio of the charge on an object to its electric potential difference.

$$C = \frac{q}{\Delta V}$$

- Capacitance is independent of the charge on an object and the electric potential difference across it.
- Capacitors are used to store charge.

Concept Mapping

42. Complete the concept map below using the following terms: *capacitance, field strength, J/C, work.*



Mastering Concepts

43. What are the two properties that a test charge must have? (21.1)

44. How is the direction of an electric field defined? (21.1)

45. What are electric field lines? (21.1)

46. How is the strength of an electric field indicated with electric field lines? (21.1)

47. Draw some of the electric field lines between each of the following. (21.1)
- **a.** two like charges of equal magnitude
- **b.** two unlike charges of equal magnitude
- **c.** a positive charge and a negative charge having twice the magnitude of the positive charge
- **d.** two oppositely charged parallel plates

48. In **Figure 21-15,** where do the electric field lines leave the positive charge end? (21.1)

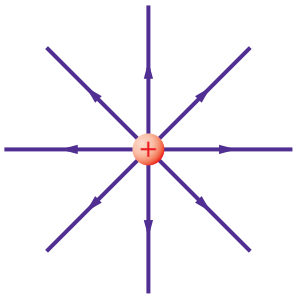


Figure 21-15

49. What SI unit is used to measure electric potential energy? What SI unit is used to measure electric potential difference? (21.2)

50. Define *volt* in terms of the change in potential energy of a charge moving in an electric field. (21.2)

51. Why does a charged object lose its charge when it is touched to the ground? (21.2)

52. A charged rubber rod that is placed on a table maintains its charge for some time. Why is the charged rod not discharged immediately? (21.2)

53. A metal box is charged. Compare the concentration of charge at the corners of the box to the charge concentration on the sides of the box. (21.2)

54. Computers Delicate parts in electronic equipment, such as those pictured in **Figure 21-16,** are contained within a metal box inside a plastic case. Why? (21.2)



Figure 21-16

Applying Concepts

55. What happens to the strength of an electric field when the charge on the test charge is halved?

56. Does it require more energy or less energy to move a constant positive charge through an increasing electric field?

57. What will happen to the electric potential energy of a charged particle in an electric field when the particle is released and free to move?

58. Figure 21-17 shows three spheres with charges of equal magnitude, with their signs as shown. Spheres *y* and *z* are held in place, but sphere *x* is free to move. Initially, sphere *x* is equidistant from spheres *y* and *z*. Choose the path that sphere *x* will begin to follow. Assume that no other forces are acting on the spheres.

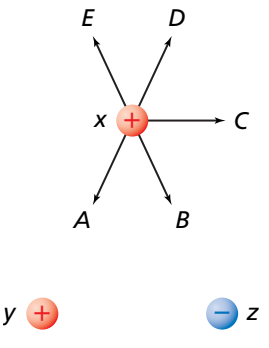


Figure 21-17

59. What is the unit of electric potential difference in terms of m, kg, s, and C?

60. What do the electric field lines look like when the electric field has the same strength at all points in a region?

61. **Millikan Oil-Drop Experiment** When doing a Millikan oil-drop experiment, it is best to work with drops that have small charges. Therefore, when the electric field is turned on, should you try to find drops that are moving rapidly or slowly? Explain.

62. Two oil drops are held motionless in a Millikan oil-drop experiment.
- **a.** Can you be sure that the charges are the same?
- **b.** The ratios of which two properties of the oil drops have to be equal?

63. José and Sue are standing on an insulating platform and holding hands when they are given a charge, as in **Figure 21-18.** José is larger than Sue. Who has the larger amount of charge, or do they both have the same amount?



Figure 21-18

64. Which has a larger capacitance, an aluminum sphere with a 1-cm diameter or one with a 10-cm diameter?

65. How can you store different amounts of charge in a capacitor?

Mastering Problems

21.1 Creating and Measuring Electric Fields

The charge of an electron is -1.60×10^{-19} C.

66. What charge exists on a test charge that experiences a force of 1.4×10^{-8} N at a point where the electric field intensity is 5.0×10^{-4} N/C?

67. A positive charge of 1.0×10^{-5} C, shown in **Figure 21-19,** experiences a force of 0.30 N when it is located at a certain point. What is the electric field intensity at that point?

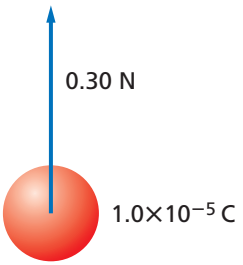


Figure 21-19

68. A test charge experiences a force of 0.30 N on it when it is placed in an electric field intensity of 4.5×10^{5} N/C. What is the magnitude of the charge?

69. The electric field in the atmosphere is about 150 N/C downward.
- **a.** What is the direction of the force on a negatively charged particle?
- **b.** Find the electric force on an electron with charge -1.6×10^{-19} C.
- **c.** Compare the force in part **b** with the force of gravity on the same electron (mass = 9.1×10^{-31} kg).

70. Carefully sketch each of the following.
- **a.** the electric field produced by a $+1.0$-μC charge
- **b.** the electric field resulting from a $+2.0$-μC charge (Make the number of field lines proportional to the change in charge.)

71. A positive test charge of 6.0×10^{-6} C is placed in an electric field of 50.0-N/C intensity, as in **Figure 21-20.** What is the strength of the force exerted on the test charge?

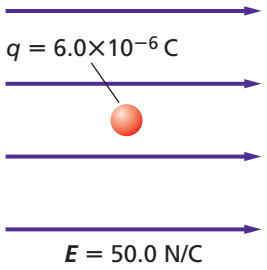


Figure 21-20

72. Charges X, Y, and Z all are equidistant from each other. X has a $+1.0$-μC charge, Y has a $+2.0$-μC charge, and Z has a small negative charge.
- **a.** Draw an arrow representing the force on charge Z.
- **b.** Charge Z now has a small positive charge on it. Draw an arrow representing the force on it.

73. In a television picture tube, electrons are accelerated by an electric field having a value of 1.00×10^{5} N/C.
a. Find the force on an electron.
b. If the field is constant, find the acceleration of the electron (mass = 9.11×10^{-31} kg).

74. What is the electric field strength 20.0 cm from a point charge of 8.0×10^{-7} C?

75. The nucleus of a lead atom has a charge of 82 protons.
a. What are the direction and magnitude of the electric field at 1.0×10^{-10} m from the nucleus?
b. What are the direction and magnitude of the force exerted on an electron located at this distance?

21.2 Applications of Electric Fields

76. If 120 J of work is performed to move 2.4 C of charge from the positive plate to the negative plate shown in **Figure 21-21,** what potential difference exists between the plates?

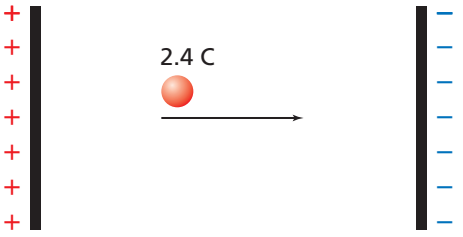


Figure 21-21

77. How much work is done to transfer 0.15 C of charge through an electric potential difference of 9.0 V?

78. An electron is moved through an electric potential difference of 450 V. How much work is done on the electron?

79. A 12-V battery does 1200 J of work transferring charge. How much charge is transferred?

80. The electric field intensity between two charged plates is 1.5×10^{3} N/C. The plates are 0.060 m apart. What is the electric potential difference, in volts, between the plates?

81. A voltmeter indicates that the electric potential difference between two plates is 70.0 V. The plates are 0.020 m apart. What electric field intensity exists between them?

82. A capacitor that is connected to a 45.0-V source contains 90.0 μC of charge. What is the capacitor's capacitance?

83. What electric potential difference exists across a 5.4-μF capacitor that has a charge of 8.1×10^{-4} C?

84. The oil drop shown in **Figure 21-22** is negatively charged and weighs 4.5×10^{-15} N. The drop is suspended in an electric field intensity of 5.6×10^{3} N/C.
a. What is the charge on the drop?
b. How many excess electrons does it carry?

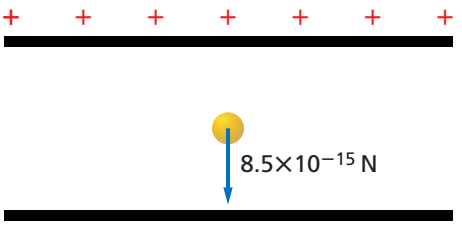


Figure 21-22

85. What is the charge on a 15.0-pF capacitor when it is connected across a 45.0-V source?

86. A force of 0.065 N is required to move a charge of 37 μC a distance of 25 cm in a uniform electric field, as in **Figure 21-23.** What is the size of the electric potential difference between the two points?

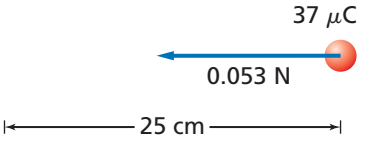


Figure 21-23

87. Photoflash The energy stored in a capacitor with capacitance C, and an electric potential difference, ΔV, is represented by $W = \frac{1}{2}C\Delta V^{2}$. One application of this is in the electronic photoflash of a strobe light, like the one in **Figure 21-24.** In such a unit, a capacitor of 10.0 μF is charged to 3.0×10^{2} V. Find the energy stored.



Figure 21-24

88. Suppose it took 25 s to charge the capacitor in problem 87.
a. Find the average power required to charge the capacitor in this time.
b. When this capacitor is discharged through the strobe lamp, it transfers all its energy in 1.0×10^{-4} s. Find the power delivered to the lamp.
c. How is such a large amount of power possible?

89. **Lasers** Lasers are used to try to produce controlled fusion reactions. These lasers require brief pulses of energy that are stored in large rooms filled with capacitors. One such room has a capacitance of 61×10^{-3} F charged to a potential difference of 10.0 kV.
- **a.** Given that $W = \frac{1}{2}C\Delta V^2$, find the energy stored in the capacitors.
- **b.** The capacitors are discharged in 10 ns (1.0×10^{-8} s). What power is produced?
- **c.** If the capacitors are charged by a generator with a power capacity of 1.0 kW, how many seconds will be required to charge the capacitors?

Mixed Review

90. How much work does it take to move 0.25 μC between two parallel plates that are 0.40 cm apart if the field between the plates is 6400 N/C?

91. How much charge is stored on a 0.22-μF parallel plate capacitor if the plates are 1.2 cm apart and the electric field between them is 2400 N/C?

92. Two identical small spheres, 25 cm apart, carry equal but opposite charges of 0.060 µC, as in **Figure 21-25.** If the potential difference between them is 300 V, what is the capacitance of the system?

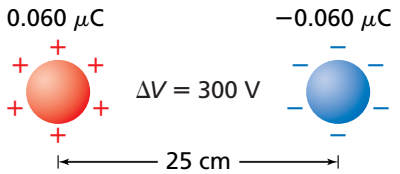


Figure 21-25

93. The plates of a 0.047 μF capacitor are 0.25 cm apart and are charged to a potential difference of 120 V. How much charge is stored on the capacitor?

94. What is the strength of the electric field between the plates of the capacitor in Problem 93 above?

95. An electron is placed between the plates of the capacitor in Problem 93 above, as in **Figure 21-26.** What force is exerted on that electron?

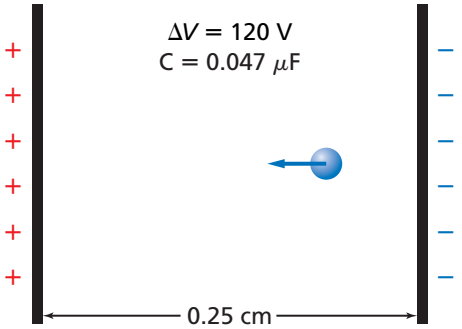


Figure 21-26

96. How much work would it take to move an additional 0.010 μC between the plates at 120 V in Problem 93?

97. The graph in **Figure 21-27** represents the charge stored in a capacitor as the charging potential increases. What does the slope of the line represent?

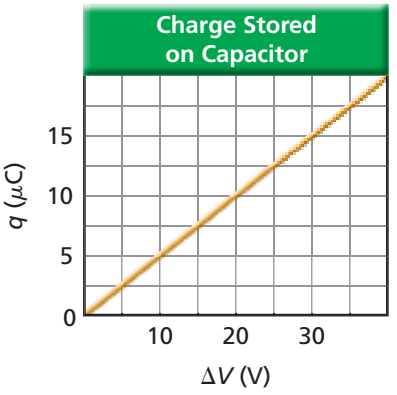


Figure 21-27

98. What is the capacitance of the capacitor represented by Figure 21-27?

99. What does the area under the graph line in Figure 21-27 represent?

100. How much work is required to charge the capacitor in problem 98 to a potential difference of 25 V?

101. The work found in Problem 100 above is not equal to $q\Delta V$. Why not?

102. Graph the electric field strength near a positive point charge as a function of distance from it.

103. Where is the field of a point charge equal to zero?

104. What is the electric field strength at a distance of zero meters from a point charge? Is there such a thing as a true point charge?

Thinking Critically

105. **Apply Concepts** Although a lightning rod is designed to carry charge safely to the ground, its primary purpose is to prevent lightning from striking in the first place. How does it do that?

106. **Analyze and Conclude** In an early set of experiments in 1911, Millikan observed that the following measured charges could appear on a single oil drop. What value of elementary charge can be deduced from these data?
- **a.** 6.563×10^{-19} C
- **b.** 8.204×10^{-19} C
- **c.** 11.50×10^{-19} C
- **d.** 13.13×10^{-19} C
- **e.** 16.48×10^{-19} C
- **f.** 18.08×10^{-19} C
- **g.** 19.71×10^{-19} C
- **h.** 22.89×10^{-19} C
- **i.** 26.13×10^{-19} C

107. Analyze and Conclude Two small spheres, A and B, lie on the *x*-axis, as in **Figure 21-28.** Sphere A has a charge of $+3.00\times10^{-6}$ C. Sphere B is 0.800 m to the right of sphere A and has a charge of -5.00×10^{-6} C. Find the magnitude and direction of the electric field strength at a point above the *x*-axis that would form the apex of an equilateral triangle with spheres A and B.

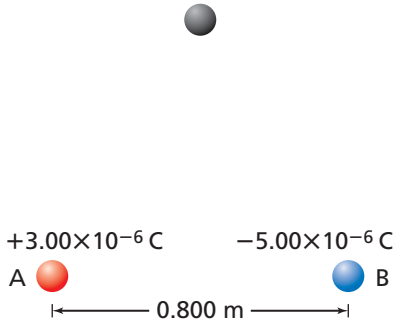


Figure 21-28

108. Analyze and Conclude In an ink-jet printer, drops of ink are given a certain amount of charge before they move between two large, parallel plates. The purpose of the plates is to deflect the charges so that they are stopped by a gutter and do not reach the paper. This is shown in **Figure 21-29.** The plates are 1.5-cm long and have an electric field of $E = 1.2\times10^{6}$ N/C between them. Drops with a mass $m = 0.10$ ng, and a charge $q = 1.0\times10^{-16}$ C, are moving horizontally at a speed, $v = 15$ m/s, parallel to the plates. What is the vertical displacement of the drops when they leave the plates? To answer this question, complete the following steps.

a. What is the vertical force on the drops?
b. What is their vertical acceleration?
c. How long are they between the plates?
d. How far are they displaced?

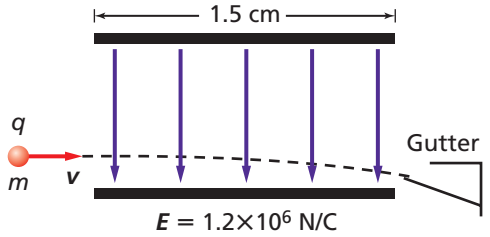


Figure 21-29

109. Apply Concepts Suppose the Moon had a net negative charge equal to $-q$, and Earth had a net positive charge equal to $+10q$. What value of q would yield the same magnitude of force that you now attribute to gravity?

Writing in Physics

110. Choose the name of an electric unit, such as coulomb, volt, or farad, and research the life and work of the scientist for whom it was named. Write a brief essay on this person and include a discussion of the work that justified the honor of having a unit named for him.

Cumulative Review

111. Michelson measured the speed of light by sending a beam of light to a mirror on a mountain 35 km away. (Chapter 16)

a. How long does it take light to travel the distance to the mountain and back?
b. Assume that Michelson used a rotating octagon with a mirror on each face of the octagon. Also assume that the light reflects from one mirror, travels to the other mountain, reflects off of a fixed mirror on that mountain, and returns to the rotating mirrors. If the rotating mirror has advanced so that when the light returns, it reflects off of the next mirror in the rotation, how fast is the mirror rotating?
c. If each mirror has a mass of 1.0×10^{1} g and rotates in a circle with an average radius of 1.0×10^{1} cm, what is the approximate centripetal force needed to hold the mirror while is it rotating?

112. Mountain Scene You can see an image of a distant mountain in a smooth lake just as you can see a mountain biker next to the lake because light from each strikes the surface of the lake at about the same angle of incidence and is reflected to your eyes. If the lake is about 100 m in diameter, the reflection of the top of the mountain is about in the middle of the lake, the mountain is about 50 km away from the lake, and you are about 2 m tall, then approximately how high above the lake does the top of the mountain reach? (Chapter 17)

113. A converging lens has a focal length of 38.0 cm. If it is placed 60.0 cm from an object, at what distance from the lens will the image be? (Chapter 18)

114. A force, *F*, is measured between two charges, *Q* and *q*, separated by a distance, *r*. What would the new force be for each of the following? (Chapter 20)

a. *r* is tripled
b. *Q* is tripled
c. both *r* and *Q* are tripled
d. both *r* and *Q* are doubled
e. all three, *r*, *Q*, and *q*, are tripled

Multiple Choice

1. Why is an electric field measured only by a small test charge?

Ⓐ so the charge doesn't disturb the field
Ⓑ because small charges have small momentum
Ⓒ so its size doesn't nudge the charge to be measured aside
Ⓓ because an electron always is used as the test charge and electrons are small

2. A force of 14 N exists on charge q, which is 2.1×10^{-9} C. What is the magnitude of the electric field?

Ⓐ 0.15×10^{-9} N/C Ⓒ 29×10^{-9} N/C
Ⓑ 6.7×10^{-9} N/C Ⓓ 6.7×10^{9} N/C

3. A positive test charge of 8.7 μC experiences a force of 8.1×10^{-6} N at an angle of 24° N of E. What are the magnitude and direction of the electric field strength at the location of the test charge?

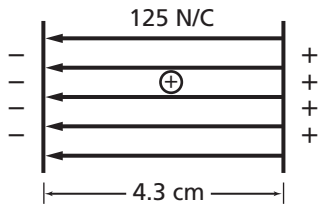
Ⓐ 7.0×10^{-8} N/C, 24° N of E
Ⓑ 1.7×10^{-6} N/C, 24° S of W
Ⓒ 1.1×10^{-3} N/C, 24° W of S
Ⓓ 9.3×10^{-1} N/C, 24° N of E

4. What is the potential difference between two plates that are 18 cm apart with a field of 4.8×10^{3} N/C?

Ⓐ 27 V Ⓒ 0.86 kV
Ⓑ 86 V Ⓓ 27 kV

5. How much work is done on a proton to move it from the negative plate to a positive plate 4.3 cm away if the field is 125 N/C?

Ⓐ 5.5×10^{-23} J Ⓒ 1.1×10^{-16} J
Ⓑ 8.6×10^{-19} J Ⓓ 5.4 J

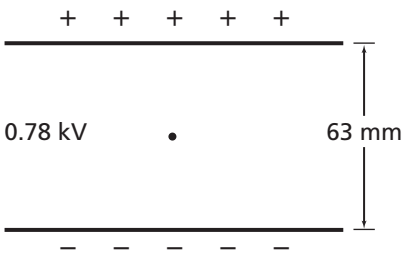


6. How was the magnitude of the field in Millikan's oil-drop experiment determined?

Ⓐ using a measurable electromagnet
Ⓑ from the electric potential between the plates
Ⓒ from the magnitude of the charge
Ⓓ by an electrometer

7. In an oil drop experiment, a drop with a weight of 1.9×10^{-14} N was suspended motionless when the potential difference between the plates that were 63 mm apart was 0.78 kV. What was the charge on the drop?

Ⓐ -1.5×10^{-18} C Ⓒ -1.2×10^{-15} C
Ⓑ -3.9×10^{-16} C Ⓓ -9.3×10^{-13} C



8. A capacitor has a capacitance of 0.093 μF. If the charge on the capacitor is 58 μC, what is the electrical potential difference?

Ⓐ 5.4×10^{-12} V Ⓒ 6.2×10^{2} V
Ⓑ 1.6×10^{-6} V Ⓓ 5.4×10^{3} V

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